

```

{ 'heisenberg-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': False,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'heisenberg_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 '
'X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + '
'1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 6.16720116017694,
6.053414910848409,
6.077121337791871,
6.207207195486473,
6.262374330970613,
5.986895693279655,
6.205368029344304,
6.088664307040117,
6.103832976424956,
6.272477585483895],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],

```

```
'status': False,  
'value': 0},
```

```
'noise_level': [0, 0, 0, 0],  
'noise_profile': { 'noise_on_init_param': {
```

```
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],
```

```
    'status': False,  
    'type': 'dephasing'},
```

```
    'odd_wires': 4},
```

```
'optimized_minimum_cost': [ -6.619825343373176,  
                             -6.869318772840936,  
                             -6.5087775704967585,  
                             -7.1560679492790396,  
                             -5.981413153350314,  
                             -7.048298049430781,  
                             -6.747425000505078,  
                             -7.036568265783533,  
                             -6.711138717392998,  
                             -6.584576110804561],
```

```
'optimized_parameters': [ [ 0.01871413497894055,  
                             0.018714134978940724,  
                             0.2613379173642502,  
                             0.26133791736425005,  
                             0.6318216555020777,  
                             0.8372961654276051,  
                             1.821968513401251,  
                             3.1474951432390372,  
                             3.1474951432390372,  
                             3.1485170671950535,  
                             3.148517067195054,  
                             3.9814416284807077,  
                             3.9814416284807073,  
                             3.981441628480708,  
                             4.856140958800601,  
                             4.8561409588006015,  
                             4.8561409588006015,  
                             5.754795754733104,  
                             5.754795754733105,  
                             6.303270741932773,  
                             7.392681786700986,  
                             7.513290762899488,  
                             8.03209503507907,  
                             8.03209503507907,  
                             8.535813710002731,  
                             8.54624263805951,  
                             9.443687821611801,  
                             9.503087561223712,  
                             10.313956361986106,  
                             10.900532627949254,  
                             2.6461639258839518,  
                             1.5417589283675894,  
                             0.2257956920933908,  
                             -1.0768335736951873,  
                             0.8613207946208413,  
                             2.1849773957616625,  
                             -0.9107734594639508,  
                             -2.1667382648752667,  
                             1.867688722281462,  
                             2.3668974556947022,  
                             0.13339305568895185,  
                             -1.4570319604035509,  
                             1.3777202351900353,  
                             3.4700392631056327,  
                             -0.9559545761912553,  
                             0.4368026830493454,  
                             1.3334925579084722,  
                             -0.4156350852698361,  
                             -1.4414158014372636,  
                             0.7393914946360534,  
                             0.3010801567914183,  
                             0.36212277206485,  
                             -1.1765017516736327,  
                             -0.5545301330477921,
```

0.17858637974936872,
-0.27306390316354434,
1.2204707357225755,
0.10264508982433834,
1.311297488937931,
0.022638960119600232,
1.2224185268837013,
0.09419012847959232,
-0.9858808684001222,
1.7667365936947288,
2.0543865297241832,
-1.313309276179505,
0.6052943491453872,
2.920794362822403,
0.36729426823797706,
-0.4907693755345126,
-0.4907652929618582,
1.773301393432159,
1.0324005805505643,
0.15690855137959606,
0.17275788677592632,
1.6408842546175209,
1.1810448498033803,
0.5094988282552109,
-0.8827869902210956,
1.2283442273595282,
0.6657750092218239,
0.5822857875828481,
0.3981602545186293,
2.2986803263433013,
0.20507142489757735,
1.166074901081292,
-0.35093625013481605,
-0.4603785745816099,
-1.2308941818850956,
-2.0204446136516436,
0.7427864901575015,
0.49035804674601163,
-0.17818011332340386,
2.306444847843727,
0.8423842147773574,
0.3763774472076569,
-0.8127513813727298,
-0.6393080466279192,
-0.606760192596074,
1.239376081712239,
1.4934740200078889,
0.45014851381382026,
0.16024116637379804,
0.5909714786810217,
-0.23164751524086005,
-0.5232030433099774,
0.294735067059697,
0.674054302120824,
0.1291960398503005,
0.724266435798732,
0.6177233678090197,
0.8069752318018093,
-0.09640123299025632,
1.7506436772618832,
2.1071360750039685,
0.0898164078382077,
0.45058491103669474,
-0.8122072218131493,
-1.300894827674866,
2.0270032652414245,
2.3174633740346895,
-0.4244649110516584,
-0.922127126878378,
-0.128757892981865,
-0.05614505811514483,
-0.31034827471285004,
1.1560780386513783,
0.9427392383108351,
3.464248310365335,

1.1861860813777607,
0.6548919301302338,
-0.47688935032578766,
0.31878656040010656,
1.4181187106551563,
0.5034359418789143,
0.06789612685201406,
1.2910094025873902,
0.6395363733901798,
0.8068834648629276,
0.20002934026866853,
1.123820073617421,
0.5141860951115667,
-0.15567109011068703,
1.389800114049256,
1.3314801123479834,
0.5620442654487701,
-1.002959512005978,
1.0830163724627513,
2.3672930390807605,
-1.322930394436911],
[9.119557350302325e-19,
0.25959468526947127,
0.6593100126785064,
0.9010070143348925,
2.864384142587381,
2.864384142587381,
2.8643841425873813,
2.864384142587381,
2.8643841425873813,
2.8643841425873813,
4.6531454467618625,
4.6531454467618625,
5.19853052504346,
5.19853052504346,
6.184870510697861,
6.184870510697861,
7.044770226594686,
7.107215243020832,
7.686493340293053,
7.686493340293053,
7.686493340293053,
7.686493340293053,
8.066147968701737,
8.483745150563264,
8.490894072830061,
8.7108807618689,
8.7108807618689,
8.90963636528622,
9.449700779737539,
10.622376926466515,
1.4198721365577263,
2.0532215860895215,
-0.6671090661407124,
-0.1784736484260618,
2.452830737154847,
0.6445964871090722,
-0.5699863409114161,
-0.11908529766890505,
0.11072609611018341,
0.5517633838461717,
0.258947519950729,
0.2370221833218279,
-0.08827498348755955,
1.5529545851776476,
1.4270685728467551,
-1.297406543763459,
0.21671603620015034,
0.8053327735799074,
1.2240347618351852,
0.19490372040109893,
0.4463632351173059,
0.5723077007901808,
0.7291097022188683,
-0.9340621080403456,

0.7814401198421698,
1.335934708679698,
1.6419357207705991,
-0.32432865447217823,
-0.5119906202275178,
0.4069043309823562,
-0.9892467176495091,
0.2984840383067449,
0.8657043381513185,
0.3531089935489972,
1.3451552159797637,
-0.6265940619359847,
-0.48694952223842436,
0.8839372319696429,
-0.7937473350851352,
-0.6284369925286846,
0.988187027480221,
0.14204093422108516,
-0.7928447811689399,
-0.02575893659255237,
0.6034077764970499,
-1.0594098028179662,
-0.5497734939439616,
0.025642354245823534,
-0.20869881122054,
-1.2359715138124214,
-0.2946232864432856,
0.1124699418659181,
0.8079624623340408,
0.3058649198413362,
0.16875024197938412,
-0.4319850590361839,
0.9532996200659013,
0.9710216361923732,
0.3268179051781202,
-0.9107075661609054,
-0.5341988035286006,
0.13994538780187468,
0.9868907551406109,
0.05744643111324586,
-1.1377247833961386,
0.5139551019130977,
0.5537607458906798,
0.2287706491940902,
-0.44737160486571353,
-0.22916413274711533,
-0.08417189284299845,
-1.3242046442500788,
1.6166580464751512,
0.8079048681248637,
1.3145961695384563,
-0.18697892770553287,
0.7313225410666617,
-1.041621052087622,
0.2643819048238151,
1.1433173955584817,
1.1335781234696722,
-0.8444186654603157,
0.6239242025717586,
1.3684750637982472,
0.3938801567657146,
0.2134741750368225,
-0.5543571548158037,
0.675866319898802,
0.5827634828639766,
0.08356178045592735,
0.7945300964131254,
0.5755633000801618,
0.035712817174237056,
0.8694548740378415,
-0.8796227630992239,
-0.724563085043669,
0.28387680573365254,
0.6162107997557882,
0.18804506353720024,

0.13573104516153978,
0.4121054765005755,
0.24383414393551647,
2.1177520472698306,
0.2513566624255866,
0.1835043860190697,
-1.315482809191855,
-0.0799177390764499,
0.7017032401307175,
-0.20268293353480266,
0.014920329005843978,
1.422493387523363,
-1.3230470412127697,
-0.8499409925206722,
-1.9452530607024414,
0.9453797675102356,
-1.2987244862149754,
-0.31790218075907056,
-1.4580404827053768,
0.7015060173886379,
-1.9161725887237533],
[1.1127990762108912,
1.1143426527141589,
1.77370013920208,
1.9993547683714867,
1.9993547683714867,
2.001672770722739,
2.001672770722739,
2.497477077662224,
2.497477077662224,
3.412019182336653,
4.153064085946221,
4.153064085946221,
4.153064085946221,
4.158592712098033,
4.189306073689003,
4.856266090926401,
4.860230532882879,
5.928763246664035,
6.074154358531821,
6.551844343127418,
6.551844343127418,
7.366351465372161,
8.010206307388854,
8.49993655633506,
8.49993655633506,
9.359509952123227,
9.969968384422238,
10.513762742024985,
10.513762742024985,
10.640701635795219,
1.595436628312185,
-0.3811088004495,
-0.8909401299341706,
-0.3821976726093325,
2.209455823932932,
0.4084507894487605,
-1.3491711508388657,
0.18219394232884736,
-0.1991395495968878,
1.3877045909644812,
1.2810472453057546,
-1.8198938592553289,
0.691696952821756,
1.9651275722497656,
0.8952970332383311,
-1.572888015515471,
0.10018127814164592,
2.5087678810448395,
0.2303656982914484,
0.6644753240981718,
-0.2940767603661997,
0.9937376992458091,
0.19118749930842235,
-1.7309245646223925,

0.6920377423752287,
1.0669653712016844,
-0.2725348409017723,
0.2108196720828406,
0.7600208432998801,
-0.39468875007311516,
-1.6426053139934083,
0.22140594406577463,
0.3798758679922964,
-0.13579541355890495,
-0.502502519286503,
-0.07503549665980291,
-0.24005906561438198,
0.2999706370689256,
0.4291869988824075,
0.49459447229554165,
-0.2693528321546647,
1.3233447406041847,
0.010128993489178412,
-0.7350427372075996,
0.588999296326396,
-0.03887378376443501,
0.7842242805141085,
-1.017856875421701,
-0.050420029711616104,
1.7478460277120142,
0.46693496239216137,
-1.390597925219579,
-0.1527982976709753,
1.3561114516684873,
-0.11782202169904106,
-0.6172298563496877,
0.2337872361631034,
1.5343227540197375,
-1.246330933330494,
0.2671789790306671,
-0.5231899648601245,
-0.2993536711140956,
-0.3794897720952542,
0.5953851600991978,
-0.016488955527838106,
-0.04239234937973134,
-0.8530294973472092,
0.09298474140637386,
-0.6610063165341049,
-0.7501983006092152,
-0.23667128690009429,
-0.49322661069162854,
0.2817510492822266,
0.8804151571222927,
0.25487104244438136,
-0.5104439984528633,
0.8221175331336923,
0.42372453837884605,
1.1270985005003262,
0.10209111179644431,
0.4047521854484605,
-0.9498909052734678,
0.8591561814039869,
-0.4549855289146429,
-0.7085768506420614,
0.2460272461737723,
-0.08852438964027515,
-0.28170619044977985,
0.18782561055277286,
-0.3648963858755429,
-0.37968971213994723,
0.35188735669812166,
1.8917682407126328,
0.37786421566408035,
0.7636381567370941,
0.07799046336192586,
-0.23205705900380685,
-1.186712365635249,
-0.10791936865692263,

1.6876574701547749,
0.03719295338640027,
-0.062311995058818626,
-0.5208937011021405,
0.4557021320307782,
-2.7664261049727545,
-0.8770083117671578,
0.9937005253370358,
1.250538444528313,
0.6461816519581883,
1.6953467436382619,
2.0814745138972848,
-0.7151854978123715,
-1.6567720673097688,
-1.1703575007457885,
0.44660262232395087,
0.012526409819125981,
-0.0348982634801462,
2.0503923727756113,
1.154948415623314,
0.9176857519192531],
[3.101411838661581e-05,
0.00028167581912982354,
0.002108388113032011,
0.0021083881149554745,
0.002110224126734301,
0.013404723029560417,
0.44387140902527444,
0.4998584267335595,
2.519337827669618,
2.592441150742434,
3.279416160731671,
4.179752140912941,
4.179752161353468,
5.661115424822064,
6.166122332331656,
6.1661223340249105,
6.166122334025239,
7.1531688903854835,
7.165666077614885,
7.257231636090788,
7.810595355153735,
8.056713547662405,
8.059562409019513,
9.326100481695063,
9.326100481704493,
9.781728007060766,
9.888676215095867,
10.278005103664306,
10.559163217755993,
10.559486031108996,
1.2851036574167212,
-0.72547345631182,
-1.4718960991273107,
-1.209251703975404,
-0.016838854839331462,
1.1653703713624246,
-0.21930660603454502,
-0.6325267011020643,
0.27235598773055514,
1.3431283556950424,
0.29725144436322054,
-1.2158342217245894,
0.8631495871791491,
0.12397167154542266,
1.116880444349936,
-0.5886908967334331,
0.37424943307839126,
-0.4246401431552608,
-0.052554167142824744,
-1.151521387506872,
0.45352480044418075,
0.2968389539105363,
-0.36994249285514086,
0.029824637432390475,

0.7039777013639383,
1.0545235270667865,
0.5948019323555648,
0.9185753062546861,
-0.3475338326570484,
1.2642221406528609,
-0.6424417079073889,
-1.8836355388168116,
-0.1226587707935595,
-0.751178954303451,
1.2759264830517028,
-0.7303074304229613,
1.0498129361868127,
0.29592762783390236,
0.4716900159260768,
-0.20829913915791434,
-0.8217397159362575,
0.8503209585884646,
-0.49309125036010926,
0.07716668431879356,
-1.7384586268468483,
0.5748268283601387,
-0.01480830786786586,
-0.03742465163157027,
-0.9568191257597272,
0.07510834673857922,
0.021240686884975563,
0.4240852295635142,
0.10265729129130313,
-0.812822916440861,
-0.14268033459407103,
-0.3753169735052697,
-0.7175759512497852,
0.03918867047505918,
-0.7186687987691884,
-1.6942314888856695,
1.268466372526145,
0.3288584725277509,
-1.242111559182382,
1.113325215651688,
0.40176863135675694,
0.15033001800176832,
-0.8493210333167794,
-0.7302387336933833,
0.5331942268313786,
0.8016954161004247,
0.11571359434030207,
0.9513684680035512,
-0.26463380223931005,
0.5608071632066808,
-0.40528112642254077,
-0.6754638378880663,
1.5778663953317416,
0.16647468944212254,
0.24883728001741243,
-0.9973732709393777,
0.3409424132427824,
-1.113466076369018,
0.15519481593579493,
0.641216026706457,
0.8449849609678267,
1.911891894634777,
-0.34738047822667395,
1.5893587533055515,
0.4884297002665817,
-0.25227237110126166,
1.5183173129023069,
0.5103598279259709,
1.4687346401745565,
2.532539737478952,
-1.540460532405588,
-0.8294616308793386,
-0.2412174307882085,
2.6406875676770416,
1.2361221502323787,

0.13408461910385386,
0.270490641328192,
-0.5308435068885323,
-0.8511283733806999,
0.26595037966666163,
-0.8831519048460094,
-0.4997472202494575,
2.276782887198034,
1.5876131255720232,
1.1044306497561618,
0.22518259496424625,
-0.12906375147756488,
2.4499067189894523,
0.5016533011795579,
-2.1824706389019832,
0.9088257454879881,
-1.3873459886882669,
0.43069183082779305,
0.5835382283652092,
1.9517705176455047,
0.639974902940826],
[0.8405962708734059,
0.8409366166472388,
1.2852356970263128,
1.9656905972413197,
2.6709433729539978,
2.670943372994037,
2.678176776414865,
2.738130595852174,
2.9440618301573322,
2.944061837308021,
3.5946493033065674,
3.5946493033066593,
3.5946493377836535,
3.5946493377949182,
3.6475297284551167,
4.423523257877865,
4.423523257879113,
4.744549401355183,
5.030736177428605,
5.030766895303107,
5.030766896066702,
5.030798272122615,
6.102563406307042,
6.3550093593541375,
6.35500935935416,
7.56205183282469,
8.453276420655834,
9.985009441062077,
10.30999329905164,
11.043022236178203,
2.1889940156552368,
1.249116821544238,
-0.6605760096192125,
-2.1135047950110155,
2.129245868952901,
0.511519543367031,
-0.8638463209571715,
-0.20202526205492455,
-0.5988641121015101,
2.145161094485569,
1.4958876569154136,
0.9003565348629307,
0.8986066176810353,
-0.0859890065364003,
-0.08547563138217676,
-0.6842205219399397,
1.347771315836974,
1.9235660701575357,
0.5300781712046526,
0.9684639862422223,
1.8099194640434915,
0.6789385897860352,
0.46838913395608867,
-1.9553664340235406,

0.15424587849578802,
0.7464909631769433,
0.28409739802447187,
0.5646186506463579,
-0.0688718084707713,
1.3403239673426846,
0.7803011173906335,
-1.2634531934792825,
-0.403184603220626,
0.9911656930801824,
-0.27952114699970115,
-1.17647727881294,
0.6160770530384655,
0.7910105296614427,
-0.26329493021464934,
-0.5093546557195857,
-0.078946955577166,
1.6118855098996396,
-0.8560858484853413,
0.43713047898075813,
0.15951625123225024,
1.1017815594363278,
-0.5112214984829941,
0.8616309140244529,
-0.2539134574216648,
0.6900329022420518,
0.7767771669699943,
1.3495069152164292,
-0.10718702519877486,
0.23011810567539243,
-0.4233144560883979,
1.0277501391374537,
-0.13408439221848256,
-1.2122213001358895,
-0.9574798764432794,
-1.272419025076084,
0.6697485644230207,
0.09252521117768966,
0.8443957240867669,
-0.5978394384474206,
0.4330892667970026,
0.07884699311842308,
1.140608340632257,
0.07864060845749296,
1.1489940789117206,
0.02443540527852337,
-1.0034700299434305,
-0.1412123681218965,
0.42135897985221993,
0.00946489374514043,
0.37971823276367245,
0.6021380541926072,
1.71390465753914,
-0.5080953537535351,
-1.7394955866645057,
-0.55775760641077,
-1.6305248910687546,
0.4704683857374631,
2.1610569951035585,
-0.11166064159089376,
0.47170657685906686,
0.07305148434226297,
0.6150465742311105,
-0.5141382282093759,
-0.8714158035322278,
-0.543899342278249,
-0.22729288133654413,
-1.1237376131773689,
0.5077284416312242,
-0.02493907277109512,
-0.061410284019419926,
1.7654158813927403,
0.24648921891280573,
0.026991979318021532,
-0.4476437051863693,

1.0288055112299304,
0.2791879596887287,
0.552260553667315,
0.5446649167338916,
-1.1781972761427792,
0.24145179880489284,
-0.6620357948168841,
-0.48109037039574104,
0.4693605610272632,
1.5881294256793275,
-0.6259573215794967,
0.502524529722019,
0.9011500652440121,
0.31953241444072733,
-0.0763403773509461,
0.8504085952834295,
-0.9720571531313896,
2.085636878242165,
0.7499605974794749,
1.9701875686083887,
1.4162728880693862],
[0.36934729750713446,
0.3693472975071345,
0.4408171698279066,
0.7641387276708653,
0.7641387276708653,
1.2405539686841192,
1.9615715228812416,
2.180937952046039,
2.956248658909984,
2.956248658909984,
3.615620941947,
4.0363267766272015,
4.283148746851272,
4.283148746851272,
4.287727890051584,
4.294687143195195,
4.959378804296946,
4.959378804296946,
5.299283731795155,
6.02722363939295,
6.706058929046358,
6.706058929046358,
8.58973622967867,
8.58973622967867,
8.735021968997517,
9.724832411320188,
9.882458613321258,
10.054210085109215,
10.144098772484403,
11.057581665321372,
0.6138165123929756,
0.45636276738375425,
-1.6225613647488533,
0.8571182758975707,
1.411989147942658,
1.6605353912366554,
0.31855217609147096,
1.1904862760406203,
1.4473424068486804,
0.687246532966651,
-0.3805095501565646,
-1.4931296153328852,
2.326224921001399,
0.9788391575979964,
-0.08794459312808878,
-1.606604838553076,
0.045095161702097514,
1.1926561845584958,
0.9624943300259794,
0.8526934578403719,
0.3595367653801707,
0.6208577142062074,
0.23723466831561657,
0.8137494677209746,

0.017380156001790208,
-0.7727421772857405,
-0.7696652964277857,
0.28558266237504587,
1.2665997402365978,
0.030324559296025642,
-1.0215734949860629,
-0.435015538681086,
0.09892690725060405,
-0.021280714063487676,
0.6605217667305735,
-0.48357943656762575,
0.23903870372504285,
-0.20353372422303415,
0.40182288676040395,
0.36859613731879964,
-1.0350820651690253,
0.0783547441763707,
-0.5235573738556314,
0.20634909129668547,
-0.22551355647851357,
0.1673560398509943,
0.7468649723369892,
1.0994749531926022,
-0.2018797749925425,
-0.577779100669964,
-0.6205566643688822,
-0.25020034171796784,
-0.24121553611864016,
0.687677741000175,
-0.6022485096507909,
-0.7648342894907719,
2.68891232458234,
-0.038914232418390225,
1.516345449400868,
0.7772620643765826,
-0.5809695972554458,
-0.16788488824728048,
-0.11974201627038453,
-0.31585900129180533,
0.9730861233839377,
-0.23174660705736103,
0.10129760163123996,
-0.38814603215785604,
0.20257130727462175,
-0.5242689321123916,
0.5757755417303894,
-0.4147685482642834,
0.5155383127778999,
-0.9069003556538804,
-0.3931316447570312,
-0.22854166467415718,
0.08849924429856736,
0.11802474333811182,
-0.6697697401245005,
-0.4497368814763953,
0.8371559036446393,
-0.27762346391800535,
1.129889136554746,
-0.7313980725752758,
1.7322939714189498,
-0.30841564461339865,
0.08815569274790189,
-0.45821949503742376,
1.3903832187168985,
2.7423681670027595,
-0.709192498697989,
0.38042285795749525,
0.9330002997189369,
1.5220073637023084,
-1.9071482092674334,
-1.4201112861485827,
-0.18587844516118163,
-0.728966506004035,
2.779449113015483,

-0.5840054192970229,
2.1502692645150225,
-0.4443255851345571,
1.5289006083178085,
0.05130509171668746,
-1.6556175117860825,
-0.5876167642044237,
1.5144905946024154,
1.6252191316647524,
-0.7878469829469384,
0.4020569928547576,
0.833940688908704,
1.2564138083018812,
-1.9521172256438166,
-0.3660865019791046,
-0.7569121463795273,
0.08857539049178031,
0.04440194030774997,
1.346246034983794,
1.5664381297937569,
1.6926501995283607],
[0.07932463514989117,
0.07933390228500832,
0.6712683622930262,
1.8057579746144639,
2.2332189550395363,
2.2332189550395363,
2.2333175311875997,
2.2333175311875997,
2.5803200362154985,
3.149742042876931,
3.2745115489857257,
3.946611320928826,
3.946611320928826,
4.527074130340705,
4.527074130340705,
5.9153316417928945,
6.27481936788149,
6.2748193678814905,
6.718992557341518,
6.964319085694909,
6.964360455572235,
7.61675802651207,
7.61675802651207,
7.61675802651207,
8.031230813321006,
9.561856352158596,
9.835291253191413,
9.958392502186584,
10.272929617553823,
11.17699585622391,
2.3714062527042183,
0.008507065719361826,
-1.7490339115302889,
-0.9098574742643126,
3.2594955351617316,
2.2425950423007475,
-0.8261382715647944,
0.4893495058771166,
0.29218298380822827,
0.7749189883526413,
0.0732829317784662,
-0.427680076829418,
0.835806375311316,
1.2252687690883595,
0.5943555800897335,
0.41209106000270895,
1.8808578438044583,
0.8934024527009041,
-0.5658257233318801,
-0.24310864449817618,
1.2970958730834994,
1.2397558998854112,
-0.13432853133680148,
-0.33892864650006427,

1.2500942843462377,
1.2463490630241996,
0.4505745589505527,
-2.888971827979111,
1.391085002142137,
1.3755760556021903,
0.7798831597382527,
-0.6932853439946646,
0.13475625995390494,
-0.48377623448510626,
-0.4751082457272343,
-0.9134219715133307,
0.5602537728773588,
0.1457020102747123,
2.094813129086275,
0.16050162220250241,
0.255533199882997,
0.5798962545827221,
-0.3296339137986921,
0.20407592924333015,
1.002364909727297,
0.8996033431107227,
-0.40087766178798867,
0.21850323438685998,
-0.3148827979921678,
0.5255859647152507,
-0.981835244220926,
0.18194510004086287,
2.491369267725132,
0.32340553765254526,
-1.422371307906928,
1.3350586578994175,
-0.19038703199979035,
-0.18759028526199026,
0.31727468639934,
-0.13719279505644447,
0.8870810962833492,
0.5605043977550492,
-0.45997587805687895,
0.6203449810744726,
0.35343801179058787,
-0.29094576977122605,
1.0749874898125031,
-2.3258043801847514,
-0.023837190424521523,
0.8602839037993392,
-0.3076595227850746,
2.083827131184913,
-0.709910688132804,
1.9396556568539463,
-0.7118178630647494,
1.024487978597129,
-0.035154872964647726,
0.6794001499523024,
0.6455217590344408,
1.2833952770953792,
1.7594723933960992,
2.366727526539801,
-0.28632630179908214,
1.356168747751164,
0.31201643292476877,
0.364096860526536,
0.8226195660291765,
0.2699337047887902,
-0.4803548322886457,
0.7089202255488762,
1.9081826927626873,
-0.3868588206474107,
-0.05728681029210421,
-0.6103514866875638,
-2.670376487838684,
0.03980060533677062,
-1.0774346470258729,
1.4309826927584666,
1.4923854024295125,

0.4230703239257903,
1.1568773489185937,
0.7815638389040719,
-0.4834118045846874,
1.1499199224441905,
0.01202048539239927,
0.3105021058015101,
-0.3749329298855971,
2.2071268409248224,
0.7242633662153387,
0.6404704587343512,
-0.30896311918370867,
0.6474915553174321,
-0.6471036454486253,
-1.0139549843186435,
2.7326032351157084,
0.4640689770354461,
0.7197232245440613,
1.099219858753326,
0.5370670409426629,
2.2362623976980602],
[0.018174692163319142,
0.02381340297275515,
0.02421332579520929,
0.024525182595047955,
0.4460556740257204,
0.4463126705844747,
0.8080995635599163,
1.5479347906990655,
1.547935096587381,
2.726547871299545,
3.3242466002093267,
3.324246600509028,
4.111398089568711,
4.145752980013426,
4.146049797332313,
6.415277102843476,
6.572144069759118,
6.927329335261905,
6.927379209146919,
6.933957897577306,
7.907817926834528,
8.01227391181071,
8.377715493176042,
8.574692491681018,
8.665468291409324,
9.338599502827458,
9.57363802367466,
10.233446160908231,
10.471009909188941,
11.163878412926959,
2.236671766008264,
0.35159542629328217,
0.8146718981296579,
-1.5439346128384686,
0.8972133735648259,
-0.0553493415862735,
-1.090899451761626,
-2.2352667548818665,
1.2209528829332086,
0.5204242802804685,
-0.618288383541334,
-0.3171787171591956,
-0.025836377068857316,
1.3596603169147203,
0.4695190032736128,
0.11760891446657133,
0.44944359330584466,
-1.4358459977495694,
0.7103123867655211,
1.6252846466436086,
0.17089579517121128,
1.0480108474391834,
-0.21482373854468306,
-0.5796520218220037,

-0.39701726293440376,
-0.15484503866231408,
0.5161603909940105,
-0.08958972382743205,
-0.5178549594803129,
0.23176676286013279,
0.030062676987165922,
-1.5076956114966404,
-0.5464208818685263,
0.0502122423742188,
1.6977256264031593,
0.8909956552917536,
0.4009810515503771,
0.3307875796941022,
-0.5725943763367822,
0.23702814500484895,
-0.15434680984627044,
-1.2435315402871425,
-0.5149922580540566,
0.30951994519916565,
0.5096095319131432,
0.30303640166579543,
-0.3239136690540364,
-0.7206962091038183,
0.8555191973102813,
1.0120763935686161,
0.04024592288077597,
-0.2886549630459275,
-0.3277156141890048,
-0.5588571150564244,
1.5118849277062045,
-0.644801476310519,
0.35391159762671204,
0.036581088484196625,
0.07324701372485032,
-0.19410908530178353,
1.5993036761438029,
0.10651551669568386,
0.32822741091828495,
0.39824063300111046,
0.6674060102702002,
2.0756693329629017,
-1.5421800768661706,
-0.49667839050246243,
0.0124564376427262,
0.4601453458726383,
0.38665655977671154,
-0.3280159653083997,
1.37660238729691,
0.5459980296904141,
-0.10248779795861306,
0.3193368932413526,
0.022394510397351762,
-0.9106037248563306,
-1.8448022296193607,
-0.2878663593630255,
0.3486991485685759,
0.6803232619964643,
0.473978523083063,
0.7630725375206788,
0.8596671356435096,
-0.47717853860533954,
-1.1511176506358933,
0.044726686440788394,
2.206461610291793,
-0.251613145100058,
0.3827507786112393,
-1.0837839025823046,
0.5215057436278849,
-0.6433936043135513,
1.0423632753231085,
-1.5853099693823727,
-0.4899100779852182,
0.6771990537468359,
1.636801051947153,

0.09520300124188852,
0.7384792012405127,
-0.17322471627475994,
1.5991844229586945,
1.501804750971724,
0.5057839423702514,
0.16164961971359962,
2.292178045044038,
1.3450921233724473,
0.6682426373209176,
0.5914595140792696,
-0.5621151585806636,
1.4649155202102295,
-0.9414482790819952,
-0.10805380420514368,
0.9307303586657315,
2.4106406607028164,
-0.01284340130889141,
0.41649266055337153,
1.314622528108089,
2.7227723366977616],
[0.007743638893136335,
0.013160125498327223,
0.608035339315491,
1.107635474685494,
1.1648862034449325,
1.852802815648869,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.586998837285682,
2.6225868976549798,
3.0707103569201006,
3.0707103569201006,
4.1228521273997165,
4.138888702236146,
4.138888702236146,
5.347444125894556,
5.347444125894556,
5.412363261255616,
6.659592716290906,
7.287740867223119,
8.540473259892655,
8.81303185082801,
9.303384987416969,
9.767355442925265,
9.939116248646888,
9.97261625652093,
11.225779420142475,
11.480121962431193,
1.6482545183670085,
-0.013015976711505003,
-1.0388467926799678,
-0.8652539511107419,
1.9084118027497516,
-0.9394653223441272,
-1.7468041006561723,
-1.5142947156306645,
0.2590641138232099,
1.3773414247538143,
-0.6292953946653098,
-0.8862298412863635,
0.3042825725334,
1.123835887892568,
0.9314084077434932,
-2.9280573641387413,
2.0853053971786153,
-0.34681890463267223,
-0.04690984704002986,
0.8070574318910052,
-0.5779942015518676,
1.0090669446225737,
-0.1775287190554287,
-1.6842820449103186,

0.763376948317048,
0.3384712786913095,
1.7367769407755378,
-1.3580490459136343,
-0.8837296315105077,
-0.8963079645873018,
-0.18784202847085701,
-1.4055444367955956,
0.671829961387558,
1.8019702343361095,
0.30718376850684487,
0.08135901917238793,
-0.18472697951808345,
0.5668536919232505,
-1.5958337467143746,
0.002509488020009276,
1.0849426355423382,
0.7452955666922372,
0.9398973688185909,
0.16370191516119634,
-0.20185726086923075,
0.5303635124624408,
0.4908517160520784,
-0.4888169221995994,
0.9032694192338024,
-0.8158468897545128,
0.7026292391012148,
0.011827947942572397,
0.6138675828667193,
0.1834450857087751,
-0.9915622527095793,
-0.5829508078463853,
1.064065245998552,
0.49061771989414976,
0.23606904394636888,
-0.20435452332134307,
-1.0928372780788398,
0.057443193557461136,
0.10260811255105017,
-0.8555456605404742,
-1.096930933544511,
0.04006347013812424,
0.7151909132583844,
0.4885582422451459,
0.09828549627753237,
-0.41856423316032376,
-1.5659826249535125,
0.35337608728639364,
0.5013401269099331,
0.275554280225739,
0.123838948347921,
1.1982351112905236,
-0.5571464296579999,
0.227611732218496,
0.4903765072246047,
0.21638817013741035,
0.5329904920964728,
0.023244037871050734,
0.15105119714954549,
0.5432645474025147,
-1.6089466195523248,
0.5580452428696917,
0.2980389035466241,
-0.7189872564709854,
0.2941454207257381,
2.0652774083354823,
-3.197423241640053,
1.2171902372756198,
-0.9312606307980903,
0.7138860186292121,
0.7458573347434515,
-1.0460906183029233,
-0.4042108821911456,
-0.5284604323913634,
1.1561662773370494,

-1.0010987456125164,
-1.286779340316877,
-0.7935901981914314,
-0.6758512775621703,
-0.253355636189949,
1.083705550488119,
-1.5951444189401682,
-0.2763950378070173,
1.9938759834766908,
0.1467137702454683,
0.18970870490254518,
-0.6591405102572232,
2.3883322313766877,
-0.5426139401957691,
1.4573385975452764,
0.7440623559719126,
1.9382959176617307,
0.07233875596199366,
-1.2256407906907247,
0.8075799371504198,
1.6375675375176653],
[-2.670381383900579e-16,
0.07474127433216288,
0.9423639350936485,
0.9423639350936486,
0.9423639350936487,
0.9423639350936486,
1.8776414733739433,
2.6093592361765396,
2.6549976526560917,
2.6549976526560917,
4.309323876617324,
4.812226043223447,
5.3735992861643656,
5.998589378822616,
6.041219527369439,
6.041219527369439,
6.041219527369439,
6.455755583840882,
7.6346582000153935,
8.456131775478068,
8.456131775478068,
8.456131775478068,
9.265231361717003,
9.265231361717003,
9.265231361717003,
9.265231361717003,
10.025925584440094,
10.402247132343707,
10.703459031001074,
10.703459031001074,
3.3569835366016436,
0.9215266459605492,
-2.040479701749094,
-1.3412114880744723,
1.3650402871975202,
2.328108797667404,
-1.3186160481421627,
0.4833318128451331,
2.239983409823259,
1.0329578572589242,
-0.1517540238142264,
-1.4974261238923248,
1.7473459056580782,
1.4519306881870349,
-2.206161402631766,
-0.8907223077237956,
2.3813682348978937,
0.5973875310190014,
-0.8910649708758184,
0.26277461789671996,
1.6008335914847227,
3.025884654690384,
0.8431565773802462,
-1.3891693904439881,

3.086427239689084,
-0.3399924211336387,
0.9616731269325118,
-0.4538264391900641,
1.6045391274841239,
1.2066545032742897,
2.5203408436794805,
-0.7822247170003326,
0.5795340175268288,
1.1137644213878175,
1.338413506379888,
-1.170792708483999,
0.7895425377479481,
1.929612997870515,
2.1013047486982743,
-0.1747799610123698,
0.34124409150219404,
0.1345366524340121,
0.12829425701000113,
-0.07853692563407341,
0.10525428958729864,
2.4094888168745765,
0.6357560012375729,
0.3483402865911379,
-2.7204359945096264,
0.7864668566884239,
2.710084659416198,
-0.42133711298349386,
-0.7606725606109204,
4.024573300631783,
1.0416894631362523,
0.549877653228643,
-0.23095740430710715,
1.2081110571007485,
-0.35409975752454176,
0.9082778411762691,
0.9011042810290772,
2.2659344214863575,
0.030960161362888827,
1.709001611830215,
0.7901365980951575,
-0.25541821293373157,
-0.5438725550461083,
-1.6272180586430949,
2.7727467118019176,
2.937339318913365,
0.16303685557629694,
1.9109318898942156,
-0.41619598488045445,
-0.186609722459416,
0.14598277516592972,
-0.08246623139297415,
2.8463994197955897,
1.599311519948962,
-0.3139472278021089,
0.8671768693326729,
1.2547286400858166,
1.2488092065429963,
-0.4426042076320244,
-0.26147177110889996,
1.5139530957520613,
1.698807756116421,
0.7729868493874114,
1.30698727115783,
0.15584206991884653,
1.7938264286277585,
-0.19656274733701168,
0.033815913473199004,
3.683537194772181,
1.3989993033480204,
0.3819269315951613,
1.7115449870773622,
0.11877637359815167,
-0.3580223266367696,
2.5683966853471936,

```

1.1633017601186986,
0.7010591039102597,
0.6321456093464749,
1.2531601332271327,
1.460291237936634,
-0.0031513754098698096,
-0.7920192885995154,
1.112751285243459,
1.2193221700281838,
1.0214928812662405,
-0.12719592003733096,
1.6086786238146749,
1.3655040824209228,
-1.4212254487699023,
-1.4031310724327033,
2.140187196374229,
-1.788444809286011,
0.36492229278601657,
0.8263664226867995,
2.5133575298607385,
0.292333075844064]],
'run_time_sec': 34196.97281765938}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
                                     'layer': 30,
                                     'ugate': { 'coefficients': { 'bn': [ 0,
                                                                           0,
                                                                           0,
                                                                           0,
                                                                           0,
                                                                           0],
                                                                           'cn': [ 0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5],
                                                                           'r': 0},
                                     'time': {'max': 10.0, 'min': 0.0},
                                     'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.723632677978758,
                             0.8013713721964207,
                             0.8013713721964207,
                             0.8013713721964206,
                             1.7866251780297595,
                             1.786625178029759,
                             2.0864992088537258,
                             2.7277814684323256,
                             2.7277814684323256,
                             3.2780082387901137,
                             3.2780082387901137,
                             4.125286968934425,
                             4.125286968934425,
                             6.2058132923329055,
                             6.495923744510521,
                             6.7707474398596865,
                             6.7707474398596865,
                             7.238549805010759,
                             7.238549805010759,
                             7.424016758372523,
                             7.424016758372523,
                             7.45123520722395,
                             8.286067277272798,
                             9.005302817051474,
                             9.005302817051472,
                             9.005302817051474,
                             9.468112593488378,
                             9.980769303066321,
                             10.029230447532314,
                             10.496216102492953,
                             1.6849607035180754,
                             -0.17702247224764014,
                             -1.2741911423346686,
                             -2.844813020435471,
                             2.6306502527193634,

```

1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,

```

-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,

```



```

        'method': 'richardson',
        'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
        '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
        '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
        'Y 1 + 0.5 Z 0 Z 1 + '
        '0.5 X 1 X 2 + 0.5 Y 1 '
        'Y 2 + 0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + 0.5 Y 2 '
        'Y 3 + 0.5 Z 2 Z 3 + '
        '0.5 X 3 X 4 + 0.5 Y 3 '
        'Y 4 + 0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + 0.5 Y 4 '
        'Y 5 + 0.5 Z 4 Z 5 + '
        '0.5 X 5 X 6 + 0.5 Y 5 '
        'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.126749766362046],
        [12, 6, 0, 0, -4.835388535288831]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 7.123104572296143}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,

```

```

'status': False,
'value': 0},

```

```

'status': False,
'value': 0},

```

3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,
9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,

```
-0.972400173321991,  
2.8186480601422346,  
0.26977522672128323,  
-0.43357532748811417,  
-0.1398888669570251,  
0.5703082715904539,  
1.1288698319231192,  
0.5103090181408015,  
0.9751915834601443,  
1.3322237635112688,  
0.1151655141844926,  
-0.009281838713691305,  
-0.6214752845525063,  
-1.5805807688467428,  
-0.10117261020635139,  
0.07763614048447905,  
1.3788004377565666,  
1.8240844635205977,  
0.5421982630966959,  
0.41635499438864665,  
0.6019503022394472,  
-0.13876425465218306,  
-1.389335520601386,  
-0.11160302566710978,  
1.9532500107835935,  
3.1783952012172545,  
1.2324746355871234,  
-0.3300299009483176,  
2.0563071170406455,  
0.5070328560575449,  
-1.4244738939854995,  
-1.0849327697753446,  
2.012447413284947,  
0.7321644003415494,  
2.1697243109783226,  
2.3939500038986976,  
1.5095457334141147,  
0.4771497809888841,  
0.7264385677911265,  
-1.2750578122129246,  
-0.3527900448343432,  
1.4172346289410647,  
0.8432803814903324,  
3.051114458214373,  
-0.3334344329994917,  
2.17934906384921,  
1.266420876986024,  
2.7644241701095216,  
0.636538971425229,  
-1.847398851714829,  
-0.013702660156780564,  
2.331176430726721,  
1.7725302577647513,  
0.5106846778295356,  
1.0449783439357938,  
1.1157484211342301,  
-0.3758082507859313,  
-0.5755462941925489,  
3.6338269047911145,  
1.1686374374905806,  
-1.25881396302495,  
1.2372510547976072,  
3.5862251252548343,  
1.88315539412517]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],  
                  'status': True,  
                  'type': 'dephasing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,
```

[illegible]

```
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.0008444707970882174,
0.0011376501074209704,
0.5478052879830236,
0.5478052879830236,
0.5478052879830237,
0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,
9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
```

0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,

```

-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.344797923049557],
                             [12, 6, 0, 0, -4.988789667979189]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
```

```

                                0.0,
                                0.0],
                                'status': True,
                                'type': 'dephasing'},
                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}},
                                'run_time_sec': 6.4181132316589355}},
                                { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,

```


1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,

```

-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '

```

```

'2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
'5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.390073643790839],
[12, 6, 0, 0, -5.048002042316807]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 5.683874130249023}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.3358069569174437,
0.33751316628129896,
0.37613413155379394,
0.9037599675903363,
0.941040799157583,
1.6105111871436553,
1.7638163741391246,
1.7638163741391246,
3.06849123809687,
3.600239571878297,
4.175689787149765,
4.175689787149765,
5.544091982519477,
5.544091982519477,

```

6.308463651138072,
6.308463651138072,
6.308463651138072,
6.3184465106806345,
6.521816085515529,
6.521816085515529,
7.288786933299201,
7.54709747254392,
7.597867588518987,
7.962268849194133,
7.991785735138846,
8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,

```

-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,

```

[illegible]

```
                                0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'heisenberg'}},
'init_param': { 'value': [ 2.1921898489539642e-17,
                             1.3966237144281936e-17,
                             5.875629127222605e-17,
                             1.0041795331735186,
                             1.0493846336909358,
                             2.3268506812305976,
                             3.014247762257123,
                             3.5587330689470917,
                             3.598019824659659,
                             3.598019824659659,
                             3.598019824659659,
                             4.594861139703674,
                             5.132217790928667,
                             5.132217790928667,
                             6.816072831791245,
                             6.906986102298665,
                             7.2472760308271145,
                             7.2472760308271145,
                             8.200086671450705,
                             8.200086671450705,
                             8.200086671450705,
                             8.200086671450705,
                             8.942806427307234,
                             9.150430959012654,
                             9.150430959012654,
                             9.436537073101752,
                             9.664513899938346,
                             10.186874850218704,
                             10.186874850218702,
                             10.784457886918862,
                             1.4923280250442645,
                             -0.2249463453233081,
                             0.03240277963395433,
                             0.031811746673685456,
                             1.8117023364465294,
                             -0.20170292630031428,
                             0.21975366136216332,
                             0.17798084128170438,
                             0.15372773772227344,
                             0.9274008024787889,
                             0.4001492174566485,
                             0.7146224592116278,
                             1.5995558838041692,
                             0.5527977709299036,
                             -1.1764350892263153,
                             0.46503179152183083,
                             1.695729635964121,
                             1.1136467790441111,
                             -1.746792733239722,
                             -0.14004161870441295,
                             0.32452227687295954,
                             0.6844673620662513,
                             0.0854148711216103,
                             0.07337284978077696,
                             0.03946382901684809,
                             0.38955412394420164,
                             0.48394719071539255,
                             0.16206823911430795,
                             -0.8516299678716804,
                             0.04915630664847796,
                             0.7718109282613742,
                             -0.2065447357739375,
                             -0.14165427311861,
                             0.11805421787298728,
                             1.0598635939215315,
```

0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,


```

0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.181142575648604],
                             [12, 6, 0, 0, -4.947722861409576]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
```


-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,

```

0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '

```

```

'Y 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Y 1 '
'Y 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.871225062671609],
                             [12, 6, 0, 0, -4.462319254461681]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 5.998746395111084}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.26700956978337703,
                           0.2670095697833769,
                           0.7897559248250247,
                           1.60994438629722,
                           1.6240137789724642,
                           1.9029927716215564,
                           2.754961581608366,
                           2.754961581608366,
                           4.416760838293873,
                           4.472730549723469,
                           4.472730549723469,
                           4.472730549723469,
                           4.9018244764195,
                           4.9018244764195,
                           6.142575905365314,
                           6.99918307198117,
                           6.99918307198117,

```

7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,
9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,

```
0.2941548416592414,  
0.5858620932416747,  
0.7670403629229918,  
0.30554255161337057,  
-0.45658754906270327,  
-0.42843355002838807,  
-0.3023769889698021,  
0.007053602739091254,  
0.6545125121375878,  
-1.3230514751857634,  
0.5525700851182778,  
0.6077856841225572,  
0.3193555486658347,  
1.0972045202793819,  
-0.9608463705403721,  
-0.8800113323184338,  
1.5893473527453552,  
-0.17930075995350292,  
0.13243112332175283,  
2.218523374768228,  
0.3265682752545504,  
0.8010356407066903,  
-0.26882287505987873,  
1.9214544754255154,  
1.3539481795361321,  
-0.5909164412323823,  
0.3965669693797851,  
-2.2493204845536354,  
0.19587180737910695,  
0.11919500443011696,  
1.0895825531254497,  
-0.8262207903952641,  
-0.9954080023703576,  
-1.0169640919417637,  
0.6255474806084755,  
-1.2785460468142644,  
1.8640192664022799,  
0.26835529578172257,  
0.6968836525252171,  
1.4078350308699439,  
0.8234537653883477,  
0.7818882402859949,  
0.946183618997053,  
1.904140472055704,  
-0.5964775479033396,  
1.6130219764165838,  
0.03420262362389698,  
0.2971461141757466,  
0.38494337865180756,  
1.8373477360582482,  
0.5875634587735682,  
0.4088919615500518,  
0.4770455671450362,  
1.9125631892854644,  
0.4190740079559775,  
0.8595568972619075,  
0.4240513002790927,  
2.6162509667045732]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],  
                  'status': True,  
                  'type': 'dephasing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0,  
                                           1.0],  
                                'cn': [ 0.5,  
                                         0.5,  
                                         0.5,  
                                         0.5,
```

[illegible]


```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
```

0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,

```

-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#7'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X '
                                '2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X '
                                '5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 '
                                                         'Y 1 + 0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + 0.5 Y 1 '
                                                         'Y 2 + 0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + 0.5 Y 3 '
                                                         'Y 4 + 0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + 0.5 Y 4 '
                                                         'Y 5 + 0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + 0.5 Y 5 '
                                                         'Y 6 + 0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.143320656704589],
                             [12, 6, 0, 0, -4.62093405872061]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],

```

```
'status': False,  
'value': 0},
```

```
'identity_factors': [1, 1, 0, 0],  
'noise_level': [12, 6, 0, 0],  
'noise_profile': { 'noise_on_init_param': {  
  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],  
    'status': True,  
    'type': 'dephasing'},  
    'odd_wires': 4}],  
    'run_time_sec': 5.660057306289673}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                           'layer': 30,  
                           'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                                           'cn': [ 0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5,  
                                                                0.5],  
                                                           'r': 0},  
                           'time': {'max': 10.0, 'min': 0.0},  
                           'type': 'heisenberg'}},  
    'init_param': { 'value': [ 0.05909938957877547,  
                                0.05909939012498705,  
                                0.4766831071513548,  
                                0.4767050288044724,  
                                0.476729910017027,  
                                0.47672998527996446,  
                                2.99469059821818,  
                                3.0000368876553107,  
                                3.03481022556516,  
                                3.0348102255651677,  
                                3.0960777044795975,  
                                3.0960777044802072,  
                                3.096077704551025,  
                                5.243967325820749,  
                                5.24396732593715,  
                                5.696137660948558,  
                                5.6961376610109316,  
                                6.622089050039713,  
                                6.65393172347244,  
                                7.224334835522734,  
                                7.698532497695266,  
                                7.80231633505566,  
                                8.26676599072812,  
                                9.142506968904602,  
                                9.143106298076189,  
                                9.481224923876267,  
                                9.547418340766493,  
                                10.421309029539785,  
                                11.175654656469316,  
                                11.175654850900237,  
                                3.362844107326414,  
                                0.29831395836496843,  
                                -3.1839955728525164,  
                                -2.274863578730854,  
                                -0.3088418730608092,  
                                0.6979840126794992,  
                                0.4650589650223805,  
                                -0.5403997868775179,  
                                -0.4207789901056385,  
                                -0.6821445865591288,  
                                1.2179172859744154,  
                                -1.1069270923796861,  
                                -0.38365670293222115,  
                                2.3779178278712347,
```

0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,

[illegible]

```

'0.5 X 2 X 3 + 0.5 Y 2 '
'Y 3 + 0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + 0.5 Y 3 '
'Y 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Y 4 '
'Y 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Y 5 '
'Y 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.425213401717635],
                             [12, 6, 0, 0, -5.161355570345689]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [4, 2, 0, 0],
    'noise_profile': { 'noise_on_init_param': {
      'noise_prob': [ 0.001,
                     0.001,
                     0.0,
                     0.0],
      'status': True,
      'type': 'dephasing'},
    'odd_wires': 4},
    { 'gates_num': [12, 3, 1, 0],
      'identity_factors': [1, 1, 0, 0],
      'noise_level': [12, 6, 0, 0],
      'noise_profile': { 'noise_on_init_param': {
        'noise_prob': [ 0.001,
                       0.001,
                       0.0,
                       0.0],
        'status': True,
        'type': 'dephasing'},
        'odd_wires': 4}],
      'run_time_sec': 6.1168904304504395}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
  'layer': 30,
  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
    'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'heisenberg'}},
  'init_param': {'value': 'random'},
  'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
    'noise_prob': [0.001, 0.001, 0.0, 0.0],
    'status': True,
    'type': 'dephasing'},
  'nqubits': 7,
  'observable': { 'coefficients': { 'bn': [ 1.0,
      1.0,
      1.0,
      1.0,
      1.0,
      1.0,
      1.0],
    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
    'r': 1},
    'def': 'ising'},
  'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
    'file_name_prefix': 'heisenberg_noise_free_time_evo'},
  'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]],
  'run': 'vqe',
  'state': 'dmatrix',
  'vqe': { 'iteration': 10,
    'optimization': { 'algorithm': 'SLSQP',
      'constraint': True,
      'status': True}},
  'zne': { 'data_points': None,

```

```

        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
        '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
        '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Y 0 Y 1 + '
        '0.5 Z 0 Z 1 + 0.5 X 1 X 2 + '
        '0.5 Y 1 Y 2 + 0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + 0.5 Y 2 Y 3 + '
        '0.5 Z 2 Z 3 + 0.5 X 3 X 4 + '
        '0.5 Y 3 Y 4 + 0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + 0.5 Y 4 Y 5 + '
        '0.5 Z 4 Z 5 + 0.5 X 5 X 6 + '
        '0.5 Y 5 Y 6 + 0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 6.146913233691134,
        5.993838002068908,
        5.949850054037247,
        6.210098036642341,
        6.152155884142049,
        6.148463718261344,
        6.155325453778133,
        6.006166196398716,
        6.005219627807008,
        6.0413306719300985],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': { 'status':
        'value':
        'noise_prob': [ 0.001,
        0.001,
        0.0,
        0.0],
        'status': True,
        'type': 'dephasing'},
        'odd_wires': 4},
        'optimized_minimum_cost': [ -6.523058281579068,
        -6.1267497663620825,
        -6.42521340171773,
        -6.39007364379089,
        -6.344797923049613,
        -5.871225062671643,
        -6.143320656704609,
        -6.2758943446211894,
        -6.3403937569891395,
        -6.18114257564869],
        'optimized_parameters': [ [ 0.3358069569174437,
        0.33751316628129896,
        0.37613413155379394,
        0.9037599675903363,
        0.941040799157583,
        1.6105111871436553,
        1.7638163741391246,
        1.7638163741391246,
        3.06849123809687,
        3.600239571878297,
        4.175689787149765,
        4.175689787149765,
        5.544091982519477,
        5.544091982519477,
        6.308463651138072,
        6.308463651138072,
        6.308463651138072,
        6.3184465106806345,
        6.521816085515529,
        6.521816085515529,
        7.288786933299201,
        7.54709747254392,
        7.597867588518987,
        7.962268849194133,
        7.991785735138846,

```

False,
0},

8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,

0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428],
[0.723632677978758,
0.8013713721964207,
0.8013713721964207,
0.8013713721964206,
1.7866251780297595,
1.786625178029759,
2.0864992088537258,
2.7277814684323256,
2.7277814684323256,
3.2780082387901137,
3.2780082387901137,
4.125286968934425,
4.125286968934425,
6.2058132923329055,
6.495923744510521,
6.7707474398596865,
6.7707474398596865,
7.238549805010759,
7.238549805010759,
7.424016758372523,
7.424016758372523,
7.45123520722395,
8.286067277272798,
9.005302817051474,
9.005302817051472,

9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,

1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826],
[0.05909938957877547,
0.05909939012498705,
0.4766831071513548,
0.4767050288044724,
0.476729910017027,
0.47672998527996446,
2.99469059821818,
3.0000368876553107,
3.03481022556516,
3.0348102255651677,
3.0960777044795975,
3.0960777044802072,
3.096077704551025,
5.243967325820749,
5.24396732593715,
5.696137660948558,
5.6961376610109316,
6.622089050039713,
6.65393172347244,
7.224334835522734,
7.698532497695266,
7.80231633505566,
8.26676599072812,
9.142506968904602,
9.143106298076189,

9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,

-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522],
[0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,

9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,

-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048],
[0.0008444707970882174,
0.0011376501074209704,
0.5478052879830236,
0.5478052879830236,
0.5478052879830237,
0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,

9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,

0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.014422377633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475],
[0.17583917736716262,
0.17583917736716265,
0.9725326222396186,
0.9725326222396184,
1.154796264329948,
2.3531638611520562,
3.742172656987498,
3.810257862376578,
3.8745068996857803,
3.8745068996857803,
3.8745068996857803,
4.643045833818014,
4.643045833818014,
4.643045833818013,
4.643045833818014,
4.643045833818014,
4.711671630422156,
5.405427388054719,
5.405427388054719,
7.4602327607935,
7.610173264016766,
7.858440336330247,
7.968223945319675,
8.51525712638587,
8.54015380483253,

8.797039458545115,
8.797039458545115,
8.858280642077405,
9.30423425746159,
11.6502765461574,
1.4243949815721635,
1.853782837605556,
0.616412956408157,
-0.43476006789942184,
1.5440059756230975,
1.1700457070837984,
-1.8512161685606496,
0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,

0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613],
[0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,

8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,

-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182],
[1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,

9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,

-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517],
[0.26700956978337703,
0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,

9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,

0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732],
[2.1921898489539642e-17,
1.3966237144281936e-17,
5.875629127222605e-17,
1.0041795331735186,
1.0493846336909358,
2.3268506812305976,
3.014247762257123,
3.5587330689470917,
3.598019824659659,
3.598019824659659,
3.598019824659659,
4.594861139703674,
5.132217790928667,
5.132217790928667,
6.816072831791245,
6.906986102298665,
7.2472760308271145,
7.2472760308271145,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.942806427307234,
9.150430959012654,
9.150430959012654,

9.436537073101752,
9.664513899938346,
10.186874850218704,
10.186874850218702,
10.784457886918862,
1.4923280250442645,
-0.2249463453233081,
0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,

```

-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'run_time_sec': 42850.28223609924}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.3358069569174437,
0.33751316628129896,
0.37613413155379394,
0.9037599675903363,
0.941040799157583,
1.6105111871436553,
1.7638163741391246,
1.7638163741391246,
3.06849123809687,
3.600239571878297,
4.175689787149765,
4.175689787149765,

```

5.544091982519477,
5.544091982519477,
6.308463651138072,
6.308463651138072,
6.308463651138072,
6.3184465106806345,
6.521816085515529,
6.521816085515529,
7.288786933299201,
7.54709747254392,
7.597867588518987,
7.962268849194133,
7.991785735138846,
8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,


```

'cn': [0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
    'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#1'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
            'optimization': { 'algorithm': 'SLSQP',
                              'constraint': True,
                              'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -6.523058281578953],
                              [12, 6, 0, 0, -5.228944859106495]],
            'degree': 0,
            'method': 'richardson',
            'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
                'run_time_sec': 4.3392181396484375e-05,
                'zne_values': { 'degree': 0,
                                'extrapolated_value': -7.170114992815181,
                                'others': { 'beta_coefficients': [1.5, -0.5],
                                              'cost_richardson_zne': 2.5,
                                              'sorted_expectation_vals': [
-6.523058281578953,
-5.228944859106495],
                                'sorted_noise': [6, 18]],
                                'sampled data': [ [4, 2, 0, 0, -6.523058281578953],
                                                  [12, 6, 0, 0, -5.228944859106495]],
                                'sampling': 'default'}}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                              'time': {'max': 10.0, 'min': 0.0},
                              'type': 'heisenberg'}}},
      'init_param': { 'value': [ 2.1921898489539642e-17,
                                1.3966237144281936e-17,
                                5.875629127222605e-17,
                                1.0041795331735186,
                                1.0493846336909358,
                                2.3268506812305976,
                                3.014247762257123,
                                3.5587330689470917,
                                3.598019824659659,
                                3.598019824659659,
                                3.598019824659659,
                                4.594861139703674,
                                5.132217790928667,
                                5.132217790928667,
                                6.816072831791245,
                                6.906986102298665,
                                7.2472760308271145,
                                7.2472760308271145,
                                8.200086671450705,
                                8.200086671450705,
                                8.200086671450705,
                                8.200086671450705,
                                8.942806427307234,
                                9.150430959012654,
                                9.150430959012654,
                                9.436537073101752,
                                9.664513899938346,
                                10.186874850218704,
                                10.186874850218702,
                                10.784457886918862,
                                1.4923280250442645,
                                -0.2249463453233081,

```

0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,


```

1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noiseefree_time_evo_dephasing_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.181142575648604],
                          [12, 6, 0, 0, -4.947722861409576]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.910064697265625e-05,

```

```

'zne_values': { 'degree': 0,
                'extrapolated_value': -6.797852432768117,
                'others': { 'beta_coefficients': [1.5, -0.5],
                           'cost_richardson_zne': 2.5,
                           'sorted_expectation_vals': [
-6.181142575648604,
-4.947722861409576],

                           'sorted_noise': [6, 18]],
                'sampled data': [ [4, 2, 0, 0, -6.181142575648604],
                                   [12, 6, 0, 0, -4.947722861409576]],
                'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                       'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                       'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.05909938957877547,
                             0.05909939012498705,
                             0.4766831071513548,
                             0.4767050288044724,
                             0.476729910017027,
                             0.47672998527996446,
                             2.99469059821818,
                             3.0000368876553107,
                             3.0348102255651677,
                             3.0348102255651677,
                             3.0960777044795975,
                             3.0960777044802072,
                             3.096077704551025,
                             5.243967325820749,
                             5.24396732593715,
                             5.696137660948558,
                             5.6961376610109316,
                             6.622089050039713,
                             6.65393172347244,
                             7.224334835522734,
                             7.698532497695266,
                             7.80231633505566,
                             8.26676599072812,
                             9.142506968904602,
                             9.143106298076189,
                             9.481224923876267,
                             9.547418340766493,
                             10.421309029539785,
                             11.175654656469316,
                             11.175654850900237,
                             3.362844107326414,
                             0.29831395836496843,
                             -3.1839955728525164,
                             -2.274863578730854,
                             -0.3088418730608092,
                             0.6979840126794992,
                             0.4650589650223805,
                             -0.5403997868775179,
                             -0.4207789901056385,
                             -0.6821445865591288,
                             1.2179172859744154,
                             -1.1069270923796861,
                             -0.38365670293222115,
                             2.3779178278712347,
                             0.4759901226592819,
                             -0.927969640150436,
                             -0.6955516255591223,
                             1.2922090119966039,
                             -0.10607902133440596,
                             0.5543353597795015,
                             -0.4302086089464882,
                             1.4798439267149097,

```

-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,

```

0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#3'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.425213401717635],
[12, 6, 0, 0, -5.161355570345689]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.361701965332031e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.057142317403609,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-6.425213401717635,
-5.161355570345689],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.425213401717635],
[12, 6, 0, 0, -5.161355570345689]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],

```

```

        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'heisenberg'},
'init_param': { 'value': [ 1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,
9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
```

2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,
-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,

```

1.2372510547976072,
3.5862251252548343,
1.88315539412517]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.2758943446211575],
[12, 6, 0, 0, -4.972375668379767]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 7.05718994140625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.927653682741853,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-6.2758943446211575,
-4.972375668379767],
'sorted_noise': [6, 18]],
'sampled data': [ [4, 2, 0, 0, -6.2758943446211575],
[12, 6, 0, 0, -4.972375668379767]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.723632677978758,
0.8013713721964207,
0.8013713721964207,
0.8013713721964206,
1.7866251780297595,
1.786625178029759,
2.0864992088537258,
2.7277814684323256,
2.7277814684323256,
3.2780082387901137,
3.2780082387901137,
4.125286968934425,
4.125286968934425,
6.2058132923329055,
6.495923744510521,
6.7707474398596865,
6.7707474398596865,

```

7.238549805010759,
7.238549805010759,
7.424016758372523,
7.424016758372523,
7.45123520722395,
8.286067277272798,
9.005302817051474,
9.005302817051472,
9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,


```
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
```

```

'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#2'},
  'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
  'run': 'zne',
  'state': 'dmatrix',
  'vqe': { 'iteration': 1,
    'optimization': { 'algorithm': 'SLSQP',
      'constraint': True,
      'status': False}},
  'zne': { 'data_points': [ [4, 2, 0, 0, -6.126749766362046],
    [12, 6, 0, 0, -4.835388535288831]],
    'degree': 0,
    'method': 'richardson',
    'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
  'run_time_sec': 3.838539123535156e-05,
  'zne_values': { 'degree': 0,
    'extrapolated_value': -6.772430381898653,
    'others': { 'beta_coefficients': [1.5, -0.5],
      'cost_richardson_zne': 2.5,
      'sorted_expectation_vals': [
-6.126749766362046,
-4.835388535288831],

      'sorted_noise': [6, 18]],
      'sampled data': [ [4, 2, 0, 0, -6.126749766362046],
        [12, 6, 0, 0, -4.835388535288831]],
      'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
  'layer': 30,
  'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
    'cn': [ 0.5,
      0.5,
      0.5,
      0.5,
      0.5,
      0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'heisenberg'}},
  'init_param': { 'value': [ 0.26700956978337703,
    0.2670095697833769,
    0.7897559248250247,
    1.60994438629722,
    1.6240137789724642,
    1.9029927716215564,
    2.754961581608366,
    2.754961581608366,
    4.416760838293873,
    4.472730549723469,
    4.472730549723469,
    4.472730549723469,
    4.9018244764195,
    4.9018244764195,
    6.142575905365314,
    6.99918307198117,
    6.99918307198117,
    7.3452528051326675,
    7.478132650082225,
    8.520612438756238,
    8.520612438756238,
    8.991228072316343,
    9.227791963806121,
    9.7884313312372,
    9.7884313312372,
    9.7884313312372,
    10.06004176448028,
    11.030427598537905,
    11.030427598537905,
    11.500583935823688,
    1.3913972994109516,
    0.07774370637080774,
    -0.49382826121080187,
    1.73722078706814,
    1.1477124735700583,
    -0.01244613489207998,
    0.1104474230926492,

```

-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,

```

0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.3403937569890845],
[12, 6, 0, 0, -4.976052575965789]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.5762786865234375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.022564347500733,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [

```

-6.3403937569890845,
-4.976052575965789],

```
        'sorted_noise': [6, 18]],  
        'sampled data': [ [4, 2, 0, 0, -6.3403937569890845],  
                          [12, 6, 0, 0, -4.976052575965789]],  
        'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],  
                                                         'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                         'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}},  
  'init_param': { 'value': [ 0.17583917736716262,  
                             0.17583917736716265,  
                             0.9725326222396186,  
                             0.9725326222396184,  
                             1.154796264329948,  
                             2.3531638611520562,  
                             3.742172656987498,  
                             3.810257862376578,  
                             3.8745068996857803,  
                             3.8745068996857803,  
                             3.8745068996857803,  
                             3.8745068996857803,  
                             4.643045833818014,  
                             4.643045833818014,  
                             4.643045833818013,  
                             4.643045833818014,  
                             4.643045833818014,  
                             4.711671630422156,  
                             5.405427388054719,  
                             5.405427388054719,  
                             7.4602327607935,  
                             7.610173264016766,  
                             7.858440336330247,  
                             7.968223945319675,  
                             8.51525712638587,  
                             8.54015380483253,  
                             8.797039458545115,  
                             8.797039458545115,  
                             8.858280642077405,  
                             9.30423425746159,  
                             11.6502765461574,  
                             1.4243949815721635,  
                             1.853782837605556,  
                             0.616412956408157,  
                             -0.43476006789942184,  
                             1.5440059756230975,  
                             1.1700457070837984,  
                             -1.8512161685606496,  
                             0.5034873196770298,  
                             -1.3667011208822544,  
                             1.5586288186942996,  
                             0.6807721504395611,  
                             -0.5694108741752331,  
                             0.15882528136634905,  
                             2.2872584636590165,  
                             -1.3655797366869253,  
                             0.5428984525730501,  
                             -0.5905673445154871,  
                             0.2347743623798973,  
                             -0.44300125429382003,  
                             -0.3419704070408729,  
                             -0.557684730924058,  
                             -0.39393239001581876,  
                             0.2907206177517228,  
                             -0.20455476793263844,  
                             0.9506883178672908,  
                             -0.5484133147843897,  
                             0.13971335509755736,
```

-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,

```

1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.871225062671609],
[12, 6, 0, 0, -4.462319254461681]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 5.412101745605469e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.575677966776573,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [
-5.871225062671609,
-4.462319254461681],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -5.871225062671609],
[12, 6, 0, 0, -4.462319254461681]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6486934998593055,
0.6706921527624933,

```

0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,


```
0.43356714924634276,  
-0.8591483084450674,  
-0.18461893041919494,  
1.116992572824263,  
1.1669294382352948,  
0.029792610192383724,  
0.8391977011137817,  
-0.31345005435646217,  
0.18250807811893774,  
0.1692196481359008,  
-0.8984153824952316,  
-0.8025661732867799,  
-0.4459950806048063,  
-0.6388250625028883,  
-0.16920658797384605,  
-0.6596078743198966,  
0.13675425086946338,  
0.37060376839848896,  
-0.02320319905123886,  
-0.758554011380563,  
0.2855477607473741,  
0.40986471148811776,  
-0.47064746228161686,  
-0.5695276821086359,  
-0.23798089255567295,  
0.021208599954383085,  
-0.957792978850957,  
-0.3892960105133825,  
-0.07165870922911506,  
0.6373978530836436,  
0.11583373543812793,  
-0.6869467612152537,  
-0.109279803260324,  
-0.7618543036104211,  
-0.28370326239549154,  
-0.2212046842298352,  
-0.04751119916340401,  
0.20449386754071522,  
0.705432158326348,  
-0.1477392252159656,  
-0.7161855054209953,  
0.14441137676467267,  
1.3906449907875076,  
1.3299068108993992,  
0.43548394087882114,  
0.10987629965719209,  
0.26908108246362716,  
1.0150638572902744,  
0.04483602986273151,  
-0.153649795167794,  
0.044240654262805976,  
0.3417710011612185,  
0.06331633335604989,  
2.075122314076187,  
1.004062083901195,  
0.7792740260615444,  
1.4928465702619802,  
0.7274077578302571,  
0.5108906894057771,  
0.6929155972389786,  
-0.38025768754944894,  
-0.16052687515696545,  
-0.2996322099808623,  
0.07791949448737949,  
-1.421471362119057,  
-1.238198529035781,  
-0.5725940454397166,  
0.79842254320011,  
1.1392600222135925,  
-0.6816896404490814,  
-0.3035432394760949,  
0.26895925341115173,  
1.7196471710373182]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
```

```

        'status': True,
        'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
                                                1.0,
                                                1.0,
                                                1.0,
                                                1.0,
                                                1.0,
                                                1.0],
                                      'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                      'r': 1},
                  'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#7'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -6.143320656704589],
                              [12, 6, 0, 0, -4.62093405872061]],
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
                'run_time_sec': 3.719329833984375e-05,
                'zne_values': { 'degree': 0,
                                'extrapolated_value': -6.9045139556965776,
                                'others': { 'beta_coefficients': [1.5, -0.5],
                                              'cost_richardson_zne': 2.5,
                                              'sorted_expectation_vals': [
-6.143320656704589,
-4.62093405872061],
                                              'sorted_noise': [6, 18]],
                                'sampled data': [ [4, 2, 0, 0, -6.143320656704589],
                                                  [12, 6, 0, 0, -4.62093405872061]],
                                'sampling': 'default'}}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                              'time': {'max': 10.0, 'min': 0.0},
                              'type': 'heisenberg'}}},
      'init_param': { 'value': [ 0.013671733667398815,
                                0.013671733667398838,
                                0.40762796768685644,
                                0.7598808855555792,
                                0.7614715630840724,
                                0.7614715630840723,
                                0.7792763091267767,
                                1.6629027237260914,
                                1.9378940282792974,
                                3.350993251000224,
                                4.102190293916363,
                                4.285616461674835,
                                4.285616461674835,
                                5.265551146051114,
                                5.272209322381249,
                                5.559550971186632,
                                5.621065732928831,
                                6.49189669389436,
                                6.49189669389436,
                                6.901458013958807,
                                6.901458013958807,
                                7.453720513962848,

```

7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,

```

0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

-6.390073643790839,
-5.048002042316807],

```
'optimization': { 'algorithm': 'SLSQP',  
                  'constraint': True,  
                  'status': False}},  
'zne': { 'data_points': [ [4, 2, 0, 0, -6.390073643790839],  
                           [12, 6, 0, 0, -5.048002042316807]],  
        'degree': 0,  
        'method': 'richardson',  
        'sampling': 'default'}},  
'output': { 'exact_sol': -8.566772233505624,  
            'run_time_sec': 4.506111145019531e-05,  
            'zne_values': { 'degree': 0,  
                           'extrapolated_value': -7.061109444527856,  
                           'others': { 'beta_coefficients': [1.5, -0.5],  
                                         'cost_richardson_zne': 2.5,  
                                         'sorted_expectation_vals': [  
  
                                             'sorted_noise': [6, 18]],  
                                             'sampled data': [ [4, 2, 0, 0, -6.390073643790839],  
                                                                [12, 6, 0, 0, -5.048002042316807]],  
                                             'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],  
                                                         'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                         'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}},  
  'init_param': { 'value': [ 0.0008444707970882174,  
                              0.0011376501074209704,  
                              0.5478052879830236,  
                              0.5478052879830236,  
                              0.5478052879830237,  
                              0.784723345688809,  
                              0.784723345688809,  
                              0.784723345688809,  
                              1.9248523134122373,  
                              3.1827054999432645,  
                              3.5310059925523927,  
                              3.5310059925523927,  
                              5.052695732413606,  
                              5.052695732413606,  
                              6.136548648234324,  
                              6.136548648234324,  
                              6.785869019003799,  
                              6.785869019003799,  
                              7.198041513907379,  
                              7.3330889202217895,  
                              8.496582874867597,  
                              8.496582874867597,  
                              8.496739633755235,  
                              8.496739633755235,  
                              8.496741833065137,  
                              9.22890579273074,  
                              9.930471725964209,  
                              10.003370383780338,  
                              10.15144599090014,  
                              10.665059744564415,  
                              2.9193389895672706,  
                              2.873179484562977,  
                              0.24817648473142762,  
                              -1.1978615116990414,  
                              0.5013428170881429,  
                              -0.14698874126247163,  
                              -0.1346344592985022,  
                              -0.18688791288547063,  
                              0.30396646405908945,  
                              -0.3893332461107723,  
                              -2.4132158231627026,  
                              -1.3025870666843387,
```

2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,

```

0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.014422377633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.344797923049557],
                           [12, 6, 0, 0, -4.988789667979189]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.173683166503906e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.022802050584741,
                            'others': { 'beta_coefficients': [1.5, -0.5],
                                         'cost_richardson_zne': 2.5,
                                         'sorted_expectation_vals': [
-6.344797923049557,
-4.988789667979189],
'sorted_noise': [6, 18]],
'sampled data': [ [4, 2, 0, 0, -6.344797923049557],
                  [12, 6, 0, 0, -4.988789667979189]],

```

```

'sampling': 'default'}}}}],
'heisenberg-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': False,
'type': 'dephasing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'heisenberg_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 '
'+ 1 Z 1 + 1 X 2 X 3 + 1 Z 2 + '
'1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 '
'+ 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
'1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '

```



```
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 6.16720116017694,
                                      6.053414910848409,
                                      6.077121337791871,
                                      6.207207195486473,
                                      6.262374330970613,
                                      5.986895693279655,
                                      6.205368029344304,
                                      6.088664307040117,
                                      6.103832976424956,
                                      6.272477585483895],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [0, 0, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status':
False,
'type':
'dephasing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.619825343373176,
                             -6.869318772840936,
                             -6.5087775704967585,
                             -7.1560679492790396,
                             -5.981413153350314,
                             -7.048298049430781,
                             -6.747425000505078,
                             -7.036568265783533,
                             -6.711138717392998,
```

```
-6.584576110804561],
'optimized_parameters': [ [ 0.01871413497894055,
0.018714134978940724,
0.2613379173642502,
0.26133791736425005,
0.6318216555020777,
0.8372961654276051,
1.821968513401251,
3.1474951432390372,
3.1474951432390372,
3.1485170671950535,
3.148517067195054,
3.9814416284807077,
3.9814416284807073,
3.981441628480708,
4.856140958800601,
4.8561409588006015,
4.8561409588006015,
5.754795754733104,
5.754795754733105,
6.303270741932773,
7.392681786700986,
7.513290762899488,
8.03209503507907,
8.03209503507907,
8.535813710002731,
8.54624263805951,
9.443687821611801,
9.503087561223712,
10.313956361986106,
10.900532627949254,
2.6461639258839518,
1.5417589283675894,
0.2257956920933908,
-1.0768335736951873,
0.8613207946208413,
2.1849773957616625,
-0.9107734594639508,
-2.1667382648752667,
1.867688722281462,
2.3668974556947022,
0.13339305568895185,
-1.4570319604035509,
1.3777202351900353,
3.4700392631056327,
-0.9559545761912553,
0.4368026830493454,
1.3334925579084722,
-0.4156350852698361,
-1.4414158014372636,
0.7393914946360534,
0.3010801567914183,
0.362122777206485,
-1.1765017516736327,
-0.5545301330477921,
0.17858637974936872,
-0.27306390316354434,
1.2204707357225755,
0.10264508982433834,
1.311297488937931,
0.022638960119600232,
1.2224185268837013,
0.09419012847959232,
-0.9858808684001222,
1.7667365936947288,
2.0543865297241832,
-1.313309276179505,
0.6052943491453872,
2.920794362822403,
0.36729426823797706,
-0.4907693755345126,
-0.4907652929618582,
1.773301393432159,
1.0324005805505643,
0.15690855137959606,
```

0.17275788677592632,
1.6408842546175209,
1.1810448498033803,
0.5094988282552109,
-0.8827869902210956,
1.2283442273595282,
0.6657750092218239,
0.5822857875828481,
0.3981602545186293,
2.2986803263433013,
0.20507142489757735,
1.166074901081292,
-0.35093625013481605,
-0.4603785745816099,
-1.2308941818850956,
-2.0204446136516436,
0.7427864901575015,
0.49035804674601163,
-0.17818011332340386,
2.306444847843727,
0.8423842147773574,
0.3763774472076569,
-0.8127513813727298,
-0.6393080466279192,
-0.606760192596074,
1.239376081712239,
1.4934740200078889,
0.45014851381382026,
0.16024116637379804,
0.5909714786810217,
-0.23164751524086005,
-0.5232030433099774,
0.294735067059697,
0.674054302120824,
0.1291960398503005,
0.724266435798732,
0.6177233678090197,
0.8069752318018093,
-0.09640123299025632,
1.7506436772618832,
2.1071360750039685,
0.0898164078382077,
0.45058491103669474,
-0.8122072218131493,
-1.300894827674866,
2.0270032652414245,
2.3174633740346895,
-0.4244649110516584,
-0.922127126878378,
-0.128757892981865,
-0.05614505811514483,
-0.31034827471285004,
1.1560780386513783,
0.9427392383108351,
3.464248310365335,
1.1861860813777607,
0.6548919301302338,
-0.47688935032578766,
0.31878656040010656,
1.4181187106551563,
0.5034359418789143,
0.06789612685201406,
1.2910094025873902,
0.6395363733901798,
0.8068834648629276,
0.20002934026866853,
1.123820073617421,
0.5141860951115667,
-0.15567109011068703,
1.389800114049256,
1.3314801123479834,
0.5620442654487701,
-1.002959512005978,
1.0830163724627513,
2.3672930390807605,

-1.322930394436911],
[9.119557350302325e-19,
0.25959468526947127,
0.6593100126785064,
0.9010070143348925,
2.864384142587381,
2.864384142587381,
2.864384142587381,
2.864384142587381,
2.864384142587381,
2.864384142587381,
4.6531454467618625,
4.6531454467618625,
5.19853052504346,
5.19853052504346,
6.184870510697861,
6.184870510697861,
7.044770226594686,
7.107215243020832,
7.686493340293053,
7.686493340293053,
7.686493340293053,
7.686493340293053,
8.066147968701737,
8.483745150563264,
8.490894072830061,
8.7108807618689,
8.7108807618689,
8.90963636528622,
9.449700779737539,
10.622376926466515,
1.4198721365577263,
2.0532215860895215,
-0.6671090661407124,
-0.1784736484260618,
2.452830737154847,
0.6445964871090722,
-0.5699863409114161,
-0.11908529766890505,
0.11072609611018341,
0.5517633838461717,
0.258947519950729,
0.2370221833218279,
-0.08827498348755955,
1.5529545851776476,
1.4270685728467551,
-1.297406543763459,
0.21671603620015034,
0.8053327735799074,
1.2240347618351852,
0.19490372040109893,
0.4463632351173059,
0.5723077007901808,
0.7291097022188683,
-0.9340621080403456,
0.7814401198421698,
1.335934708679698,
1.6419357207705991,
-0.32432865447217823,
-0.5119906202275178,
0.4069043309823562,
-0.9892467176495091,
0.2984840383067449,
0.8657043381513185,
0.3531089935489972,
1.3451552159797637,
-0.6265940619359847,
-0.48694952223842436,
0.8839372319696429,
-0.7937473350851352,
-0.6284369925286846,
0.988187027480221,
0.14204093422108516,
-0.7928447811689399,
-0.02575893659255237,

0.6034077764970499,
-1.0594098028179662,
-0.5497734939439616,
0.025642354245823534,
-0.20869881122054,
-1.2359715138124214,
-0.2946232864432856,
0.1124699418659181,
0.8079624623340408,
0.3058649198413362,
0.16875024197938412,
-0.4319850590361839,
0.9532996200659013,
0.9710216361923732,
0.3268179051781202,
-0.9107075661609054,
-0.5341988035286006,
0.13994538780187468,
0.9868907551406109,
0.05744643111324586,
-1.1377247833961386,
0.5139551019130977,
0.5537607458906798,
0.2287706491940902,
-0.44737160486571353,
-0.22916413274711533,
-0.08417189284299845,
-1.3242046442500788,
1.6166580464751512,
0.8079048681248637,
1.3145961695384563,
-0.18697892770553287,
0.7313225410666617,
-1.041621052087622,
0.2643819048238151,
1.1433173955584817,
1.1335781234696722,
-0.8444186654603157,
0.6239242025717586,
1.3684750637982472,
0.3938801567657146,
0.2134741750368225,
-0.5543571548158037,
0.675866319898802,
0.5827634828639766,
0.08356178045592735,
0.7945300964131254,
0.5755633000801618,
0.035712817174237056,
0.8694548740378415,
-0.8796227630992239,
-0.724563085043669,
0.28387680573365254,
0.6162107997557882,
0.18804506353720024,
0.13573104516153978,
0.4121054765005755,
0.24383414393551647,
2.1177520472698306,
0.2513566624255866,
0.1835043860190697,
-1.315482809191855,
-0.0799177390764499,
0.7017032401307175,
-0.20268293353480266,
0.014920329005843978,
1.422493387523363,
-1.3230470412127697,
-0.8499409925206722,
-1.9452530607024414,
0.9453797675102356,
-1.2987244862149754,
-0.31790218075907056,
-1.4580404827053768,
0.7015060173886379,

-1.9161725887237533],
[1.1127990762108912,
1.1143426527141589,
1.77370013920208,
1.9993547683714867,
1.9993547683714867,
2.001672770722739,
2.001672770722739,
2.497477077662224,
2.497477077662224,
3.412019182336653,
4.153064085946221,
4.153064085946221,
4.153064085946221,
4.158592712098033,
4.189306073689003,
4.856266090926401,
4.860230532882879,
5.928763246664035,
6.074154358531821,
6.551844343127418,
6.551844343127418,
7.366351465372161,
8.010206307388854,
8.49993655633506,
8.49993655633506,
9.359509952123227,
9.969968384422238,
10.513762742024985,
10.513762742024985,
10.640701635795219,
1.595436628312185,
-0.3811088004495,
-0.8909401299341706,
-0.3821976726093325,
2.209455823932932,
0.4084507894487605,
-1.3491711508388657,
0.18219394232884736,
-0.1991395495968878,
1.3877045909644812,
1.2810472453057546,
-1.8198938592553289,
0.691696952821756,
1.9651275722497656,
0.8952970332383311,
-1.572888015515471,
0.10018127814164592,
2.5087678810448395,
0.2303656982914484,
0.6644753240981718,
-0.2940767603661997,
0.9937376992458091,
0.19118749930842235,
-1.7309245646223925,
0.6920377423752287,
1.0669653712016844,
-0.2725348409017723,
0.2108196720828406,
0.7600208432998801,
-0.39468875007311516,
-1.6426053139934083,
0.22140594406577463,
0.3798758679922964,
-0.13579541355890495,
-0.502502519286503,
-0.07503549665980291,
-0.24005906561438198,
0.2999706370689256,
0.4291869988824075,
0.49459447229554165,
-0.2693528321546647,
1.3233447406041847,
0.010128993489178412,
-0.7350427372075996,

0.588999296326396,
-0.03887378376443501,
0.7842242805141085,
-1.017856875421701,
-0.050420029711616104,
1.7478460277120142,
0.46693496239216137,
-1.390597925219579,
-0.1527982976709753,
1.3561114516684873,
-0.11782202169904106,
-0.6172298563496877,
0.2337872361631034,
1.5343227540197375,
-1.246330933330494,
0.2671789790306671,
-0.5231899648601245,
-0.2993536711140956,
-0.3794897720952542,
0.5953851600991978,
-0.016488955527838106,
-0.04239234937973134,
-0.8530294973472092,
0.09298474140637386,
-0.6610063165341049,
-0.7501983006092152,
-0.23667128690009429,
-0.49322661069162854,
0.2817510492822266,
0.8804151571222927,
0.25487104244438136,
-0.5104439984528633,
0.8221175331336923,
0.42372453837884605,
1.1270985005003262,
0.10209111179644431,
0.4047521854484605,
-0.9498909052734678,
0.8591561814039869,
-0.4549855289146429,
-0.7085768506420614,
0.2460272461737723,
-0.08852438964027515,
-0.28170619044977985,
0.18782561055277286,
-0.3648963858755429,
-0.37968971213994723,
0.35188735669812166,
1.8917682407126328,
0.37786421566408035,
0.7636381567370941,
0.07799046336192586,
-0.23205705900380685,
-1.186712365635249,
-0.10791936865692263,
1.6876574701547749,
0.03719295338640027,
-0.062311995058818626,
-0.5208937011021405,
0.4557021320307782,
-2.7664261049727545,
-0.8770083117671578,
0.9937005253370358,
1.250538444528313,
0.6461816519581883,
1.6953467436382619,
2.0814745138972848,
-0.7151854978123715,
-1.6567720673097688,
-1.1703575007457885,
0.44660262232395087,
0.012526409819125981,
-0.0348982634801462,
2.0503923727756113,
1.154948415623314,

0.9176857519192531],
[3.101411838661581e-05,
0.00028167581912982354,
0.002108388113032011,
0.0021083881149554745,
0.002110224126734301,
0.013404723029560417,
0.44387140902527444,
0.4998584267335595,
2.519337827669618,
2.592441150742434,
3.279416160731671,
4.179752140912941,
4.179752161353468,
5.661115424822064,
6.166122332331656,
6.1661223340249105,
6.166122334025239,
7.1531688903854835,
7.165666077614885,
7.257231636090788,
7.810595355153735,
8.056713547662405,
8.059562409019513,
9.326100481695063,
9.326100481704493,
9.781728007060766,
9.888676215095867,
10.278005103664306,
10.559163217755993,
10.559486031108996,
1.2851036574167212,
-0.72547345631182,
-1.4718960991273107,
-1.209251703975404,
-0.016838854839331462,
1.1653703713624246,
-0.21930660603454502,
-0.6325267011020643,
0.27235598773055514,
1.3431283556950424,
0.29725144436322054,
-1.2158342217245894,
0.8631495871791491,
0.12397167154542266,
1.116880444349936,
-0.5886908967334331,
0.37424943307839126,
-0.4246401431552608,
-0.052554167142824744,
-1.151521387506872,
0.45352480044418075,
0.2968389539105363,
-0.36994249285514086,
0.029824637432390475,
0.7039777013639383,
1.0545235270667865,
0.5948019323555648,
0.9185753062546861,
-0.3475338326570484,
1.2642221406528609,
-0.6424417079073889,
-1.8836355388168116,
-0.1226587707935595,
-0.751178954303451,
1.2759264830517028,
-0.7303074304229613,
1.0498129361868127,
0.29592762783390236,
0.4716900159260768,
-0.20829913915791434,
-0.8217397159362575,
0.8503209585884646,
-0.49309125036010926,
0.07716668431879356,

-1.7384586268468483,
0.5748268283601387,
-0.01480830786786586,
-0.03742465163157027,
-0.9568191257597272,
0.07510834673857922,
0.021240686884975563,
0.4240852295635142,
0.10265729129130313,
-0.812822916440861,
-0.14268033459407103,
-0.3753169735052697,
-0.7175759512497852,
0.03918867047505918,
-0.7186687987691884,
-1.6942314888856695,
1.268466372526145,
0.3288584725277509,
-1.242111559182382,
1.113325215651688,
0.40176863135675694,
0.15033001800176832,
-0.8493210333167794,
-0.7302387336933833,
0.5331942268313786,
0.8016954161004247,
0.11571359434030207,
0.9513684680035512,
-0.26463380223931005,
0.5608071632066808,
-0.40528112642254077,
-0.6754638378880663,
1.5778663953317416,
0.16647468944212254,
0.24883728001741243,
-0.9973732709393777,
0.3409424132427824,
-1.113466076369018,
0.15519481593579493,
0.641216026706457,
0.8449849609678267,
1.911891894634777,
-0.34738047822667395,
1.5893587533055515,
0.4884297002665817,
-0.25227237110126166,
1.5183173129023069,
0.5103598279259709,
1.4687346401745565,
2.532539737478952,
-1.540460532405588,
-0.8294616308793386,
-0.2412174307882085,
2.6406875676770416,
1.2361221502323787,
0.13408461910385386,
0.270490641328192,
-0.5308435068885323,
-0.8511283733806999,
0.26595037966666163,
-0.8831519048460094,
-0.4997472202494575,
2.276782887198034,
1.5876131255720232,
1.1044306497561618,
0.22518259496424625,
-0.12906375147756488,
2.4499067189894523,
0.5016533011795579,
-2.1824706389019832,
0.9088257454879881,
-1.3873459886882669,
0.43069183082779305,
0.5835382283652092,
1.9517705176455047,

0.639974902940826],
[0.8405962708734059,
0.8409366166472388,
1.2852356970263128,
1.9656905972413197,
2.6709433729539978,
2.670943372994037,
2.678176776414865,
2.738130595852174,
2.9440618301573322,
2.944061837308021,
3.5946493033065674,
3.5946493033066593,
3.5946493377836535,
3.5946493377949182,
3.6475297284551167,
4.423523257877865,
4.423523257879113,
4.744549401355183,
5.030736177428605,
5.030766895303107,
5.030766896066702,
5.030798272122615,
6.102563406307042,
6.3550093593541375,
6.35500935935416,
7.56205183282469,
8.453276420655834,
9.985009441062077,
10.30999329905164,
11.043022236178203,
2.1889940156552368,
1.249116821544238,
-0.6605760096192125,
-2.1135047950110155,
2.129245868952901,
0.511519543367031,
-0.8638463209571715,
-0.20202526205492455,
-0.5988641121015101,
2.145161094485569,
1.4958876569154136,
0.9003565348629307,
0.8986066176810353,
-0.0859890065364003,
-0.08547563138217676,
-0.6842205219399397,
1.347771315836974,
1.9235660701575357,
0.5300781712046526,
0.9684639862422223,
1.8099194640434915,
0.6789385897860352,
0.46838913395608867,
-1.9553664340235406,
0.15424587849578802,
0.7464909631769433,
0.28409739802447187,
0.5646186506463579,
-0.0688718084707713,
1.3403239673426846,
0.7803011173906335,
-1.2634531934792825,
-0.403184603220626,
0.9911656930801824,
-0.27952114699970115,
-1.17647727881294,
0.6160770530384655,
0.7910105296614427,
-0.26329493021464934,
-0.5093546557195857,
-0.078946955577166,
1.6118855098996396,
-0.8560858484853413,
0.43713047898075813,

0.15951625123225024,
1.1017815594363278,
-0.5112214984829941,
0.8616309140244529,
-0.2539134574216648,
0.6900329022420518,
0.7767771669699943,
1.3495069152164292,
-0.10718702519877486,
0.23011810567539243,
-0.4233144560883979,
1.0277501391374537,
-0.13408439221848256,
-1.2122213001358895,
-0.9574798764432794,
-1.272419025076084,
0.6697485644230207,
0.09252521117768966,
0.8443957240867669,
-0.5978394384474206,
0.4330892667970026,
0.07884699311842308,
1.140608340632257,
0.07864060845749296,
1.1489940789117206,
0.02443540527852337,
-1.0034700299434305,
-0.1412123681218965,
0.42135897985221993,
0.00946489374514043,
0.37971823276367245,
0.6021380541926072,
1.71390465753914,
-0.5080953537535351,
-1.7394955866645057,
-0.55775760641077,
-1.6305248910687546,
0.4704683857374631,
2.1610569951035585,
-0.11166064159089376,
0.47170657685906686,
0.07305148434226297,
0.6150465742311105,
-0.5141382282093759,
-0.8714158035322278,
-0.543899342278249,
-0.22729288133654413,
-1.1237376131773689,
0.5077284416312242,
-0.02493907277109512,
-0.061410284019419926,
1.7654158813927403,
0.24648921891280573,
0.026991979318021532,
-0.4476437051863693,
1.0288055112299304,
0.2791879596887287,
0.552260553667315,
0.5446649167338916,
-1.1781972761427792,
0.24145179880489284,
-0.6620357948168841,
-0.48109037039574104,
0.4693605610272632,
1.5881294256793275,
-0.6259573215794967,
0.502524529722019,
0.9011500652440121,
0.31953241444072733,
-0.0763403773509461,
0.8504085952834295,
-0.9720571531313896,
2.085636878242165,
0.7499605974794749,
1.9701875686083887,

1.4162728880693862],
[0.36934729750713446,
0.3693472975071345,
0.4408171698279066,
0.7641387276708653,
0.7641387276708653,
1.2405539686841192,
1.9615715228812416,
2.180937952046039,
2.956248658909984,
2.956248658909984,
3.615620941947,
4.0363267766272015,
4.283148746851272,
4.283148746851272,
4.287727890051584,
4.294687143195195,
4.959378804296946,
4.959378804296946,
5.299283731795155,
6.02722363939295,
6.706058929046358,
6.706058929046358,
8.58973622967867,
8.58973622967867,
8.735021968997517,
9.724832411320188,
9.882458613321258,
10.054210085109215,
10.144098772484403,
11.057581665321372,
0.6138165123929756,
0.45636276738375425,
-1.6225613647488533,
0.8571182758975707,
1.411989147942658,
1.6605353912366554,
0.31855217609147096,
1.1904862760406203,
1.4473424068486804,
0.687246532966651,
-0.3805095501565646,
-1.4931296153328852,
2.326224921001399,
0.9788391575979964,
-0.08794459312808878,
-1.606604838553076,
0.045095161702097514,
1.1926561845584958,
0.9624943300259794,
0.8526934578403719,
0.3595367653801707,
0.6208577142062074,
0.23723466831561657,
0.8137494677209746,
0.017380156001790208,
-0.7727421772857405,
-0.7696652964277857,
0.28558266237504587,
1.2665997402365978,
0.030324559296025642,
-1.0215734949860629,
-0.435015538681086,
0.09892690725060405,
-0.021280714063487676,
0.6605217667305735,
-0.48357943656762575,
0.23903870372504285,
-0.20353372422303415,
0.40182288676040395,
0.36859613731879964,
-1.0350820651690253,
0.0783547441763707,
-0.5235573738556314,
0.20634909129668547,

-0.22551355647851357,
0.1673560398509943,
0.7468649723369892,
1.0994749531926022,
-0.2018797749925425,
-0.577779100669964,
-0.6205566643688822,
-0.25020034171796784,
-0.24121553611864016,
0.687677741000175,
-0.6022485096507909,
-0.7648342894907719,
2.68891232458234,
-0.038914232418390225,
1.516345449400868,
0.7772620643765826,
-0.5809695972554458,
-0.16788488824728048,
-0.11974201627038453,
-0.31585900129180533,
0.9730861233839377,
-0.23174660705736103,
0.10129760163123996,
-0.38814603215785604,
0.20257130727462175,
-0.5242689321123916,
0.5757755417303894,
-0.4147685482642834,
0.5155383127778999,
-0.9069003556538804,
-0.3931316447570312,
-0.22854166467415718,
0.08849924429856736,
0.11802474333811182,
-0.6697697401245005,
-0.4497368814763953,
0.8371559036446393,
-0.27762346391800535,
1.129889136554746,
-0.7313980725752758,
1.7322939714189498,
-0.30841564461339865,
0.08815569274790189,
-0.45821949503742376,
1.3903832187168985,
2.7423681670027595,
-0.709192498697989,
0.38042285795749525,
0.9330002997189369,
1.5220073637023084,
-1.9071482092674334,
-1.4201112861485827,
-0.18587844516118163,
-0.728966506004035,
2.779449113015483,
-0.5840054192970229,
2.1502692645150225,
-0.4443255851345571,
1.5289006083178085,
0.05130509171668746,
-1.6556175117860825,
-0.5876167642044237,
1.5144905946024154,
1.6252191316647524,
-0.7878469829469384,
0.4020569928547576,
0.833940688908704,
1.2564138083018812,
-1.9521172256438166,
-0.3660865019791046,
-0.7569121463795273,
0.08857539049178031,
0.04440194030774997,
1.346246034983794,
1.5664381297937569,

1.6926501995283607],
[0.07932463514989117,
0.07933390228500832,
0.6712683622930262,
1.8057579746144639,
2.2332189550395363,
2.2332189550395363,
2.2333175311875997,
2.2333175311875997,
2.5803200362154985,
3.149742042876931,
3.2745115489857257,
3.946611320928826,
3.946611320928826,
4.527074130340705,
4.527074130340705,
5.9153316417928945,
6.27481936788149,
6.2748193678814905,
6.718992557341518,
6.964319085694909,
6.964360455572235,
7.61675802651207,
7.61675802651207,
7.61675802651207,
8.031230813321006,
9.561856352158596,
9.835291253191413,
9.958392502186584,
10.272929617553823,
11.17699585622391,
2.3714062527042183,
0.008507065719361826,
-1.7490339115302889,
-0.9098574742643126,
3.2594955351617316,
2.2425950423007475,
-0.8261382715647944,
0.4893495058771166,
0.29218298380822827,
0.7749189883526413,
0.0732829317784662,
-0.427680076829418,
0.835806375311316,
1.2252687690883595,
0.5943555800897335,
0.41209106000270895,
1.8808578438044583,
0.8934024527009041,
-0.5658257233318801,
-0.24310864449817618,
1.2970958730834994,
1.2397558998854112,
-0.13432853133680148,
-0.33892864650006427,
1.2500942843462377,
1.2463490630241996,
0.4505745589505527,
-2.888971827979111,
1.391085002142137,
1.3755760556021903,
0.7798831597382527,
-0.6932853439946646,
0.13475625995390494,
-0.48377623448510626,
-0.4751082457272343,
-0.9134219715133307,
0.5602537728773588,
0.1457020102747123,
2.094813129086275,
0.16050162220250241,
0.255533199882997,
0.5798962545827221,
-0.3296339137986921,
0.20407592924333015,

1.002364909727297,
0.8996033431107227,
-0.40087766178798867,
0.21850323438685998,
-0.3148827979921678,
0.5255859647152507,
-0.981835244220926,
0.18194510004086287,
2.491369267725132,
0.32340553765254526,
-1.422371307906928,
1.3350586578994175,
-0.19038703199979035,
-0.18759028526199026,
0.31727468639934,
-0.13719279505644447,
0.8870810962833492,
0.5605043977550492,
-0.45997587805687895,
0.6203449810744726,
0.35343801179058787,
-0.29094576977122605,
1.0749874898125031,
-2.3258043801847514,
-0.023837190424521523,
0.8602839037993392,
-0.3076595227850746,
2.083827131184913,
-0.709910688132804,
1.9396556568539463,
-0.7118178630647494,
1.024487978597129,
-0.035154872964647726,
0.6794001499523024,
0.6455217590344408,
1.2833952770953792,
1.7594723933960992,
2.366727526539801,
-0.28632630179908214,
1.356168747751164,
0.31201643292476877,
0.364096860526536,
0.8226195660291765,
0.2699337047887902,
-0.4803548322886457,
0.7089202255488762,
1.9081826927626873,
-0.3868588206474107,
-0.05728681029210421,
-0.6103514866875638,
-2.670376487838684,
0.03980060533677062,
-1.0774346470258729,
1.4309826927584666,
1.4923854024295125,
0.4230703239257903,
1.1568773489185937,
0.7815638389040719,
-0.4834118045846874,
1.1499199224441905,
0.01202048539239927,
0.3105021058015101,
-0.3749329298855971,
2.2071268409248224,
0.7242633662153387,
0.6404704587343512,
-0.30896311918370867,
0.6474915553174321,
-0.6471036454486253,
-1.0139549843186435,
2.7326032351157084,
0.4640689770354461,
0.7197232245440613,
1.099219858753326,
0.5370670409426629,

2.2362623976980602],
[0.018174692163319142,
0.02381340297275515,
0.02421332579520929,
0.024525182595047955,
0.4460556740257204,
0.4463126705844747,
0.8080995635599163,
1.5479347906990655,
1.547935096587381,
2.726547871299545,
3.3242466002093267,
3.324246600509028,
4.111398089568711,
4.145752980013426,
4.146049797332313,
6.415277102843476,
6.572144069759118,
6.927329335261905,
6.927379209146919,
6.933957897577306,
7.907817926834528,
8.01227391181071,
8.377715493176042,
8.574692491681018,
8.665468291409324,
9.338599502827458,
9.57363802367466,
10.233446160908231,
10.471009909188941,
11.163878412926959,
2.236671766008264,
0.35159542629328217,
0.8146718981296579,
-1.5439346128384686,
0.8972133735648259,
-0.0553493415862735,
-1.090899451761626,
-2.2352667548818665,
1.2209528829332086,
0.5204242802804685,
-0.618288383541334,
-0.3171787171591956,
-0.025836377068857316,
1.3596603169147203,
0.4695190032736128,
0.11760891446657133,
0.44944359330584466,
-1.4358459977495694,
0.7103123867655211,
1.6252846466436086,
0.17089579517121128,
1.0480108474391834,
-0.21482373854468306,
-0.5796520218220037,
-0.39701726293440376,
-0.15484503866231408,
0.5161603909940105,
-0.08958972382743205,
-0.5178549594803129,
0.23176676286013279,
0.030062676987165922,
-1.5076956114966404,
-0.5464208818685263,
0.0502122423742188,
1.6977256264031593,
0.8909956552917536,
0.4009810515503771,
0.3307875796941022,
-0.5725943763367822,
0.23702814500484895,
-0.15434680984627044,
-1.2435315402871425,
-0.5149922580540566,
0.30951994519916565,

0.5096095319131432,
0.30303640166579543,
-0.3239136690540364,
-0.7206962091038183,
0.8555191973102813,
1.0120763935686161,
0.04024592288077597,
-0.2886549630459275,
-0.3277156141890048,
-0.5588571150564244,
1.5118849277062045,
-0.644801476310519,
0.35391159762671204,
0.036581088484196625,
0.07324701372485032,
-0.19410908530178353,
1.5993036761438029,
0.10651551669568386,
0.32822741091828495,
0.39824063300111046,
0.6674060102702002,
2.0756693329629017,
-1.5421800768661706,
-0.49667839050246243,
0.0124564376427262,
0.4601453458726383,
0.38665655977671154,
-0.3280159653083997,
1.37660238729691,
0.5459980296904141,
-0.10248779795861306,
0.3193368932413526,
0.022394510397351762,
-0.9106037248563306,
-1.8448022296193607,
-0.2878663593630255,
0.3486991485685759,
0.6803232619964643,
0.473978523083063,
0.7630725375206788,
0.8596671356435096,
-0.47717853860533954,
-1.1511176506358933,
0.044726686440788394,
2.206461610291793,
-0.251613145100058,
0.3827507786112393,
-1.0837839025823046,
0.5215057436278849,
-0.6433936043135513,
1.0423632753231085,
-1.5853099693823727,
-0.4899100779852182,
0.6771990537468359,
1.636801051947153,
0.09520300124188852,
0.7384792012405127,
-0.17322471627475994,
1.5991844229586945,
1.501804750971724,
0.5057839423702514,
0.16164961971359962,
2.292178045044038,
1.3450921233724473,
0.6682426373209176,
0.5914595140792696,
-0.5621151585806636,
1.4649155202102295,
-0.9414482790819952,
-0.10805380420514368,
0.9307303586657315,
2.4106406607028164,
-0.01284340130889141,
0.41649266055337153,
1.314622528108089,

2.7227723366977616],
[0.007743638893136335,
0.013160125498327223,
0.608035339315491,
1.107635474685494,
1.1648862034449325,
1.852802815648869,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.586998837285682,
2.6225868976549798,
3.0707103569201006,
3.0707103569201006,
4.1228521273997165,
4.138888702236146,
4.138888702236146,
5.347444125894556,
5.347444125894556,
5.412363261255616,
6.659592716290906,
7.287740867223119,
8.540473259892655,
8.81303185082801,
9.303384987416969,
9.767355442925265,
9.939116248646888,
9.97261625652093,
11.225779420142475,
11.480121962431193,
1.6482545183670085,
-0.013015976711505003,
-1.0388467926799678,
-0.8652539511107419,
1.9084118027497516,
-0.9394653223441272,
-1.7468041006561723,
-1.5142947156306645,
0.2590641138232099,
1.3773414247538143,
-0.6292953946653098,
-0.8862298412863635,
0.3042825725334,
1.123835887892568,
0.9314084077434932,
-2.9280573641387413,
2.0853053971786153,
-0.34681890463267223,
-0.04690984704002986,
0.8070574318910052,
-0.5779942015518676,
1.0090669446225737,
-0.1775287190554287,
-1.6842820449103186,
0.763376948317048,
0.3384712786913095,
1.7367769407755378,
-1.3580490459136343,
-0.8837296315105077,
-0.8963079645873018,
-0.18784202847085701,
-1.4055444367955956,
0.671829961387558,
1.8019702343361095,
0.30718376850684487,
0.08135901917238793,
-0.18472697951808345,
0.5668536919232505,
-1.5958337467143746,
0.002509488020009276,
1.0849426355423382,
0.7452955666922372,
0.9398973688185909,
0.16370191516119634,

-0.20185726086923075,
0.5303635124624408,
0.4908517160520784,
-0.4888169221995994,
0.9032694192338024,
-0.8158468897545128,
0.7026292391012148,
0.011827947942572397,
0.6138675828667193,
0.1834450857087751,
-0.9915622527095793,
-0.5829508078463853,
1.064065245998552,
0.49061771989414976,
0.23606904394636888,
-0.20435452332134307,
-1.0928372780788398,
0.057443193557461136,
0.10260811255105017,
-0.8555456605404742,
-1.096930933544511,
0.04006347013812424,
0.7151909132583844,
0.4885582422451459,
0.09828549627753237,
-0.41856423316032376,
-1.5659826249535125,
0.35337608728639364,
0.5013401269099331,
0.275554280225739,
0.123838948347921,
1.1982351112905236,
-0.5571464296579999,
0.227611732218496,
0.4903765072246047,
0.21638817013741035,
0.5329904920964728,
0.023244037871050734,
0.15105119714954549,
0.5432645474025147,
-1.6089466195523248,
0.5580452428696917,
0.2980389035466241,
-0.7189872564709854,
0.2941454207257381,
2.0652774083354823,
-3.197423241640053,
1.2171902372756198,
-0.9312606307980903,
0.7138860186292121,
0.7458573347434515,
-1.0460906183029233,
-0.4042108821911456,
-0.5284604323913634,
1.1561662773370494,
-1.0010987456125164,
-1.286779340316877,
-0.7935901981914314,
-0.6758512775621703,
-0.253355636189949,
1.083705550488119,
-1.5951444189401682,
-0.2763950378070173,
1.9938759834766908,
0.1467137702454683,
0.18970870490254518,
-0.6591405102572232,
2.3883322313766877,
-0.5426139401957691,
1.4573385975452764,
0.7440623559719126,
1.9382959176617307,
0.07233875596199366,
-1.2256407906907247,
0.8075799371504198,

1.6375675375176653],
[-2.670381383900579e-16,
0.07474127433216288,
0.9423639350936485,
0.9423639350936486,
0.9423639350936487,
0.9423639350936486,
1.8776414733739433,
2.6093592361765396,
2.6549976526560917,
2.6549976526560917,
4.309323876617324,
4.812226043223447,
5.3735992861643656,
5.998589378822616,
6.041219527369439,
6.041219527369439,
6.041219527369439,
6.455755583840882,
7.6346582000153935,
8.456131775478068,
8.456131775478068,
8.456131775478068,
9.265231361717003,
9.265231361717003,
9.265231361717003,
9.265231361717003,
10.025925584440094,
10.402247132343707,
10.703459031001074,
10.703459031001074,
3.3569835366016436,
0.9215266459605492,
-2.040479701749094,
-1.3412114880744723,
1.3650402871975202,
2.328108797667404,
-1.3186160481421627,
0.4833318128451331,
2.239983409823259,
1.0329578572589242,
-0.1517540238142264,
-1.4974261238923248,
1.7473459056580782,
1.4519306881870349,
-2.206161402631766,
-0.8907223077237956,
2.3813682348978937,
0.5973875310190014,
-0.8910649708758184,
0.26277461789671996,
1.6008335914847227,
3.025884654690384,
0.8431565773802462,
-1.3891693904439881,
3.086427239689084,
-0.3399924211336387,
0.9616731269325118,
-0.4538264391900641,
1.6045391274841239,
1.2066545032742897,
2.5203408436794805,
-0.7822247170003326,
0.5795340175268288,
1.1137644213878175,
1.338413506379888,
-1.170792708483999,
0.7895425377479481,
1.929612997870515,
2.1013047486982743,
-0.1747799610123698,
0.34124409150219404,
0.1345366524340121,
0.12829425701000113,
-0.07853692563407341,

0.10525428958729864,
2.4094888168745765,
0.6357560012375729,
0.3483402865911379,
-2.7204359945096264,
0.7864668566884239,
2.710084659416198,
-0.42133711298349386,
-0.7606725606109204,
4.024573300631783,
1.0416894631362523,
0.549877653228643,
-0.23095740430710715,
1.2081110571007485,
-0.35409975752454176,
0.9082778411762691,
0.9011042810290772,
2.2659344214863575,
0.030960161362888827,
1.709001611830215,
0.7901365980951575,
-0.25541821293373157,
-0.5438725550461083,
-1.6272180586430949,
2.7727467118019176,
2.937339318913365,
0.16303685557629694,
1.9109318898942156,
-0.41619598488045445,
-0.186609722459416,
0.14598277516592972,
-0.08246623139297415,
2.8463994197955897,
1.599311519948962,
-0.3139472278021089,
0.8671768693326729,
1.2547286400858166,
1.2488092065429963,
-0.4426042076320244,
-0.26147177110889996,
1.5139530957520613,
1.698807756116421,
0.7729868493874114,
1.30698727115783,
0.15584206991884653,
1.7938264286277585,
-0.19656274733701168,
0.033815913473199004,
3.683537194772181,
1.3989993033480204,
0.3819269315951613,
1.7115449870773622,
0.11877637359815167,
-0.3580223266367696,
2.5683966853471936,
1.1633017601186986,
0.7010591039102597,
0.6321456093464749,
1.2531601332271327,
1.460291237936634,
-0.0031513754098698096,
-0.7920192885995154,
1.112751285243459,
1.2193221700281838,
1.0214928812662405,
-0.12719592003733096,
1.6086786238146749,
1.3655040824209228,
-1.4212254487699023,
-1.4031310724327033,
2.140187196374229,
-1.788444809286011,
0.36492229278601657,
0.8263664226867995,
2.5133575298607385,

```
0.292333075844064]],
'run_time_sec': 34196.97281765938}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [
```

```
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
```

```
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}}],
'init_param': { 'value': [ 0.3358069569174437,
0.33751316628129896,
0.37613413155379394,
0.9037599675903363,
0.941040799157583,
1.6105111871436553,
1.7638163741391246,
1.7638163741391246,
3.06849123809687,
3.600239571878297,
4.175689787149765,
4.175689787149765,
5.544091982519477,
5.544091982519477,
6.308463651138072,
6.308463651138072,
6.308463651138072,
6.3184465106806345,
6.521816085515529,
6.521816085515529,
7.288786933299201,
7.54709747254392,
7.597867588518987,
7.962268849194133,
7.991785735138846,
8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
```

0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,

```

1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '

```



```

'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.523058281578953],
                              [12, 6, 0, 0, -5.228944859106493],
                              [ 20,
                                10,
                                0,
                                0,
                                -4.1973377330967105]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [ 0,
                                              0,
                                              0,
                                              0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

```

```

'noise_prob': [ 0.001,
               0.001,
               0.0,
               0.0],

True,

'dephasing'},

'noise_on_init_param': { 'status': False,
                        'value': 0},

'noise_prob': [ 0.001,
               0.001,
               0.0,
               0.0],

True,

'dephasing'},

'odd_wires': 4}},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                       2,
                       0,
                       0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 8.358182191848755}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0,
                                                             0],
                                                           'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.0008444707970882174,
                           0.0011376501074209704,
                           0.5478052879830236,
                           0.5478052879830236,
                           0.5478052879830237,
                           0.784723345688809,
                           0.784723345688809,
                           0.784723345688809,
                           1.9248523134122373,
                           3.1827054999432645,
                           3.5310059925523927,
                           3.5310059925523927,
                           5.052695732413606,
                           5.052695732413606,
                           6.136548648234324,
                           6.136548648234324,
                           6.785869019003799,
                           6.785869019003799,
                           7.198041513907379,
                           7.3330889202217895,
                           8.496582874867597,
                           8.496582874867597,
                           8.496739633755235,
                           8.496739633755235,
                           8.496741833065137,
                           9.22890579273074,
                           9.930471725964209,

```

10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,

```

-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,

```

```

        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                '1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
                                             '0 X 1 '
                                             '+ 0.5 '
                                             'Y 0 Y '
                                             '1 + '
                                             '0.5 Z '
                                             '0 Z 1 '
                                             '+ 0.5 '
                                             'X 1 X '
                                             '2 + '
                                             '0.5 Y '
                                             '1 Y 2 '
                                             '+ 0.5 '
                                             'Z 1 Z '
                                             '2 + '
                                             '0.5 X '
                                             '2 X 3 '
                                             '+ 0.5 '
                                             'Y 2 Y '
                                             '3 + '
                                             '0.5 Z '
                                             '2 Z 3 '
                                             '+ 0.5 '
                                             'X 3 X '
                                             '4 + '
                                             '0.5 Y '
                                             '3 Y 4 '
                                             '+ 0.5 '
                                             'Z 3 Z '
                                             '4 + '
                                             '0.5 X '
                                             '4 X 5 '
                                             '+ 0.5 '
                                             'Y 4 Y '
                                             '5 + '
                                             '0.5 Z '
                                             '4 Z 5 '
                                             '+ 0.5 '
                                             'X 5 X '
                                             '6 + '
                                             '0.5 Y '
                                             '5 Y 6 '
                                             '+ 0.5 '
                                             'Z 5 Z '
                                             '6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.344797923049557],
                             [12, 6, 0, 0, -4.988789667979189],
                             [ 20,
                              10,
                              0,
                              0,
                              -3.915584148696589]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],

```

```
'noise_on_init_param': { 'status': False,  
    'value': 0},  
'noise_prob': [ 0.001,  
    0.001,  
    0.0,  
    0.0],  
  
True,  
  
'dephasing'},  
  
'noise_on_init_param': { 'status': False,  
    'value': 0},  
'noise_prob': [ 0.001,  
    0.001,  
    0.0,  
    0.0],  
  
True,  
  
'dephasing'},  
  
'noise_on_init_param': { 'status': False,  
    'value': 0},  
'noise_prob': [ 0.001,  
    0.001,  
    0.0,  
    0.0],  
  
True,  
  
'dephasing'},  
  
0.5,  
0.5,  
0.5,  
0.5,  
0.5,  
0.5],  
  
'identity_factors': [ 0,  
    0,  
    0,  
    0],  
'noise_level': [4, 2, 0, 0],  
'noise_profile': {  
  
    'status':  
  
    'type':  
  
    'odd_wires': 4},  
{ 'gates_num': [12, 3, 1, 0],  
    'identity_factors': [ 1,  
        1,  
        0,  
        0],  
    'noise_level': [12, 6, 0, 0],  
    'noise_profile': {  
  
        'status':  
  
        'type':  
  
        'odd_wires': 4},  
{ 'gates_num': [20, 5, 1, 0],  
    'identity_factors': [ 2,  
        2,  
        0,  
        0],  
    'noise_level': [20, 10, 0, 0],  
    'noise_profile': {  
  
        'status':  
  
        'type':  
  
        'odd_wires': 4}},  
    'run_time_sec': 10.613636016845703}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
    'layer': 30,  
    'ugate': { 'coefficients': { 'bn': [ 0,  
        0,  
        0,  
        0,  
        0,  
        0,  
        0],  
        'cn': [  
  
            'r': 0},  
    'time': { 'max': 10.0,  
        'min': 0.0},  
    'type': 'heisenberg'}},  
    'init_param': { 'value': [ 0.26700956978337703,
```

0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,
9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,

0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732]],

'noise_profile': { 'noise_on_init_param': {

'status': False,

'value':

0},

'noise_prob': [0.001,
0.001,
0.0,
0.0],

'status': True,
'type': 'dephasing'},

'nqubits': 7,

'observable': { 'coefficients': { 'bn': [1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},

'def': 'ising'},

'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':

'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#9'},

'redundant': { 'identity_factors': [[0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},

'run': 'redundant',

'state': 'dmatrix',

'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},

'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},

'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',

'time_evolution_gate_hamiltonian_string': ['0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '

```

'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.340393756989085],
[12, 6, 0, 0, -4.976052575965789],
[ 20,
10,
0,
0,
-3.898762699309893]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],

'status':

```

```
True,
'dephasing'},
                                'type':
                                'odd_wires': 4}},
                                'run_time_sec': 10.062668561935425}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
                                'r': 0},
                                'time': { 'max': 10.0,
                                           'min': 0.0},
                                'type': 'heisenberg'}}},
'init_param': { 'value': [ 1.7202079031049283e-17,
                           -2.594562831748091e-19,
                           0.6068914425074802,
                           1.1321115349854645,
                           1.2806026785178235,
                           1.5886494702821472,
                           2.45154545857325,
                           2.4515454585732503,
                           2.4575334297772264,
                           2.4575334297772264,
                           3.2128626039678436,
                           3.2128626039678436,
                           4.612853662117567,
                           5.000904423260963,
                           5.000904423260963,
                           5.1471756720956385,
                           6.340371796976545,
                           6.340371796976545,
                           6.640884886785207,
                           6.640884886785207,
                           6.640884886785207,
                           8.598101965665498,
                           8.622498407967232,
                           8.622498407967234,
                           9.078776950212898,
                           9.461537111509045,
                           9.461537111509045,
                           11.193126234466078,
                           11.194773036768451,
                           11.23425035230001,
                           3.6783033545989348,
                           2.4269657917911354,
                           -0.05771344168509165,
                           -1.2030573768957538,
                           3.7213804138473705,
                           1.924827615988215,
                           1.056479566226862,
                           -1.5473494967520907,
                           4.398805627479834,
                           0.8338358403171965,
                           -0.8099255104678582,
                           -0.45767862837458095,
                           2.861078492572404,
                           3.8652468681120005,
                           -2.887845767517789,
                           -2.805755195373484,
                           4.0587596872172185,
                           -0.6699049461158488,
                           0.21333648034183758,
                           0.4524745343676716,
```

2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,
-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,

```

-1.27505781222129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '

```

```

'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.2758943446211575],
[12, 6, 0, 0, -4.972375668379768],
[ 20,
10,
0,
0,
-3.9295998440454545]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':
True,
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'noise_profile': {
    'status':
    'type':
    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                       2,
                       0,
                       0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
    'status':
    'type':
    'odd_wires': 4}},
'run_time_sec': 9.814876317977905}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
                                                                'r': 0},
                          'time': { 'max': 10.0,
                                    'min': 0.0},
                          'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.05909938957877547,
                             0.05909939012498705,
                             0.4766831071513548,
                             0.4767050288044724,
                             0.476729910017027,
                             0.47672998527996446,
                             2.99469059821818,
                             3.0000368876553107,
                             3.03481022556516,
                             3.0348102255651677,
                             3.0960777044795975,
                             3.0960777044802072,
                             3.096077704551025,
                             5.243967325820749,
                             5.24396732593715,
                             5.696137660948558,
                             5.6961376610109316,
                             6.622089050039713,
                             6.65393172347244,
                             7.224334835522734,
                             7.698532497695266,
                             7.80231633505566,
                             8.26676599072812,
                             9.142506968904602,

```

9.143106298076189,
9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,


```

0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

```

```
'r': 1},  
      'def': 'ising'},  
    'output': { 'draw': { 'fig_dpi': 100,  
                          'status': False,  
                          'type': 'png'},  
               'file_name_prefix':  
  
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#3'},  
  'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
                                       [1, 1, 0, 0],  
                                       [2, 2, 0, 0]]},  
  'run': 'redundant',  
  'state': 'dmatrix',  
  'vqe': { 'iteration': 1,  
           'optimization': { 'algorithm': 'SLSQP',  
                             'constraint': True,  
                             'status': False}},  
  'zne': { 'data_points': None,  
          'degree': 0,  
          'method': 'richardson',  
          'sampling': 'default'}},  
  'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '  
                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '  
                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '  
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '  
                                '1 Z 5 + 1 Z 6',  
             'time_evolution_gate_hamiltonian_string': [ '0.5 X '  
                                                         '+ 0.5 '  
                                                         'Y 0 Y '  
                                                         '1 + '  
                                                         '0.5 Z '  
                                                         '0 Z 1 '  
                                                         '+ 0.5 '  
                                                         'X 1 X '  
                                                         '2 + '  
                                                         '0.5 Y '  
                                                         '1 Y 2 '  
                                                         '+ 0.5 '  
                                                         'Z 1 Z '  
                                                         '2 + '  
                                                         '0.5 X '  
                                                         '2 X 3 '  
                                                         '+ 0.5 '  
                                                         'Y 2 Y '  
                                                         '3 + '  
                                                         '0.5 Z '  
                                                         '2 Z 3 '  
                                                         '+ 0.5 '  
                                                         'X 3 X '  
                                                         '4 + '  
                                                         '0.5 Y '  
                                                         '3 Y 4 '  
                                                         '+ 0.5 '  
                                                         'Z 3 Z '  
                                                         '4 + '  
                                                         '0.5 X '  
                                                         '4 X 5 '  
                                                         '+ 0.5 '  
                                                         'Y 4 Y '  
                                                         '5 + '  
                                                         '0.5 Z '  
                                                         '4 Z 5 '  
                                                         '+ 0.5 '  
                                                         'X 5 X '  
                                                         '6 + '  
                                                         '0.5 Y '  
                                                         '5 Y 6 '  
                                                         '+ 0.5 '  
                                                         'Z 5 Z '  
                                                         '6']]],  
            'output': { 'data_points': [ [4, 2, 0, 0, -6.425213401717636],  
                                         [12, 6, 0, 0, -5.16135557034569],  
                                         [ 20,  
                                          10,  
                                           0,
```

```

0,
-4.151192033326441]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [ 0,
                                         0,
                                         0,
                                         0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
               0.001,
               0.0,
               0.0],

True,

'dephasing'},

                    'status':

                    'type':

                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                      1,
                      0,
                      0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
               0.001,
               0.0,
               0.0],

True,

'dephasing'},

                    'status':

                    'type':

                    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                      2,
                      0,
                      0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
               0.001,
               0.0,
               0.0],

True,

'dephasing'},

                    'status':

                    'type':

                    'odd_wires': 4}},
'run_time_sec': 9.246371746063232}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

                        'r': 0},
                        'time': { 'max': 10.0,

```

```
        'min': 0.0},
    'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6486934998593055,
    0.6706921527624933,
    0.6915479354144154,
    0.6915479354144154,
    0.7822900063342887,
    0.7822900063342887,
    0.7822900063342887,
    0.7969446550492483,
    2.1484559222076447,
    2.662825799299848,
    3.431953071624279,
    3.431953071624279,
    4.035233416575914,
    4.035233416575914,
    4.035233416575914,
    4.568117148704361,
    5.039539225310202,
    5.039539225310202,
    5.534018322321666,
    5.534018322321666,
    6.483210707785691,
    6.490896899688113,
    7.182799214542672,
    7.182799214542672,
    7.559584008067621,
    8.43635332601195,
    8.43635332601195,
    9.091482769474323,
    9.091482769474323,
    10.51189306045493,
    0.4036051624048159,
    0.6580269923177141,
    2.030106683293408,
    -1.6087595289666783,
    1.1238448060114463,
    0.48219161744750516,
    -0.52647875832763,
    -0.7770680009741181,
    0.9584923266732629,
    1.4081968681145547,
    -1.286760308434132,
    -1.1643597658261824,
    0.43102549223791814,
    0.4762851226867799,
    -1.5375915208743731,
    -0.32604860635808525,
    0.9126587772562,
    0.4716164439155413,
    -1.185137911370813,
    0.0010435150849372951,
    1.362703657319786,
    0.15168618448664317,
    0.013738542048390118,
    -1.2488843079530334,
    0.5953067453023578,
    -0.7375945018490428,
    1.6938673074929198,
    0.2332524998863778,
    0.7631553129407195,
    0.8722488841401685,
    -0.6689205221218597,
    0.6584128273939948,
    -0.5866714617908798,
    -0.5382377239349249,
    0.07947113792533414,
    -0.638100867011655,
    -0.19922015844116167,
    0.5814496123293298,
    0.41157361684622124,
    -0.7649732243319444,
    0.562344908659212,
    0.37006719775147345,
    0.9681106498419162,
```

0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,

```

0.26895925341115173,
1.7196471710373182]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '

```

```

'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.1433206567045895],
[12, 6, 0, 0, -4.62093405872061],
[ 20,
10,
0,
0,
-3.4426712688431165]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':
True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':
True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':
True,
'type':
'dephasing'},
'odd_wires': 4}

```

```
0.0,  
0.0],  
  
True,  
  
'dephasing'},
```

'status':

'type':

```
        'odd_wires': 4]],  
        'run_time_sec': 9.943228960037231}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                           'layer': 30,  
                           'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                     'cn': [
```

```
0.5,  
0.5,  
0.5,  
0.5,  
0.5,  
0.5],
```

```
        'r': 0},  
        'time': { 'max': 10.0,  
                  'min': 0.0},  
        'type': 'heisenberg'}},  
'init_param': { 'value': [ 0.723632677978758,  
                             0.8013713721964207,  
                             0.8013713721964207,  
                             0.8013713721964206,  
                             1.7866251780297595,  
                             1.786625178029759,  
                             2.0864992088537258,  
                             2.7277814684323256,  
                             2.7277814684323256,  
                             3.2780082387901137,  
                             3.2780082387901137,  
                             4.125286968934425,  
                             4.125286968934425,  
                             6.2058132923329055,  
                             6.495923744510521,  
                             6.7707474398596865,  
                             6.7707474398596865,  
                             7.238549805010759,  
                             7.238549805010759,  
                             7.424016758372523,  
                             7.424016758372523,  
                             7.45123520722395,  
                             8.286067277272798,  
                             9.005302817051474,  
                             9.005302817051472,  
                             9.005302817051474,  
                             9.468112593488378,  
                             9.980769303066321,  
                             10.029230447532314,  
                             10.496216102492953,  
                             1.6849607035180754,  
                             -0.17702247224764014,  
                             -1.2741911423346686,  
                             -2.844813020435471,  
                             2.6306502527193634,  
                             1.1115711285155645,  
                             -1.0604767208931998,  
                             -1.2378494371437496,  
                             -0.4278587412008179,  
                             3.0662166892278284,  
                             -0.052478476294015586,  
                             -0.4305161219629684,  
                             0.9820485095264653,  
                             2.3466906714085924,  
                             -0.025291075467548628,  
                             -0.02025122162742965,  
                             2.3009633618597958,
```


2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,

```

1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '

```

```

'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.126749766362046],
[12, 6, 0, 0, -4.835388535288831],
[ 20,
10,
0,
0,
-3.8102426495131967]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':
True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1,
1]
}
]
}

```

```

0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'status':

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 9.739906311035156}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.17583917736716262,
                            0.17583917736716265,
                            0.9725326222396186,
                            0.9725326222396184,
                            1.154796264329948,
                            2.3531638611520562,
                            3.742172656987498,
                            3.810257862376578,
                            3.8745068996857803,
                            3.8745068996857803,
                            3.8745068996857803,
                            3.8745068996857803,
                            4.643045833818014,
                            4.643045833818014,
                            4.643045833818013,
                            4.643045833818014,
                            4.643045833818014,
                            4.711671630422156,
                            5.405427388054719,
                            5.405427388054719,
                            7.4602327607935,
                            7.610173264016766,

```

7.858440336330247,
7.968223945319675,
8.51525712638587,
8.54015380483253,
8.797039458545115,
8.797039458545115,
8.858280642077405,
9.30423425746159,
11.6502765461574,
1.4243949815721635,
1.853782837605556,
0.616412956408157,
-0.43476006789942184,
1.5440059756230975,
1.1700457070837984,
-1.8512161685606496,
0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,

```
'noise_profile': { 'noise_on_init_param': {
```

```
'value':
```

```
'status': False,
```

 $\emptyset\}$

```
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
```

```
'status': True,
'type': 'dephasing'},
```

```
'nqubits': 7,
```

```
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
```

[illegible]

```
[ 20,  
    10,  
    0,  
    0,  
    -3.3828918005869286]],  
'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                      'identity_factors': [ 0,  
                                             0,  
                                             0,  
                                             0],  
                      'noise_level': [4, 2, 0, 0],  
                      'noise_profile': {  
  
True,                                     'status':  
  
                                         'type':  
  
                                         'odd_wires': 4},  
{ 'gates_num': [12, 3, 1, 0],  
  'identity_factors': [ 1,  
                        1,  
                        0,  
                        0],  
  'noise_level': [12, 6, 0, 0],  
  'noise_profile': {  
  
'noise_on_init_param':{ 'status': False,  
                          'value': 0},  
'noise_prob': [ 0.001,  
                 0.001,  
                 0.0,  
                 0.0],  
  
True,                                     'status':  
  
                                         'type':  
  
                                         'odd_wires': 4},  
{ 'gates_num': [20, 5, 1, 0],  
  'identity_factors': [ 2,  
                        2,  
                        0,  
                        0],  
  'noise_level': [20, 10, 0, 0],  
  'noise_profile': {  
  
'noise_on_init_param': { 'status': False,  
                          'value': 0},  
'noise_prob': [ 0.001,  
                 0.001,  
                 0.0,  
                 0.0],  
  
True,                                     'status':  
  
                                         'type':  
  
                                         'odd_wires': 4}},  
      'run_time_sec': 8.81235408782959}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                'cn': [  
0.5,  
0.5,  
0.5,  
0.5,  
0.5,
```


0.5],

```
        'r': 0},
    'time': { 'max': 10.0,
              'min': 0.0},
    'type': 'heisenberg'}},
'init_param': { 'value': [ 2.1921898489539642e-17,
1.3966237144281936e-17,
5.875629127222605e-17,
1.0041795331735186,
1.0493846336909358,
2.3268506812305976,
3.014247762257123,
3.5587330689470917,
3.598019824659659,
3.598019824659659,
3.598019824659659,
4.594861139703674,
5.132217790928667,
5.132217790928667,
6.816072831791245,
6.906986102298665,
7.2472760308271145,
7.2472760308271145,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.942806427307234,
9.150430959012654,
9.150430959012654,
9.436537073101752,
9.664513899938346,
10.186874850218704,
10.186874850218702,
10.784457886918862,
1.4923280250442645,
-0.2249463453233081,
0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
```

0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,

```

1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '

```

```
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.181142575648604],
[12, 6, 0, 0, -4.947722861409576],
[ 20,
10,
0,
0,
-3.9537204583695584]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
```

```
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
```

```
True,
'dephasing'},
```

```
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
```

```
'status':
'type':
'odd_wires': 4]],
'run_time_sec': 9.161455869674683}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [
'r': 0},
'time': { 'max': 10.0,
'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
```

-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,

```

-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}}},
'status': False,
0},

```

```

'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
          '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
          '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
          '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
          '1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
      '0 X 1 '
      '+ 0.5 '
      'Y 0 Y '
      '1 + '
      '0.5 Z '
      '0 Z 1 '
      '+ 0.5 '
      'X 1 X '
      '2 + '
      '0.5 Y '
      '1 Y 2 '
      '+ 0.5 '
      'Z 1 Z '
      '2 + '
      '0.5 X '
      '2 X 3 '
      '+ 0.5 '
      'Y 2 Y '
      '3 + '
      '0.5 Z '
      '2 Z 3 '
      '+ 0.5 '
      'X 3 X '
      '4 + '
      '0.5 Y '
      '3 Y 4 '
      '+ 0.5 '
      'Z 3 Z '
      '4 + '
      '0.5 X '
      '4 X 5 '
      '+ 0.5 '
      'Y 4 Y '
      '5 + '
      '0.5 Z '
      '4 Z 5 '
      '+ 0.5 '
      'X 5 X '
      '6 + '
      '0.5 Y '
      '5 Y 6 '
      '+ 0.5 '
      'Z 5 Z '
      '6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.39007364379084],
                             [12, 6, 0, 0, -5.048002042316808],
                             [ 20,
                              10,
                              0,
                              0,
                              -3.9807470593727396]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [ 0,
                                           0,
                                           0,
                                           0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'status':
True,

'type':
'dephasing'},

'odd_wires': 4},

```


[illegible]

```

                                0.5],
                                'r': 1},
                                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                        'status': True,
                        'type': 'png'},
            'file_name_prefix':

'heisenberg_noisefree_time_evo'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': True}},
'zne': { 'data_points': None,
         'degree': 1,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                  'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                  '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                  'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 '
                                                         '+ 0.5 Y 0 Y '
                                                         '1 + 0.5 Z 0 '
                                                         'Z 1 + 0.5 X '
                                                         '1 X 2 + 0.5 '
                                                         'Y 1 Y 2 + '
                                                         '0.5 Z 1 Z 2 '
                                                         '+ 0.5 X 2 X '
                                                         '3 + 0.5 Y 2 '
                                                         'Y 3 + 0.5 Z '
                                                         '2 Z 3 + 0.5 '
                                                         'X 3 X 4 + '
                                                         '0.5 Y 3 Y 4 '
                                                         '+ 0.5 Z 3 Z '
                                                         '4 + 0.5 X 4 '
                                                         'X 5 + 0.5 Y '
                                                         '4 Y 5 + 0.5 '
                                                         'Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 '
                                                         '+ 0.5 Y 5 Y '
                                                         '6 + 0.5 Z 5 '
                                                         'Z 6']},

'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 6.146913233691134,
                                      5.993838002068908,
                                      5.949850054037247,
                                      6.210098036642341,
                                      6.152155884142049,
                                      6.148463718261344,
                                      6.155325453778133,
                                      6.006166196398716,
                                      6.005219627807008,
                                      6.0413306719300985],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
'optimized_minimum_cost': [ -6.523058281579068,
                             -6.1267497663620825,
                             -6.42521340171773,

```

```
-6.39007364379089,  
-6.344797923049613,  
-5.871225062671643,  
-6.143320656704609,  
-6.2758943446211894,  
-6.3403937569891395,  
-6.18114257564869],  
'optimized_parameters': [ [ 0.3358069569174437,  
0.33751316628129896,  
0.37613413155379394,  
0.9037599675903363,  
0.941040799157583,  
1.6105111871436553,  
1.7638163741391246,  
1.7638163741391246,  
3.06849123809687,  
3.600239571878297,  
4.175689787149765,  
4.175689787149765,  
5.544091982519477,  
5.544091982519477,  
6.308463651138072,  
6.308463651138072,  
6.308463651138072,  
6.3184465106806345,  
6.521816085515529,  
6.521816085515529,  
7.288786933299201,  
7.54709747254392,  
7.597867588518987,  
7.962268849194133,  
7.991785735138846,  
8.619190388880893,  
8.629220288817223,  
10.432794468229133,  
10.549342687924028,  
11.465905039453537,  
1.5708348469088267,  
-0.3283220618427727,  
0.03691181582691629,  
-2.5608652259642914,  
1.464587567475586,  
2.208272980722778,  
-0.3337202412707043,  
-0.9570571070157553,  
0.20098684515796417,  
-0.8867183197035234,  
-0.00908592189328417,  
-1.4638778850057466,  
2.6033615081338537,  
0.3763013231634618,  
-2.697434246081451,  
0.09985560627390955,  
0.6973075775075135,  
-0.5785886913208023,  
-0.13253830432587832,  
-0.003645075822409154,  
0.648250634168174,  
0.24074433115353022,  
-0.7536468259038906,  
0.08331860927339882,  
-0.5730961456214154,  
-0.20775836607022918,  
1.858859760589094,  
-0.3428004889774629,  
0.7409414856192459,  
0.34884447713535405,  
0.6678090018369852,  
-0.6382699051282278,  
0.1547862218012797,  
-0.17320932448753618,  
-0.2953648953179735,  
-0.02314845214565176,  
0.24480308430392833,  
-0.20823570419213822,
```

-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,

-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428],
[0.723632677978758,
0.8013713721964207,
0.8013713721964207,
0.8013713721964206,
1.7866251780297595,
1.786625178029759,
2.0864992088537258,
2.7277814684323256,
2.7277814684323256,
3.2780082387901137,
3.2780082387901137,
4.125286968934425,
4.125286968934425,
6.2058132923329055,
6.495923744510521,
6.7707474398596865,
6.7707474398596865,
7.238549805010759,
7.238549805010759,
7.424016758372523,
7.424016758372523,
7.45123520722395,
8.2860672727272798,
9.005302817051474,
9.005302817051472,
9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,

0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,

-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826],
[0.05909938957877547,
0.05909939012498705,
0.4766831071513548,
0.4767050288044724,
0.476729910017027,
0.47672998527996446,
2.99469059821818,
3.0000368876553107,
3.03481022556516,
3.0348102255651677,
3.0960777044795975,
3.0960777044802072,
3.096077704551025,
5.243967325820749,
5.24396732593715,
5.696137660948558,
5.6961376610109316,
6.622089050039713,
6.65393172347244,
7.224334835522734,
7.698532497695266,
7.80231633505566,
8.26676599072812,
9.142506968904602,
9.143106298076189,
9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,

0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,

0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522],
[0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,

0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,

-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048],
[0.0008444707970882174,
0.0011376501074209704,
0.5478052879830236,
0.5478052879830236,
0.5478052879830237,
0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,
9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,

0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,

-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475],
[0.17583917736716262,
0.17583917736716265,
0.9725326222396186,
0.9725326222396184,
1.154796264329948,
2.3531638611520562,
3.742172656987498,
3.810257862376578,
3.8745068996857803,
3.8745068996857803,
3.8745068996857803,
4.643045833818014,
4.643045833818014,
4.643045833818013,
4.643045833818014,
4.643045833818014,
4.711671630422156,
5.405427388054719,
5.405427388054719,
7.4602327607935,
7.610173264016766,
7.858440336330247,
7.968223945319675,
8.51525712638587,
8.54015380483253,
8.797039458545115,
8.797039458545115,
8.858280642077405,
9.30423425746159,
11.6502765461574,
1.4243949815721635,
1.853782837605556,
0.616412956408157,
-0.43476006789942184,
1.5440059756230975,
1.1700457070837984,
-1.8512161685606496,
0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,

0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,

1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613],
[0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,

0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,

-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182],
[1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,
9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,

1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,
-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,

-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517],
[0.26700956978337703,
0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,
9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,

-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,

0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732],
[2.1921898489539642e-17,
1.3966237144281936e-17,
5.875629127222605e-17,
1.0041795331735186,
1.0493846336909358,
2.3268506812305976,
3.014247762257123,
3.5587330689470917,
3.598019824659659,
3.598019824659659,
3.598019824659659,
4.594861139703674,
5.132217790928667,
5.132217790928667,
6.816072831791245,
6.906986102298665,
7.2472760308271145,
7.2472760308271145,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.942806427307234,
9.150430959012654,
9.150430959012654,
9.436537073101752,
9.664513899938346,
10.186874850218704,
10.186874850218702,
10.784457886918862,
1.4923280250442645,
-0.2249463453233081,
0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,

0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,

```
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'run_time_sec': 42850.28223609924}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                   'layer': 30,
                                   'ugate': { 'coefficients': { 'bn': [ 0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    'r': 0}},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'heisenberg'}}},
          { 'value': [ 0.723632677978758,
                        0.8013713721964207,
                        0.8013713721964207,
                        0.8013713721964206,
                        1.7866251780297595,
                        1.786625178029759,
                        2.0864992088537258,
                        2.7277814684323256,
                        2.7277814684323256,
                        3.2780082387901137,
                        3.2780082387901137,
                        4.125286968934425,
                        4.125286968934425,
                        6.2058132923329055,
                        6.495923744510521,
                        6.7707474398596865,
                        6.7707474398596865,
                        7.238549805010759,
                        7.238549805010759,
                        7.424016758372523,
                        7.424016758372523,
                        7.45123520722395,
                        8.286067277272798,
                        9.005302817051474,
                        9.005302817051472,
                        9.005302817051474,
                        9.468112593488378,
                        9.980769303066321,
                        10.029230447532314,
                        10.496216102492953,
                        1.6849607035180754,
                        -0.17702247224764014,
                        -1.2741911423346686,
                        -2.844813020435471,
                        2.6306502527193634,
                        1.1115711285155645,
                        -1.0604767208931998,
                        -1.2378494371437496,
                        -0.4278587412008179,
                        3.0662166892278284,
                        -0.052478476294015586,
                        -0.4305161219629684,
                        0.9820485095264653,
                        2.3466906714085924,
                        -0.025291075467548628,
                        -0.02025122162742965,
                        2.3009633618597958,
                        2.636290816480209,
                        -1.3375867987805241,
```

-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,


```

-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noiseefree_time_evo_dephasing_ric3_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.126749766362046],
[ 12,
6,
0,
0,
-4.835388535288831],
[ 20,
10,
0,
0,
-4.835388535288831],
[ 28,
14,
0,
0,
-4.835388535288831],
[ 36,
18,
0,
0,
-4.835388535288831],
[ 44,
22,
0,
0,
-4.835388535288831],
[ 52,
26,
0,
0,
-4.835388535288831],
[ 60,
30,
0,
0,
-4.835388535288831],
[ 68,
34,
0,
0,
-4.835388535288831],
[ 76,
38,
0,
0,
-4.835388535288831],
[ 84,
42,
0,
0,
-4.835388535288831],
[ 92,
46,
0,
0,
-4.835388535288831],
[ 100,
50,
0,
0,
-4.835388535288831],
[ 108,
54,
0,
0,
-4.835388535288831],
[ 116,
58,
0,
0,
-4.835388535288831],
[ 124,
62,
0,
0,
-4.835388535288831],
[ 132,
66,
0,
0,
-4.835388535288831],
[ 140,
70,
0,
0,
-4.835388535288831],
[ 148,
74,
0,
0,
-4.835388535288831],
[ 156,
78,
0,
0,
-4.835388535288831],
[ 164,
82,
0,
0,
-4.835388535288831],
[ 172,
86,
0,
0,
-4.835388535288831],
[ 180,
90,
0,
0,
-4.835388535288831],
[ 188,
94,
0,
0,
-4.835388535288831],
[ 196,
98,
0,
0,
-4.835388535288831],
[ 204,
102,
0,
0,
-4.835388535288831],
[ 212,
106,
0,
0,
-4.835388535288831],
[ 220,
110,
0,
0,
-4.835388535288831],
[ 228,
114,
0,
0,
-4.835388535288831],
[ 236,
118,
0,
0,
-4.835388535288831],
[ 244,
122,
0,
0,
-4.835388535288831],
[ 252,
126,
0,
0,
-4.835388535288831],
[ 260,
130,
0,
0,
-4.835388535288831],
[ 268,
134,
0,
0,
-4.835388535288831],
[ 276,
138,
0,
0,
-4.835388535288831],
[ 284,
142,
0,
0,
-4.835388535288831],
[ 292,
146,
0,
0,
-4.835388535288831],
[ 300,
150,
0,
0,
-4.835388535288831],
[ 308,
154,
0,
0,
-4.835388535288831],
[ 316,
158,
0,
0,
-4.835388535288831],
[ 324,
162,
0,
0,
-4.835388535288831],
[ 332,
166,
0,
0,
-4.835388535288831],
[ 340,
170,
0,
0,
-4.835388535288831],
[ 348,
174,
0,
0,
-4.835388535288831],
[ 356,
178,
0,
0,
-4.835388535288831],
[ 364,
182,
0,
0,
-4.835388535288831],
[ 372,
186,
0,
0,
-4.835388535288831],
[ 380,
190,
0,
0,
-4.835388535288831],
[ 388,
194,
0,
0,
-4.835388535288831],
[ 396,
198,
0,
0,
-4.835388535288831],
[ 404,
202,
0,
0,
-4.835388535288831],
[ 412,
206,
0,
0,
-4.835388535288831],
[ 420,
210,
0,
0,
-4.835388535288831],
[ 428,
214,
0,
0,
-4.835388535288831],
[ 436,
218,
0,
0,
-4.835388535288831],
[ 444,
222,
0,
0,
-4.835388535288831],
[ 452,
226,
0,
0,
-4.835388535288831],
[ 460,
230,
0,
0,
-4.835388535288831],
[ 468,
234,
0,
0,
-4.835388535288831],
[ 476,
238,
0,
0,
-4.835388535288831],
[ 484,
242,
0,
0,
-4.835388535288831],
[ 492,
246,
0,
0,
-4.835388535288831],
[ 500,
250,
0,
0,
-4.835388535288831],
[ 508,
254,
0,
0,
-4.835388535288831],
[ 516,
258,
0,
0,
-4.835388535288831],
[ 524,
262,
0,
0,
-4.835388535288831],
[ 532,
266,
0,
0,
-4.835388535288831],
[ 540,
270,
0,
0,
-4.835388535288831],
[ 548,
274,
0,
0,
-4.835388535288831],
[ 556,
278,
0,
0,
-4.835388535288831],
[ 564,
282,
0,
0,
-4.835388535288831],
[ 572,
286,
0,
0,
-4.835388535288831],
[ 580,
290,
0,
0,
-4.835388535288831],
[ 588,
294,
0,
0,
-4.835388535288831],
[ 596,
298,
0,
0,
-4.835388535288831],
[ 604,
302,
0,
0,
-4.835388535288831],
[ 612,
306,
0,
0,
-4.835388535288831],
[ 620,
310,
0,
0,
-4.835388535288831],
[ 628,
314,
0,
0,
-4.835388535288831],
[ 636,
318,
0,
0,
-4.835388535288831],
[ 644,
322,
0,
0,
-4.835388535288831],
[ 652,
326,
0,
0,
-4.835388535288831],
[ 660,
330,
0,
0,
-4.835388535288831],
[ 668,
334,
0,
0,
-4.835388535288831],
[ 676,
338,
0,
0,
-4.835388535288831],
[ 684,
342,
0,
0,
-4.83538
```

```

0,
-3.8102426495131967]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.57763671875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.872261136385245,
'others': { 'beta_coeffieicients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.126749766362046,
-4.835388535288831,
-3.8102426495131967],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
2,
0,
0,
-6.126749766362046],
[ 12,
6,
0,
0,
-4.835388535288831],
[ 20,
10,
0,
0,
-3.8102426495131967]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.05909938957877547,
0.05909939012498705,
0.4766831071513548,
0.4767050288044724,
0.476729910017027,
0.47672998527996446,
2.99469059821818,
3.0000368876553107,
3.03481022556516,
3.0348102255651677,
3.0960777044795975,
3.0960777044802072,
3.096077704551025,
5.243967325820749,
5.24396732593715,
5.696137660948558,
5.6961376610109316,
6.622089050039713,
6.65393172347244,
7.224334835522734,
7.698532497695266,
7.80231633505566,

```

8.26676599072812,
9.142506968904602,
9.143106298076189,
9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,

```

0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                1.0,
                                1.0,
                                1.0,
                                1.0,
                                1.0,
                                1.0],
                                'cn': [ 0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5],
                                'r': 1},
'def': 'ising'},

```

False,


```
0.5,  
0.5],  
      'r': 0},  
      'time': {'max': 10.0, 'min': 0.0},  
      'type': 'heisenberg'}},  
'init_param': { 'value': [ 0.0008444707970882174,  
                             0.0011376501074209704,  
                             0.5478052879830236,  
                             0.5478052879830236,  
                             0.5478052879830237,  
                             0.784723345688809,  
                             0.784723345688809,  
                             0.784723345688809,  
                             1.9248523134122373,  
                             3.1827054999432645,  
                             3.5310059925523927,  
                             3.5310059925523927,  
                             5.052695732413606,  
                             5.052695732413606,  
                             6.136548648234324,  
                             6.136548648234324,  
                             6.785869019003799,  
                             6.785869019003799,  
                             7.198041513907379,  
                             7.3330889202217895,  
                             8.496582874867597,  
                             8.496582874867597,  
                             8.496739633755235,  
                             8.496739633755235,  
                             8.496741833065137,  
                             9.22890579273074,  
                             9.930471725964209,  
                             10.003370383780338,  
                             10.15144599090014,  
                             10.665059744564415,  
                             2.9193389895672706,  
                             2.873179484562977,  
                             0.24817648473142762,  
                             -1.1978615116990414,  
                             0.5013428170881429,  
                             -0.14698874126247163,  
                             -0.1346344592985022,  
                             -0.18688791288547063,  
                             0.30396646405908945,  
                             -0.3893332461107723,  
                             -2.4132158231627026,  
                             -1.3025870666843387,  
                             2.486320477273128,  
                             -0.6536413182583717,  
                             1.1565986435504452,  
                             -1.1840220246209,  
                             0.8322463862945216,  
                             0.686475242636196,  
                             0.9748739365898487,  
                             -2.032798889096391,  
                             1.2977623784552696,  
                             -0.21362311033600742,  
                             0.9967979178359492,  
                             -0.7714825960035517,  
                             0.05140148475667656,  
                             1.2463395942761013,  
                             0.6604875776774448,  
                             1.0077664010977474,  
                             -1.5169046363817273,  
                             2.1374704261116553,  
                             -0.3986317675051551,  
                             0.9979173289768921,  
                             0.5109160756321709,  
                             -0.5699092281256025,  
                             0.06259493290058689,  
                             0.09572934677993615,  
                             -0.1276069942134142,  
                             0.5134766913530642,  
                             0.20814668833639585,  
                             -0.07283729618715824,
```

-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,

False,

```
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.344797923049557],
[ 12,
6,
0,
0,
-4.988789667979189],
[ 20,
10,
0,
0,
-3.915584148696589]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00022673606872558594,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.128853076505154,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.344797923049557,
-4.988789667979189,
-3.915584148696589],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
2,
```



```

0,
0,
-6.344797923049557],
[ 12,
6,
0,
0,
-4.988789667979189],
[ 20,
10,
0,
0,
-3.915584148696589]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,
9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,

```

3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,
-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,

2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517]],

```
'noise_profile': { 'noise_on_init_param': { 'status':
```

```
'value': 0},
```

```
'noise_prob': [0.001, 0.001, 0.0, 0.0],
```

```
'status': True,
```

```
'type': 'dephasing'},
```

```
'nqubits': 7,
```

```
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
```

```
'def': 'ising'},
```

```
'output': { 'draw': { 'fig_dpi': 100,
```

```
'status': False,
```

```
'type': 'png'},
```

```
'file_name_prefix':
```

```
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#8'},
```

```
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
```

```
'run': 'zne',
```

```
'state': 'dmatrix',
```

```
'vqe': { 'iteration': 1,
```

```
'optimization': { 'algorithm': 'SLSQP',
```

```
'constraint': True,
```

```
'zne': { 'data_points': [ [ 4,
```

```

0,
0,
-4.972375668379768],
[ 20,
10,
0,
0,
-3.9295998440454545]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.4345855712890625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.025432252207006,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
2,
0,
0,
-6.2758943446211575],
[ 12,
6,
0,
0,
-4.972375668379768],
[ 20,
10,
0,
0,
-3.9295998440454545]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.3358069569174437,
0.33751316628129896,
0.37613413155379394,
0.9037599675903363,
0.941040799157583,
1.6105111871436553,
1.7638163741391246,
1.7638163741391246,
3.06849123809687,
3.600239571878297,
4.175689787149765,
4.175689787149765,
5.544091982519477,
5.544091982519477,
6.308463651138072,
6.308463651138072,

```

6.308463651138072,
6.3184465106806345,
6.521816085515529,
6.521816085515529,
7.288786933299201,
7.54709747254392,
7.597867588518987,
7.962268849194133,
7.991785735138846,
8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,

```

-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,

```

False,

```

0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.523058281578953],
[ 12,
6,
0,
0,
-5.228944859106493],
[ 20,
10,
0,
0,
-4.1973377330967105]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.2438507080078125e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.268554853988688,
'others': { 'beta_coeffieicients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.523058281578953,
-5.228944859106493,
-4.1973377330967105],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.523058281578953],
[ 12,
6,
0,
0,
-5.228944859106493],
[ 20,
10,
0,
0,
-4.1973377330967105]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
```

```
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.26700956978337703,
0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,
9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
```


-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,

False,

'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric3_sample#9'},

0.375],

5.21875,

```
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           -6.340393756989085],
                        [ 12,
                           6,
                           0,
                           -4.976052575965789],
                        [ 20,
                           10,
                           0,
                           -3.898762699309893]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.3392181396484375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.130208586638509,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                'cost_richardson_zne':
'sorted_expectation_vals': [
```

-6.340393756989085,
-4.976052575965789,
-3.898762699309893],

```
        'sorted_noise': [6, 18, 30]],  
        'sampled data': [ [ 4,  
                             2,  
                             0,  
                             0,  
                             -6.340393756989085],  
                          [ 12,  
                             6,  
                             0,  
                             0,  
                             -4.976052575965789],  
                          [ 20,  
                             10,  
                             0,  
                             0,  
                             -3.898762699309893]],  
        'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                                           'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                           'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}},  
  'init_param': { 'value': [ 0.17583917736716262,  
                              0.17583917736716265,  
                              0.9725326222396186,  
                              0.9725326222396184,  
                              1.154796264329948,  
                              2.3531638611520562,  
                              3.742172656987498,  
                              3.810257862376578,  
                              3.8745068996857803,  
                              3.8745068996857803,  
                              3.8745068996857803,  
                              4.643045833818014,  
                              4.643045833818014,  
                              4.643045833818013,  
                              4.643045833818014,  
                              4.643045833818014,  
                              4.711671630422156,  
                              5.405427388054719,  
                              5.405427388054719,  
                              7.4602327607935,  
                              7.610173264016766,  
                              7.858440336330247,  
                              7.968223945319675,  
                              8.51525712638587,  
                              8.54015380483253,  
                              8.797039458545115,  
                              8.797039458545115,  
                              8.858280642077405,  
                              9.30423425746159,  
                              11.6502765461574,  
                              1.4243949815721635,  
                              1.853782837605556,  
                              0.616412956408157,  
                              -0.43476006789942184,  
                              1.5440059756230975,  
                              1.1700457070837984,  
                              -1.8512161685606496,
```

0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,

```

0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,

```

```

2,
0,
0,
-5.871225062671609],
[ 12,
6,
0,
0,
-4.462319254461681],
[ 20,
10,
0,
0,
-3.3828918005869286]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.8623809814453125e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.699232349652263,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-5.871225062671609,
-4.462319254461681,
-3.3828918005869286],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-5.871225062671609],
[ 12,
6,
0,
0,
-4.462319254461681],
[ 20,
10,
0,
0,
-3.3828918005869286]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,

```

3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,

```

0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,

```

False,


```

1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.1433206567045895],
[ 12,
6,
0,
0,
-4.62093405872061],
[ 20,
10,
0,
0,
-3.4426712688431165]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.2438507080078125e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.0335603837365115,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.1433206567045895,
-4.62093405872061,
-3.4426712688431165],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.1433206567045895],
[ 12,
6,
0,
0,
-4.62093405872061],
[ 20,
10,
0,
0,
-3.4426712688431165]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,

```

```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
```

-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,

```

2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': { 'status':
False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric3_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.39007364379084],
[ 12,
6,
0,
0,
-5.048002042316808],
[ 20,
10,
0,
0,
-3.9807470593727396]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.220008850097656e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.164165676476593,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne':

```

5.21875,

-6.39007364379084,
-5.048002042316808,
-3.9807470593727396],

'sorted_expectation_vals': [

'sorted_noise': [6, 18, 30]],

'sampled data': [[4,
2,
0,
0,
-6.39007364379084],

[12,
6,
0,
0,
-5.048002042316808],

[20,
10,
0,
0,
-3.9807470593727396]],

'sampling': 'default']]],

{ 'config': { 'ansatz': { 'gateset': 1,

'layer': 30,

'ugate': { 'coefficients': { 'bn': [0,

0,

0,

0,

0,

0],

'cn': [0.5,

0.5,

0.5,

0.5,

0.5,

0.5],

'r': 0},

'time': {'max': 10.0, 'min': 0.0},

'type': 'heisenberg']]],

'init_param': { 'value': [2.1921898489539642e-17,

1.3966237144281936e-17,

5.875629127222605e-17,

1.0041795331735186,

1.0493846336909358,

2.3268506812305976,

3.014247762257123,

3.5587330689470917,

3.598019824659659,

3.598019824659659,

3.598019824659659,

4.594861139703674,

5.132217790928667,

5.132217790928667,

6.816072831791245,

6.906986102298665,

7.2472760308271145,

7.2472760308271145,

8.200086671450705,

8.200086671450705,

8.200086671450705,

8.200086671450705,

8.942806427307234,

9.150430959012654,

9.150430959012654,

9.436537073101752,

9.664513899938346,

10.186874850218704,

10.186874850218702,

10.784457886918862,

1.4923280250442645,

-0.2249463453233081,

0.03240277963395433,

0.031811746673685456,

1.8117023364465294,

-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,

```

1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.1385824590558777,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1}},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noiseefree_time_evo_dephasing_ric3_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```

[illegible]


```

0.0,
0.0],
'status': False,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1}},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'heisenberg_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 '
'+ 1 Z 1 + 1 X 2 X 3 + 1 Z 2 + '
'1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 '
'+ 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
'1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '

```

```

'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']},

'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 6.16720116017694,
                                      6.053414910848409,
                                      6.077121337791871,
                                      6.207207195486473,
                                      6.262374330970613,
                                      5.986895693279655,
                                      6.205368029344304,
                                      6.088664307040117,
                                      6.103832976424956,
                                      6.272477585483895],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [0, 0, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status':

'type':

'odd_wires': 4},
'optimized_minimum_cost': [ -6.619825343373176,
                             -6.869318772840936,
                             -6.5087775704967585,
                             -7.1560679492790396,
                             -5.981413153350314,
                             -7.048298049430781,
                             -6.747425000505078,
                             -7.036568265783533,
                             -6.711138717392998,
                             -6.584576110804561],
'optimized_parameters': [ [ 0.01871413497894055,
                             0.018714134978940724,
                             0.2613379173642502,
                             0.26133791736425005,
                             0.6318216555020777,
                             0.8372961654276051,
                             1.821968513401251,
                             3.1474951432390372,
                             3.1474951432390372,
                             3.1485170671950535,
                             3.148517067195054,
                             3.9814416284807077,
                             3.9814416284807073,
                             3.981441628480708,
                             4.856140958800601,
                             4.8561409588006015,
                             4.8561409588006015,
                             5.754795754733104,
                             5.754795754733105,
                             6.303270741932773,
                             7.392681786700986,
                             7.513290762899488,
                             8.03209503507907,
                             8.03209503507907,
                             8.535813710002731,
                             8.54624263805951,
```

9.443687821611801,
9.503087561223712,
10.313956361986106,
10.900532627949254,
2.6461639258839518,
1.5417589283675894,
0.2257956920933908,
-1.0768335736951873,
0.8613207946208413,
2.1849773957616625,
-0.9107734594639508,
-2.1667382648752667,
1.867688722281462,
2.3668974556947022,
0.13339305568895185,
-1.4570319604035509,
1.3777202351900353,
3.4700392631056327,
-0.9559545761912553,
0.4368026830493454,
1.3334925579084722,
-0.4156350852698361,
-1.4414158014372636,
0.7393914946360534,
0.3010801567914183,
0.362122777206485,
-1.1765017516736327,
-0.5545301330477921,
0.17858637974936872,
-0.27306390316354434,
1.2204707357225755,
0.10264508982433834,
1.311297488937931,
0.022638960119600232,
1.2224185268837013,
0.09419012847959232,
-0.9858808684001222,
1.7667365936947288,
2.0543865297241832,
-1.313309276179505,
0.6052943491453872,
2.920794362822403,
0.36729426823797706,
-0.4907693755345126,
-0.4907652929618582,
1.773301393432159,
1.0324005805505643,
0.15690855137959606,
0.17275788677592632,
1.6408842546175209,
1.1810448498033803,
0.5094988282552109,
-0.8827869902210956,
1.2283442273595282,
0.6657750092218239,
0.5822857875828481,
0.3981602545186293,
2.2986803263433013,
0.20507142489757735,
1.166074901081292,
-0.35093625013481605,
-0.4603785745816099,
-1.2308941818850956,
-2.0204446136516436,
0.7427864901575015,
0.49035804674601163,
-0.17818011332340386,
2.306444847843727,
0.8423842147773574,
0.3763774472076569,
-0.8127513813727298,
-0.6393080466279192,
-0.606760192596074,
1.239376081712239,
1.4934740200078889,

0.45014851381382026,
0.16024116637379804,
0.5909714786810217,
-0.23164751524086005,
-0.5232030433099774,
0.294735067059697,
0.674054302120824,
0.1291960398503005,
0.724266435798732,
0.6177233678090197,
0.8069752318018093,
-0.09640123299025632,
1.7506436772618832,
2.1071360750039685,
0.0898164078382077,
0.45058491103669474,
-0.8122072218131493,
-1.300894827674866,
2.0270032652414245,
2.3174633740346895,
-0.4244649110516584,
-0.922127126878378,
-0.128757892981865,
-0.05614505811514483,
-0.31034827471285004,
1.1560780386513783,
0.9427392383108351,
3.464248310365335,
1.1861860813777607,
0.6548919301302338,
-0.47688935032578766,
0.31878656040010656,
1.4181187106551563,
0.5034359418789143,
0.06789612685201406,
1.2910094025873902,
0.6395363733901798,
0.8068834648629276,
0.20002934026866853,
1.123820073617421,
0.5141860951115667,
-0.15567109011068703,
1.389800114049256,
1.3314801123479834,
0.5620442654487701,
-1.002959512005978,
1.0830163724627513,
2.3672930390807605,
-1.322930394436911],
[9.119557350302325e-19,
0.25959468526947127,
0.6593100126785064,
0.9010070143348925,
2.864384142587381,
2.864384142587381,
2.8643841425873813,
2.864384142587381,
2.8643841425873813,
2.8643841425873813,
4.6531454467618625,
4.6531454467618625,
5.19853052504346,
5.19853052504346,
6.184870510697861,
6.184870510697861,
7.044770226594686,
7.107215243020832,
7.686493340293053,
7.686493340293053,
7.686493340293053,
7.686493340293053,
8.066147968701737,
8.483745150563264,
8.490894072830061,
8.7108807618689,

8.7108807618689,
8.90963636528622,
9.449700779737539,
10.622376926466515,
1.4198721365577263,
2.0532215860895215,
-0.6671090661407124,
-0.1784736484260618,
2.452830737154847,
0.6445964871090722,
-0.5699863409114161,
-0.11908529766890505,
0.11072609611018341,
0.5517633838461717,
0.258947519950729,
0.2370221833218279,
-0.08827498348755955,
1.5529545851776476,
1.4270685728467551,
-1.297406543763459,
0.21671603620015034,
0.8053327735799074,
1.2240347618351852,
0.19490372040109893,
0.4463632351173059,
0.5723077007901808,
0.7291097022188683,
-0.9340621080403456,
0.7814401198421698,
1.335934708679698,
1.6419357207705991,
-0.32432865447217823,
-0.5119906202275178,
0.4069043309823562,
-0.9892467176495091,
0.2984840383067449,
0.8657043381513185,
0.3531089935489972,
1.3451552159797637,
-0.6265940619359847,
-0.48694952223842436,
0.8839372319696429,
-0.7937473350851352,
-0.6284369925286846,
0.988187027480221,
0.14204093422108516,
-0.7928447811689399,
-0.02575893659255237,
0.6034077764970499,
-1.0594098028179662,
-0.5497734939439616,
0.025642354245823534,
-0.20869881122054,
-1.2359715138124214,
-0.2946232864432856,
0.1124699418659181,
0.8079624623340408,
0.3058649198413362,
0.16875024197938412,
-0.4319850590361839,
0.9532996200659013,
0.9710216361923732,
0.3268179051781202,
-0.9107075661609054,
-0.5341988035286006,
0.13994538780187468,
0.9868907551406109,
0.05744643111324586,
-1.1377247833961386,
0.5139551019130977,
0.5537607458906798,
0.2287706491940902,
-0.44737160486571353,
-0.22916413274711533,
-0.08417189284299845,

-1.3242046442500788,
1.6166580464751512,
0.8079048681248637,
1.3145961695384563,
-0.18697892770553287,
0.7313225410666617,
-1.041621052087622,
0.2643819048238151,
1.1433173955584817,
1.1335781234696722,
-0.8444186654603157,
0.6239242025717586,
1.3684750637982472,
0.3938801567657146,
0.2134741750368225,
-0.5543571548158037,
0.675866319898802,
0.5827634828639766,
0.08356178045592735,
0.7945300964131254,
0.5755633000801618,
0.035712817174237056,
0.8694548740378415,
-0.8796227630992239,
-0.724563085043669,
0.28387680573365254,
0.6162107997557882,
0.18804506353720024,
0.13573104516153978,
0.4121054765005755,
0.24383414393551647,
2.1177520472698306,
0.2513566624255866,
0.1835043860190697,
-1.315482809191855,
-0.0799177390764499,
0.7017032401307175,
-0.20268293353480266,
0.014920329005843978,
1.422493387523363,
-1.3230470412127697,
-0.8499409925206722,
-1.9452530607024414,
0.9453797675102356,
-1.2987244862149754,
-0.31790218075907056,
-1.4580404827053768,
0.7015060173886379,
-1.9161725887237533],
[1.1127990762108912,
1.1143426527141589,
1.77370013920208,
1.9993547683714867,
1.9993547683714867,
2.001672770722739,
2.001672770722739,
2.497477077662224,
2.497477077662224,
3.412019182336653,
4.153064085946221,
4.153064085946221,
4.153064085946221,
4.158592712098033,
4.189306073689003,
4.856266090926401,
4.860230532882879,
5.928763246664035,
6.074154358531821,
6.551844343127418,
6.551844343127418,
7.366351465372161,
8.010206307388854,
8.49993655633506,
8.49993655633506,
9.359509952123227,

9.969968384422238,
10.513762742024985,
10.513762742024985,
10.640701635795219,
1.595436628312185,
-0.3811088004495,
-0.8909401299341706,
-0.3821976726093325,
2.209455823932932,
0.4084507894487605,
-1.3491711508388657,
0.18219394232884736,
-0.1991395495968878,
1.3877045909644812,
1.2810472453057546,
-1.8198938592553289,
0.691696952821756,
1.9651275722497656,
0.8952970332383311,
-1.572888015515471,
0.10018127814164592,
2.5087678810448395,
0.2303656982914484,
0.6644753240981718,
-0.2940767603661997,
0.9937376992458091,
0.19118749930842235,
-1.7309245646223925,
0.6920377423752287,
1.0669653712016844,
-0.2725348409017723,
0.2108196720828406,
0.7600208432998801,
-0.39468875007311516,
-1.6426053139934083,
0.22140594406577463,
0.3798758679922964,
-0.13579541355890495,
-0.502502519286503,
-0.07503549665980291,
-0.24005906561438198,
0.2999706370689256,
0.4291869988824075,
0.49459447229554165,
-0.2693528321546647,
1.3233447406041847,
0.010128993489178412,
-0.7350427372075996,
0.588999296326396,
-0.03887378376443501,
0.7842242805141085,
-1.017856875421701,
-0.050420029711616104,
1.7478460277120142,
0.46693496239216137,
-1.390597925219579,
-0.1527982976709753,
1.3561114516684873,
-0.11782202169904106,
-0.6172298563496877,
0.2337872361631034,
1.5343227540197375,
-1.246330933330494,
0.2671789790306671,
-0.5231899648601245,
-0.2993536711140956,
-0.3794897720952542,
0.5953851600991978,
-0.016488955527838106,
-0.04239234937973134,
-0.8530294973472092,
0.09298474140637386,
-0.6610063165341049,
-0.7501983006092152,
-0.23667128690009429,

-0.49322661069162854,
0.2817510492822266,
0.8804151571222927,
0.25487104244438136,
-0.5104439984528633,
0.8221175331336923,
0.42372453837884605,
1.1270985005003262,
0.10209111179644431,
0.4047521854484605,
-0.9498909052734678,
0.8591561814039869,
-0.4549855289146429,
-0.7085768506420614,
0.2460272461737723,
-0.08852438964027515,
-0.28170619044977985,
0.18782561055277286,
-0.3648963858755429,
-0.37968971213994723,
0.35188735669812166,
1.8917682407126328,
0.37786421566408035,
0.7636381567370941,
0.07799046336192586,
-0.23205705900380685,
-1.186712365635249,
-0.10791936865692263,
1.6876574701547749,
0.03719295338640027,
-0.062311995058818626,
-0.5208937011021405,
0.4557021320307782,
-2.7664261049727545,
-0.8770083117671578,
0.9937005253370358,
1.250538444528313,
0.6461816519581883,
1.6953467436382619,
2.0814745138972848,
-0.7151854978123715,
-1.6567720673097688,
-1.1703575007457885,
0.44660262232395087,
0.012526409819125981,
-0.0348982634801462,
2.0503923727756113,
1.154948415623314,
0.9176857519192531],
[3.101411838661581e-05,
0.00028167581912982354,
0.002108388113032011,
0.0021083881149554745,
0.002110224126734301,
0.013404723029560417,
0.44387140902527444,
0.4998584267335595,
2.519337827669618,
2.592441150742434,
3.279416160731671,
4.179752140912941,
4.179752161353468,
5.661115424822064,
6.166122332331656,
6.1661223340249105,
6.166122334025239,
7.1531688903854835,
7.165666077614885,
7.257231636090788,
7.810595355153735,
8.056713547662405,
8.059562409019513,
9.326100481695063,
9.326100481704493,
9.781728007060766,

9.888676215095867,
10.278005103664306,
10.559163217755993,
10.559486031108996,
1.2851036574167212,
-0.72547345631182,
-1.4718960991273107,
-1.209251703975404,
-0.016838854839331462,
1.1653703713624246,
-0.21930660603454502,
-0.6325267011020643,
0.27235598773055514,
1.3431283556950424,
0.29725144436322054,
-1.2158342217245894,
0.8631495871791491,
0.12397167154542266,
1.116880444349936,
-0.5886908967334331,
0.37424943307839126,
-0.4246401431552608,
-0.052554167142824744,
-1.151521387506872,
0.45352480044418075,
0.2968389539105363,
-0.36994249285514086,
0.029824637432390475,
0.7039777013639383,
1.0545235270667865,
0.5948019323555648,
0.9185753062546861,
-0.3475338326570484,
1.2642221406528609,
-0.6424417079073889,
-1.8836355388168116,
-0.1226587707935595,
-0.751178954303451,
1.2759264830517028,
-0.7303074304229613,
1.0498129361868127,
0.29592762783390236,
0.4716900159260768,
-0.20829913915791434,
-0.8217397159362575,
0.8503209585884646,
-0.49309125036010926,
0.07716668431879356,
-1.7384586268468483,
0.5748268283601387,
-0.01480830786786586,
-0.03742465163157027,
-0.9568191257597272,
0.07510834673857922,
0.021240686884975563,
0.4240852295635142,
0.10265729129130313,
-0.812822916440861,
-0.14268033459407103,
-0.3753169735052697,
-0.7175759512497852,
0.03918867047505918,
-0.7186687987691884,
-1.6942314888856695,
1.268466372526145,
0.3288584725277509,
-1.242111559182382,
1.113325215651688,
0.40176863135675694,
0.15033001800176832,
-0.8493210333167794,
-0.7302387336933833,
0.5331942268313786,
0.8016954161004247,
0.11571359434030207,

0.9513684680035512,
-0.26463380223931005,
0.5608071632066808,
-0.40528112642254077,
-0.6754638378880663,
1.5778663953317416,
0.16647468944212254,
0.24883728001741243,
-0.9973732709393777,
0.3409424132427824,
-1.113466076369018,
0.15519481593579493,
0.641216026706457,
0.8449849609678267,
1.911891894634777,
-0.34738047822667395,
1.5893587533055515,
0.4884297002665817,
-0.25227237110126166,
1.5183173129023069,
0.5103598279259709,
1.4687346401745565,
2.532539737478952,
-1.540460532405588,
-0.8294616308793386,
-0.2412174307882085,
2.6406875676770416,
1.2361221502323787,
0.13408461910385386,
0.270490641328192,
-0.5308435068885323,
-0.8511283733806999,
0.26595037966666163,
-0.8831519048460094,
-0.4997472202494575,
2.276782887198034,
1.5876131255720232,
1.1044306497561618,
0.22518259496424625,
-0.12906375147756488,
2.4499067189894523,
0.5016533011795579,
-2.1824706389019832,
0.9088257454879881,
-1.3873459886882669,
0.43069183082779305,
0.5835382283652092,
1.9517705176455047,
0.639974902940826],
[0.8405962708734059,
0.8409366166472388,
1.2852356970263128,
1.9656905972413197,
2.6709433729539978,
2.670943372994037,
2.678176776414865,
2.738130595852174,
2.9440618301573322,
2.944061837308021,
3.5946493033065674,
3.5946493033066593,
3.5946493377836535,
3.5946493377949182,
3.6475297284551167,
4.423523257877865,
4.423523257879113,
4.744549401355183,
5.030736177428605,
5.030766895303107,
5.030766896066702,
5.030798272122615,
6.102563406307042,
6.3550093593541375,
6.35500935935416,
7.56205183282469,

8.453276420655834,
9.985009441062077,
10.30999329905164,
11.043022236178203,
2.1889940156552368,
1.249116821544238,
-0.6605760096192125,
-2.1135047950110155,
2.129245868952901,
0.511519543367031,
-0.8638463209571715,
-0.20202526205492455,
-0.5988641121015101,
2.145161094485569,
1.4958876569154136,
0.9003565348629307,
0.8986066176810353,
-0.0859890065364003,
-0.08547563138217676,
-0.6842205219399397,
1.347771315836974,
1.9235660701575357,
0.5300781712046526,
0.9684639862422223,
1.8099194640434915,
0.6789385897860352,
0.46838913395608867,
-1.9553664340235406,
0.15424587849578802,
0.7464909631769433,
0.28409739802447187,
0.5646186506463579,
-0.0688718084707713,
1.3403239673426846,
0.7803011173906335,
-1.2634531934792825,
-0.403184603220626,
0.9911656930801824,
-0.27952114699970115,
-1.17647727881294,
0.6160770530384655,
0.7910105296614427,
-0.26329493021464934,
-0.5093546557195857,
-0.078946955577166,
1.6118855098996396,
-0.8560858484853413,
0.43713047898075813,
0.15951625123225024,
1.1017815594363278,
-0.5112214984829941,
0.8616309140244529,
-0.2539134574216648,
0.6900329022420518,
0.7767771669699943,
1.3495069152164292,
-0.10718702519877486,
0.23011810567539243,
-0.4233144560883979,
1.0277501391374537,
-0.13408439221848256,
-1.2122213001358895,
-0.9574798764432794,
-1.272419025076084,
0.6697485644230207,
0.09252521117768966,
0.8443957240867669,
-0.5978394384474206,
0.4330892667970026,
0.07884699311842308,
1.140608340632257,
0.07864060845749296,
1.1489940789117206,
0.02443540527852337,
-1.0034700299434305,

-0.1412123681218965,
0.42135897985221993,
0.00946489374514043,
0.37971823276367245,
0.6021380541926072,
1.71390465753914,
-0.5080953537535351,
-1.7394955866645057,
-0.55775760641077,
-1.6305248910687546,
0.4704683857374631,
2.1610569951035585,
-0.11166064159089376,
0.47170657685906686,
0.07305148434226297,
0.6150465742311105,
-0.5141382282093759,
-0.8714158035322278,
-0.543899342278249,
-0.22729288133654413,
-1.1237376131773689,
0.5077284416312242,
-0.02493907277109512,
-0.061410284019419926,
1.7654158813927403,
0.24648921891280573,
0.026991979318021532,
-0.4476437051863693,
1.0288055112299304,
0.2791879596887287,
0.552260553667315,
0.5446649167338916,
-1.1781972761427792,
0.24145179880489284,
-0.6620357948168841,
-0.48109037039574104,
0.4693605610272632,
1.5881294256793275,
-0.6259573215794967,
0.502524529722019,
0.9011500652440121,
0.31953241444072733,
-0.0763403773509461,
0.8504085952834295,
-0.9720571531313896,
2.085636878242165,
0.7499605974794749,
1.9701875686083887,
1.4162728880693862],
[0.36934729750713446,
0.3693472975071345,
0.4408171698279066,
0.7641387276708653,
0.7641387276708653,
1.2405539686841192,
1.9615715228812416,
2.180937952046039,
2.956248658909984,
2.956248658909984,
3.615620941947,
4.0363267766272015,
4.283148746851272,
4.283148746851272,
4.287727890051584,
4.294687143195195,
4.959378804296946,
4.959378804296946,
5.299283731795155,
6.02722363939295,
6.706058929046358,
6.706058929046358,
8.58973622967867,
8.58973622967867,
8.735021968997517,
9.724832411320188,

9.882458613321258,
10.054210085109215,
10.144098772484403,
11.057581665321372,
0.6138165123929756,
0.45636276738375425,
-1.6225613647488533,
0.8571182758975707,
1.411989147942658,
1.6605353912366554,
0.31855217609147096,
1.1904862760406203,
1.4473424068486804,
0.687246532966651,
-0.3805095501565646,
-1.4931296153328852,
2.326224921001399,
0.9788391575979964,
-0.08794459312808878,
-1.606604838553076,
0.045095161702097514,
1.1926561845584958,
0.9624943300259794,
0.8526934578403719,
0.3595367653801707,
0.6208577142062074,
0.23723466831561657,
0.8137494677209746,
0.017380156001790208,
-0.7727421772857405,
-0.7696652964277857,
0.28558266237504587,
1.2665997402365978,
0.030324559296025642,
-1.0215734949860629,
-0.435015538681086,
0.09892690725060405,
-0.021280714063487676,
0.6605217667305735,
-0.48357943656762575,
0.23903870372504285,
-0.20353372422303415,
0.40182288676040395,
0.36859613731879964,
-1.0350820651690253,
0.0783547441763707,
-0.5235573738556314,
0.20634909129668547,
-0.22551355647851357,
0.1673560398509943,
0.7468649723369892,
1.0994749531926022,
-0.2018797749925425,
-0.577779100669964,
-0.6205566643688822,
-0.25020034171796784,
-0.24121553611864016,
0.687677741000175,
-0.6022485096507909,
-0.7648342894907719,
2.68891232458234,
-0.038914232418390225,
1.516345449400868,
0.7772620643765826,
-0.5809695972554458,
-0.16788488824728048,
-0.11974201627038453,
-0.31585900129180533,
0.9730861233839377,
-0.23174660705736103,
0.10129760163123996,
-0.38814603215785604,
0.20257130727462175,
-0.5242689321123916,
0.5757755417303894,

-0.4147685482642834,
0.5155383127778999,
-0.9069003556538804,
-0.3931316447570312,
-0.22854166467415718,
0.08849924429856736,
0.11802474333811182,
-0.6697697401245005,
-0.4497368814763953,
0.8371559036446393,
-0.27762346391800535,
1.129889136554746,
-0.7313980725752758,
1.7322939714189498,
-0.30841564461339865,
0.08815569274790189,
-0.45821949503742376,
1.3903832187168985,
2.7423681670027595,
-0.709192498697989,
0.38042285795749525,
0.9330002997189369,
1.5220073637023084,
-1.9071482092674334,
-1.4201112861485827,
-0.18587844516118163,
-0.728966506004035,
2.779449113015483,
-0.5840054192970229,
2.1502692645150225,
-0.4443255851345571,
1.5289006083178085,
0.05130509171668746,
-1.6556175117860825,
-0.5876167642044237,
1.5144905946024154,
1.6252191316647524,
-0.7878469829469384,
0.4020569928547576,
0.833940688908704,
1.2564138083018812,
-1.9521172256438166,
-0.3660865019791046,
-0.7569121463795273,
0.08857539049178031,
0.04440194030774997,
1.346246034983794,
1.5664381297937569,
1.6926501995283607],
[0.07932463514989117,
0.07933390228500832,
0.6712683622930262,
1.8057579746144639,
2.2332189550395363,
2.2332189550395363,
2.2333175311875997,
2.2333175311875997,
2.5803200362154985,
3.149742042876931,
3.2745115489857257,
3.946611320928826,
3.946611320928826,
4.527074130340705,
4.527074130340705,
5.9153316417928945,
6.27481936788149,
6.2748193678814905,
6.718992557341518,
6.964319085694909,
6.964360455572235,
7.61675802651207,
7.61675802651207,
7.61675802651207,
8.031230813321006,
9.561856352158596,

9.835291253191413,
9.958392502186584,
10.272929617553823,
11.17699585622391,
2.3714062527042183,
0.008507065719361826,
-1.7490339115302889,
-0.9098574742643126,
3.2594955351617316,
2.2425950423007475,
-0.8261382715647944,
0.4893495058771166,
0.29218298380822827,
0.7749189883526413,
0.0732829317784662,
-0.427680076829418,
0.835806375311316,
1.2252687690883595,
0.5943555800897335,
0.41209106000270895,
1.8808578438044583,
0.8934024527009041,
-0.5658257233318801,
-0.24310864449817618,
1.2970958730834994,
1.2397558998854112,
-0.13432853133680148,
-0.33892864650006427,
1.2500942843462377,
1.2463490630241996,
0.4505745589505527,
-2.888971827979111,
1.391085002142137,
1.3755760556021903,
0.7798831597382527,
-0.6932853439946646,
0.13475625995390494,
-0.48377623448510626,
-0.4751082457272343,
-0.9134219715133307,
0.5602537728773588,
0.1457020102747123,
2.094813129086275,
0.16050162220250241,
0.255533199882997,
0.5798962545827221,
-0.3296339137986921,
0.20407592924333015,
1.002364909727297,
0.8996033431107227,
-0.40087766178798867,
0.21850323438685998,
-0.3148827979921678,
0.5255859647152507,
-0.981835244220926,
0.18194510004086287,
2.491369267725132,
0.32340553765254526,
-1.422371307906928,
1.3350586578994175,
-0.19038703199979035,
-0.18759028526199026,
0.31727468639934,
-0.13719279505644447,
0.8870810962833492,
0.5605043977550492,
-0.45997587805687895,
0.6203449810744726,
0.35343801179058787,
-0.29094576977122605,
1.0749874898125031,
-2.3258043801847514,
-0.023837190424521523,
0.8602839037993392,
-0.3076595227850746,

2.083827131184913,
-0.709910688132804,
1.9396556568539463,
-0.7118178630647494,
1.024487978597129,
-0.035154872964647726,
0.6794001499523024,
0.6455217590344408,
1.2833952770953792,
1.7594723933960992,
2.366727526539801,
-0.28632630179908214,
1.356168747751164,
0.31201643292476877,
0.364096860526536,
0.8226195660291765,
0.2699337047887902,
-0.4803548322886457,
0.7089202255488762,
1.9081826927626873,
-0.3868588206474107,
-0.05728681029210421,
-0.6103514866875638,
-2.670376487838684,
0.03980060533677062,
-1.0774346470258729,
1.4309826927584666,
1.4923854024295125,
0.4230703239257903,
1.1568773489185937,
0.7815638389040719,
-0.4834118045846874,
1.1499199224441905,
0.01202048539239927,
0.3105021058015101,
-0.3749329298855971,
2.2071268409248224,
0.7242633662153387,
0.6404704587343512,
-0.30896311918370867,
0.6474915553174321,
-0.6471036454486253,
-1.0139549843186435,
2.7326032351157084,
0.4640689770354461,
0.7197232245440613,
1.099219858753326,
0.5370670409426629,
2.2362623976980602],
[0.018174692163319142,
0.02381340297275515,
0.02421332579520929,
0.024525182595047955,
0.4460556740257204,
0.4463126705844747,
0.8080995635599163,
1.5479347906990655,
1.547935096587381,
2.726547871299545,
3.3242466002093267,
3.324246600509028,
4.111398089568711,
4.145752980013426,
4.146049797332313,
6.415277102843476,
6.572144069759118,
6.927329335261905,
6.927379209146919,
6.933957897577306,
7.907817926834528,
8.01227391181071,
8.377715493176042,
8.574692491681018,
8.665468291409324,
9.338599502827458,

9.57363802367466,
10.233446160908231,
10.471009909188941,
11.163878412926959,
2.236671766008264,
0.35159542629328217,
0.8146718981296579,
-1.5439346128384686,
0.8972133735648259,
-0.0553493415862735,
-1.090899451761626,
-2.2352667548818665,
1.2209528829332086,
0.5204242802804685,
-0.618288383541334,
-0.3171787171591956,
-0.025836377068857316,
1.3596603169147203,
0.4695190032736128,
0.11760891446657133,
0.44944359330584466,
-1.4358459977495694,
0.7103123867655211,
1.6252846466436086,
0.17089579517121128,
1.0480108474391834,
-0.21482373854468306,
-0.5796520218220037,
-0.39701726293440376,
-0.15484503866231408,
0.5161603909940105,
-0.08958972382743205,
-0.5178549594803129,
0.23176676286013279,
0.030062676987165922,
-1.5076956114966404,
-0.5464208818685263,
0.0502122423742188,
1.6977256264031593,
0.8909956552917536,
0.4009810515503771,
0.3307875796941022,
-0.5725943763367822,
0.23702814500484895,
-0.15434680984627044,
-1.2435315402871425,
-0.5149922580540566,
0.30951994519916565,
0.5096095319131432,
0.30303640166579543,
-0.3239136690540364,
-0.7206962091038183,
0.8555191973102813,
1.0120763935686161,
0.04024592288077597,
-0.2886549630459275,
-0.3277156141890048,
-0.5588571150564244,
1.5118849277062045,
-0.644801476310519,
0.35391159762671204,
0.036581088484196625,
0.07324701372485032,
-0.19410908530178353,
1.5993036761438029,
0.10651551669568386,
0.32822741091828495,
0.39824063300111046,
0.6674060102702002,
2.0756693329629017,
-1.5421800768661706,
-0.49667839050246243,
0.0124564376427262,
0.4601453458726383,
0.38665655977671154,

-0.3280159653083997,
1.37660238729691,
0.5459980296904141,
-0.10248779795861306,
0.3193368932413526,
0.022394510397351762,
-0.9106037248563306,
-1.8448022296193607,
-0.2878663593630255,
0.3486991485685759,
0.6803232619964643,
0.473978523083063,
0.7630725375206788,
0.8596671356435096,
-0.47717853860533954,
-1.1511176506358933,
0.044726686440788394,
2.206461610291793,
-0.251613145100058,
0.3827507786112393,
-1.0837839025823046,
0.5215057436278849,
-0.6433936043135513,
1.0423632753231085,
-1.5853099693823727,
-0.4899100779852182,
0.6771990537468359,
1.636801051947153,
0.09520300124188852,
0.7384792012405127,
-0.17322471627475994,
1.5991844229586945,
1.501804750971724,
0.5057839423702514,
0.16164961971359962,
2.292178045044038,
1.3450921233724473,
0.6682426373209176,
0.5914595140792696,
-0.5621151585806636,
1.4649155202102295,
-0.9414482790819952,
-0.10805380420514368,
0.9307303586657315,
2.4106406607028164,
-0.01284340130889141,
0.41649266055337153,
1.314622528108089,
2.7227723366977616],
[0.007743638893136335,
0.013160125498327223,
0.608035339315491,
1.107635474685494,
1.1648862034449325,
1.852802815648869,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.167992149172641,
2.586998837285682,
2.6225868976549798,
3.0707103569201006,
3.0707103569201006,
4.1228521273997165,
4.138888702236146,
4.138888702236146,
5.347444125894556,
5.347444125894556,
5.412363261255616,
6.659592716290906,
7.287740867223119,
8.540473259892655,
8.81303185082801,
9.303384987416969,
9.767355442925265,

9.939116248646888,
9.97261625652093,
11.225779420142475,
11.480121962431193,
1.6482545183670085,
-0.013015976711505003,
-1.0388467926799678,
-0.8652539511107419,
1.9084118027497516,
-0.9394653223441272,
-1.7468041006561723,
-1.5142947156306645,
0.2590641138232099,
1.3773414247538143,
-0.6292953946653098,
-0.8862298412863635,
0.3042825725334,
1.123835887892568,
0.9314084077434932,
-2.9280573641387413,
2.0853053971786153,
-0.34681890463267223,
-0.04690984704002986,
0.8070574318910052,
-0.5779942015518676,
1.0090669446225737,
-0.1775287190554287,
-1.6842820449103186,
0.763376948317048,
0.3384712786913095,
1.7367769407755378,
-1.3580490459136343,
-0.8837296315105077,
-0.8963079645873018,
-0.18784202847085701,
-1.4055444367955956,
0.671829961387558,
1.8019702343361095,
0.30718376850684487,
0.08135901917238793,
-0.18472697951808345,
0.5668536919232505,
-1.5958337467143746,
0.002509488020009276,
1.0849426355423382,
0.7452955666922372,
0.9398973688185909,
0.16370191516119634,
-0.20185726086923075,
0.5303635124624408,
0.4908517160520784,
-0.4888169221995994,
0.9032694192338024,
-0.8158468897545128,
0.7026292391012148,
0.011827947942572397,
0.6138675828667193,
0.1834450857087751,
-0.9915622527095793,
-0.5829508078463853,
1.064065245998552,
0.49061771989414976,
0.23606904394636888,
-0.20435452332134307,
-1.0928372780788398,
0.057443193557461136,
0.10260811255105017,
-0.8555456605404742,
-1.096930933544511,
0.04006347013812424,
0.7151909132583844,
0.4885582422451459,
0.09828549627753237,
-0.41856423316032376,
-1.5659826249535125,

0.35337608728639364,
0.5013401269099331,
0.2755554280225739,
0.123838948347921,
1.1982351112905236,
-0.5571464296579999,
0.227611732218496,
0.4903765072246047,
0.21638817013741035,
0.5329904920964728,
0.023244037871050734,
0.15105119714954549,
0.5432645474025147,
-1.6089466195523248,
0.5580452428696917,
0.2980389035466241,
-0.7189872564709854,
0.2941454207257381,
2.0652774083354823,
-3.197423241640053,
1.2171902372756198,
-0.9312606307980903,
0.7138860186292121,
0.7458573347434515,
-1.0460906183029233,
-0.4042108821911456,
-0.5284604323913634,
1.1561662773370494,
-1.0010987456125164,
-1.286779340316877,
-0.7935901981914314,
-0.6758512775621703,
-0.253355636189949,
1.083705550488119,
-1.5951444189401682,
-0.2763950378070173,
1.9938759834766908,
0.1467137702454683,
0.18970870490254518,
-0.6591405102572232,
2.3883322313766877,
-0.5426139401957691,
1.4573385975452764,
0.7440623559719126,
1.9382959176617307,
0.07233875596199366,
-1.2256407906907247,
0.8075799371504198,
1.6375675375176653],
[-2.670381383900579e-16,
0.07474127433216288,
0.9423639350936485,
0.9423639350936486,
0.9423639350936487,
0.9423639350936486,
1.8776414733739433,
2.6093592361765396,
2.6549976526560917,
2.6549976526560917,
4.309323876617324,
4.812226043223447,
5.3735992861643656,
5.998589378822616,
6.041219527369439,
6.041219527369439,
6.041219527369439,
6.455755583840882,
7.6346582000153935,
8.456131775478068,
8.456131775478068,
8.456131775478068,
9.265231361717003,
9.265231361717003,
9.265231361717003,
9.265231361717003,

10.025925584440094,
10.402247132343707,
10.703459031001074,
10.703459031001074,
3.3569835366016436,
0.9215266459605492,
-2.040479701749094,
-1.3412114880744723,
1.3650402871975202,
2.328108797667404,
-1.3186160481421627,
0.4833318128451331,
2.239983409823259,
1.0329578572589242,
-0.1517540238142264,
-1.4974261238923248,
1.7473459056580782,
1.4519306881870349,
-2.206161402631766,
-0.8907223077237956,
2.3813682348978937,
0.5973875310190014,
-0.8910649708758184,
0.26277461789671996,
1.6008335914847227,
3.025884654690384,
0.8431565773802462,
-1.3891693904439881,
3.086427239689084,
-0.3399924211336387,
0.9616731269325118,
-0.4538264391900641,
1.6045391274841239,
1.2066545032742897,
2.5203408436794805,
-0.7822247170003326,
0.5795340175268288,
1.1137644213878175,
1.338413506379888,
-1.170792708483999,
0.7895425377479481,
1.929612997870515,
2.1013047486982743,
-0.1747799610123698,
0.34124409150219404,
0.1345366524340121,
0.12829425701000113,
-0.07853692563407341,
0.10525428958729864,
2.4094888168745765,
0.6357560012375729,
0.3483402865911379,
-2.7204359945096264,
0.7864668566884239,
2.710084659416198,
-0.42133711298349386,
-0.7606725606109204,
4.024573300631783,
1.0416894631362523,
0.549877653228643,
-0.23095740430710715,
1.2081110571007485,
-0.35409975752454176,
0.9082778411762691,
0.9011042810290772,
2.2659344214863575,
0.030960161362888827,
1.709001611830215,
0.7901365980951575,
-0.25541821293373157,
-0.5438725550461083,
-1.6272180586430949,
2.7727467118019176,
2.937339318913365,
0.16303685557629694,

```
'run_time_sec': 34196.97281765938}}],  
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,  
                                         'layer': 30,  
                                         'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                                   0,  
                                                                                   0,  
                                                                                   0,  
                                                                                   0,  
                                                                                   0],  
                                         'cn': [
```

```

        'r': 0},
        'time': { 'max': 10.0,
                   'min': 0.0},
        'type': 'heisenberg'}},
'init_param': { 'value': [ 0.0008444707970882174,
                             0.0011376501074209704,
                             0.5478052879830236,
                             0.5478052879830236,
                             0.5478052879830237,

```

0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,
9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,

-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],

'noise_profile': { 'noise_on_init_param': {

'value':

'status': False,

0},

'noise_prob': [0.001,


```

0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '

```

```

'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.344797923049556],
[12, 6, 0, 0, -4.988789667979189],
[20, 10, 0, 0, -3.915584148696589],
[ 28,
14,
0,
0,
-3.058304699492839]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':

```

'dephasing'},

'noise_on_init_param': { 'status': False,
 'value': 0},
'noise_prob': [0.001,
 0.001,
 0.0,
 0.0],

True,

'dephasing'},

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
 'identity_factors': [3,
 3,
 0,
 0],
 'noise_level': [28, 14, 0, 0],
 'noise_profile': {

'status':

'type':

'odd_wires': 4}},
 'run_time_sec': 14.272984743118286}},
{ 'config': { 'ansatz': { 'gateset': 1,
 'layer': 30,
 'ugate': { 'coefficients': { 'bn': [0,
 0,
 0,
 0,
 0,
 0,
 0],
 'cn': [

'r': 0},
 'time': { 'max': 10.0,
 'min': 0.0},
 'type': 'heisenberg'}}},
'init_param': { 'value': [0.3358069569174437,
 0.33751316628129896,
 0.37613413155379394,
 0.9037599675903363,
 0.941040799157583,
 1.6105111871436553,
 1.7638163741391246,
 1.7638163741391246,
 3.06849123809687,
 3.600239571878297,
 4.175689787149765,
 4.175689787149765,
 5.544091982519477,
 5.544091982519477,
 6.308463651138072,
 6.308463651138072,
 6.308463651138072,
 6.3184465106806345,
 6.521816085515529,
 6.521816085515529,
 7.288786933299201,
 7.54709747254392,
 7.597867588518987,
 7.962268849194133,
 7.991785735138846,
 8.619190388880893,
 8.629220288817223,
 10.432794468229133,
 10.549342687924028,
 11.465905039453537,
 1.5708348469088267,
 -0.3283220618427727,
 0.03691181582691629,
 -2.5608652259642914,

1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,

```

-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],

```



```
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},
```

0,
0,
0,
0,
0],

'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [0.26700956978337703,
 0.2670095697833769,
 0.7897559248250247,
 1.60994438629722,
 1.6240137789724642,
 1.9029927716215564,
 2.754961581608366,
 2.754961581608366,
 4.416760838293873,
 4.472730549723469,
 4.472730549723469,
 4.472730549723469,
 4.9018244764195,
 4.9018244764195,
 6.142575905365314,
 6.99918307198117,
 6.99918307198117,
 7.3452528051326675,
 7.478132650082225,
 8.520612438756238,
 8.520612438756238,
 8.991228072316343,
 9.227791963806121,
 9.7884313312372,
 9.7884313312372,
 9.7884313312372,
 10.06004176448028,
 11.030427598537905,
 11.030427598537905,
 11.500583935823688,
 1.3913972994109516,
 0.07774370637080774,
 -0.49382826121080187,
 1.73722078706814,
 1.1477124735700583,
 -0.01244613489207998,
 0.1104474230926492,
 -0.4185268033829736,
 1.470187851781374,
 0.2683710961451789,
 -0.4350955069921617,
 0.47701522697148346,
 -0.12059167163697146,
 -0.29388493308433694,
 0.7760009417155218,
 0.2157710734488978,
 1.2495404382304187,
 -0.3273973942211235,
 -0.7657986587430435,
 -0.29167448156050724,
 1.0175732352511992,
 -0.26691063020915523,
 -0.9498568893096563,
 -0.7761349266449543,
 -0.03917968596759244,
 0.9253714376815375,
 0.10438717726675471,
 0.119240158066098,
 -0.8869854257116645,

0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,

```

0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '

```

```

'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.340393756989085],
[12, 6, 0, 0, -4.976052575965789],
[20, 10, 0, 0, -3.898762699309893],
[ 28,
14,
0,
0,
-3.0403762362524778]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],

```

```

True,
'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,
'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,
'dephasing'},

'odd_wires': 4}},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 15.029438972473145}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
           'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.6486934998593055,
                           0.6706921527624933,
                           0.6915479354144154,
                           0.6915479354144154,
                           0.7822900063342887,
                           0.7822900063342887,
                           0.7822900063342887,
                           0.7969446550492483,
                           2.1484559222076447,
                           2.662825799299848,
                           3.431953071624279,
                           3.431953071624279,
                           4.035233416575914,

```

4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,

```

-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,

```

```

1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '

```

```

'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.143320656704589],
[12, 6, 0, 0, -4.62093405872061],
[ 20,
10,
0,
0,
-3.4426712688431165],
[ 28,
14,
0,
0,
-2.5225763020917817]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [ 1,
1,
0,
0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [ 2,
2,
0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
True,
'dephasing'},
'status':
'type':
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [ 3,

```



```
3,
0,
0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
True,
'status':
'type':
'dephasing'},
'odd_wires': 4}},
'run_time_sec': 12.831977367401123}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.05909938957877547,
                           0.05909939012498705,
                           0.4766831071513548,
                           0.4767050288044724,
                           0.476729910017027,
                           0.47672998527996446,
                           2.99469059821818,
                           3.0000368876553107,
                           3.03481022556516,
                           3.0348102255651677,
                           3.0960777044795975,
                           3.0960777044802072,
                           3.096077704551025,
                           5.243967325820749,
                           5.24396732593715,
                           5.696137660948558,
                           5.6961376610109316,
                           6.622089050039713,
                           6.65393172347244,
                           7.224334835522734,
                           7.698532497695266,
                           7.80231633505566,
                           8.26676599072812,
                           9.142506968904602,
                           9.143106298076189,
                           9.481224923876267,
                           9.547418340766493,
                           10.421309029539785,
                           11.175654656469316,
                           11.175654850900237,
                           3.362844107326414,
                           0.29831395836496843,
                           -3.1839955728525164,
                           -2.274863578730854,
                           -0.3088418730608092,
                           0.6979840126794992,
                           0.4650589650223805,
                           -0.5403997868775179,
```

-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,

```

-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'status': False,
0},
0},

```

[illegible]

[illegible]

0.5,
0.5,
0.5,
0.5],

```
        'r': 0},  
        'time': { 'max': 10.0,  
                   'min': 0.0},  
        'type': 'heisenberg'}},  
    'init_param': { 'value': [ 2.1921898489539642e-17,  
                                1.3966237144281936e-17,  
                                5.875629127222605e-17,  
                                1.0041795331735186,  
                                1.0493846336909358,  
                                2.3268506812305976,  
                                3.014247762257123,  
                                3.5587330689470917,  
                                3.598019824659659,  
                                3.598019824659659,  
                                3.598019824659659,  
                                4.594861139703674,  
                                5.132217790928667,  
                                5.132217790928667,  
                                6.816072831791245,  
                                6.906986102298665,  
                                7.2472760308271145,  
                                7.2472760308271145,  
                                8.200086671450705,  
                                8.200086671450705,  
                                8.200086671450705,  
                                8.200086671450705,  
                                8.942806427307234,  
                                9.150430959012654,  
                                9.150430959012654,  
                                9.436537073101752,  
                                9.664513899938346,  
                                10.186874850218704,  
                                10.186874850218702,  
                                10.784457886918862,  
                                1.4923280250442645,  
                                -0.2249463453233081,  
                                0.03240277963395433,  
                                0.031811746673685456,  
                                1.8117023364465294,  
                                -0.20170292630031428,  
                                0.21975366136216332,  
                                0.17798084128170438,  
                                0.15372773772227344,  
                                0.9274008024787889,  
                                0.4001492174566485,  
                                0.7146224592116278,  
                                1.5995558838041692,  
                                0.5527977709299036,  
                                -1.1764350892263153,  
                                0.46503179152183083,  
                                1.695729635964121,  
                                1.1136467790441111,  
                                -1.746792733239722,  
                                -0.14004161870441295,  
                                0.32452227687295954,  
                                0.6844673620662513,  
                                0.0854148711216103,  
                                0.07337284978077696,  
                                0.03946382901684809,  
                                0.38955412394420164,  
                                0.48394719071539255,  
                                0.16206823911430795,  
                                -0.8516299678716804,  
                                0.04915630664847796,  
                                0.7718109282613742,  
                                -0.2065447357739375,  
                                -0.14165427311861,  
                                0.11805421787298728,  
                                1.0598635939215315,  
                                0.39576809919914385,  
                                -0.6055477801146832,
```

0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,

```

-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '

```



```

'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.181142575648604],
                              [12, 6, 0, 0, -4.947722861409576],
                              [20, 10, 0, 0, -3.953720458369559],
                              [ 28,
                                14,
                                0,
                                0,
                                -3.1457163005887416]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [ 0,
                                           0,
                                           0,
                                           0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'status':
'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'status':
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,

```

```

0,
0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'odd_wires': 4}},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'status':

'type':

'odd_wires': 4}},
'run_time_sec': 14.704960584640503}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 0.723632677978758,
                            0.8013713721964207,
                            0.8013713721964207,
                            0.8013713721964206,
                            1.7866251780297595,
                            1.786625178029759,
                            2.0864992088537258,
                            2.7277814684323256,
                            2.7277814684323256,
                            3.2780082387901137,
                            3.2780082387901137,
                            4.125286968934425,
                            4.125286968934425,
                            6.2058132923329055,
                            6.495923744510521,
                            6.7707474398596865,
                            6.7707474398596865,
                            7.238549805010759,
                            7.238549805010759,
                            7.424016758372523,
                            7.424016758372523,

```

7.45123520722395,
8.286067277272798,
9.005302817051474,
9.005302817051472,
9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,

```
-0.9052301036303075,  
-0.7478544743067898,  
1.440838637841082,  
-0.3509099808044935,  
1.2358346145311443,  
-0.6571877561823198,  
0.18396549243569166,  
0.5572253610451431,  
-0.08245783604551615,  
-1.227171845284735,  
-0.6560441239508517,  
2.6593298322359935,  
0.41277334619182954,  
0.31722506904944375,  
-0.6896496257584998,  
1.020969683655719,  
-0.4213531010246796,  
-0.6028714639415698,  
-0.32068790108967526,  
0.9359829761321473,  
0.2722611321195526,  
0.7279956983784606,  
2.105530837572665,  
0.6301085819517248,  
-0.4741842530657499,  
-0.41210701140497913,  
1.1418366036045013,  
0.5540480699420616,  
-1.0638929508149726,  
2.9650032716304366,  
0.13103138388576777,  
-0.18446871100115916,  
-0.20115179595309238,  
-0.7685797513964847,  
0.5993305389810505,  
1.1977914772421903,  
-0.07742797176395931,  
-0.36846685866459333,  
-0.1992471400523476,  
-1.5504150718346672,  
0.2491762098499702,  
-0.892147097942845,  
-0.6207645529629603,  
-0.6674451932252692,  
2.1498335055935183,  
2.152070265954198,  
-0.2994969538531137,  
-0.16704777910826452,  
0.35924103266084867,  
2.6159801365080866,  
-1.6995077279131865,  
1.2838418662740392,  
1.5880628185621348,  
2.6023475449592826]],  
'noise_profile': { 'noise_on_init_param': {  
  
    'value':  
  
        'noise_prob': [ 0.001,  
                        0.001,  
                        0.0,  
                        0.0],  
        'status': True,  
        'type': 'dephasing'},  
    'nqubits': 7,  
    'observable': { 'coefficients': { 'bn': [ 1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0,  
                                              1.0],  
                                          'cn': [ 0.5,  
                                                  0.5,  
                                                  0.5,
```

```

0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                '1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X '
                                                         '0 X 1 '
                                                         '+ 0.5 '
                                                         'Y 0 Y '
                                                         '1 + '
                                                         '0.5 Z '
                                                         '0 Z 1 '
                                                         '+ 0.5 '
                                                         'X 1 X '
                                                         '2 + '
                                                         '0.5 Y '
                                                         '1 Y 2 '
                                                         '+ 0.5 '
                                                         'Z 1 Z '
                                                         '2 + '
                                                         '0.5 X '
                                                         '2 X 3 '
                                                         '+ 0.5 '
                                                         'Y 2 Y '
                                                         '3 + '
                                                         '0.5 Z '
                                                         '2 Z 3 '
                                                         '+ 0.5 '
                                                         'X 3 X '
                                                         '4 + '
                                                         '0.5 Y '
                                                         '3 Y 4 '
                                                         '+ 0.5 '
                                                         'Z 3 Z '
                                                         '4 + '
                                                         '0.5 X '
                                                         '4 X 5 '
                                                         '+ 0.5 '
                                                         'Y 4 Y '
                                                         '5 + '
                                                         '0.5 Z '
                                                         '4 Z 5 '
                                                         '+ 0.5 '
                                                         'X 5 X '
                                                         '6 + '
                                                         '0.5 Y '
                                                         '5 Y 6 '
                                                         '+ 0.5 '
                                                         'Z 5 Z '
                                                         '6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.126749766362046],

```

```

[12, 6, 0, 0, -4.835388535288831],
[ 20,
  10,
  0,
  0,
  -3.8102426495131967],
[ 28,
  14,
  0,
  0,
  -2.988701014248795]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [ 0,
                                          0,
                                          0,
                                          0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

                                'status':

                                'type':

                    'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

```

True,

0.5,

```
init_param': { 'value': [ 0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,
9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
0.5676146077203189,
-1.30828022074471,
```

-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,


```

0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '

```

```

'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '
'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.3900736437908385],
[12, 6, 0, 0, -5.048002042316807],
[ 20,
10,
0,
0,
-3.9807470593727396],
[ 28,
14,
0,
0,
-3.1247321790461613]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [ 0,
0,
0,
0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status':

```

```
True,  
    'dephasing'},  
  
    'noise_on_init_param': { 'status': False,  
                             'value': 0},  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],  
  
True,  
    'dephasing'},  
  
    'noise_on_init_param': { 'status': False,  
                             'value': 0},  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],  
  
True,  
    'dephasing'},  
  
    'noise_on_init_param': { 'status': False,  
                             'value': 0},  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],  
  
True,  
    'dephasing'},  
  
    'noise_on_init_param': { 'status': False,  
                             'value': 0},  
    'noise_prob': [ 0.001,  
                    0.001,  
                    0.0,  
                    0.0],  
  
True,  
    'dephasing'},  
  
    'odd_wires': 4}},  
    'run_time_sec': 17.87613034248352}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                     'cn': [
```

```
        'r': 0},
    'time': { 'max': 10.0,
              'min': 0.0},
    'type': 'heisenberg'}},
'init_param': { 'value': [ 0.17583917736716262,
                           0.17583917736716265,
                           0.9725326222396186,
                           0.9725326222396184,
                           1.154796264329948,
                           2.3531638611520562,
                           3.742172656987498,
                           3.810257862376578,
                           3.8745068996857803,
                           3.8745068996857803,
                           3.8745068996857803,
                           4.643045833818014,
                           4.643045833818014,
                           4.643045833818013,
                           4.643045833818014,
                           4.643045833818014,
                           4.711671630422156,
                           5.405427388054719,
                           5.405427388054719,
                           7.4602327607935,
                           7.610173264016766,
                           7.858440336330247,
                           7.968223945319675,
                           8.51525712638587,
                           8.54015380483253,
                           8.797039458545115,
                           8.797039458545115,
                           8.858280642077405,
                           9.30423425746159,
                           11.6502765461574,
                           1.4243949815721635,
                           1.853782837605556,
                           0.616412956408157,
                           -0.43476006789942184,
                           1.5440059756230975,
                           1.1700457070837984,
                           -1.8512161685606496,
                           0.5034873196770298,
                           -1.3667011208822544,
                           1.5586288186942996,
                           0.6807721504395611,
                           -0.5694108741752331,
                           0.15882528136634905,
                           2.2872584636590165,
                           -1.3655797366869253,
                           0.5428984525730501,
                           -0.5905673445154871,
                           0.2347743623798973,
                           -0.44300125429382003,
                           -0.3419704070408729,
                           -0.557684730924058,
                           -0.39393239001581876,
                           0.2907206177517228,
                           -0.20455476793263844,
                           0.9506883178672908,
                           -0.5484133147843897,
                           0.13971335509755736,
                           -0.8702011448235134,
                           -0.8865241771432285,
                           3.6566903393961887,
                           1.9040162201640494,
                           1.1703036584804183,
                           -1.612571874409814,
                           1.5937439406218687,
                           -0.7966362559752604,
                           1.6639456273768476,
                           -0.07788431093380854,
                           1.6494646253000458,
                           0.26682023241866526,
                           0.6098831056897478,
                           0.6434791877968039,
```

-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,

```

1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
'2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
'4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X '
'0 X 1 '
'+ 0.5 '
'Y 0 Y '
'1 + '
'0.5 Z '
'0 Z 1 '
'+ 0.5 '
'X 1 X '
'2 + '
'0.5 Y '
'1 Y 2 '
'+ 0.5 '
'Z 1 Z '
'2 + '
'0.5 X '
'2 X 3 '
'+ 0.5 '
'Y 2 Y '
'3 + '
'0.5 Z '

```

```

'2 Z 3 '
'+ 0.5 '
'X 3 X '
'4 + '
'0.5 Y '
'3 Y 4 '
'+ 0.5 '
'Z 3 Z '
'4 + '
'0.5 X '
'4 X 5 '
'+ 0.5 '
'Y 4 Y '
'5 + '
'0.5 Z '
'4 Z 5 '
'+ 0.5 '
'X 5 X '
'6 + '
'0.5 Y '
'5 Y 6 '
'+ 0.5 '
'Z 5 Z '
'6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.871225062671609],
                              [12, 6, 0, 0, -4.462319254461681],
                              [20, 10, 0, 0, -3.382891800586928],
                              [ 28,
                                14,
                                0,
                                0,
                                -2.547698159867475]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [ 0,
                                              0,
                                              0,
                                              0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                        1,
                        0,
                        0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

'status':

True,

'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                        2,
                        0,
                        0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                        3,
                        0,
                        0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

'odd_wires': 4}},
'run_time_sec': 14.192391633987427}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [

0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

'r': 0},
'time': { 'max': 10.0,
          'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 1.7202079031049283e-17,
                           -2.594562831748091e-19,
                           0.6068914425074802,
                           1.1321115349854645,
                           1.2806026785178235,
                           1.5886494702821472,
                           2.45154545857325,
                           2.4515454585732503,
                           2.4575334297772264,
                           2.4575334297772264,
                           2.4575334297772264,
                           3.2128626039678436,
                           3.2128626039678436,
                           4.612853662117567,
                           5.000904423260963,
                           5.000904423260963,
                           5.1471756720956385,
                           6.340371796976545,
                           6.340371796976545,
                           6.640884886785207,
                           6.640884886785207,
                           6.640884886785207,
                           8.598101965665498,
                           8.622498407967232,
                           8.622498407967234,
                           9.078776950212898,

```


9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,

```

-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517]],
'noise_profile': { 'noise_on_init_param': {
'value':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'status': False,
0},

```

[illegible]

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

True,

'dephasing'},

```

```

'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],

```

```

0,
-3.9295998440454545],
[ 28,
 14,
 0,
 0,
-3.0888624551018555]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [ 0,
                                           0,
                                           0,
                                           0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [ 1,
                      1,
                      0,
                      0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [ 2,
                      2,
                      0,
                      0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [ 3,
                      3,
                      0,
                      0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

```

'status':

'type':

'status':

'type':

'status':

'type':

'status':

```

True,
' dephasing'},
' type':
' odd_wires': 4}},
' run_time_sec': 13.9875967502594}}],
' vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
' layer': 30,
' ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
' cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
' r': 0},
' time': {'max': 10.0, 'min': 0.0},
' type': 'heisenberg'}}},
' init_param': {'value': 'random'},
' noise_profile': { 'noise_on_init_param': { 'status':
' value': 0},
' noise_prob': [0.001, 0.001, 0.0, 0.0],
' status': True,
' type': 'dephasing'}},
' nqubits': 7,
' observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
' cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
' r': 1},
' def': 'ising'}},
' output': { 'draw': { 'fig_dpi': 100,
' status': True,
' type': 'png'},
' file_name_prefix':
' heisenberg_noise_free_time_evo'},
' redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 1]]},
' run': 'vqe',
' state': 'dmatrix',
' vqe': { 'iteration': 10,
' optimization': { 'algorithm': 'SLSQP',
' constraint': True,
' status': True}},
' zne': { 'data_points': None,
' degree': 1,
' method': 'richardson',
' sampling': 'default'}},
' others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
' Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
' 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
' X 5 X 6 + 1 Z 5 + 1 Z 6',
' time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 '
' + 0.5 Y 0 Y '
' 1 + 0.5 Z 0 '
' Z 1 + 0.5 X '
' 1 X 2 + 0.5 '
' Y 1 Y 2 + '
' 0.5 Z 1 Z 2 '
' + 0.5 X 2 X '

```

```

'3 + 0.5 Y 2 '
'Y 3 + 0.5 Z '
'2 Z 3 + 0.5 '
'X 3 X 4 + '
'0.5 Y 3 Y 4 '
'+ 0.5 Z 3 Z '
'4 + 0.5 X 4 '
'X 5 + 0.5 Y '
'4 Y 5 + 0.5 '
'Z 4 Z 5 + '
'0.5 X 5 X 6 '
'+ 0.5 Y 5 Y '
'6 + 0.5 Z 5 '
'Z 6']],
'output': { 'exact_sol': -8.56677223350563,
  'initial_cost_history': [ 6.146913233691134,
    5.993838002068908,
    5.949850054037247,
    6.210098036642341,
    6.152155884142049,
    6.148463718261344,
    6.155325453778133,
    6.006166196398716,
    6.005219627807008,
    6.0413306719300985],
  'noise_details': { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [4, 2, 0, 0],
    'noise_profile': {
      'noise_prob': [
        0.001,
        0.001,
        0.0],
        'status': True,
        'type':
          'odd_wires': 4},
      'optimized_minimum_cost': [ -6.523058281579068,
        -6.1267497663620825,
        -6.42521340171773,
        -6.39007364379089,
        -6.344797923049613,
        -5.871225062671643,
        -6.143320656704609,
        -6.2758943446211894,
        -6.3403937569891395,
        -6.18114257564869],
      'optimized_parameters': [ [ 0.3358069569174437,
        0.33751316628129896,
        0.37613413155379394,
        0.9037599675903363,
        0.941040799157583,
        1.6105111871436553,
        1.7638163741391246,
        1.7638163741391246,
        3.06849123809687,
        3.600239571878297,
        4.175689787149765,
        4.175689787149765,
        5.544091982519477,
        5.544091982519477,
        6.308463651138072,
        6.308463651138072,
        6.308463651138072,
        6.3184465106806345,
        6.521816085515529,
        6.521816085515529,
        7.288786933299201,
        7.54709747254392,
        7.597867588518987,
        7.962268849194133,
        7.991785735138846,
```

8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,
-0.3650459287261044,

0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428],
[0.723632677978758,
0.8013713721964207,
0.8013713721964207,
0.8013713721964206,
1.7866251780297595,
1.786625178029759,
2.0864992088537258,
2.7277814684323256,
2.7277814684323256,
3.2780082387901137,
3.2780082387901137,
4.125286968934425,
4.125286968934425,
6.2058132923329055,
6.495923744510521,
6.7707474398596865,
6.7707474398596865,
7.238549805010759,
7.238549805010759,
7.424016758372523,
7.424016758372523,
7.45123520722395,
8.286067277272798,
9.005302817051474,
9.005302817051472,

9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,

1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,
1.5880628185621348,
2.6023475449592826],
[0.05909938957877547,
0.05909939012498705,
0.4766831071513548,
0.4767050288044724,
0.476729910017027,
0.47672998527996446,
2.99469059821818,
3.0000368876553107,
3.03481022556516,
3.0348102255651677,
3.0960777044795975,
3.0960777044802072,
3.096077704551025,
5.243967325820749,
5.24396732593715,
5.696137660948558,
5.6961376610109316,
6.622089050039713,
6.65393172347244,
7.224334835522734,
7.698532497695266,
7.80231633505566,
8.26676599072812,
9.142506968904602,
9.143106298076189,

9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,

-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522],
[0.013671733667398815,
0.013671733667398838,
0.40762796768685644,
0.7598808855555792,
0.7614715630840724,
0.7614715630840723,
0.7792763091267767,
1.6629027237260914,
1.9378940282792974,
3.350993251000224,
4.102190293916363,
4.285616461674835,
4.285616461674835,
5.265551146051114,
5.272209322381249,
5.559550971186632,
5.621065732928831,
6.49189669389436,
6.49189669389436,
6.901458013958807,
6.901458013958807,
7.453720513962848,
7.453720513962848,
7.638729004736868,
8.981314635749522,

9.036582294014133,
9.298591258404642,
10.774856261068669,
10.985435664973474,
11.405689613341023,
3.402049333572153,
-0.09204740692141915,
-0.005385703162861625,
-0.12744857538380094,
-0.3481194894790832,
-0.032922919116929295,
-0.8213554647609927,
-1.68109752492848,
1.933784512728038,
-0.1999747293100837,
-0.06905142559770702,
0.17251731253523298,
-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,

-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,
2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048],
[0.0008444707970882174,
0.0011376501074209704,
0.5478052879830236,
0.5478052879830236,
0.5478052879830237,
0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,

9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,
-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,

0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475],
[0.17583917736716262,
0.17583917736716265,
0.9725326222396186,
0.9725326222396184,
1.154796264329948,
2.3531638611520562,
3.742172656987498,
3.810257862376578,
3.8745068996857803,
3.8745068996857803,
3.8745068996857803,
4.643045833818014,
4.643045833818014,
4.643045833818013,
4.643045833818014,
4.643045833818014,
4.711671630422156,
5.405427388054719,
5.405427388054719,
7.4602327607935,
7.610173264016766,
7.858440336330247,
7.968223945319675,
8.51525712638587,
8.54015380483253,

8.797039458545115,
8.797039458545115,
8.858280642077405,
9.30423425746159,
11.6502765461574,
1.4243949815721635,
1.853782837605556,
0.616412956408157,
-0.43476006789942184,
1.5440059756230975,
1.1700457070837984,
-1.8512161685606496,
0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,

0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,
1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613],
[0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,
3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,

8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,
0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,

-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182],
[1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,
2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,

9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,
0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,

-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517],
[0.26700956978337703,
0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,

9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,

0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732],
[2.1921898489539642e-17,
1.3966237144281936e-17,
5.875629127222605e-17,
1.0041795331735186,
1.0493846336909358,
2.3268506812305976,
3.014247762257123,
3.5587330689470917,
3.598019824659659,
3.598019824659659,
3.598019824659659,
4.594861139703674,
5.132217790928667,
5.132217790928667,
6.816072831791245,
6.906986102298665,
7.2472760308271145,
7.2472760308271145,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.942806427307234,
9.150430959012654,
9.150430959012654,

9.436537073101752,
9.664513899938346,
10.186874850218704,
10.186874850218702,
10.784457886918862,
1.4923280250442645,
-0.2249463453233081,
0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,
0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,

```

-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,
-0.05994450656840446,
1.295326344233161,
1.643528772065058,
0.6078294171847716,
-0.5414822669865806,
0.9476987879530461,
-0.5918060843747338,
0.052115348982246026,
0.17303729547159033,
0.8356162346646744,
0.6175568577642787,
0.5665086177732088,
-0.2976089155494843,
0.6784374599036339,
0.8882560669191352,
1.7967642990525536,
0.8822189067565646,
1.6135024149712653,
0.7013175670070841,
2.0615253680033234]],
'run_time_sec': 42850.28223609924}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 1.7202079031049283e-17,
-2.594562831748091e-19,
0.6068914425074802,
1.1321115349854645,
1.2806026785178235,
1.5886494702821472,

```

2.45154545857325,
2.4515454585732503,
2.4575334297772264,
2.4575334297772264,
3.2128626039678436,
3.2128626039678436,
4.612853662117567,
5.000904423260963,
5.000904423260963,
5.1471756720956385,
6.340371796976545,
6.340371796976545,
6.640884886785207,
6.640884886785207,
6.640884886785207,
8.598101965665498,
8.622498407967232,
8.622498407967234,
9.078776950212898,
9.461537111509045,
9.461537111509045,
11.193126234466078,
11.194773036768451,
11.23425035230001,
3.6783033545989348,
2.4269657917911354,
-0.05771344168509165,
-1.2030573768957538,
3.7213804138473705,
1.924827615988215,
1.056479566226862,
-1.5473494967520907,
4.398805627479834,
0.8338358403171965,
-0.8099255104678582,
-0.45767862837458095,
2.861078492572404,
3.8652468681120005,
-2.887845767517789,
-2.805755195373484,
4.0587596872172185,
-0.6699049461158488,
0.21333648034183758,
0.4524745343676716,
2.986432468734924,
0.3439325002550967,
-0.8275045443760665,
0.13067451225114587,
2.238488698389344,
-0.7017786785426589,
1.2978317219394626,
-1.301485631967723,
3.965835074906498,
0.8969824578766956,
2.7952734104214465,
-0.7742154013266447,
1.5675465154968056,
2.0464253264269097,
1.8800069620035462,
-1.8044011720919966,
0.9434521417873769,
2.7275648876259666,
1.488415118705842,
-0.737507651078596,
-0.645648781385225,
2.712515098674733,
2.3858616806735897,
-0.6332194623409646,
0.7242797471713182,
3.040367706889497,
-0.31974435408472995,
0.4095160551701009,
0.2497813960768828,
0.8700029141395006,
-0.09253761630265683,

0.0021906014498840436,
-0.05942235585046714,
3.0569057148483503,
-0.3466039731736911,
-0.08625294612714683,
-0.972400173321991,
2.818648060142346,
0.26977522672128323,
-0.43357532748811417,
-0.1398888669570251,
0.5703082715904539,
1.1288698319231192,
0.5103090181408015,
0.9751915834601443,
1.3322237635112688,
0.1151655141844926,
-0.009281838713691305,
-0.6214752845525063,
-1.5805807688467428,
-0.10117261020635139,
0.07763614048447905,
1.3788004377565666,
1.8240844635205977,
0.5421982630966959,
0.41635499438864665,
0.6019503022394472,
-0.13876425465218306,
-1.389335520601386,
-0.11160302566710978,
1.9532500107835935,
3.1783952012172545,
1.2324746355871234,
-0.3300299009483176,
2.0563071170406455,
0.5070328560575449,
-1.4244738939854995,
-1.0849327697753446,
2.012447413284947,
0.7321644003415494,
2.1697243109783226,
2.3939500038986976,
1.5095457334141147,
0.4771497809888841,
0.7264385677911265,
-1.2750578122129246,
-0.3527900448343432,
1.4172346289410647,
0.8432803814903324,
3.051114458214373,
-0.3334344329994917,
2.17934906384921,
1.266420876986024,
2.7644241701095216,
0.636538971425229,
-1.847398851714829,
-0.013702660156780564,
2.331176430726721,
1.7725302577647513,
0.5106846778295356,
1.0449783439357938,
1.1157484211342301,
-0.3758082507859313,
-0.5755462941925489,
3.6338269047911145,
1.1686374374905806,
-1.25881396302495,
1.2372510547976072,
3.5862251252548343,
1.88315539412517]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0},

'noise_prob': [0.001, 0.001, 0.0, 0.0],

'status': True,

'type': 'dephasing'},

False,

```

'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -6.2758943446211575],
                        [ 12,
                           6,
                           0,
                           0,
                           -4.972375668379767],
                        [ 20,
                           10,
                           0,
                           0,
                           -3.9295998440454545],
                        [ 28,
                           14,
                           0,
                           0,
                           -3.0888624551018555]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 7.557868957519531e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.043777382368372,
                           'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
11.390625,
-6.2758943446211575,
-4.972375668379767,
-3.9295998440454545,
-3.0888624551018555],
'cost_richardson_zne':
'sorted_expectation_vals': [
'sorted_noise': [ 6,
                  18,
                  30,
                  42]},
'sampled data': [ [ 4,

```

```

2,
0,
0,
-6.2758943446211575],
[ 12,
6,
0,
0,
-4.972375668379767],
[ 20,
10,
0,
0,
-3.9295998440454545],
[ 28,
14,
0,
0,
-3.0888624551018555]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.0008444707970882174,
0.0011376501074209704,
0.5478052879830236,
0.5478052879830236,
0.5478052879830237,
0.784723345688809,
0.784723345688809,
0.784723345688809,
1.9248523134122373,
3.1827054999432645,
3.5310059925523927,
3.5310059925523927,
5.052695732413606,
5.052695732413606,
6.136548648234324,
6.136548648234324,
6.785869019003799,
6.785869019003799,
7.198041513907379,
7.3330889202217895,
8.496582874867597,
8.496582874867597,
8.496739633755235,
8.496739633755235,
8.496741833065137,
9.22890579273074,
9.930471725964209,
10.003370383780338,
10.15144599090014,
10.665059744564415,
2.9193389895672706,
2.873179484562977,
0.24817648473142762,
-1.1978615116990414,
0.5013428170881429,
-0.14698874126247163,
-0.1346344592985022,

```

-0.18688791288547063,
0.30396646405908945,
-0.3893332461107723,
-2.4132158231627026,
-1.3025870666843387,
2.486320477273128,
-0.6536413182583717,
1.1565986435504452,
-1.1840220246209,
0.8322463862945216,
0.686475242636196,
0.9748739365898487,
-2.032798889096391,
1.2977623784552696,
-0.21362311033600742,
0.9967979178359492,
-0.7714825960035517,
0.05140148475667656,
1.2463395942761013,
0.6604875776774448,
1.0077664010977474,
-1.5169046363817273,
2.1374704261116553,
-0.3986317675051551,
0.9979173289768921,
0.5109160756321709,
-0.5699092281256025,
0.06259493290058689,
0.09572934677993615,
-0.1276069942134142,
0.5134766913530642,
0.20814668833639585,
-0.07283729618715824,
-1.572605595336637,
0.046834826805737875,
0.9754117887783622,
-0.33049799730083496,
-0.5177265259797135,
-0.04355189602132581,
0.14976458188191247,
0.25592632582729447,
0.9304650429250412,
-0.3249924553930563,
-0.9782247783234811,
0.5608311563544173,
0.47985335249742045,
-0.330399671604431,
-0.8520140047652908,
-0.10054671379827794,
0.04505475925207872,
-0.23886462292775137,
-0.06253899689152312,
-0.5367365159989609,
0.11081168579196325,
0.7895828215662509,
0.001664050802353895,
-0.023296885636891907,
0.4236278032398412,
0.13185849296844476,
0.6925025040153038,
0.7959349783086688,
-0.24452249453754032,
-0.1961780227944993,
0.18114635146463365,
-0.5690361952966106,
-0.8652566665636684,
1.1722520540580956,
-0.08429437547626617,
0.38809452735168,
-0.12772850571008293,
0.8254705326002666,
2.728435066689564,
0.2762008465630276,
0.32283065693323815,
0.29724178038654964,

```
1.590761448212885,
-2.542678559046311,
1.8426140885602824,
-0.7536427909760223,
0.06657859520733085,
0.09240922143572869,
0.25333258594571073,
1.3323200052961734,
1.0415674472460652,
-1.3158991164150335,
0.5790778967786893,
-0.027086512138488276,
-0.4523808433268666,
0.8941348650986891,
0.2993948163110877,
0.9945737025449459,
1.4729122791178781,
0.47756161698756283,
1.8984204359050885,
0.3009555819949919,
1.0276401924150993,
-1.08923180856859,
-0.39766839114152513,
-0.04192134427478414,
1.0556908333623896,
-1.4936687805293032,
-1.0144223777633155,
-0.496197450304602,
0.6189941883918804,
-0.2917583185167685,
3.446963714204082,
-0.8070478397284955,
0.08143180094223462,
1.6464671071125407,
0.15347286071303512,
1.2250081014240108,
0.05581094766683211,
0.9405683335283475]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
```


[illegible]

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.26700956978337703,
0.2670095697833769,
0.7897559248250247,
1.60994438629722,
1.6240137789724642,
1.9029927716215564,
2.754961581608366,
2.754961581608366,
4.416760838293873,
4.472730549723469,
4.472730549723469,
4.472730549723469,
4.9018244764195,
4.9018244764195,
6.142575905365314,
6.99918307198117,
6.99918307198117,
7.3452528051326675,
7.478132650082225,
8.520612438756238,
8.520612438756238,
8.991228072316343,
9.227791963806121,
9.7884313312372,
9.7884313312372,
9.7884313312372,
10.06004176448028,
11.030427598537905,
11.030427598537905,
11.500583935823688,
1.3913972994109516,
0.07774370637080774,
-0.49382826121080187,
1.73722078706814,
1.1477124735700583,
-0.01244613489207998,
0.1104474230926492,
-0.4185268033829736,
1.470187851781374,
0.2683710961451789,
-0.4350955069921617,
0.47701522697148346,
-0.12059167163697146,
-0.29388493308433694,
0.7760009417155218,
0.2157710734488978,
1.2495404382304187,
-0.3273973942211235,
-0.7657986587430435,
-0.29167448156050724,
1.0175732352511992,
-0.26691063020915523,
-0.9498568893096563,
-0.7761349266449543,
-0.03917968596759244,
0.9253714376815375,
0.10438717726675471,
0.119240158066098,
-0.8869854257116645,
0.04307801423033725,
0.06347331663464305,
-0.30105647297315635,
-0.09889278852928217,
-0.3652919068964209,
-0.0693406009004909,
-0.4647505796661569,
0.22928728839598264,
0.004811057975813908,
```

-0.05466429107309376,
1.028863770675019,
0.16879402166298368,
-1.1055840341912755,
0.37883052217910285,
0.020578527093579314,
0.02031617143340724,
1.3356628439106515,
0.24029821405041826,
-0.8280329987357055,
-0.03478678790209601,
0.4032911416795608,
1.8211062640760638,
-0.5955129579806626,
-0.12389825475476993,
0.15078479774791417,
0.25606626819121847,
-0.5141688497431932,
0.6061088230694278,
0.46841078155670723,
-0.6184620199507063,
0.756637551737996,
1.7329747111109826,
0.05692839943403828,
0.2941548416592414,
0.5858620932416747,
0.7670403629229918,
0.30554255161337057,
-0.45658754906270327,
-0.42843355002838807,
-0.3023769889698021,
0.007053602739091254,
0.6545125121375878,
-1.3230514751857634,
0.5525700851182778,
0.6077856841225572,
0.3193555486658347,
1.0972045202793819,
-0.9608463705403721,
-0.8800113323184338,
1.5893473527453552,
-0.17930075995350292,
0.13243112332175283,
2.218523374768228,
0.3265682752545504,
0.8010356407066903,
-0.26882287505987873,
1.9214544754255154,
1.3539481795361321,
-0.5909164412323823,
0.3965669693797851,
-2.2493204845536354,
0.19587180737910695,
0.11919500443011696,
1.0895825531254497,
-0.8262207903952641,
-0.9954080023703576,
-1.0169640919417637,
0.6255474806084755,
-1.2785460468142644,
1.8640192664022799,
0.26835529578172257,
0.6968836525252171,
1.4078350308699439,
0.8234537653883477,
0.7818882402859949,
0.946183618997053,
1.904140472055704,
-0.5964775479033396,
1.6130219764165838,
0.03420262362389698,
0.2971461141757466,
0.38494337865180756,
1.8373477360582482,
0.5875634587735682,

False,

```
0.4088919615500518,
0.4770455671450362,
1.9125631892854644,
0.4190740079559775,
0.8595568972619075,
0.4240513002790927,
2.6162509667045732]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noise_free_time_evo_dephasing_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -6.340393756989085],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.976052575965789],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.898762699309893],
                          [ 28,
                           14,
                           0,
                           0,
                           -3.0403762362524778]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.413459777832031e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.151504802503796,
                           'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
```

```
-0.3125],  
11.390625,  
-6.340393756989085,  
-4.976052575965789,  
-3.898762699309893,  
-3.0403762362524778],
```

```
'cost_richardson_zne':  
'sorted_expectation_vals': [  
  
    'sorted_noise': [ 6,  
                     18,  
                     30,  
                     42]}],  
    'sampled data': [ [ 4,  
                       2,  
                       0,  
                       0,  
                       -6.340393756989085],  
                      [ 12,  
                       6,  
                       0,  
                       0,  
                       -4.976052575965789],  
                      [ 20,  
                       10,  
                       0,  
                       0,  
                       -3.898762699309893],  
                      [ 28,  
                       14,  
                       0,  
                       0,  
                       -3.0403762362524778]]],  
    'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0,  
                                                             0],  
                                                             'cn': [ 0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5,  
                                                             0.5],  
                                                             'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}}},  
  'init_param': { 'value': [ 0.3358069569174437,  
                             0.33751316628129896,  
                             0.37613413155379394,  
                             0.9037599675903363,  
                             0.941040799157583,  
                             1.6105111871436553,  
                             1.7638163741391246,  
                             1.7638163741391246,  
                             3.06849123809687,  
                             3.600239571878297,  
                             4.175689787149765,  
                             4.175689787149765,  
                             5.544091982519477,  
                             5.544091982519477,  
                             6.308463651138072,  
                             6.308463651138072,  
                             6.308463651138072,  
                             6.3184465106806345,  
                             6.521816085515529,  
                             6.521816085515529,  
                             7.288786933299201,  
                             7.54709747254392,  
                             7.597867588518987,  
                             7.962268849194133,
```

7.991785735138846,
8.619190388880893,
8.629220288817223,
10.432794468229133,
10.549342687924028,
11.465905039453537,
1.5708348469088267,
-0.3283220618427727,
0.03691181582691629,
-2.5608652259642914,
1.464587567475586,
2.208272980722778,
-0.3337202412707043,
-0.9570571070157553,
0.20098684515796417,
-0.8867183197035234,
-0.00908592189328417,
-1.4638778850057466,
2.6033615081338537,
0.3763013231634618,
-2.697434246081451,
0.09985560627390955,
0.6973075775075135,
-0.5785886913208023,
-0.13253830432587832,
-0.003645075822409154,
0.648250634168174,
0.24074433115353022,
-0.7536468259038906,
0.08331860927339882,
-0.5730961456214154,
-0.20775836607022918,
1.858859760589094,
-0.3428004889774629,
0.7409414856192459,
0.34884447713535405,
0.6678090018369852,
-0.6382699051282278,
0.1547862218012797,
-0.17320932448753618,
-0.2953648953179735,
-0.02314845214565176,
0.24480308430392833,
-0.20823570419213822,
-0.39307400959737193,
-0.10025250214034016,
1.6727519415472074,
-0.9747413627922913,
-0.3833767883715735,
0.048011939803101665,
1.18312232276534,
-0.14428967118116512,
0.36563575326495207,
0.12119576008368955,
0.4387535829489819,
0.9681038210810762,
-0.6691274598635867,
-0.3521753787453993,
-0.355012662953856,
-0.20576613832679122,
0.1140651512187755,
0.5153298337016836,
-0.39555821358839643,
-0.04716458735412639,
0.816517053493595,
-0.24857467769432498,
0.7027118663260605,
-0.027602198169676975,
-0.4051834986620014,
-1.8516570656368516,
0.5820873061837243,
0.7126583334669851,
-0.11066526397524724,
-1.3306139702670972,
0.6774223306964047,

```

-0.3650459287261044,
0.4283437000286837,
-0.06653200080212002,
0.5675137187755209,
0.8389187220813384,
0.018917198943831585,
0.09191245072124354,
-0.526396234817696,
-0.7816472128996359,
-0.44136460802163247,
-0.4369989493239652,
0.293075842227442,
-0.08524656443417457,
-0.6429606894490985,
-2.410698527055141,
1.0120137481126148,
0.8564893555858171,
-0.13410206690609555,
-0.9845670359114355,
1.526893886889543,
-0.7686684244601373,
-1.0943277418547575,
2.0842448547688535,
1.3753261950807358,
1.7080497020352088,
-0.7565848251808305,
1.3444474246686076,
0.9108065385428427,
-1.3419466460375384,
1.2964907448683802,
0.8331395661011098,
1.4601752953966671,
-0.7731535534825779,
0.7435978213651214,
0.40470144975051525,
0.7060090619127611,
-0.5434857345297852,
0.6026047646877396,
-0.21577981479695774,
-0.025695167927078984,
-0.20540702789842594,
1.0167330871418905,
-0.872845590481918,
0.20800958153274315,
-0.5165304385134258,
-0.2697015375489779,
0.9243191151157019,
0.5666128931848439,
0.93497763777263,
0.08307103938152688,
1.3587069870377428]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,

```

False,

```

        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#1'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0],
                                              [3, 3, 0, 0]]},

        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},

        'zne': { 'data_points': [ [ 4,
                                     2,
                                     0,
                                     0,
                                     -6.523058281578953],
                                   [ 12,
                                     6,
                                     0,
                                     0,
                                     -5.228944859106494],
                                   [ 20,
                                     10,
                                     0,
                                     0,
                                     -4.1973377330967105],
                                   [ 28,
                                     14,
                                     0,
                                     0,
                                     -3.367027452945099]],

                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}}},
        'output': { 'exact_sol': -8.566772233505624,
                    'run_time_sec': 4.363059997558594e-05,
                    'zne_values': { 'degree': 0,
                                    'extrapolated_value': -7.287682807302595,
                                    'others': { 'beta_coeffieicients': [
2.1875,
-2.1875,
1.3125,
-0.3125],

11.390625,

-6.523058281578953,
-5.228944859106494,
-4.1973377330967105,
-3.367027452945099],

                                'cost_richardson_zne':

                                'sorted_expectation_vals': [

                                'sorted_noise': [ 6,
                                                    18,
                                                    30,
                                                    42]},

                    'sampled data': [ [ 4,
                                         2,
                                         0,
                                         0,
                                         -6.523058281578953],
                                       [ 12,
                                         6,
                                         0,
                                         0,
                                         -5.228944859106494],
                                       [ 20,
                                         10,
                                         0,
                                         0,
                                         -4.1973377330967105],
                                       [ 28,
                                         14,
                                         0,
                                         0,
                                         ]

```



```

-3.367027452945099]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}}},
'init_param': { 'value': [ 2.1921898489539642e-17,
1.3966237144281936e-17,
5.875629127222605e-17,
1.0041795331735186,
1.0493846336909358,
2.3268506812305976,
3.014247762257123,
3.5587330689470917,
3.598019824659659,
3.598019824659659,
3.598019824659659,
4.594861139703674,
5.132217790928667,
5.132217790928667,
6.816072831791245,
6.906986102298665,
7.2472760308271145,
7.2472760308271145,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.200086671450705,
8.942806427307234,
9.150430959012654,
9.150430959012654,
9.436537073101752,
9.664513899938346,
10.186874850218704,
10.186874850218702,
10.784457886918862,
1.4923280250442645,
-0.2249463453233081,
0.03240277963395433,
0.031811746673685456,
1.8117023364465294,
-0.20170292630031428,
0.21975366136216332,
0.17798084128170438,
0.15372773772227344,
0.9274008024787889,
0.4001492174566485,
0.7146224592116278,
1.5995558838041692,
0.5527977709299036,
-1.1764350892263153,
0.46503179152183083,
1.695729635964121,
1.1136467790441111,
-1.746792733239722,
-0.14004161870441295,
0.32452227687295954,
0.6844673620662513,
0.0854148711216103,
0.07337284978077696,
0.03946382901684809,

```

0.38955412394420164,
0.48394719071539255,
0.16206823911430795,
-0.8516299678716804,
0.04915630664847796,
0.7718109282613742,
-0.2065447357739375,
-0.14165427311861,
0.11805421787298728,
1.0598635939215315,
0.39576809919914385,
-0.6055477801146832,
0.15174535642921344,
0.37347096289724246,
0.7012360268729083,
0.3542709759268278,
0.4882231311332355,
-0.5718535566609203,
-0.5141748323784885,
0.6302639648026501,
0.38620406200914825,
-1.1120561171735972,
-0.4242993251495611,
0.6960617189877939,
-0.20395802510096012,
-1.5267965024740064,
0.37206855518718707,
0.3903205558108704,
-0.15806125440222468,
-0.7393855239189431,
-0.2473222700357295,
-0.6335812955372796,
0.4834513595433696,
-0.507920609072258,
0.1291144531948748,
-0.6846710196076853,
0.43797341781861193,
0.900419905764507,
0.9335535636585514,
0.08879625773059918,
0.9617014761808407,
0.4188749955823149,
0.641642866783994,
-1.0759696241657866,
-0.11568695136579071,
-0.5835036749708086,
0.0014576767655774953,
0.9393727866577569,
1.4182731605702288,
0.3230862629286186,
0.05737300168803934,
1.388953564819239,
1.3825242261157324,
-0.4964523061875459,
0.2936961440882872,
1.860334311180735,
0.7411583726992121,
0.2679210558840293,
-0.14275215320012302,
0.4698844290903277,
-0.07226660787052742,
0.3803826953929657,
-1.411422251891662,
2.176823885566318,
-0.16675444695024147,
0.49149867066267733,
0.9203309434282454,
-0.5873330160605318,
0.6468270759602834,
1.5084139769303189,
-0.30670176180569,
-0.46474267067817565,
-0.8083323211008838,
0.13858245905587877,
0.15377127171399946,

[illegible]

```

0,
-3.1457163005887416]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.839897155761719e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.904327382573938,
'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.181142575648604,
-4.947722861409576,
-3.953720458369559,
-3.1457163005887416],
'sorted_noise': [ 6,
18,
30,
42]},
'sampled data': [ [ 4,
2,
0,
0,
-6.181142575648604],
[ 12,
6,
0,
0,
-4.947722861409576],
[ 20,
10,
0,
0,
-3.953720458369559],
[ 28,
14,
0,
0,
-3.1457163005887416]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.6486934998593055,
0.6706921527624933,
0.6915479354144154,
0.6915479354144154,
0.7822900063342887,
0.7822900063342887,
0.7822900063342887,
0.7969446550492483,
2.1484559222076447,
2.662825799299848,
3.431953071624279,

```

3.431953071624279,
4.035233416575914,
4.035233416575914,
4.035233416575914,
4.568117148704361,
5.039539225310202,
5.039539225310202,
5.534018322321666,
5.534018322321666,
6.483210707785691,
6.490896899688113,
7.182799214542672,
7.182799214542672,
7.559584008067621,
8.43635332601195,
8.43635332601195,
9.091482769474323,
9.091482769474323,
10.51189306045493,
0.4036051624048159,
0.6580269923177141,
2.030106683293408,
-1.6087595289666783,
1.1238448060114463,
0.48219161744750516,
-0.52647875832763,
-0.7770680009741181,
0.9584923266732629,
1.4081968681145547,
-1.286760308434132,
-1.1643597658261824,
0.43102549223791814,
0.4762851226867799,
-1.5375915208743731,
-0.32604860635808525,
0.9126587772562,
0.4716164439155413,
-1.185137911370813,
0.0010435150849372951,
1.362703657319786,
0.15168618448664317,
0.013738542048390118,
-1.2488843079530334,
0.5953067453023578,
-0.7375945018490428,
1.6938673074929198,
0.2332524998863778,
0.7631553129407195,
0.8722488841401685,
-0.6689205221218597,
0.6584128273939948,
-0.5866714617908798,
-0.5382377239349249,
0.07947113792533414,
-0.638100867011655,
-0.19922015844116167,
0.5814496123293298,
0.41157361684622124,
-0.7649732243319444,
0.562344908659212,
0.37006719775147345,
0.9681106498419162,
0.36638640066147987,
-0.7951387872640513,
0.45941103993764054,
-0.2267572852324774,
0.43356714924634276,
-0.8591483084450674,
-0.18461893041919494,
1.116992572824263,
1.1669294382352948,
0.029792610192383724,
0.8391977011137817,
-0.31345005435646217,
0.18250807811893774,

```

0.1692196481359008,
-0.8984153824952316,
-0.8025661732867799,
-0.4459950806048063,
-0.6388250625028883,
-0.16920658797384605,
-0.6596078743198966,
0.13675425086946338,
0.37060376839848896,
-0.02320319905123886,
-0.758554011380563,
0.2855477607473741,
0.40986471148811776,
-0.47064746228161686,
-0.5695276821086359,
-0.23798089255567295,
0.021208599954383085,
-0.957792978850957,
-0.3892960105133825,
-0.07165870922911506,
0.6373978530836436,
0.11583373543812793,
-0.6869467612152537,
-0.109279803260324,
-0.7618543036104211,
-0.28370326239549154,
-0.2212046842298352,
-0.04751119916340401,
0.20449386754071522,
0.705432158326348,
-0.1477392252159656,
-0.7161855054209953,
0.14441137676467267,
1.3906449907875076,
1.3299068108993992,
0.43548394087882114,
0.10987629965719209,
0.26908108246362716,
1.0150638572902744,
0.04483602986273151,
-0.153649795167794,
0.044240654262805976,
0.3417710011612185,
0.06331633335604989,
2.075122314076187,
1.004062083901195,
0.7792740260615444,
1.4928465702619802,
0.7274077578302571,
0.5108906894057771,
0.6929155972389786,
-0.38025768754944894,
-0.16052687515696545,
-0.2996322099808623,
0.07791949448737949,
-1.421471362119057,
-1.238198529035781,
-0.5725940454397166,
0.79842254320011,
1.1392600222135925,
-0.6816896404490814,
-0.3035432394760949,
0.26895925341115173,
1.7196471710373182]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                1.0,
                                1.0,
                                1.0,

```

False,

```
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [ 4,
2,
0,
0,
-6.143320656704589],
[ 12,
6,
0,
0,
-4.62093405872061],
[ 20,
10,
0,
0,
-3.4426712688431165],
[ 28,
14,
0,
0,
-2.5225763020917817]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00010585784912109375,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.060421629042862,
'others': { 'beta_coeffieicients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.143320656704589,
-4.62093405872061,
-3.4426712688431165,
-2.5225763020917817],
'sorted_noise': [ 6,
18,
30,
42]},
'sampled data': [ [ 4,
2,
0,
0,
-6.143320656704589],
[ 12,
```

```

        6,
        0,
        0,
        -4.62093405872061],
    [ 20,
      10,
      0,
      0,
      -3.4426712688431165],
    [ 28,
      14,
      0,
      0,
      -2.5225763020917817]],
    'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'heisenberg'}}},
  'init_param': { 'value': [ 0.013671733667398815,
                              0.013671733667398838,
                              0.40762796768685644,
                              0.7598808855555792,
                              0.7614715630840724,
                              0.7614715630840723,
                              0.7792763091267767,
                              1.6629027237260914,
                              1.9378940282792974,
                              3.350993251000224,
                              4.102190293916363,
                              4.285616461674835,
                              4.285616461674835,
                              5.265551146051114,
                              5.272209322381249,
                              5.559550971186632,
                              5.621065732928831,
                              6.49189669389436,
                              6.49189669389436,
                              6.901458013958807,
                              6.901458013958807,
                              7.453720513962848,
                              7.453720513962848,
                              7.638729004736868,
                              8.981314635749522,
                              9.036582294014133,
                              9.298591258404642,
                              10.774856261068669,
                              10.985435664973474,
                              11.405689613341023,
                              3.402049333572153,
                              -0.09204740692141915,
                              -0.005385703162861625,
                              -0.12744857538380094,
                              -0.3481194894790832,
                              -0.032922919116929295,
                              -0.8213554647609927,
                              -1.68109752492848,
                              1.933784512728038,
                              -0.1999747293100837,
                              -0.06905142559770702,
                              0.17251731253523298,

```


-0.29209950757543574,
1.9513629766062992,
-0.5676146077203189,
-1.30828022074471,
-0.39448602558973367,
2.8898568639190474,
0.7013622886452797,
0.26544077065537275,
0.6092585136402319,
0.39577135630450067,
0.6467620437646749,
0.21576418945272974,
1.5307712738023358,
0.036118123631889515,
-0.4522715145702613,
0.23873726563257947,
-0.5048956742472485,
-1.0930780792284351,
0.3032753479197677,
-0.7352108075519095,
2.388361488538045,
-0.26021473348059354,
-0.15069319270481707,
0.027894641785103223,
-0.8264639451986096,
0.35484877308180596,
0.3634605390797818,
-0.6782786306119145,
-0.4584373707969238,
-0.02554073317719059,
1.368791882618868,
0.22287848313248318,
-0.2583655693641652,
0.816978308583948,
0.013147606797621707,
0.05698163495924544,
-0.10079294572026389,
-0.031444469177844374,
-0.7293984794648443,
0.2748832371107343,
-0.6405508135788557,
-0.34649786169568175,
-0.7129088100865897,
-0.17647106725668124,
0.5283722459340723,
0.07619761644401915,
-0.7919447064807553,
-0.10064108207705522,
-0.06871701920974893,
-0.6704013223808513,
-0.10305747446551641,
-0.44685019184739355,
0.47585671024510207,
-0.5220226715241708,
-0.2280447320934421,
0.2936108455269187,
-0.15405793128572018,
0.5990729015407984,
-0.19291559657627058,
-0.9983329413888067,
0.06183078156007772,
-0.39888138991817884,
0.10015032721026125,
0.26145493022070637,
0.24697998275786284,
2.309703946957057,
-0.636761752972196,
0.20283595531231904,
-0.34422565238172687,
-1.811812618680636,
-0.49932305549368694,
-0.005354026261784876,
1.1618635049710246,
-0.2797857231732124,
-3.139209341650586,

```

2.0637348100204087,
-0.5177237968866337,
-0.15688312220828537,
0.2587523624435701,
0.49135592716377213,
0.43691541247309595,
1.2894314878973545,
1.3005295207704686,
1.495226836554953,
0.9996022744756439,
-1.5874790912892243,
2.2762238566718955,
0.5367949469391156,
1.1760708004626512,
0.9710849792074049,
0.6980586712660793,
2.72315991893461,
1.2971833278243776,
0.09659115597174431,
0.09342881414470793,
1.3325478190287798,
-1.0112906801034072,
1.0158358800365146,
2.393856578358819,
1.3824348835579319,
0.4676902275435639,
-4.840732545310297,
0.7878781030390952,
-0.28202203140023785,
-0.5161784186146438,
-2.320181258265768,
-0.18967728316611845,
-0.7476120256380048]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                             2,
                             0,
                             0,
                             -6.3900736437908385],

```

```

[ 12,
  6,
  0,
  0,
  -5.048002042316807],
[ 20,
  10,
  0,
  0,
  -3.9807470593727396],
[ 28,
  14,
  0,
  0,
  -3.1247321790461613]],
  'degree': 0,
  'method': 'richardson',
  'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
  'run_time_sec': 5.841255187988281e-05,
  'zne_values': { 'degree': 0,
    'extrapolated_value': -7.184033337699239,
    'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
11.390625,

-6.3900736437908385,
-5.048002042316807,
-3.9807470593727396,
-3.1247321790461613],

'cost_richardson_zne':

'sorted_expectation_vals': [

'sorted_noise': [ 6,
                  18,
                  30,
                  42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.3900736437908385],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.048002042316807],
                  [ 20,
                    10,
                    0,
                    0,
                    -3.9807470593727396],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.1247321790461613]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
  'layer': 30,
  'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},

```

```
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.723632677978758,
0.8013713721964207,
0.8013713721964207,
0.8013713721964206,
1.7866251780297595,
1.786625178029759,
2.0864992088537258,
2.7277814684323256,
2.7277814684323256,
3.2780082387901137,
3.2780082387901137,
4.125286968934425,
4.125286968934425,
6.2058132923329055,
6.495923744510521,
6.7707474398596865,
6.7707474398596865,
7.238549805010759,
7.238549805010759,
7.424016758372523,
7.424016758372523,
7.45123520722395,
8.286067277272798,
9.005302817051474,
9.005302817051472,
9.005302817051474,
9.468112593488378,
9.980769303066321,
10.029230447532314,
10.496216102492953,
1.6849607035180754,
-0.17702247224764014,
-1.2741911423346686,
-2.844813020435471,
2.6306502527193634,
1.1115711285155645,
-1.0604767208931998,
-1.2378494371437496,
-0.4278587412008179,
3.0662166892278284,
-0.052478476294015586,
-0.4305161219629684,
0.9820485095264653,
2.3466906714085924,
-0.025291075467548628,
-0.02025122162742965,
2.3009633618597958,
2.636290816480209,
-1.3375867987805241,
-0.06984817387757897,
0.8379766315051282,
2.9385638522351325,
0.3757787757676439,
-0.5976763830250241,
1.952966768288683,
-0.18392843022993377,
-0.06945125900955239,
-0.46790539601389963,
1.0666476796127082,
-0.45002074862148034,
0.485756726904901,
0.7028527569311613,
-0.8841998733674782,
0.5790627163409446,
0.24900370863916496,
-1.3998930789319726,
0.571424852181861,
4.1551639391971555,
0.977010759596294,
0.33290175228802893,
-0.054203288453907456,
2.484325073603772,
-0.06532957451035779,
```

0.7093120717910988,
0.6483260114745797,
-0.7137589563592477,
-0.9970068698829373,
0.25936949570895534,
0.5261290573132944,
-0.17449910346983083,
-0.3473491308138607,
0.256089362893497,
0.7703235035289624,
2.4662145578092742,
-0.3255635113638319,
-0.7993346591979026,
-0.27656756639458474,
-0.6197164225594639,
1.461100687024406,
-0.5325789028291374,
1.3026671869564812,
0.6659764681771295,
0.05091100558174678,
0.6853605998913553,
0.14650643821436643,
0.35354380566246224,
-0.9052301036303075,
-0.7478544743067898,
1.440838637841082,
-0.3509099808044935,
1.2358346145311443,
-0.6571877561823198,
0.18396549243569166,
0.5572253610451431,
-0.08245783604551615,
-1.227171845284735,
-0.6560441239508517,
2.6593298322359935,
0.41277334619182954,
0.31722506904944375,
-0.6896496257584998,
1.020969683655719,
-0.4213531010246796,
-0.6028714639415698,
-0.32068790108967526,
0.9359829761321473,
0.2722611321195526,
0.7279956983784606,
2.105530837572665,
0.6301085819517248,
-0.4741842530657499,
-0.41210701140497913,
1.1418366036045013,
0.5540480699420616,
-1.0638929508149726,
2.9650032716304366,
0.13103138388576777,
-0.18446871100115916,
-0.20115179595309238,
-0.7685797513964847,
0.5993305389810505,
1.1977914772421903,
-0.07742797176395931,
-0.36846685866459333,
-0.1992471400523476,
-1.5504150718346672,
0.2491762098499702,
-0.892147097942845,
-0.6207645529629603,
-0.6674451932252692,
2.1498335055935183,
2.152070265954198,
-0.2994969538531137,
-0.16704777910826452,
0.35924103266084867,
2.6159801365080866,
-1.6995077279131865,
1.2838418662740392,

False,

```
1.5880628185621348,
2.6023475449592826]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -6.126749766362046],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.835388535288831],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.8102426495131967],
                          [ 28,
                           14,
                           0,
                           0,
                           -2.988701014248795]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.2438507080078125e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.891827103505979,
                           'others': { 'beta_coeffieicients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
11.390625,
-6.126749766362046,
```

-4.835388535288831,
-3.8102426495131967,
-2.988701014248795],

```
      'sorted_noise': [ 6,  
                        18,  
                        30,  
                        42]],  
    'sampled data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -6.126749766362046],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -4.835388535288831],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -3.8102426495131967],  
                      [ 28,  
                        14,  
                        0,  
                        0,  
                        -2.988701014248795]],  
    'sampling': 'default']}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0],  
                                                              'cn': [ 0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5],  
                                                              'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'heisenberg'}},  
  'init_param': { 'value': [ 0.17583917736716262,  
                              0.17583917736716265,  
                              0.9725326222396186,  
                              0.9725326222396184,  
                              1.154796264329948,  
                              2.3531638611520562,  
                              3.742172656987498,  
                              3.810257862376578,  
                              3.8745068996857803,  
                              3.8745068996857803,  
                              3.8745068996857803,  
                              4.643045833818014,  
                              4.643045833818014,  
                              4.643045833818013,  
                              4.643045833818014,  
                              4.643045833818014,  
                              4.711671630422156,  
                              5.405427388054719,  
                              5.405427388054719,  
                              7.4602327607935,  
                              7.610173264016766,  
                              7.858440336330247,  
                              7.968223945319675,  
                              8.51525712638587,  
                              8.54015380483253,  
                              8.797039458545115,  
                              8.797039458545115,  
                              8.858280642077405,  
                              9.30423425746159,
```

11.6502765461574,
1.4243949815721635,
1.853782837605556,
0.616412956408157,
-0.43476006789942184,
1.5440059756230975,
1.1700457070837984,
-1.8512161685606496,
0.5034873196770298,
-1.3667011208822544,
1.5586288186942996,
0.6807721504395611,
-0.5694108741752331,
0.15882528136634905,
2.2872584636590165,
-1.3655797366869253,
0.5428984525730501,
-0.5905673445154871,
0.2347743623798973,
-0.44300125429382003,
-0.3419704070408729,
-0.557684730924058,
-0.39393239001581876,
0.2907206177517228,
-0.20455476793263844,
0.9506883178672908,
-0.5484133147843897,
0.13971335509755736,
-0.8702011448235134,
-0.8865241771432285,
3.6566903393961887,
1.9040162201640494,
1.1703036584804183,
-1.612571874409814,
1.5937439406218687,
-0.7966362559752604,
1.6639456273768476,
-0.07788431093380854,
1.6494646253000458,
0.26682023241866526,
0.6098831056897478,
0.6434791877968039,
-0.3642625162283491,
-0.6005046410459869,
1.150908039509824,
1.047623616881836,
-1.88935920087463,
-0.6383293560704371,
-0.21589295131960512,
1.1626516759119074,
-1.7337024940961776,
0.2745871369074163,
1.180976730275941,
0.3380372191722413,
-0.7295020077567086,
-1.1475444911604376,
-0.41606699516184503,
0.34479967563611297,
-1.8756981188290198,
0.04866246560567376,
0.012484099153236356,
-1.7301439515486168,
-0.05331100044365285,
-1.7420104478147471,
0.27369646350096233,
0.7847986223655491,
0.164946635698038,
0.5222048030486672,
-0.2489517328000029,
-0.23891464867152398,
-0.5559464400375377,
0.09062663365782511,
-1.1031138787335897,
1.4016845007634189,
-0.12334323864201413,


```

1.0541054865353667,
0.1131731022351882,
1.1334016336326587,
1.0445896472322274,
-0.3384477056788242,
0.8562300424264522,
-1.185951257251967,
-0.28195201407806436,
0.49489672690633446,
2.1691644499599145,
-0.10611204631619445,
-0.29316710611678704,
0.4244142056378012,
1.4232556043268565,
-0.43855569637576686,
0.4481402106296697,
-0.8366912353597888,
-0.17900284718440188,
0.4012908890737685,
1.9617681531822835,
2.696806378221078,
-0.29935581729750876,
0.249494490518143,
2.1195342002515956,
-0.5001087380784989,
0.43210977848166987,
0.04333786836680723,
1.0322878414371686,
1.1119974543192979,
-0.26136285079486304,
-0.8969318919082698,
0.7031359208122375,
0.4583643546082401,
1.1321706860962937,
0.1470460759922183,
0.11572603306386549,
-0.6786010146777492,
0.37642235223899184,
0.792302711212988,
1.5225714245834412,
0.6193853672704486,
1.6538082045246512,
1.0911911905391962,
1.0888363065520819,
1.3075011415833737,
2.7405602323600613]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],

```

```

[2, 2, 0, 0],
[3, 3, 0, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -5.871225062671609],
                          [ 12,
                           6,
                           0,
                           0,
                           -4.462319254461681],
                          [ 20,
                           10,
                           0,
                           0,
                           -3.382891800586928],
                          [ 28,
                           14,
                           0,
                           0,
                           -2.547698159867475]],

         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},

'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.245208740234375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.725871268770975,
                           'others': { 'beta_coeffieicients': [

2.1875,
-2.1875,
1.3125,
-0.3125],

11.390625,

-5.871225062671609,
-4.462319254461681,
-3.382891800586928,
-2.547698159867475],

'cost_richardson_zne':

'sorted_expectation_vals': [

'sorted_noise': [ 6,
                  18,
                  30,
                  42]}},

'sampled data': [ [ 4,
                   2,
                   0,
                   0,
                   -5.871225062671609],
                  [ 12,
                   6,
                   0,
                   0,
                   -4.462319254461681],
                  [ 20,
                   10,
                   0,
                   0,
                   -3.382891800586928],
                  [ 28,
                   14,
                   0,
                   0,
                   -2.547698159867475]],

'sampling': 'default'}},

{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,

```

```
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'heisenberg'}},
'init_param': { 'value': [ 0.05909938957877547,
0.05909939012498705,
0.4766831071513548,
0.4767050288044724,
0.476729910017027,
0.47672998527996446,
2.99469059821818,
3.0000368876553107,
3.03481022556516,
3.0348102255651677,
3.0960777044795975,
3.0960777044802072,
3.096077704551025,
5.243967325820749,
5.24396732593715,
5.696137660948558,
5.6961376610109316,
6.622089050039713,
6.65393172347244,
7.224334835522734,
7.698532497695266,
7.80231633505566,
8.26676599072812,
9.142506968904602,
9.143106298076189,
9.481224923876267,
9.547418340766493,
10.421309029539785,
11.175654656469316,
11.175654850900237,
3.362844107326414,
0.29831395836496843,
-3.1839955728525164,
-2.274863578730854,
-0.3088418730608092,
0.6979840126794992,
0.4650589650223805,
-0.5403997868775179,
-0.4207789901056385,
-0.6821445865591288,
1.2179172859744154,
-1.1069270923796861,
-0.38365670293222115,
2.3779178278712347,
0.4759901226592819,
-0.927969640150436,
-0.6955516255591223,
1.2922090119966039,
-0.10607902133440596,
0.5543353597795015,
-0.4302086089464882,
1.4798439267149097,
-0.3093276153211663,
1.714412867449583,
-0.18884068113132443,
-0.1325469313246992,
0.03502941932668827,
0.05129162795554355,
0.003190516145023343,
0.010396746926301235,
```

-0.36540696654143495,
-0.5785274388950852,
2.738392476633112,
-0.381595738653264,
1.1894079117852934,
0.10877444053911303,
-0.7972905480625686,
-1.0478348537850688,
0.2798485526741883,
-0.4274204195839525,
-0.48479620588573813,
1.986721851729558,
0.3727109480291794,
-0.9999024859659025,
-0.9001340058505058,
1.059192624806707,
0.5500998699824982,
0.8164935105159342,
0.23246116191399568,
1.5829804590134011,
-0.7035846125833406,
-2.245759711895512,
-0.31792549274216875,
1.6300127330969527,
0.07371355303908503,
-0.23043599427986267,
0.03669525096895327,
-1.2487795935665424,
-0.22444287505209085,
0.03260098380380376,
0.15426094303909776,
-0.15607515377906878,
-2.4039452742432754,
0.5777111231875812,
-0.47410318462012296,
-1.3084220057881686,
-0.2670279930054443,
0.6586758433941311,
0.519919725659534,
0.7788261497169975,
-0.7359144565103657,
0.26798924420019987,
-0.07549437218845514,
-0.12918238774279284,
0.3241799178428526,
0.20082313670495355,
-1.0445169957404505,
1.0131270532523022,
0.037808962222118346,
0.013425310172117042,
3.1324778901275305,
-0.4886964313870908,
-0.38549338714439446,
-0.21618705222743279,
1.4128502098242202,
1.3724483982038045,
0.6088565122591526,
-3.1235123751877847,
0.8039157375019335,
-0.02808098165723019,
-0.9663532420543466,
0.40449045476262857,
1.1791960234919567,
-0.7788063484927917,
0.09071852797557509,
-0.9772493598709,
1.384424737779476,
0.24347682394682063,
2.2487975299533223,
-0.8658214946635034,
-0.45501077217496616,
0.9150231144109652,
0.7382517509084283,
0.12424754469362319,
0.13398478884809817,

False,

```
1.3358953438583405,
0.6477072222130105,
0.18008468335004657,
-0.7194830727480847,
2.669925212365636,
-0.502211740134513,
1.7261222864835546,
-2.02052756373994,
0.3633351903141943,
1.1648987979966707,
0.36638517704268964,
-0.4709684542950341,
-0.2607092234067287,
-0.3086261015591958,
1.2798207278969522]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_heisenberg_noisefree_time_evo_dephasing_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [ 4,
                           2,
                           0,
                           0,
                           -6.425213401717636],
                          [ 12,
                           6,
                           0,
                           0,
                           -5.16135557034569],
                          [ 20,
                           10,
                           0,
                           0,
                           -4.151192033326441],
                          [ 28,
                           14,
                           0,
                           0,
                           -3.3368564728119017]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}}
```

```

'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.38690185546875e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.170360902113367,
                            'others': { 'beta_coefficients': [
2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne':
'sorted_expectation_vals': [
-6.425213401717636,
-5.16135557034569,
-4.151192033326441,
-3.3368564728119017],
'sorted_noise': [ 6,
                  18,
                  30,
                  42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.425213401717636],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.16135557034569],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.151192033326441],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.3368564728119017]],
'sampling': 'default'}}]],
'ising-ric2': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                                                         'layer': 30,
                                                         'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                                 'cn': [ 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5,
                                                                 0.5],
                                                                 'r': 0},
                                                         'time': {'max': 10.0, 'min': 0.0},
                                                         'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': False,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                           'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                           'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'ising_noise_off'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,

```

```

        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
        'zne': { 'data_points': None,
                  'degree': 1,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                          '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                          '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
                    'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X '
                                                                '2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
                                                                '2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
                                                                '1 Z 5 + 1 Z 6']}},
        'output': { 'exact_sol': -8.56677223350563,
                    'initial_cost_history': [ 4.217815007292055,
                                              3.394484740359831,
                                              3.3211044240247634,
                                              3.65155648872765,
                                              4.031781430865161,
                                              2.5147460401592765,
                                              2.635251011361358,
                                              2.6229212283850565,
                                              3.6175377967590236,
                                              3.125720978907869],
                    'noise_details': { 'gates_num': [4, 1, 1, 0],
                                        'identity_factors': [0, 0, 0, 0],
                                        'noise_level': [0, 0, 0, 0],
                                        'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                      'value':
                                                                                      'noise_prob': [ 0.001,
                                                                                      0.001,
                                                                                      0.0,
                                                                                      0.0],
                                                                                      'status': False,
                                                                                      'type': 'dephasing'},
                                        'odd_wires': 4},
                    'optimized_minimum_cost': [ -6.945492229929243,
                                              -7.1165547898808885,
                                              -7.2842760350438445,
                                              -6.724195761448022,
                                              -7.306479477026572,
                                              -7.415156948711891,
                                              -7.202776954905004,
                                              -6.836396584821816,
                                              -7.611342393541609,
                                              -7.203866848572784],
                    'optimized_parameters': [ [ 0.33159768767931425,
                                              0.3415216906329597,
                                              0.34152169063295984,
                                              0.5593316391966133,
                                              0.5593316391966133,
                                              2.351554445259766,
                                              2.9331738085905035,
                                              3.4748030853426815,
                                              3.4748030853426815,
                                              3.616855036152045,
                                              3.616855036152045,
                                              3.8672141033893017,
                                              4.423346341593822,
                                              4.423346341593821,
                                              4.845338698326811,
                                              4.845338698326812,
                                              7.020561389723981,
                                              7.020561389723981,
                                              8.059188148673007,
                                              8.486998646842132,
                                              8.48889562974299,
                                              8.48889562974299,
                                              8.48889562974299,
                                              8.852856222269512,
                                              8.852856222269512,

```

False,
0},

8.85285622226951,
9.91447562836983,
9.968000692942592,
10.549940074823725,
10.549940074823727,
0.8537391604836984,
0.9168200846380319,
0.2559354991965516,
1.0210726229552618,
1.2632715564780685,
-0.3480423461281932,
1.7009717288666621,
0.8945988480638971,
0.0342878706217857,
1.5867242856094939,
-0.46323092808777266,
0.9197703726655558,
0.7442882527479103,
0.9265871236439417,
1.007434201350355,
0.029736173082484843,
0.47331341236562946,
0.4861485561257538,
1.0394118763980624,
-2.9809358180453613,
-0.2596551057591169,
0.274814845028427,
-1.4049453096263822,
-1.0753350027554103,
-1.11413441253153,
0.9507335703621095,
0.8064528289725565,
-0.4246412780854718,
0.4472568123237856,
1.634610917427341,
0.8557019227225735,
-0.10158119911628401,
-0.31211728480663814,
-0.9970023252564638,
0.25431186671884126,
0.7248953017537308,
-0.4073131817183126,
1.6792046485134604,
0.5505815433075344,
-0.8093860553948707,
-0.6172486758497164,
-0.12685386183371894,
0.21223631022564798,
0.8168562896383363,
-0.17647756432116687,
-0.5731604235592278,
0.4333036783698612,
-0.7682262923606079,
1.6429671181235093,
0.6337476059862421,
1.67916133236013,
-0.5926217976812627,
-0.298257419735069,
-0.6583760201171601,
-0.889601924248653,
-1.3248621012262398,
0.013854140862500447,
-0.7662338030087398,
-0.6194525061041963,
0.6815602243054589,
0.08161257575833167,
-0.3072072388036838,
0.42225747459757357,
0.06754004785449946,
0.4843941525302115,
-0.1684623955114203,
0.8138047240050311,
0.052754329692318934,
-0.03499144746108355,
0.04466125788054358,

-0.028799258738440928,
0.7127927371021741,
0.7421378220215715,
-0.6858695512918895,
-0.11720668569532972,
-1.3579343673989164,
-0.9607782638271641,
2.0876158669999754,
0.030744163991020267,
-0.6062544285001135,
-0.08046454707386355,
-0.9589458574514981,
0.6218575881236439,
0.008264172331609584,
0.1779434874985719,
-0.2072304227788729,
0.9938506593352908,
-1.0471692270924964,
1.0898429645649914,
-0.22066548691854398,
0.8076838524861312,
1.7135973843396541,
0.23198771018109937,
-0.14790366150484735,
-0.7779724161623084,
-0.5740177045120107,
-0.8356765454865094,
-0.12758989923704314,
-0.3192845855088301,
0.9992686200806561,
0.18910999621207844,
-0.3626298108536852,
0.5832805029120673,
-0.8863720585918194,
-1.6851050525249511,
-0.3551820269447921,
-0.4184387520471605,
1.05736378101237,
2.031993671638238,
0.20520618587592238,
1.6363439075818225,
0.14342936783484309,
-0.6043776905728043,
0.3922986049495649,
1.4322290738810806,
1.585876902523968,
-0.2738910870161085,
1.7714953537072202,
-0.29354248742932953,
-1.2273457901372085],
[2.0562096130838231e-16,
0.03538692433604476,
0.5493527934980842,
0.5695965809115139,
1.2739811069641827,
1.2739811069641827,
1.5440077454822854,
2.6187560157598857,
2.6193651223262795,
4.47656169572244,
4.47656169572244,
4.900258790927604,
4.903975205691072,
4.965015364809818,
4.971477584375831,
4.9714775843758305,
6.285851088616883,
6.285851088616883,
6.5727281068015415,
6.914906570174498,
6.914906570174498,
7.5825149096464,
7.5825149096464,
7.5825149096464,
8.266106449462097,

8.772625056171776,
8.772625056171774,
8.772625056171774,
9.888607991704035,
10.650763302381714,
-0.25736569540817156,
-0.13749177250114622,
0.6913360970917151,
-1.0251451053141543,
0.04952583576033172,
0.6839090084427677,
1.2841797475889756,
-1.4498903904619547,
-0.6609983487872932,
-0.9195300213913112,
1.0882692755947088,
2.28444907625875,
0.5013640158162996,
1.3195089116901169,
-0.44969609005592753,
-0.19760723927349785,
0.047416784300074,
-0.2592183302467101,
-0.95267369637524,
0.2510197484008247,
0.12431427724764418,
-1.9455191869747381,
0.0834449432923055,
0.6181240781879018,
0.06318897794461549,
0.7469995321450943,
-0.13869412828035013,
1.7789335199415868,
-0.2133263169614118,
-1.747801096143808,
1.7374950822862127,
-0.1617072567646695,
-0.13415848991929505,
2.0599871163251184,
1.2617098025896833,
0.7421863291532844,
0.6134289282076183,
-0.010288112985795142,
-0.164325473683837,
-0.19747193793554596,
-0.1095664145547814,
0.3448002170188953,
-0.008329131684078827,
-0.27021098586965386,
0.5642984785825904,
0.26951720880242647,
0.9416408813291534,
0.18894282686740613,
0.6744670741290004,
0.15073677601160052,
0.7600850898072902,
0.23785895970346504,
-0.3096016709123609,
-0.29939543701383164,
-1.2773708023878871,
0.012610842713240655,
-2.001134648031791,
0.5569056601340245,
-0.022823668895979605,
-0.44069878780800564,
0.9238325880974837,
0.627132735311806,
0.833823816751152,
1.174260792549426,
-0.8145978291018492,
-1.8911045685123373,
0.03189540714483889,
-0.6114896265404774,
0.18855141943106907,
-0.5415449375701122,

0.3556432019461298,
-0.02940037290701983,
0.7847914443788936,
-0.8204017405579882,
-0.22117427936396428,
0.9668207238887914,
0.6982235282720571,
1.1161787173582038,
-1.4967922582645339,
0.6955597513963624,
-0.08409555407567708,
-0.7515653427189505,
-0.13809041364831362,
0.694805471024674,
-0.16980356694456364,
-0.7378676064228189,
-0.5510006148273209,
-0.4664642602089831,
0.3131774610690113,
0.25515679468716435,
0.01472701586112837,
0.9612508062048253,
0.3876700512851733,
0.34932481898103945,
-0.0017068592723796239,
-0.9400752962906423,
0.20680236697496335,
1.2239109612714394,
1.5295424358244767,
-0.16313500004842318,
-0.6283905359663017,
0.6007919237891753,
1.4275642137136517,
0.8373669928162908,
-0.3460569718594813,
0.30478283310349635,
0.7741593665645724,
-1.49133186355475,
-0.40261318631906845,
-0.6780231584993508,
0.2796292694824153,
0.8154646298435348,
0.6705233696972283,
0.8515977010836673,
1.38835715204434,
1.5513313401022284,
-1.9456573322270103,
0.6825855385724343,
0.3411942765245547,
-1.8463135511065893],
[-1.394844222866792e-17,
0.2294538829927996,
0.7521460514270448,
1.524448292944527,
1.699326062163185,
1.699326062163185,
1.699326062163185,
1.699326062163185,
1.699326062163185,
1.699326062163185,
2.04427958156354,
2.044279581563538,
3.760855608085718,
3.7608556080857176,
4.422362610147448,
4.422388771925709,
4.423247474933146,
5.088788131744124,
5.518902595462681,
5.518902595462682,
5.518902595462682,
5.518902595462682,
6.713884925045287,
6.713884925045287,
7.229900049876175,
7.23519041379673,

7.5137384820073985,
9.896584707205738,
9.89724831030544,
11.177528016222873,
11.333986770578415,
-0.5874618645254188,
-0.38498635294065175,
-0.14869891390826667,
-1.385574110751756,
0.3783147753212974,
0.1998192518416486,
1.8873134139140266,
-1.0608247810156872,
0.2365140987973328,
-0.9738696834833499,
1.5453125784784605,
1.8657042183850436,
1.5293157207302228,
-0.7517266227054149,
-0.8871771295479706,
-1.6950047018057137,
-0.5885280426519628,
0.04826899994478067,
-0.4330318552574016,
-1.388710608286563,
-0.08211595300211308,
-0.5283548125054784,
0.3763483175118206,
0.335884831266236,
-0.5070752646015597,
-0.2910412053619489,
0.2354568453295162,
-1.4336146483959564,
0.228779351175754,
-0.06520272691838713,
-0.8209928725245101,
0.7001970509858747,
0.5785044584879039,
-0.49932901566271715,
0.45729974286308867,
0.46052818546886504,
1.4986129468586522,
-1.4491938093059253,
0.58130192556905,
0.052769119433839204,
-0.00640289944448277,
-0.3687606063433655,
-0.006145485686113623,
-0.024597955898482345,
-1.411768598869635,
0.08097884701091772,
-0.9927225428945597,
-0.0075215312387787595,
0.010579195204485638,
-0.42659138073244207,
0.002225130694705108,
-0.17841021982057653,
0.9375108947888807,
-2.7757789566035065,
-0.010127033509228812,
-0.5146645773967682,
-0.3567686303606847,
1.4398753496798034,
-0.5703995887616833,
-0.05677408229384658,
0.957894065877575,
1.9071290282089846,
-0.6507219310887598,
-1.6162530141223683,
1.1818941119762436,
1.6868779692850424,
0.98800409304883,
-1.0587465299812662,
-1.094775277351337,
-0.2877600991535592,

1.8986470088901148,
0.8207273276904813,
1.2066175393079017,
-0.22168365465550993,
0.17258447386478676,
-1.8715762567403773,
-0.48240059727904117,
0.7145730346214052,
1.341149548855606,
1.3388694132947494,
1.1029357907173007,
1.7761192703736626,
0.389134329678578,
0.6998749042461945,
-0.4466329063075881,
0.9939411509507131,
0.5933993643264636,
-0.2238849148647213,
0.16806395587982337,
0.40153989063889134,
-0.3072506770355845,
-0.3084801740315303,
0.35020486394553546,
-0.6893875836471265,
1.1621768900261087,
-0.005422106750069378,
-0.6392006402141877,
-2.6960821965791135,
1.2350659365734071,
0.02344533464890316,
1.6191412587681537,
-0.624233381931643,
1.164388497878687,
2.2414892885807953,
2.3459185940732037,
-0.18816057823941634,
1.0703302771481322,
-0.6022027174173271,
1.5454586171163143,
0.49083778015320845,
-0.5414313403110282,
0.5943360939664153,
0.8531518824889626,
-2.0712921555535835,
-0.9507228787104378,
0.1663695492007906,
-0.4829435842745145,
-0.9852459132453085,
0.6444213968977472,
1.9304389405092637],
[0.47195138158384564,
0.4745413934062829,
0.6416788974326727,
0.6416788974326727,
1.3767805975406497,
1.3774084633016226,
1.3850075193774707,
1.385007519377471,
1.961329521314093,
1.961329521314093,
2.6299528219180073,
2.6299528219180073,
2.6462286434741538,
4.376679341839624,
5.443648648286045,
5.443648648286045,
5.443648648286044,
6.427045191271688,
6.427045191271688,
6.737104333282549,
7.345768854864493,
7.345768854864493,
7.523575375204561,
8.254985561732491,
8.914602409288278,

8.914602409288278,
8.914602409288278,
10.045438587538639,
10.045438587538639,
10.045438587538639,
-0.014294444089673569,
0.8050246736150516,
-0.4603650249158636,
3.8028784519891046,
-0.40535003035622474,
-0.20345609256189173,
0.6921728480649123,
1.2635655003066155,
-0.5816868768558133,
0.8004330576026809,
-1.2247112488961414,
-0.7226464348007927,
0.021600305535157273,
0.2422993600121551,
-0.2813766982328762,
-1.8376799098437777,
0.1842235451250743,
0.08033551390998013,
1.2509807369145618,
-0.43993928060300874,
0.02749412053256657,
0.9446032887644298,
0.5876918496487636,
-1.2193113596650038,
-0.30025121886784994,
0.40718701943488855,
-7.055881324256668e-05,
-0.3230027373880063,
-1.3595946794844804,
0.4025901861096578,
-0.331697269297153,
0.8588883241978998,
1.7499317067572346,
0.2974332602533821,
0.36978773801612186,
-0.1369546347590247,
0.17848126558694688,
0.5078793586065516,
-0.026587668618036284,
-0.32676172114586377,
0.043331505672591836,
0.7587982826624224,
0.5755592246710275,
0.5753466292739289,
0.6580990661684004,
-1.8654933062390762,
1.4954582293028102,
1.1206423350708015,
-0.6308795940680783,
-1.1220353310878195,
0.3113810383877842,
1.1285056041045296,
0.8014038814434817,
-0.3123476059423272,
-0.46649205373142577,
-1.0739566662407007,
1.7799335725034942,
0.3271320302071495,
-0.4957408509507477,
-0.8965339685216933,
0.9394687605669856,
-1.0123362977633281,
-0.8493226829357736,
1.9016506939491882,
0.39163185359556635,
1.2450374039748047,
0.6637318474595636,
1.4487854848879562,
0.8556001420669627,
0.2844708394448605,

1.19751608671342,
-0.2783699910645836,
0.2731694463477748,
-0.8573920204053973,
0.4298526210883498,
0.1620542969403055,
0.7385715014812971,
0.10161961925998468,
0.08514730828974114,
1.5393767966723602,
-0.7517708453085385,
-1.0667182547896288,
0.7653330843147801,
0.2770240276337771,
0.44649448354109117,
1.839692443661341,
0.6074846807891079,
0.5576661480775236,
-0.5735679701859718,
-0.5213243179488682,
-0.13640843594155702,
1.4993013794616976,
0.959504645257717,
-0.8869155163266706,
-1.0471786039389914,
0.6169717474797428,
0.010330090116839902,
-0.36051081047743266,
0.08240138993527565,
-1.19819214697301,
-1.2973765177480467,
-0.8056172495711347,
-0.04435763741623314,
0.14581600625591132,
0.26060016989542056,
0.07248105118801107,
0.5605125602873733,
0.7950998674453089,
1.1145979627095777,
0.00625222978811041,
-0.3867814373266647,
-0.1324550289145508,
-1.2907509838890496,
-2.272496441870613,
-0.4069714152055066,
-0.09281234013047389,
-0.02440219940061265,
-1.4477733222094868,
1.1238797279606656,
0.08808709541843764],
[0.4802107213676794,
0.5872443280976499,
1.2030235571335923,
1.6244635367917757,
1.6244635367917757,
3.26071227407772,
3.26071227407772,
3.8336805164974805,
3.8336805164974805,
4.826555535372041,
4.895860683399619,
4.953829011998897,
5.427903354045862,
5.904661066786138,
5.904661066786139,
6.1444777924663665,
6.1444777924663665,
8.501726489902856,
8.560790735911917,
8.56230393956988,
8.56230393956988,
8.56230393956988,
9.349516793328615,
9.383607475180085,
10.075280307758542,

10.10947387193881,
10.10947387193881,
10.676123926986733,
10.676123926986731,
10.676123926986731,
1.4057383265279821,
0.12082364332553383,
-0.26633169071888585,
0.8440095288231939,
-1.6788257521351209,
0.5654604704832584,
-0.8067769238497999,
-3.0754933205615647,
0.08918153353934634,
-0.44144180897560614,
-0.12233030448793467,
0.7466185674598658,
2.029541324404434,
2.6182216828178912,
-1.132391894978203,
-0.24681552984026395,
0.016701788041529435,
2.056331221347139,
0.4687565935163574,
-0.814652769544358,
1.3493029022207552,
-0.0979175823892474,
1.1060724549152727,
0.10932839153376225,
-0.021236171452369973,
-0.10240895200106154,
0.0678743319848748,
0.31105500906993255,
-1.3775189120014106,
0.1972259082837923,
0.5685166045457986,
-0.04805822649685886,
0.05656484641847995,
0.28963771198353216,
-0.0823457664175828,
0.10970044186707124,
0.4470565049690292,
0.3481580101245137,
1.429670826498759,
0.9929735218605864,
-0.118080682882002,
-0.24768077486638254,
0.14000119823560106,
-0.0033695140848767277,
0.28167924146812856,
0.33054680125209646,
0.25446310854586207,
0.4874199962063169,
-0.4292630164890844,
-0.8493517422041822,
1.10338617740252,
1.008639770575481,
-1.333715224046274,
1.5531304809990258,
-0.9099800622014602,
0.3709414533710614,
0.013272332284884182,
0.2649430125577457,
0.2281562083367391,
-0.8531201835868235,
0.3516874446864623,
-0.20156681030383022,
-0.20235738350440882,
-0.042569687456911996,
0.15952312088272655,
-1.5291183017296281,
-0.14843597874824083,
0.29841129212111156,
-1.0541286856947945,
0.1912222645636818,

-0.6584533547304796,
-0.8290997819567264,
-0.8585614987938113,
-0.925446343244433,
-0.31902871745414296,
0.059023427516986,
1.0543033158567225,
0.06361350569662501,
-0.24874942116320348,
0.14393135270630777,
-0.6226920500261081,
-0.9314119807859277,
0.08452985204639987,
-1.009402214771574,
-1.0062750028527756,
-0.28841331398033154,
0.5886312985955309,
0.25138239542645724,
0.4840684579382569,
1.66880370795529,
0.4947479443765874,
-0.5723605201777653,
-0.16848739586132902,
-1.2344957342488987,
-0.21521798449848678,
-0.061827721816023445,
-0.2248741094525013,
-0.14743116036125348,
1.3439891420398065,
0.26093100927833396,
2.340498096953527,
0.48577707139825455,
1.3703195203028469,
-0.03611016151365655,
-0.39327539415785684,
0.19418290589939505,
-1.2131007544515844,
-0.2043806706538651,
0.10177491262713155,
-0.04073135722774902,
-0.6647700988785717,
1.8155909796605325,
-0.9294726054396647,
0.06362652018359347,
-0.3885510819409685,
0.2259437203319192,
-0.29241696657905075,
1.5301954112147191,
1.6475949707589932,
-0.20840788688387885],
[0.42838040299935903,
0.43437695063037796,
0.7671204362948421,
1.4217493914297048,
1.532465319753789,
1.532465319753789,
2.7699209108087253,
2.7699209108087257,
4.156218307623238,
4.2237366972759505,
5.0429088489710905,
5.613108479440035,
5.957592523811918,
5.957592523811918,
7.200159227676843,
7.200159227676843,
7.511063978540512,
7.511063978540512,
7.5293149335701015,
8.184114862205785,
8.59759634646416,
8.59759634646416,
8.967813742088222,
8.967813742088222,
8.967813742088222,

9.731896873309175,
9.95178485601347,
10.151200227727777,
10.572249489684078,
11.21122434413465,
0.6103673192422585,
-2.2449370868204332,
-0.5531680672940967,
1.6213620933184167,
0.9025434887524114,
0.7267620669083164,
0.688558576899684,
-0.6961599873617196,
0.1412349369195377,
0.7459812495914973,
0.4704664842082417,
-0.38033435666108695,
0.35864815337369327,
-0.9120464289406482,
-0.39019902865511685,
0.8340346919183055,
-2.0544585551745573,
1.25105004616506,
-2.352820292981314,
0.4238462318210131,
0.07915902405687532,
0.20643459967058658,
0.23897101893275827,
-0.8664160098114893,
-1.685166937074754,
-0.06832039714759808,
-0.9791405090065629,
0.025203133044215387,
0.0013271421621207855,
-1.0715671968015688,
-0.008370532310604396,
0.13052855552000728,
0.497310552248094,
0.30798451910916236,
-0.6467356358635334,
-0.411683867969946,
-0.10065713602082733,
0.5062131798374145,
-0.10337024350765668,
-0.9070679783587652,
-0.6555610326122162,
2.2255239458177773,
1.727993426652873,
-1.440768657668252,
-1.3828846188724404,
-0.4461523368168654,
1.0838317458165323,
0.38330860036255543,
-1.499700854710827,
-1.3408221959756346,
0.7419236696397365,
-0.26719837498867116,
-0.2377148775462395,
2.1193168382343814,
-0.3237137921938176,
0.30565535651577447,
-0.3512550922173362,
-0.12043790779624768,
-0.7473597606659746,
0.04801624913587568,
0.0018824715948976833,
-1.0616614621946072,
0.0026415486583119922,
-0.3446629765669456,
1.7936621861690503,
0.6484630040536189,
0.911272728030746,
1.1140004969268402,
-0.7180242734781173,
0.5255116360573389,

0.21141077992540375,
-0.45008152208030955,
0.27750531007906354,
0.32764881051843897,
-0.018511194189945894,
0.3875950700305173,
-0.3325176658862957,
-1.8796967870400432,
-0.6725696299533431,
-0.016521571218157557,
-0.08350862144921116,
-0.5447867614937103,
0.09550333474849668,
0.008862454446484716,
0.22308949354372234,
0.43503110895758224,
0.08385997823794214,
0.8516706560998085,
0.37101021681963386,
-1.5090995065898887,
0.3045648040580248,
0.6581099200957676,
0.35742409273461206,
1.7397310471751102,
0.19044279487017676,
0.9986798590605289,
0.31257833320056017,
0.19093877798180586,
-0.12038962867457878,
-0.4069447684098612,
-1.0520519597232647,
-0.45448340350076705,
-0.09417833232522027,
1.3653492736312018,
0.6606235759628448,
-0.1314405098844259,
1.082951248594517,
0.2015466917595484,
0.16965874881834048,
0.2649109746606924,
0.8389747424450928,
1.4332963880188982,
0.288517036279826,
-0.13031950550882798,
0.31918733601921234,
1.3571234506254648,
1.4386307786107537,
0.4541349276361828,
-1.1895164340492679,
-0.27888567947798426],
[0.403034597497222,
0.4502729934454179,
0.5700423320933159,
0.5700423320933157,
0.6781241992806778,
1.3786879425414615,
1.4501147159778993,
2.4844307076867937,
2.4844307076867937,
3.0256317298652835,
3.0256317298652835,
4.15344004944777,
4.15344004944777,
4.2038017426165855,
4.6259251670404895,
4.6259251670404895,
4.797805401698216,
4.797805401698216,
5.52869212939408,
5.565345896572568,
7.178724808111429,
7.362080567239832,
9.32786374549076,
9.32786374549076,
9.69844688689066,

9.970706574916212,
9.970706574916212,
10.264471030354098,
10.264471030354098,
10.987148477608002,
-0.928231690167026,
0.8057374905304977,
-0.09124110656415482,
-1.2764053392323713,
-0.1699102369047808,
0.2854283542912807,
-0.38934147484457715,
1.9562479348306092,
-0.08583239283592208,
0.1375503575288643,
-0.3029413642384467,
0.3573494966906954,
0.4106808877553357,
1.0168255078067534,
1.020920504974372,
1.182081306796335,
-0.381751450929397,
0.7872502675264477,
0.11650702005295199,
-0.7041962676262727,
-0.9664718849607234,
0.0017002173560816299,
1.2143682588418563,
0.27506501793685123,
0.20402751900538993,
-0.18948284766781337,
0.003995331337594065,
1.1960049920193474,
0.5655953653806649,
0.045530409523104075,
-0.6946141334022584,
-0.014434864812455215,
0.011912663267278172,
0.3387818931591513,
0.03142887465994337,
-0.6463763503344852,
0.17750599329640465,
0.014482690293971139,
1.8640640992205775,
-0.048924475364900355,
0.07047222436652016,
0.5836409098942513,
-0.033227003016975516,
0.5638116666909924,
0.6159645449318359,
-0.28815587863252157,
1.0812127139774732,
0.379778365443053,
1.5589007157572847,
0.42162649727394796,
0.05475856745796003,
0.7192262097201122,
0.3263213716214118,
-1.3661280753618785,
-1.292135594114921,
-0.8807331583336575,
0.4025768854927813,
0.5295243646894815,
-0.8002135124103957,
0.1750760012142075,
-0.054032551513248435,
-0.40188384379722514,
0.07562722660439603,
0.9832576708267071,
-0.2899613446815228,
-0.06710629496184976,
0.7055866137794157,
-0.007144529560920351,
0.12285567358407816,
1.143563250691634,

0.15056960823466894,
1.2837012011259124,
-0.6023181038676345,
0.045174795264068736,
-0.16273344906061393,
0.06659839307873978,
-0.018202395096242784,
-0.39240523180078335,
-0.08377351665795224,
0.97309387436951,
2.220420875428843,
0.26956954819495804,
-0.9599252243572389,
-0.804945736082422,
-0.11945317743340336,
-0.8427551316765194,
0.30866551832235317,
2.4872766722318986,
-0.6424953924634126,
1.3858530207015087,
-0.28083276964757203,
0.4192227932599254,
-0.823908670962852,
-1.1126529481475467,
0.21496241595348659,
-0.08934899247793467,
0.1474411947098728,
0.5744211770388311,
-0.04170488097810379,
-0.22724449437110839,
1.9972780099267948,
-0.8591254315731605,
1.1685333425381264,
0.35810812429508854,
0.4121168481727771,
-0.15141964973712507,
-0.4596563680482823,
0.2311510501006746,
-1.1273861464736383,
0.5254398836685342,
0.8562906940888275,
0.2485294131097614,
0.5338805724034291,
0.4862154010986529,
-0.6197058235436752,
0.20248976894253998,
-0.8370526377715465,
-0.2508663863331003,
0.895805892021248,
0.966286773500402],
[0.45986682004385615,
0.587884042281194,
0.587884042281194,
0.8634874478193177,
0.8634874478193176,
2.1638996627094382,
2.6675296177894383,
2.7047736270183154,
4.061300455462818,
5.103553089959934,
5.103553089959934,
6.751801791048447,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.781371497069936,
7.781371497069936,
7.781371497069936,
8.420463205941301,
8.420463205941301,
8.420463205941301,
8.420463205941301,
9.122294773242167,
9.202160901745463,

9.814044623063202,
10.300168251850302,
10.300979681078617,
10.504460883512992,
11.02433376042467,
0.6573588347385189,
0.5828935548145123,
-0.574462871025295,
-1.546656732274899,
-0.27660055256222155,
1.273681854258797,
1.2384634856925236,
0.2379867607060117,
-0.9322977855669335,
-0.7820979498963491,
2.372805485180463,
-0.032501906954750276,
-0.7800293806349019,
0.4387330183441131,
-0.7149138216831045,
-1.1824362472270316,
0.5918930779181975,
0.829172686537006,
0.977795961853665,
-1.6128990556187608,
0.19086262573792945,
-1.3821924448334508,
-0.37777709213522187,
-0.7880705929271598,
-0.17099996546962573,
0.7254760017497719,
0.9819125877821568,
0.3047932425929136,
-0.02742999568379597,
0.5485981155871978,
0.11485583255370253,
-0.32472329363004176,
-0.6331630162839701,
1.9724401417574067,
1.7698339388439421,
-1.1254891310675255,
-1.1920329227643298,
-0.11899515772905239,
0.6149649547843763,
0.08915119991626204,
0.4168497404293414,
1.795330514128595,
0.13783861826417038,
-0.9065003531161022,
-0.3985523491151137,
-0.36816430284390833,
-1.1942463088521762,
-1.129677747629269,
0.6710845981341038,
-0.6149988458361377,
0.4665365160069161,
-0.19209648586871406,
0.013763508723981719,
-0.8135432145712183,
-0.4057592495812904,
-0.6414147049863589,
0.8876536381187553,
-0.7757241021147325,
0.40873541229116084,
0.514258938404948,
0.3722392877827026,
1.8009017308404462,
0.116225182388376,
0.2191004740219447,
0.8255374993004068,
0.8886674208207993,
-0.579134397544579,
0.06903106340432566,
-0.7849541624527424,
1.2193465265846974,

-0.5822657541432643,
-0.5730195760225829,
-0.08026565436629246,
-0.5615010238556278,
-0.24965155606323408,
0.9668325985251349,
0.32484880948626715,
-0.419910520268625,
0.9941102073040009,
0.290737643308207,
-0.05558286792755536,
-0.07023052502159025,
0.6542319702891408,
-0.7125513821501609,
0.3724489707231321,
0.6482228950077775,
0.6813557635502394,
0.5904217885770654,
0.3344272908394536,
0.9892271354161819,
-0.6202922067549016,
0.12947475126138008,
0.35715623223344434,
-2.26745180129,
-0.20006687506113488,
-0.6290028667838425,
0.08022472893529967,
-0.961248450361056,
-0.20142658286918888,
-0.26748808365437937,
0.3448273909474204,
0.6331413989953146,
-0.5916282783618806,
-0.951916970919588,
-1.150804089444678,
-0.8692217930277405,
0.2236835205843831,
-0.10881486654903144,
-0.004228302858090612,
0.16762641757787602,
1.4232995099688168,
0.17348586714896594,
1.0253598045594392,
0.3864684766177986,
1.0340175270140362,
1.1184694851102563,
1.066055569095364,
0.07092789808311199,
-0.8620685771604671,
0.46490022331063124],
[0.49845241666821427,
0.5080489367341008,
0.9738886018422865,
1.569493854319001,
1.5867336495888638,
2.22018578038962,
2.2222975784710255,
2.7355212499054673,
3.167907147345212,
3.912280465424413,
3.9324465712853787,
4.601639034821639,
4.6023794814303285,
4.622186895887535,
4.629761229768061,
5.8119318913753775,
5.81529780322803,
5.862601007224966,
5.868327135385091,
7.931887080538795,
8.400559524273383,
8.400626671117248,
8.400626709821431,
9.338300284642187,
9.338300284937413,

10.032003053056542,
10.032003053750085,
10.151775184539204,
10.15177520525735,
12.777878105202333,
-1.0652764150946168,
0.1030911905785532,
2.1938566419651506,
-1.687039127173871,
-1.2892167937928738,
0.454901922887811,
-0.10701158278386386,
-0.27610154265055814,
0.3864936366984189,
-2.1008557969724673,
0.03526686701549039,
1.2075362869846002,
0.9729996942576583,
0.011602974506752167,
-1.7901344125733514,
-0.5312859895436812,
-0.6764114285522254,
-0.568464725902531,
0.7761280206550648,
0.10820918045329357,
-1.191918699884087,
0.6710933645796882,
-0.998882902161297,
0.9668399094583318,
-0.5526879367651495,
-0.3448144391622963,
0.3546231755031622,
0.36267379522525617,
-1.3188129954125047,
0.7986540408445592,
0.5385879858279273,
1.3987311647526153,
0.17032526930657013,
-0.2765839794470922,
-0.4561318195988584,
0.7779672645593669,
-1.6618874628312856,
0.7145344738782802,
-0.8703397140730336,
0.3664383221285875,
0.09488224223418325,
0.17319381149242297,
-0.15027776082229982,
-1.4690811524212324,
-1.4342980641635596,
-0.0435375212610062,
-1.1192667984833862,
1.489571420955581,
-1.385268501544282,
0.07500036651414835,
-1.0373356270085057,
1.6055364169931783,
-1.1908916083661096,
0.47394571447103884,
-0.23948832756930136,
1.6865746730435722,
-0.07643328425683309,
-0.26016480364478584,
0.1230869402232525,
0.24188831995971777,
-0.12202515710835343,
-1.3273176630150638,
-0.8685353737050294,
0.16431995234498184,
0.5641710812415553,
-0.1851122077039956,
-0.3773492143584495,
-1.5621470223099339,
-1.4505536127106367,
0.571773409102061,

0.20129944645039902,
1.1233832799408359,
0.09021785654141615,
-0.09203212520614017,
1.4656739277180684,
1.5116863426084852,
-1.4548391160838867,
-0.8665597553970019,
-0.8311755953666669,
-1.2962978849547035,
2.7863849971356167,
-0.026842981867963016,
-0.5147225252861769,
1.0767631952186236,
0.23432587613057304,
-0.3849014581870786,
-0.5492566246326465,
-1.590706022355079,
0.2729988606074437,
-1.0566347351025576,
0.3490824852559465,
-0.08255987668621423,
2.389615689206544,
-0.5033499561524312,
0.8858613199859888,
0.002820627497544732,
0.19486994113578365,
-0.12438999158785295,
-0.2658298270496061,
-0.2763678550214429,
0.4048526015479744,
-0.06841768991626676,
-0.32117880607439586,
1.6019482252187531,
0.41890698857400605,
-0.8510923614851694,
0.16411947273918887,
0.3096085421164439,
-1.0434867263587853,
1.2153787936463343,
-0.9841260925764858,
1.5878541492278757,
0.5868781859412673,
0.26283212361760866,
-0.9081790869869905,
1.2003679274444946,
3.155405601054565,
0.19810365707195607,
0.6950762676177132,
0.9262562990824308],
[0.11310194099017501,
0.35533489757839665,
0.7274669277129748,
0.7931766663622793,
0.9215124586161831,
1.5868010098342087,
1.6938861718192633,
1.6938861718192633,
2.4953100977078537,
2.4953100977078537,
2.8524614686988468,
2.852561718095858,
3.5986528286299406,
4.15163213852048,
4.658101059037985,
4.662433860401005,
5.170840359657884,
5.170840359657884,
7.389891885603243,
7.408023950817849,
7.472384801069389,
7.816163992050368,
8.61557017602429,
9.097989314327426,
9.523961629484033,

9.523961629484033,
9.983169099900275,
10.18670132329559,
10.652310618690768,
11.298231074807717,
0.0018299791732057594,
-2.098323349499344,
-0.32648794289911637,
-0.9015207614699141,
0.16889281860224642,
-0.23801018256690964,
-0.2710900692754434,
-0.354821626365823,
0.2834027319981757,
0.5370764326800048,
0.3290793485287995,
-1.7004803192621687,
-0.8267928250682555,
0.7226363685371004,
-0.23961420559359078,
-0.9892421071296497,
-0.3062192663980587,
-1.2857004554953664,
-0.00570902688287085,
-0.7828701256611971,
0.036023661097245405,
-1.405764493898646,
0.09523715910714706,
1.0678085339341385,
-0.3477609497961018,
-1.3254201123902754,
-0.5891224027277412,
-0.14272029692701035,
0.10490796904880517,
2.930201181567992,
0.04738429480019876,
0.0461816378739841,
-0.48875992312198635,
-0.1220291693117149,
-0.4148582659190601,
0.17851659149817373,
0.4130064273610762,
-0.06315869176871276,
0.3800783279561489,
-0.02544638146603332,
2.215655886557845,
0.11710086424953603,
0.9518430371086406,
-0.12584714752332027,
0.018420823357896687,
1.2484195950702188,
0.13344161240602143,
0.09153656466168367,
0.7487460411896285,
0.4894289612390041,
0.03364631653773711,
-0.40583141994665783,
-1.508449266642364,
-0.0876621026521168,
0.16953459800120027,
0.7644948380070223,
1.6192126500413289,
-0.10749735247146877,
2.0545430524982202,
0.024857404521674214,
-0.048992600661546246,
1.151942209932795,
-0.5651424971501683,
-0.3291333996292886,
1.3277495361297842,
0.3468681089352445,
0.25355521459481833,
-1.43151363095255,
-0.6791478024501341,
-0.11073517248868558,

```

0.6626722725413648,
-1.1931469775385106,
-0.5976012599191521,
-0.6367586148577103,
-0.2241173972369649,
-1.0874289140947961,
-0.4768780848197268,
1.2286899896766026,
1.7976782920817624,
0.10434792845100443,
-0.31719050779977503,
0.7620589119443072,
-0.6377447181256041,
0.484158469702177,
-0.5854682569359358,
0.7650979634948629,
-0.6510076440579313,
1.4650942622453875,
1.0359738562191985,
0.7750682527878552,
-0.37331893064057897,
-0.1914404490007497,
0.25187298257054047,
0.370887355356332,
0.9916451969775287,
2.255013915351184,
0.564469362172986,
-1.1669103509730057,
0.42926615022397796,
0.43095576457077756,
0.7007537480246202,
3.3633287927079483,
-0.5433314326657547,
0.03986376903117299,
-1.913610430050787,
-1.183549627581873,
-1.5712622883507832,
-1.282300011205569,
-0.31197479792826754,
-2.6880651613874877,
0.33078250437153595,
0.07570451779538533,
0.1724953212076054,
-1.3944682324956643,
-0.31604217181433353,
1.821638064235333,
-0.4612503790165566,
0.19201221884571829,
-2.5536752551832254,
-3.5133198239284997]],
'run_time_sec': 34029.58519411087}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,

```

4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,

[illegible]

```

        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#3'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': None,
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 +
                                      + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 +
                                      + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
                'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                            'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                            '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                            '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                            '5 X 6 + 1 Z 5 + 1 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
                                  [12, 6, 0, 0, -5.679026952355478]],
                'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                      'identity_factors': [0, 0, 0, 0],
                                      'noise_level': [4, 2, 0, 0],
                                      'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 6.94391393661499}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'},
'init_param': { 'value': [ 0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,

```

3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,

```

-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,

```



```

1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
    'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
    'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#7'},
    'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
    'optimization': { 'algorithm': 'SLSQP',
    'constraint': True,
    'status': False}},
    'zne': { 'data_points': None,
    'degree': 0,
    'method': 'richardson',
    'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
    '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
    '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
    'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
    'X 2 + 1 Z 1 + 1 X 2 X 3 + '
    '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
    '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
    '5 X 6 + 1 Z 5 + 1 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -6.536883635090364],
    [12, 6, 0, 0, -5.491252101222686]],
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [4, 2, 0, 0],
    'noise_profile': { 'noise_on_init_param': {
'status': False,
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 6.029532432556152}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': { 'max': 10.0, 'min': 0.0},
'type': 'ising'},
'init_param': { 'value': [ 0.17096748946205537,
0.24182068788319971,

```

0.5294082221811705,
0.6436439076685059,
0.6752523962811601,
1.4661557963348293,
1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,

```
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
```

```

        'status': True,
        'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#2'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 +
        + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 +
        1 X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X
            X 2 + 1 Z 1 + 1 X 2 X 3 +
            1 Z 2 + 1 X 3 X 4 + 1 Z 3
            + 1 X 4 X 5 + 1 Z 4 + 1 X
            5 X 6 + 1 Z 5 + 1 Z 6']],
    'output': { 'data_points': [ [4, 2, 0, 0, -7.139980799817413],
        [12, 6, 0, 0, -5.986699722057316]],
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': { 'noise_on_init_param': {
            'value':
                'noise_prob': [ 0.001,
                    0.001,
                    0.0,
                    0.0],
                'status': True,
                'type': 'dephasing'},
            'odd_wires': 4},
        { 'gates_num': [12, 3, 1, 0],
            'identity_factors': [1, 1, 0, 0],
            'noise_level': [12, 6, 0, 0],
            'noise_profile': { 'noise_on_init_param': {
                'value':
                    'noise_prob': [ 0.001,
                        0.001,
                        0.0,
                        0.0],
                    'status': True,
                    'type': 'dephasing'},
                'odd_wires': 4}],
        'run_time_sec': 6.2098565101623535}},
    { 'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
            'cn': [ 0.5,
                0.5,
                0.5,
                0.5,
                0.5,
                0.5],

```

```
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}},
'init_param': { 'value': [ 0.23860472738944535,
    0.24738626223792576,
    0.368610904156845,
    0.9146619524924428,
    0.9146619524924426,
    1.6697383257267104,
    1.6697383257267104,
    2.054253047546413,
    2.3726208960540474,
    2.9146286026908905,
    2.9146286026908905,
    2.9146286026908905,
    2.9146286026908905,
    2.9146286026908905,
    3.4924659345227917,
    4.165338212258259,
    4.165338212258259,
    4.795197187789762,
    5.022734448096858,
    5.707107792388194,
    5.707107792388194,
    8.323917907910845,
    8.967030236986108,
    9.27289747465919,
    9.27289747465919,
    9.651472654450101,
    9.651472654450103,
    10.4607661524399,
    10.4607661524399,
    13.149200610156198,
    0.08980902398510449,
    0.6419619591442681,
    0.3976886140600265,
    -0.1339261955535598,
    -0.07178891234334085,
    0.20844549930040607,
    -0.6039755260047369,
    0.5345472321438205,
    0.4885623179827016,
    0.9007848202255657,
    0.29262188635397585,
    1.0311974603475478,
    -0.7987894429612461,
    -0.6323817753587336,
    -0.8923011457984407,
    -0.37936698482152614,
    0.06585808205756116,
    0.5122477185376288,
    -0.2246894945378454,
    -0.8796883548310565,
    0.4023829077269032,
    -0.6752839410958273,
    0.20069365692379137,
    0.5033852680281804,
    0.24960186092606218,
    0.058252930735255344,
    0.2566722934175659,
    -0.1810188232578722,
    -1.0819601868914308,
    0.6548412011669319,
    -0.14141206607793577,
    1.592915324309131,
    -0.8278164458051894,
    -0.3812348095673508,
    -0.9060991568034324,
    0.5864630340426413,
    -0.2612726529884887,
    -0.23456784875555411,
    0.3993109852728946,
    -1.6100891847686745,
    -0.025086921642605235,
    0.17965009471210514,
```

-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,

```

-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.823046671583829],
[12, 6, 0, 0, -5.5974585022328744]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 6.004320859909058}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,

```

```
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.3675336714235261,
0.40575472279006464,
0.6952672462769884,
2.137221581425111,
2.137221581425111,
2.4653237039513405,
2.4653237039513414,
2.671087921572092,
2.6710879215720915,
3.889448133547601,
3.889448133547601,
5.229066948311018,
5.229066948311018,
5.736515631630257,
5.736515631630257,
5.736515631630257,
8.147692740111799,
8.179590109004724,
8.179590109004724,
8.179590109004724,
8.21310311779852,
8.21310311779852,
8.752241426452148,
8.752241426452148,
8.803045108086177,
9.928410928444814,
9.928410928444814,
9.928410928444814,
9.969030315653287,
14.058841247147779,
-0.593395889833616,
1.0797089435996057,
-0.18386516671058062,
0.9172540646403935,
0.5920567467507187,
0.5030949392578172,
-0.19817311203429638,
0.8732878271906367,
1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
```


-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,

```

2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
'+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
'1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
'X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
'+ 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.274309228763829],
[12, 6, 0, 0, -6.102630979708389]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],

```

```
'run_time_sec': 5.897228240966797}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'}},
  'init_param': { 'value': [ 0.5483166258326411,
                             0.5788252343888002,
                             0.5803330366161238,
                             1.2169503585394688,
                             1.4292236971359287,
                             1.4293923120692815,
                             1.4365169928726365,
                             1.4365169928726365,
                             2.510196103669615,
                             2.510196103669615,
                             2.5544154452534937,
                             2.5544154452534937,
                             4.029649065019888,
                             4.067883071594656,
                             4.076808377533052,
                             4.1651466476623735,
                             4.656187563066811,
                             4.656187563066811,
                             5.18228186954163,
                             5.18228186954163,
                             6.808184896226372,
                             6.808184896226372,
                             6.960163700827979,
                             6.960163700827979,
                             6.960163700827979,
                             8.49759837936073,
                             8.49759837936073,
                             8.526772837307739,
                             9.456275223201612,
                             9.799125697936137,
                             -1.1324033485399458,
                             0.3066519452505097,
                             0.43856397893505916,
                             2.603476267806093,
                             0.0006914105351412576,
                             1.1085344664632693,
                             0.4735183530419211,
                             1.0969942437601548,
                             1.7819039437129416,
                             1.2220922008004673,
                             -0.516581792367389,
                             -1.8223032952581883,
                             -0.1454480817152758,
                             -1.2131062816976166,
                             -0.02593999163205417,
                             0.5898254930964741,
                             0.7377016495064371,
                             -0.18729526225026624,
                             -0.24589424731168963,
                             0.8550874647467951,
                             -0.3224945411184162,
                             -0.36832376208491285,
                             -0.881745818749188,
                             -0.4517353913610538,
                             -2.6507955691692096,
                             1.4089040411711087,
                             -1.98647805300985,
                             0.09207178440352033,
                             0.1176545491584157,
                             -0.4036806074380365,
                             0.28240629583972154,
                             -0.5198549065378171,
```

1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,

```

1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'veqe': { 'iteration': 1,
          'optimization': { 'algorithm': 'SLSQP',
                            'constraint': True,
                            'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                 '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                 '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
           'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']]},
'output': { 'data_points': [ [4, 2, 0, 0, -6.2062380598075135],
                             [12, 6, 0, 0, -5.187681916745086]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
                 0.001,
                 0.0,
                 0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,
                 0.001,
```

```

0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4]],
'run_time_sec': 6.081955671310425}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,

```

-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,

```

1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.376486414275041],
                             [12, 6, 0, 0, -5.250000083007176]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'value':

```


'status': False,

'value':

0},

'noise_prob': [0.001,
0.001,
0.0,
0.0],

'status': True,
'type': 'dephasing'},

'odd_wires': 4]],

'run_time_sec': 7.567617416381836}},

{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,

'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},

'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},

'init_param': { 'value': [7.923394728339895e-17,

0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
5.879155312872552,
5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,

0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,

```

0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.275631329374742],
                             [12, 6, 0, 0, -5.287235063864106]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': { 'noise_on_init_param': {
'status': False,
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'value':
0},

```

```

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': { 'noise_on_init_param': {
    'value':
      'noise_prob': [ 0.001,
                      0.001,
                      0.0,
                      0.0],
    'status': True,
    'type': 'dephasing'},
    'odd_wires': 4}},
  'run_time_sec': 5.542722702026367}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'ising'}},
    'init_param': { 'value': [ 0.18025020416614318,
                                0.18025020416614315,
                                0.4610778566518408,
                                0.6219220463806964,
                                0.7278326366057072,
                                0.8201944581570828,
                                1.2788460286146952,
                                1.4054470533314678,
                                1.4054470533314678,
                                2.282706833033757,
                                2.2962450119862856,
                                2.3028314628270254,
                                4.069670148635338,
                                4.069670148635338,
                                4.605143078506114,
                                4.605143078506114,
                                4.605143078506114,
                                4.712341711691788,
                                4.712341711691788,
                                4.71827378027349,
                                4.71827378027349,
                                5.585132855182541,
                                5.585132855182542,
                                6.936688988043727,
                                6.996547924306228,
                                7.327044121023376,
                                7.847751295142583,
                                7.8562202008953275,
                                8.679046525727268,
                                9.452714795876016,
                                0.01979341787597725,
                                -2.1649513485863423,
                                -0.3469207112923858,
                                -0.19054577342605242,
                                0.08943279174694625,
                                -0.40023385514982557,
                                -0.3378513267335698,
                                -0.32184385507584645,
                                0.17281592706447121,
                                0.5432446490193347,
                                -0.2468193649067236,
                                -1.491986060579654,
                                -0.6929963404149894,
                                0.030224915087063667,
                                -0.08162385956356048,
                                -1.6881840196674878,
                                -0.5438179745034216,

```

0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,

```

0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 '
                                '+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + '
                                '1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 '
                                                         'X 2 + 1 Z 1 + 1 X 2 X 3 + '
                                                         '1 Z 2 + 1 X 3 X 4 + 1 Z 3 '
                                                         '+ 1 X 4 X 5 + 1 Z 4 + 1 X '
                                                         '5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.029866062982579],
                             [12, 6, 0, 0, -5.108515955211577]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': { 'noise_on_init_param': {
'status': False,
'value':
0},
'noise_prob': [ 0.001,

```

```

0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'init_param': { 'value':
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 7.560657262802124}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.08925866669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,

```

-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,


```

-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noise_free_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3
+ 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 +
1 X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1
X 2 + 1 Z 1 + 1 X 2 X 3 +
1 Z 2 + 1 X 3 X 4 + 1 Z 3
+ 1 X 4 X 5 + 1 Z 4 + 1 X
5 X 6 + 1 Z 5 + 1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
[12, 6, 0, 0, -4.800941579254223]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],

```

```

'noise_profile': { 'noise_on_init_param': {
'status': False,
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': { 'noise_on_init_param': {
'status': False,
0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4}],
'run_time_sec': 7.055020570755005}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
'file_name_prefix': 'ising_noisefree_time_evo'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z '
'2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + '
'1 Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X '
'3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z '
'4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 4.358359699677247,
4.598584386614463,
5.205211991476004,
2.498239515374607,
3.620239544509655,
3.5202907273108752,
3.874294382863561,
2.6240487828355272,
3.466109018012848,
4.215104097495168],
'noise_details': { 'gates_num': [4, 1, 1, 0],

```

```
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -7.274309228763741,
                             -7.1399807998173275,
                             -6.7841170852099,
                             -6.206238059807503,
                             -6.275631329374697,
                             -5.70418119478636,
                             -6.53688363509032,
                             -6.029866062982587,
                             -6.376486414275039,
                             -6.823046671583788],
'optimized_parameters': [ [ 0.3675336714235261,
                             0.40575472279006464,
                             0.6952672462769884,
                             2.137221581425111,
                             2.137221581425111,
                             2.4653237039513405,
                             2.4653237039513414,
                             2.671087921572092,
                             2.6710879215720915,
                             3.889448133547601,
                             3.889448133547601,
                             5.229066948311018,
                             5.229066948311018,
                             5.736515631630257,
                             5.736515631630257,
                             5.736515631630257,
                             8.147692740111799,
                             8.179590109004724,
                             8.179590109004724,
                             8.179590109004724,
                             8.21310311779852,
                             8.21310311779852,
                             8.752241426452148,
                             8.752241426452148,
                             8.803045108086177,
                             9.928410928444814,
                             9.928410928444814,
                             9.928410928444814,
                             9.969030315653287,
                             14.058841247147779,
                             -0.593395889833616,
                             1.0797089435996057,
                             -0.18386516671058062,
                             0.9172540646403935,
                             0.5920567467507187,
                             0.5030949392578172,
                             -0.19817311203429638,
                             0.8732878271906367,
                             1.3773497870778637,
                             -0.2091442862578227,
                             0.9912881446359345,
                             2.1471684900095735,
                             1.8011951688988466,
                             0.08640841047975001,
                             0.12889510785919772,
                             0.05509015903577425,
                             0.056646924667161334,
                             -0.24683198530133132,
                             0.10458007099731792,
                             1.1863156573315168,
                             -2.4620703709527216,
                             0.7027036608466238,
                             1.7463688042058148,
                             -0.22769098390292125,
                             0.021713799462876483,
                             -0.0317703225281112,
                             0.35077834879841635,
```

0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,

0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804],
[0.17096748946205537,
0.24182068788319971,
0.5294082221811705,
0.6436439076685059,
0.6752523962811601,
1.4661557963348293,
1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,

0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,

0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993],
[4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,

-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,

-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138],
[0.5483166258326411,
0.5788252343888002,
0.5803330366161238,
1.2169503585394688,
1.4292236971359287,
1.4293923120692815,
1.4365169928726365,
1.4365169928726365,
2.510196103669615,
2.510196103669615,
2.5544154452534937,
2.5544154452534937,
4.029649065019888,
4.067883071594656,
4.076808377533052,
4.1651466476623735,
4.656187563066811,
4.656187563066811,
5.18228186954163,
5.18228186954163,
6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,

0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,

0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301],
[7.923394728339895e-17,
0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
5.879155312872552,
5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,

1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,

0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084],
[0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.08925866669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,

0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,

0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674],
[0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,

-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,

-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603],
[0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,

-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,

0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171],
[0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,

-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,

1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893],
[0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,

-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,

```

-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
    'run_time_sec': 46646.7807404995}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
    'init_param': { 'value': [ 4.68500097130382e-19,
    0.00019188574128314593,
    0.4999939126216869,
    0.571475986197529,
    0.6379337879222081,
    1.2223970983691403,
    1.405127109337375,
    1.405127109337375,
    1.405127109337375,
    3.9334167957034354,
    3.9801359916712626,
    3.9801359916712626,
    4.868547987503254,
    4.868547987503254,
    5.067947873787899,
    5.231153270915478,
    5.2311532709154775,
    6.116966983963522,
    6.116966983963522,
    7.100862078116925,
    7.100862078116925,
    7.100862078116925,
    8.658171837297239,
    8.670337232190553,
    8.670337232190553,
    8.670337232190553,
    10.099540897359793,
    10.10264280920201,
    10.102675042499124,
    11.255727095089838,
    -0.004954852195250444,
    0.49334633432738895,
    0.4083998920567888,
    0.7005599603945909,
    -0.2726710135013912,
    1.7017720535726606,
    0.31242824153108295,
    0.18213977913198362,
    -0.768972357244276,
    -1.2675908133090252,
    0.24851075547279064,
    0.758632579424417,
    0.15831380308868334,
    -0.40713384248795287,
    0.5550964450005433,
    0.49124548320800016,
    0.12908691722609594,
    0.7495294852763479,
    0.2183951382187659,

```

0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,


```

-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
[12, 6, 0, 0, -5.679026952355478]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.719329833984375e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.336662151637054,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.7841170852098625,
-5.679026952355478],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.7841170852098625],
[12, 6, 0, 0, -5.679026952355478]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,

```

2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.08925866669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,

```

-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],

```

```
'cn': [0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
[12, 6, 0, 0, -4.800941579254223]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.124641418457031e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.155801002552482,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -5.704181194786395,
-4.800941579254223],
'sorted_noise': [6, 18]},
'sampled_data': [ [4, 2, 0, 0, -5.704181194786395],
[12, 6, 0, 0, -4.800941579254223]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
```

0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,

```

0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.029866062982579],
[12, 6, 0, 0, -5.108515955211577]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.6716461181640625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.490541116868082,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.029866062982579,
-5.108515955211577],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.029866062982579],
[12, 6, 0, 0, -5.108515955211577]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,

```

```
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                              'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                              'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'ising'}},
'init_param': { 'value': [ 0.17096748946205537,
                           0.24182068788319971,
                           0.5294082221811705,
                           0.6436439076685059,
                           0.6752523962811601,
                           1.4661557963348293,
                           1.466155796334829,
                           1.5116530105229151,
                           3.783489526832599,
                           3.7842138074378933,
                           3.8804047790072995,
                           3.8804047790072995,
                           4.154125602724648,
                           4.692532374514593,
                           4.692532374514593,
                           5.064132124282894,
                           5.064132124282894,
                           5.064132124282894,
                           6.382147767045899,
                           6.751768360743096,
                           6.751768360743096,
                           6.751768360743096,
                           7.743234563929715,
                           8.1520396126177,
                           8.1520396126177,
                           8.546274908334533,
                           8.589779496078958,
                           8.589779496078958,
                           9.859248780822462,
                           13.3058320214528,
                           0.25900326422191966,
                           0.7885699864164276,
                           0.41320213613295365,
                           -1.1189869522952365,
                           0.26027865496517016,
                           0.9396191194055923,
                           -0.036590936599100087,
                           0.4526959896459153,
                           0.23545371285521352,
                           -0.08089020453927935,
                           -0.04037254840266629,
                           1.8055011929897393,
                           1.0437583322662911,
                           0.8301172091681238,
                           1.4721458579301823,
                           1.0565790032678872,
                           -0.0620204872484031,
                           0.7778737999221846,
                           -0.07568060381342757,
                           -0.4832550478503789,
                           1.561646348208689,
                           1.0276654568346077,
                           0.6741324750771829,
                           0.5058216800397286,
                           -1.0558943326715462,
                           -0.14283396260407313,
                           -0.12020755874532785,
                           0.37547144626073276,
                           -0.3549607031100976,
                           -0.10410956462681178,
                           1.5153634922337307,
                           1.9495981063680297,
                           0.31118583706033537,
                           -0.7235645457595358,
                           0.17173795551894178,
                           -0.33392096172962504,
                           0.2993290942776598,
                           -0.07002988913014692,
                           0.10352604443544179,
                           0.5331508884476422,
```

-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,


```

1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.139980799817413],
[12, 6, 0, 0, -5.986699722057316]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.457069396972656e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.716621338697461,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.139980799817413,
-5.986699722057316],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.139980799817413],
[12, 6, 0, 0, -5.986699722057316]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,

```

9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,

```

-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.536883635090364],
                          [12, 6, 0, 0, -5.491252101222686]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.933906555175781e-05,
            'zne_values': { 'degree': 0,

```

```

        'extrapolated_value': -7.059699402024204,
        'others': { 'beta_coefficients': [1.5, -0.5],
                    'cost_richardson_zne': 2.5,
                    'sorted_expectation_vals': [ -6.536883635090364,
                                                  -5.491252101222686],
                    'sorted_noise': [6, 18]},
        'sampled data': [ [4, 2, 0, 0, -6.536883635090364],
                          [12, 6, 0, 0, -5.491252101222686]],
        'sampling': 'default'}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'ising'}}},
  'init_param': { 'value': [ 0.22237247586897502,
                              0.3434883293456931,
                              0.600978934518952,
                              0.6193810427253835,
                              1.6475035333031534,
                              2.204130997587115,
                              2.4096412381325085,
                              2.515509113107616,
                              2.5155322756928205,
                              2.9680516150838403,
                              2.9680516150838403,
                              3.0576851370343725,
                              3.8016986766973213,
                              4.264365678603219,
                              4.264365678603219,
                              5.4290969626181775,
                              5.4290969626181775,
                              5.453519119574046,
                              6.855500326085536,
                              6.858194453958984,
                              6.956907552727685,
                              7.899693187867781,
                              8.578442687794452,
                              8.60387860142029,
                              9.193247925689713,
                              9.205803142575261,
                              9.205803142575261,
                              9.815391824330385,
                              9.815391824330385,
                              10.97297451152796,
                              -0.4325514790797943,
                              -1.0860755794816048,
                              -0.34917301406803075,
                              0.9554400980633908,
                              -0.3272724546684993,
                              0.13698615504934158,
                              -0.3120459104790063,
                              -0.5324325213761384,
                              1.6476833500899999,
                              2.184163963641692,
                              1.0643894246063128,
                              -2.7436302898700338,
                              -0.9203607373406255,
                              0.2960792725949677,
                              1.646667339143857,
                              -0.34154797106217816,
                              0.5971775478384601,
                              -0.8758117372527006,
                              -0.5328437036469449,
                              1.0093796179148329,
                              -0.3390771908975529,
                              -1.0516631129606762,
                              -1.899848962378923,
                              1.0171271290916915,
                              1.938798354471919,
                              -0.9913918052370154,
                              -0.5882123471692551,
                              -0.2591906358136659,
                              -0.16051793033532072,

```

-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,

```

-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.376486414275041],
[12, 6, 0, 0, -5.250000083007176]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.1484832763671875e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.939729579908972,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.376486414275041,
-5.250000083007176],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.376486414275041],
[12, 6, 0, 0, -5.250000083007176]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.5483166258326411,
0.5788252343888002,
0.5803330366161238,
1.2169503585394688,
1.4292236971359287,
1.4293923120692815,
1.4365169928726365,
1.4365169928726365,
2.510196103669615,
2.510196103669615,
2.5544154452534937,
2.5544154452534937,
4.029649065019888,
4.067883071594656,
4.076808377533052,
4.1651466476623735,

```

4.656187563066811,
4.656187563066811,
5.18228186954163,
5.18228186954163,
6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,

```

-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301]],
'noise_profile': { 'noise_on_init_param': { 'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#4'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```



```

      'optimization': { 'algorithm': 'SLSQP',
                        'constraint': True,
                        'status': False}},
      'zne': { 'data_points': [ [4, 2, 0, 0, -6.2062380598075135],
                                [12, 6, 0, 0, -5.187681916745086]],
                'degree': 0,
                'method': 'richardson',
                'sampling': 'default'}},
      'output': { 'exact_sol': -8.566772233505624,
                  'run_time_sec': 3.6716461181640625e-05,
                  'zne_values': { 'degree': 0,
                                  'extrapolated_value': -6.7155161313387275,
                                  'others': { 'beta_coeffieicients': [1.5, -0.5],
                                              'cost_richardson_zne': 2.5,
                                              'sorted_expectation_vals': [ -6.2062380598075135,
                                                                           -5.187681916745086],
                                              'sorted_noise': [6, 18]},
                                  'sampled data': [ [4, 2, 0, 0, -6.2062380598075135],
                                                    [12, 6, 0, 0, -5.187681916745086]],
                                  'sampling': 'default'}}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                          'time': {'max': 10.0, 'min': 0.0},
                                          'type': 'ising'}}},
      'init_param': { 'value': [ 7.923394728339895e-17,
                                0.44053533716557003,
                                0.44053533716557,
                                0.8633875163783187,
                                0.8807523602927853,
                                0.8807523602927853,
                                1.0224141131682343,
                                1.599513745187798,
                                2.8464451213500714,
                                2.922472045892935,
                                2.922472045892935,
                                2.9239478183172296,
                                4.400338604169817,
                                4.721305166667179,
                                4.721305166667179,
                                4.721305166667178,
                                5.011876780494166,
                                5.011876780494166,
                                5.011876780494166,
                                5.879155312872552,
                                5.879155312872551,
                                6.750159190843359,
                                6.75015919084336,
                                8.012324492482197,
                                8.012324492482197,
                                8.7704267977754,
                                9.114848260128039,
                                9.114848260128039,
                                9.87613180402372,
                                10.23973172788205,
                                0.09774272849121603,
                                0.0018601482237552494,
                                -0.09414894191588159,
                                0.03531507271404531,
                                0.01801632266215566,
                                -0.031446214752801294,
                                0.04762079993218455,
                                0.16430279282954635,
                                -1.4212045547488918,
                                -0.051616216342739524,
                                -1.2094395115841,
                                1.961457452158631,
                                -0.09882210453707145,
                                1.4772087948778743,
                                -1.07174135216323,
                                -0.1881882042247284,
                                0.9429310525529949,
                                0.06018460007044777,

```

-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,

```

0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.275631329374742],
[12, 6, 0, 0, -5.287235063864106]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 5.435943603515625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.76982946213006,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.275631329374742,
-5.287235063864106],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.275631329374742],
[12, 6, 0, 0, -5.287235063864106]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'},
'init_param': { 'value': [ 0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.275631329374742],
[12, 6, 0, 0, -5.287235063864106]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 5.435943603515625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.76982946213006,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.275631329374742,
-5.287235063864106],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -6.275631329374742],
[12, 6, 0, 0, -5.287235063864106]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'},
'init_param': { 'value': [ 0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.275631329374742],
[12, 6, 0, 0, -5.287235063864106]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 5.435943603515625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -6.76982946213006,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -6.275631329374742
```

1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,

```
0.4312357294916266,  
-0.18299603845951454,  
0.706673586721164,  
-0.7147228256515984,  
0.21227947796353308,  
0.0588792409702735,  
0.37104688015833304,  
0.5624160446643004,  
-0.9359341524655871,  
1.3022223571582927,  
-0.08087685811880535,  
-0.9026141041038307,  
0.16238954726297725,  
1.5451829819142204,  
0.7129468042556484,  
0.7230095383104325,  
0.17583341711081185,  
0.8889071634718573,  
1.3752518501200566,  
0.0640253150209717,  
1.2836939598332835,  
0.3344321550301345,  
0.3035357788500751,  
0.049516329509299586,  
0.012257236048235042,  
1.5823783745435407,  
-0.09537942982333185,  
0.5341486888465604,  
-4.067358953041359,  
-0.2870183941358629,  
0.4906845201891505,  
0.6709676538790386,  
-0.10883760977815361,  
-0.9015075659614459,  
0.5057755407774712,  
0.2603749012295441,  
0.8480384556641941,  
0.3777251280717097,  
-0.7517099448725303,  
0.5824300904263875,  
0.42464435669423206,  
-0.9724368356834788,  
-1.035580562026805,  
-0.008606395290722258,  
2.0040454226391233,  
-0.326939809664038,  
0.0009364740955449761,  
1.6853561871618157,  
0.2235885662175078,  
1.2569352560498372,  
-0.35590863169612874,  
-0.40928456866831275,  
-0.8730825626925035,  
-0.2434340162934524,  
0.014655642366467374,  
-0.07506183248897615,  
0.12344014428925512,  
0.4451771135631272,  
-1.1817851818398777,  
-0.17749779140411484,  
0.48159794071814915,  
-0.12344850129759947,  
0.0485840790320397,  
-0.8467407381542291,  
0.027429753657626434,  
0.7224917290233174,  
-0.9085522245342863,  
-0.08852023391973605,  
-3.0071097182531115,  
-1.466370256379572]],  
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],  
                    'status': True,  
                    'type': 'dephasing'},  
'nqubits': 7,
```


0.8732878271906367,
1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,

```

0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.274309228763829],
[12, 6, 0, 0, -6.102630979708389]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.170967102050781e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.860148353291549,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.274309228763829,
-6.102630979708389],
'sorted_noise': [6, 18]},
'sampled_data': [ [4, 2, 0, 0, -7.274309228763829],
[12, 6, 0, 0, -6.102630979708389]],
'sampling': 'default'}}}},
'ising-ric3-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,

```


False,

```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': False,
'type': 'dephasing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix': 'ising_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + '
'1 Z 0 + 1 X '
'1 X 2 + 1 Z '
'1 + 1 X 2 X '
'3 + 1 Z 2 + '
'1 X 3 X 4 + '
'1 Z 3 + 1 X '
'4 X 5 + 1 Z '
'4 + 1 X 5 X '
'6 + 1 Z 5 + '
'1 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
'initial_cost_history': [ 4.217815007292055,
3.394484740359831,
3.3211044240247634,
```

```
3.65155648872765,
4.031781430865161,
2.5147460401592765,
2.635251011361358,
2.6229212283850565,
3.6175377967590236,
3.125720978907869],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [0, 0, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0],
'status': False,
'type':
'dephasing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -6.945492229929243,
-7.1165547898808885,
-7.2842760350438445,
-6.724195761448022,
-7.306479477026572,
-7.415156948711891,
-7.202776954905004,
-6.836396584821816,
-7.611342393541609,
-7.203866848572784],
'optimized_parameters': [ [ 0.33159768767931425,
0.3415216906329597,
0.34152169063295984,
0.5593316391966133,
0.5593316391966133,
2.351554445259766,
2.9331738085905035,
3.4748030853426815,
3.4748030853426815,
3.616855036152045,
3.616855036152045,
3.8672141033893017,
4.423346341593822,
4.423346341593821,
4.845338698326811,
4.845338698326812,
7.020561389723981,
7.020561389723981,
8.059188148673007,
8.486998646842132,
8.48889562974299,
8.48889562974299,
8.48889562974299,
8.852856222269512,
8.852856222269512,
8.85285622226951,
9.91447562836983,
9.968000692942592,
10.549940074823725,
10.549940074823727,
0.8537391604836984,
0.9168200846380319,
0.2559354991965516,
1.0210726229552618,
1.2632715564780685,
-0.3480423461281932,
1.7009717288666621,
0.8945988480638971,
0.0342878706217857,
1.5867242856094939,
-0.46323092808777266,
0.9197703726655558,
0.7442882527479103,
```

0.9265871236439417,
1.007434201350355,
0.029736173082484843,
0.47331341236562946,
0.4861485561257538,
1.0394118763980624,
-2.9809358180453613,
-0.2596551057591169,
0.274814845028427,
-1.4049453096263822,
-1.0753350027554103,
-1.11413441253153,
0.9507335703621095,
0.8064528289725565,
-0.4246412780854718,
0.4472568123237856,
1.634610917427341,
0.8557019227225735,
-0.10158119911628401,
-0.31211728480663814,
-0.9970023252564638,
0.25431186671884126,
0.7248953017537308,
-0.4073131817183126,
1.6792046485134604,
0.5505815433075344,
-0.8093860553948707,
-0.6172486758497164,
-0.12685386183371894,
0.21223631022564798,
0.8168562896383363,
-0.17647756432116687,
-0.5731604235592278,
0.4333036783698612,
-0.7682262923606079,
1.6429671181235093,
0.6337476059862421,
1.67916133236013,
-0.5926217976812627,
-0.298257419735069,
-0.6583760201171601,
-0.889601924248653,
-1.3248621012262398,
0.013854140862500447,
-0.7662338030087398,
-0.6194525061041963,
0.6815602243054589,
0.08161257575833167,
-0.3072072388036838,
0.42225747459757357,
0.06754004785449946,
0.4843941525302115,
-0.1684623955114203,
0.8138047240050311,
0.052754329692318934,
-0.03499144746108355,
0.04466125788054358,
-0.028799258738440928,
0.7127927371021741,
0.7421378220215715,
-0.6858695512918895,
-0.11720668569532972,
-1.3579343673989164,
-0.9607782638271641,
2.0876158669999754,
0.030744163991020267,
-0.6062544285001135,
-0.08046454707386355,
-0.9589458574514981,
0.6218575881236439,
0.008264172331609584,
0.1779434874985719,
-0.2072304227788729,
0.9938506593352908,
-1.0471692270924964,

1.0898429645649914,
-0.22066548691854398,
0.8076838524861312,
1.7135973843396541,
0.23198771018109937,
-0.14790366150484735,
-0.7779724161623084,
-0.5740177045120107,
-0.8356765454865094,
-0.12758989923704314,
-0.3192845855088301,
0.9992686200806561,
0.18910999621207844,
-0.3626298108536852,
0.5832805029120673,
-0.8863720585918194,
-1.6851050525249511,
-0.3551820269447921,
-0.4184387520471605,
1.05736378101237,
2.031993671638238,
0.20520618587592238,
1.6363439075818225,
0.14342936783484309,
-0.6043776905728043,
0.3922986049495649,
1.4322290738810806,
1.585876902523968,
-0.2738910870161085,
1.7714953537072202,
-0.29354248742932953,
-1.2273457901372085],
[2.0562096130838231e-16,
0.03538692433604476,
0.5493527934980842,
0.5695965809115139,
1.2739811069641827,
1.2739811069641827,
1.5440077454822854,
2.6187560157598857,
2.6193651223262795,
4.47656169572244,
4.47656169572244,
4.900258790927604,
4.903975205691072,
4.965015364809818,
4.971477584375831,
4.9714775843758305,
6.285851088616883,
6.285851088616883,
6.5727281068015415,
6.914906570174498,
6.914906570174498,
7.5825149096464,
7.5825149096464,
7.5825149096464,
8.266106449462097,
8.772625056171776,
8.772625056171774,
8.772625056171774,
9.888607991704035,
10.650763302381714,
-0.25736569540817156,
-0.13749177250114622,
0.6913360970917151,
-1.0251451053141543,
0.04952583576033172,
0.6839090084427677,
1.2841797475889756,
-1.4498903904619547,
-0.6609983487872932,
-0.9195300213913112,
1.0882692755947088,
2.28444907625875,
0.5013640158162996,

1.3195089116901169,
-0.44969609005592753,
-0.19760723927349785,
0.047416784300074,
-0.2592183302467101,
-0.95267369637524,
0.2510197484008247,
0.12431427724764418,
-1.9455191869747381,
0.0834449432923055,
0.6181240781879018,
0.06318897794461549,
0.7469995321450943,
-0.13869412828035013,
1.7789335199415868,
-0.2133263169614118,
-1.747801096143808,
1.7374950822862127,
-0.1617072567646695,
-0.13415848991929505,
2.0599871163251184,
1.2617098025896833,
0.7421863291532844,
0.6134289282076183,
-0.010288112985795142,
-0.164325473683837,
-0.19747193793554596,
-0.1095664145547814,
0.3448002170188953,
-0.008329131684078827,
-0.27021098586965386,
0.5642984785825904,
0.26951720880242647,
0.9416408813291534,
0.18894282686740613,
0.6744670741290004,
0.15073677601160052,
0.7600850898072902,
0.23785895970346504,
-0.3096016709123609,
-0.29939543701383164,
-1.2773708023878871,
0.012610842713240655,
-2.001134648031791,
0.5569056601340245,
-0.022823668895979605,
-0.44069878780800564,
0.9238325880974837,
0.627132735311806,
0.833823816751152,
1.174260792549426,
-0.8145978291018492,
-1.8911045685123373,
0.03189540714483889,
-0.6114896265404774,
0.18855141943106907,
-0.5415449375701122,
0.3556432019461298,
-0.02940037290701983,
0.7847914443788936,
-0.8204017405579882,
-0.22117427936396428,
0.9668207238887914,
0.6982235282720571,
1.1161787173582038,
-1.4967922582645339,
0.6955597513963624,
-0.08409555407567708,
-0.7515653427189505,
-0.13809041364831362,
0.694805471024674,
-0.16980356694456364,
-0.7378676064228189,
-0.5510006148273209,
-0.4664642602089831,

0.3131774610690113,
0.25515679468716435,
0.01472701586112837,
0.9612508062048253,
0.3876700512851733,
0.34932481898103945,
-0.0017068592723796239,
-0.9400752962906423,
0.20680236697496335,
1.2239109612714394,
1.5295424358244767,
-0.16313500004842318,
-0.6283905359663017,
0.6007919237891753,
1.4275642137136517,
0.8373669928162908,
-0.3460569718594813,
0.30478283310349635,
0.7741593665645724,
-1.49133186355475,
-0.40261318631906845,
-0.6780231584993508,
0.2796292694824153,
0.8154646298435348,
0.6705233696972283,
0.8515977010836673,
1.38835715204434,
1.5513313401022284,
-1.9456573322270103,
0.6825855385724343,
0.3411942765245547,
-1.8463135511065893],
[-1.394844222866792e-17,
0.2294538829927996,
0.7521460514270448,
1.524448292944527,
1.699326062163185,
1.699326062163185,
1.699326062163185,
1.699326062163185,
1.6993260621631845,
1.6993260621631843,
2.04427958156354,
2.044279581563538,
3.760855608085718,
3.7608556080857176,
4.422362610147448,
4.422388771925709,
4.423247474933146,
5.088788131744124,
5.518902595462681,
5.518902595462682,
5.518902595462682,
5.518902595462682,
6.713884925045287,
6.713884925045287,
7.229900049876175,
7.23519041379673,
7.5137384820073985,
9.896584707205738,
9.89724831030544,
11.177528016222873,
11.333986770578415,
-0.5874618645254188,
-0.38498635294065175,
-0.14869891390826667,
-1.385574110751756,
0.3783147753212974,
0.1998192518416486,
1.8873134139140266,
-1.0608247810156872,
0.2365140987973328,
-0.9738696834833499,
1.5453125784784605,
1.8657042183850436,
1.5293157207302228,

-0.7517266227054149,
-0.8871771295479706,
-1.6950047018057137,
-0.5885280426519628,
0.04826899994478067,
-0.4330318552574016,
-1.388710608286563,
-0.08211595300211308,
-0.5283548125054784,
0.3763483175118206,
0.335884831266236,
-0.5070752646015597,
-0.2910412053619489,
0.2354568453295162,
-1.4336146483959564,
0.228779351175754,
-0.06520272691838713,
-0.8209928725245101,
0.7001970509858747,
0.5785044584879039,
-0.49932901566271715,
0.45729974286308867,
0.46052818546886504,
1.4986129468586522,
-1.4491938093059253,
0.58130192556905,
0.052769119433839204,
-0.00640289944448277,
-0.3687606063433655,
-0.006145485686113623,
-0.024597955898482345,
-1.411768598869635,
0.08097884701091772,
-0.9927225428945597,
-0.0075215312387787595,
0.010579195204485638,
-0.42659138073244207,
0.002225130694705108,
-0.17841021982057653,
0.9375108947888807,
-2.7757789566035065,
-0.010127033509228812,
-0.5146645773967682,
-0.3567686303606847,
1.4398753496798034,
-0.5703995887616833,
-0.05677408229384658,
0.957894065877575,
1.9071290282089846,
-0.6507219310887598,
-1.6162530141223683,
1.1818941119762436,
1.6868779692850424,
0.98800409304883,
-1.0587465299812662,
-1.094775277351337,
-0.2877600991535592,
1.8986470088901148,
0.8207273276904813,
1.2066175393079017,
-0.22168365465550993,
0.17258447386478676,
-1.8715762567403773,
-0.48240059727904117,
0.7145730346214052,
1.341149548855606,
1.3388694132947494,
1.1029357907173007,
1.7761192703736626,
0.389134329678578,
0.6998749042461945,
-0.4466329063075881,
0.9939411509507131,
0.5933993643264636,
-0.2238849148647213,

0.16806395587982337,
0.40153989063889134,
-0.3072506770355845,
-0.3084801740315303,
0.35020486394553546,
-0.6893875836471265,
1.1621768900261087,
-0.005422106750069378,
-0.6392006402141877,
-2.6960821965791135,
1.2350659365734071,
0.02344533464890316,
1.6191412587681537,
-0.624233381931643,
1.164388497878687,
2.2414892885807953,
2.3459185940732037,
-0.18816057823941634,
1.0703302771481322,
-0.6022027174173271,
1.5454586171163143,
0.49083778015320845,
-0.5414313403110282,
0.5943360939664153,
0.8531518824889626,
-2.0712921555535835,
-0.9507228787104378,
0.1663695492007906,
-0.4829435842745145,
-0.9852459132453085,
0.6444213968977472,
1.9304389405092637],
[0.47195138158384564,
0.4745413934062829,
0.6416788974326727,
0.6416788974326727,
1.3767805975406497,
1.3774084633016226,
1.3850075193774707,
1.385007519377471,
1.961329521314093,
1.961329521314093,
2.6299528219180073,
2.6299528219180073,
2.6462286434741538,
4.376679341839624,
5.443648648286045,
5.443648648286045,
5.443648648286044,
6.427045191271688,
6.427045191271688,
6.737104333282549,
7.345768854864493,
7.345768854864493,
7.523575375204561,
8.254985561732491,
8.914602409288278,
8.914602409288278,
8.914602409288278,
10.045438587538639,
10.045438587538639,
10.045438587538639,
-0.014294444089673569,
0.8050246736150516,
-0.4603650249158636,
3.8028784519891046,
-0.40535003035622474,
-0.20345609256189173,
0.6921728480649123,
1.2635655003066155,
-0.5816868768558133,
0.8004330576026809,
-1.2247112488961414,
-0.7226464348007927,
0.021600305535157273,

0.2422993600121551,
-0.2813766982328762,
-1.8376799098437777,
0.1842235451250743,
0.08033551390998013,
1.2509807369145618,
-0.43993928060300874,
0.02749412053256657,
0.9446032887644298,
0.5876918496487636,
-1.2193113596650038,
-0.30025121886784994,
0.40718701943488855,
-7.055881324256668e-05,
-0.3230027373880063,
-1.3595946794844804,
0.4025901861096578,
-0.331697269297153,
0.8588883241978998,
1.7499317067572346,
0.2974332602533821,
0.36978773801612186,
-0.1369546347590247,
0.17848126558694688,
0.5078793586065516,
-0.026587668618036284,
-0.32676172114586377,
0.043331505672591836,
0.7587982826624224,
0.5755592246710275,
0.5753466292739289,
0.6580990661684004,
-1.8654933062390762,
1.4954582293028102,
1.1206423350708015,
-0.6308795940680783,
-1.1220353310878195,
0.3113810383877842,
1.1285056041045296,
0.8014038814434817,
-0.3123476059423272,
-0.46649205373142577,
-1.0739566662407007,
1.7799335725034942,
0.3271320302071495,
-0.4957408509507477,
-0.8965339685216933,
0.9394687605669856,
-1.0123362977633281,
-0.8493226829357736,
1.9016506939491882,
0.39163185359556635,
1.2450374039748047,
0.6637318474595636,
1.4487854848879562,
0.8556001420669627,
0.2844708394448605,
1.19751608671342,
-0.2783699910645836,
0.2731694463477748,
-0.8573920204053973,
0.4298526210883498,
0.1620542969403055,
0.7385715014812971,
0.10161961925998468,
0.08514730828974114,
1.5393767966723602,
-0.7517708453085385,
-1.0667182547896288,
0.7653330843147801,
0.2770240276337771,
0.44649448354109117,
1.839692443661341,
0.6074846807891079,
0.5576661480775236,

-0.5735679701859718,
-0.5213243179488682,
-0.13640843594155702,
1.4993013794616976,
0.959504645257717,
-0.8869155163266706,
-1.0471786039389914,
0.6169717474797428,
0.010330090116839902,
-0.36051081047743266,
0.08240138993527565,
-1.19819214697301,
-1.2973765177480467,
-0.8056172495711347,
-0.04435763741623314,
0.14581600625591132,
0.26060016989542056,
0.07248105118801107,
0.5605125602873733,
0.7950998674453089,
1.1145979627095777,
0.00625222978811041,
-0.3867814373266647,
-0.1324550289145508,
-1.2907509838890496,
-2.272496441870613,
-0.4069714152055066,
-0.09281234013047389,
-0.02440219940061265,
-1.4477733222094868,
1.1238797279606656,
0.08808709541843764],
[0.4802107213676794,
0.5872443280976499,
1.2030235571335923,
1.6244635367917757,
1.6244635367917757,
3.26071227407772,
3.26071227407772,
3.8336805164974805,
3.8336805164974805,
4.826555535372041,
4.895860683399619,
4.953829011998897,
5.427903354045862,
5.904661066786138,
5.904661066786139,
6.1444777924663665,
6.1444777924663665,
8.501726489902856,
8.560790735911917,
8.56230393956988,
8.56230393956988,
8.56230393956988,
9.349516793328615,
9.383607475180085,
10.075280307758542,
10.10947387193881,
10.10947387193881,
10.676123926986733,
10.676123926986731,
10.676123926986731,
1.4057383265279821,
0.12082364332553383,
-0.26633169071888585,
0.8440095288231939,
-1.6788257521351209,
0.5654604704832584,
-0.8067769238497999,
-3.0754933205615647,
0.08918153353934634,
-0.44144180897560614,
-0.12233030448793467,
0.7466185674598658,
2.029541324404434,

2.6182216828178912,
-1.132391894978203,
-0.24681552984026395,
0.016701788041529435,
2.056331221347139,
0.4687565935163574,
-0.814652769544358,
1.3493029022207552,
-0.0979175823892474,
1.1060724549152727,
0.10932839153376225,
-0.021236171452369973,
-0.10240895200106154,
0.0678743319848748,
0.31105500906993255,
-1.3775189120014106,
0.1972259082837923,
0.5685166045457986,
-0.04805822649685886,
0.05656484641847995,
0.28963771198353216,
-0.0823457664175828,
0.10970044186707124,
0.4470565049690292,
0.3481580101245137,
1.429670826498759,
0.9929735218605864,
-0.118080682882002,
-0.24768077486638254,
0.14000119823560106,
-0.0033695140848767277,
0.28167924146812856,
0.33054680125209646,
0.25446310854586207,
0.4874199962063169,
-0.4292630164890844,
-0.8493517422041822,
1.10338617740252,
1.008639770575481,
-1.333715224046274,
1.5531304809990258,
-0.9099800622014602,
0.3709414533710614,
0.013272332284884182,
0.2649430125577457,
0.2281562083367391,
-0.8531201835868235,
0.3516874446864623,
-0.20156681030383022,
-0.20235738350440882,
-0.042569687456911996,
0.15952312088272655,
-1.5291183017296281,
-0.14843597874824083,
0.29841129212111156,
-1.0541286856947945,
0.1912222645636818,
-0.6584533547304796,
-0.8290997819567264,
-0.8585614987938113,
-0.925446343244433,
-0.31902871745414296,
0.059023427516986,
1.0543033158567225,
0.06361350569662501,
-0.24874942116320348,
0.14393135270630777,
-0.6226920500261081,
-0.9314119807859277,
0.08452985204639987,
-1.009402214771574,
-1.0062750028527756,
-0.28841331398033154,
0.5886312985955309,
0.25138239542645724,

0.4840684579382569,
1.66880370795529,
0.4947479443765874,
-0.5723605201777653,
-0.16848739586132902,
-1.2344957342488987,
-0.21521798449848678,
-0.061827721816023445,
-0.2248741094525013,
-0.14743116036125348,
1.3439891420398065,
0.26093100927833396,
2.340498096953527,
0.48577707139825455,
1.3703195203028469,
-0.03611016151365655,
-0.39327539415785684,
0.19418290589939505,
-1.2131007544515844,
-0.2043806706538651,
0.10177491262713155,
-0.04073135722774902,
-0.6647700988785717,
1.8155909796605325,
-0.9294726054396647,
0.06362652018359347,
-0.3885510819409685,
0.2259437203319192,
-0.29241696657905075,
1.5301954112147191,
1.6475949707589932,
-0.20840788688387885],
[0.42838040299935903,
0.43437695063037796,
0.7671204362948421,
1.4217493914297048,
1.532465319753789,
1.532465319753789,
2.7699209108087253,
2.7699209108087257,
4.156218307623238,
4.2237366972759505,
5.0429088489710905,
5.613108479440035,
5.957592523811918,
5.957592523811918,
7.200159227676843,
7.200159227676843,
7.511063978540512,
7.511063978540512,
7.5293149335701015,
8.184114862205785,
8.59759634646416,
8.59759634646416,
8.967813742088222,
8.967813742088222,
8.967813742088222,
9.731896873309175,
9.95178485601347,
10.151200227727777,
10.572249489684078,
11.21122434413465,
0.6103673192422585,
-2.2449370868204332,
-0.5531680672940967,
1.6213620933184167,
0.9025434887524114,
0.7267620669083164,
0.688558576899684,
-0.6961599873617196,
0.1412349369195377,
0.7459812495914973,
0.4704664842082417,
-0.38033435666108695,
0.35864815337369327,

-0.9120464289406482,
-0.39019902865511685,
0.8340346919183055,
-2.0544585551745573,
1.25105004616506,
-2.352820292981314,
0.4238462318210131,
0.07915902405687532,
0.20643459967058658,
0.23897101893275827,
-0.8664160098114893,
-1.685166937074754,
-0.06832039714759808,
-0.9791405090065629,
0.025203133044215387,
0.0013271421621207855,
-1.0715671968015688,
-0.008370532310604396,
0.13052855552000728,
0.497310552248094,
0.30798451910916236,
-0.6467356358635334,
-0.411683867969946,
-0.10065713602082733,
0.5062131798374145,
-0.10337024350765668,
-0.9070679783587652,
-0.6555610326122162,
2.2255239458177773,
1.727993426652873,
-1.440768657668252,
-1.3828846188724404,
-0.4461523368168654,
1.0838317458165323,
0.38330860036255543,
-1.499700854710827,
-1.3408221959756346,
0.7419236696397365,
-0.26719837498867116,
-0.2377148775462395,
2.1193168382343814,
-0.3237137921938176,
0.30565535651577447,
-0.3512550922173362,
-0.12043790779624768,
-0.7473597606659746,
0.04801624913587568,
0.0018824715948976833,
-1.0616614621946072,
0.0026415486583119922,
-0.3446629765669456,
1.7936621861690503,
0.6484630040536189,
0.911272728030746,
1.1140004969268402,
-0.7180242734781173,
0.5255116360573389,
0.21141077992540375,
-0.45008152208030955,
0.27750531007906354,
0.32764881051843897,
-0.018511194189945894,
0.3875950700305173,
-0.3325176658862957,
-1.8796967870400432,
-0.6725696299533431,
-0.016521571218157557,
-0.08350862144921116,
-0.5447867614937103,
0.09550333474849668,
0.008862454446484716,
0.22308949354372234,
0.43503110895758224,
0.08385997823794214,
0.8516706560998085,

0.37101021681963386,
-1.5090995065898887,
0.3045648040580248,
0.6581099200957676,
0.35742409273461206,
1.7397310471751102,
0.19044279487017676,
0.9986798590605289,
0.31257833320056017,
0.19093877798180586,
-0.12038962867457878,
-0.4069447684098612,
-1.0520519597232647,
-0.45448340350076705,
-0.09417833232522027,
1.3653492736312018,
0.6606235759628448,
-0.1314405098844259,
1.082951248594517,
0.2015466917595484,
0.16965874881834048,
0.2649109746606924,
0.8389747424450928,
1.4332963880188982,
0.288517036279826,
-0.13031950550882798,
0.31918733601921234,
1.3571234506254648,
1.4386307786107537,
0.4541349276361828,
-1.1895164340492679,
-0.27888567947798426],
[0.403034597497222,
0.4502729934454179,
0.5700423320933159,
0.5700423320933157,
0.6781241992806778,
1.3786879425414615,
1.4501147159778993,
2.4844307076867937,
2.4844307076867937,
3.0256317298652835,
3.0256317298652835,
4.15344004944777,
4.15344004944777,
4.2038017426165855,
4.6259251670404895,
4.6259251670404895,
4.797805401698216,
4.797805401698216,
5.52869212939408,
5.565345896572568,
7.178724808111429,
7.362080567239832,
9.32786374549076,
9.32786374549076,
9.69844688689066,
9.970706574916212,
9.970706574916212,
10.264471030354098,
10.264471030354098,
10.987148477608002,
-0.928231690167026,
0.8057374905304977,
-0.09124110656415482,
-1.2764053392323713,
-0.1699102369047808,
0.2854283542912807,
-0.38934147484457715,
1.9562479348306092,
-0.08583239283592208,
0.1375503575288643,
-0.3029413642384467,
0.3573494966906954,
0.4106808877553357,

1.0168255078067534,
1.020920504974372,
1.182081306796335,
-0.381751450929397,
0.7872502675264477,
0.11650702005295199,
-0.7041962676262727,
-0.9664718849607234,
0.0017002173560816299,
1.2143682588418563,
0.27506501793685123,
0.20402751900538993,
-0.18948284766781337,
0.003995331337594065,
1.1960049920193474,
0.5655953653806649,
0.045530409523104075,
-0.6946141334022584,
-0.014434864812455215,
0.011912663267278172,
0.3387818931591513,
0.03142887465994337,
-0.6463763503344852,
0.17750599329640465,
0.014482690293971139,
1.8640640992205775,
-0.048924475364900355,
0.07047222436652016,
0.5836409098942513,
-0.033227003016975516,
0.5638116666909924,
0.6159645449318359,
-0.28815587863252157,
1.0812127139774732,
0.379778365443053,
1.5589007157572847,
0.42162649727394796,
0.05475856745796003,
0.7192262097201122,
0.3263213716214118,
-1.3661280753618785,
-1.292135594114921,
-0.8807331583336575,
0.4025768854927813,
0.5295243646894815,
-0.8002135124103957,
0.1750760012142075,
-0.054032551513248435,
-0.40188384379722514,
0.07562722660439603,
0.9832576708267071,
-0.2899613446815228,
-0.06710629496184976,
0.7055866137794157,
-0.007144529560920351,
0.12285567358407816,
1.143563250691634,
0.15056960823466894,
1.2837012011259124,
-0.6023181038676345,
0.045174795264068736,
-0.16273344906061393,
0.06659839307873978,
-0.018202395096242784,
-0.39240523180078335,
-0.08377351665795224,
0.97309387436951,
2.220420875428843,
0.26956954819495804,
-0.9599252243572389,
-0.804945736082422,
-0.11945317743340336,
-0.8427551316765194,
0.30866551832235317,
2.4872766722318986,

-0.6424953924634126,
1.3858530207015087,
-0.28083276964757203,
0.4192227932599254,
-0.823908670962852,
-1.1126529481475467,
0.21496241595348659,
-0.08934899247793467,
0.1474411947098728,
0.5744211770388311,
-0.04170488097810379,
-0.22724449437110839,
1.9972780099267948,
-0.8591254315731605,
1.1685333425381264,
0.35810812429508854,
0.4121168481727771,
-0.15141964973712507,
-0.4596563680482823,
0.2311510501006746,
-1.1273861464736383,
0.5254398836685342,
0.8562906940888275,
0.2485294131097614,
0.5338805724034291,
0.4862154010986529,
-0.6197058235436752,
0.20248976894253998,
-0.8370526377715465,
-0.2508663863331003,
0.895805892021248,
0.966286773500402],
[0.45986682004385615,
0.587884042281194,
0.587884042281194,
0.8634874478193177,
0.8634874478193176,
2.1638996627094382,
2.6675296177894383,
2.7047736270183154,
4.061300455462818,
5.103553089959934,
5.103553089959934,
6.751801791048447,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.781371497069936,
7.781371497069936,
7.781371497069936,
8.420463205941301,
8.420463205941301,
8.420463205941301,
8.420463205941301,
9.122294773242167,
9.202160901745463,
9.814044623063202,
10.300168251850302,
10.300979681078617,
10.504460883512992,
11.02433376042467,
0.6573588347385189,
0.5828935548145123,
-0.574462871025295,
-1.546656732274899,
-0.27660055256222155,
1.273681854258797,
1.2384634856925236,
0.2379867607060117,
-0.9322977855669335,
-0.7820979498963491,
2.372805485180463,
-0.032501906954750276,
-0.7800293806349019,

0.4387330183441131,
-0.7149138216831045,
-1.1824362472270316,
0.5918930779181975,
0.829172686537006,
0.977795961853665,
-1.6128990556187608,
0.19086262573792945,
-1.3821924448334508,
-0.37777709213522187,
-0.7880705929271598,
-0.17099996546962573,
0.7254760017497719,
0.9819125877821568,
0.3047932425929136,
-0.02742999568379597,
0.5485981155871978,
0.11485583255370253,
-0.32472329363004176,
-0.6331630162839701,
1.9724401417574067,
1.7698339388439421,
-1.1254891310675255,
-1.1920329227643298,
-0.11899515772905239,
0.6149649547843763,
0.08915119991626204,
0.4168497404293414,
1.795330514128595,
0.13783861826417038,
-0.9065003531161022,
-0.3985523491151137,
-0.36816430284390833,
-1.1942463088521762,
-1.129677747629269,
0.6710845981341038,
-0.6149988458361377,
0.4665365160069161,
-0.19209648586871406,
0.013763508723981719,
-0.8135432145712183,
-0.4057592495812904,
-0.6414147049863589,
0.8876536381187553,
-0.7757241021147325,
0.40873541229116084,
0.514258938404948,
0.3722392877827026,
1.8009017308404462,
0.116225182388376,
0.2191004740219447,
0.8255374993004068,
0.8886674208207993,
-0.579134397544579,
0.06903106340432566,
-0.7849541624527424,
1.2193465265846974,
-0.5822657541432643,
-0.5730195760225829,
-0.08026565436629246,
-0.5615010238556278,
-0.24965155606323408,
0.9668325985251349,
0.32484880948626715,
-0.419910520268625,
0.9941102073040009,
0.290737643308207,
-0.05558286792755536,
-0.07023052502159025,
0.6542319702891408,
-0.7125513821501609,
0.3724489707231321,
0.6482228950077775,
0.6813557635502394,
0.5904217885770654,

0.3344272908394536,
0.9892271354161819,
-0.6202922067549016,
0.12947475126138008,
0.35715623223344434,
-2.26745180129,
-0.20006687506113488,
-0.6290028667838425,
0.08022472893529967,
-0.961248450361056,
-0.20142658286918888,
-0.26748808365437937,
0.3448273909474204,
0.6331413989953146,
-0.5916282783618806,
-0.951916970919588,
-1.150804089444678,
-0.8692217930277405,
0.2236835205843831,
-0.10881486654903144,
-0.004228302858090612,
0.16762641757787602,
1.4232995099688168,
0.17348586714896594,
1.0253598045594392,
0.3864684766177986,
1.0340175270140362,
1.1184694851102563,
1.066055569095364,
0.07092789808311199,
-0.8620685771604671,
0.46490022331063124],
[0.49845241666821427,
0.5080489367341008,
0.9738886018422865,
1.569493854319001,
1.5867336495888638,
2.22018578038962,
2.2222975784710255,
2.7355212499054673,
3.167907147345212,
3.912280465424413,
3.9324465712853787,
4.601639034821639,
4.6023794814303285,
4.622186895887535,
4.629761229768061,
5.8119318913753775,
5.81529780322803,
5.862601007224966,
5.868327135385091,
7.931887080538795,
8.400559524273383,
8.400626671117248,
8.400626709821431,
9.338300284642187,
9.338300284937413,
10.032003053056542,
10.032003053750085,
10.151775184539204,
10.15177520525735,
12.777878105202333,
-1.0652764150946168,
0.1030911905785532,
2.1938566419651506,
-1.687039127173871,
-1.2892167937928738,
0.454901922887811,
-0.10701158278386386,
-0.27610154265055814,
0.3864936366984189,
-2.1008557969724673,
0.03526686701549039,
1.2075362869846002,
0.9729996942576583,

0.011602974506752167,
-1.7901344125733514,
-0.5312859895436812,
-0.6764114285522254,
-0.568464725902531,
0.7761280206550648,
0.10820918045329357,
-1.191918699884087,
0.6710933645796882,
-0.998882902161297,
0.9668399094583318,
-0.5526879367651495,
-0.3448144391622963,
0.3546231755031622,
0.36267379522525617,
-1.3188129954125047,
0.7986540408445592,
0.5385879858279273,
1.3987311647526153,
0.17032526930657013,
-0.2765839794470922,
-0.4561318195988584,
0.7779672645593669,
-1.6618874628312856,
0.7145344738782802,
-0.8703397140730336,
0.3664383221285875,
0.09488224223418325,
0.17319381149242297,
-0.15027776082229982,
-1.4690811524212324,
-1.4342980641635596,
-0.0435375212610062,
-1.1192667984833862,
1.489571420955581,
-1.385268501544282,
0.07500036651414835,
-1.0373356270085057,
1.6055364169931783,
-1.1908916083661096,
0.47394571447103884,
-0.23948832756930136,
1.6865746730435722,
-0.07643328425683309,
-0.26016480364478584,
0.1230869402232525,
0.24188831995971777,
-0.12202515710835343,
-1.3273176630150638,
-0.8685353737050294,
0.16431995234498184,
0.5641710812415553,
-0.1851122077039956,
-0.3773492143584495,
-1.5621470223099339,
-1.4505536127106367,
0.571773409102061,
0.20129944645039902,
1.1233832799408359,
0.09021785654141615,
-0.09203212520614017,
1.4656739277180684,
1.5116863426084852,
-1.4548391160838867,
-0.8665597553970019,
-0.8311755953666669,
-1.2962978849547035,
2.7863849971356167,
-0.026842981867963016,
-0.5147225252861769,
1.0767631952186236,
0.23432587613057304,
-0.3849014581870786,
-0.5492566246326465,
-1.590706022355079,

0.2729988606074437,
-1.0566347351025576,
0.3490824852559465,
-0.08255987668621423,
2.389615689206544,
-0.5033499561524312,
0.8858613199859888,
0.002820627497544732,
0.19486994113578365,
-0.12438999158785295,
-0.2658298270496061,
-0.2763678550214429,
0.4048526015479744,
-0.06841768991626676,
-0.32117880607439586,
1.6019482252187531,
0.41890698857400605,
-0.8510923614851694,
0.16411947273918887,
0.3096085421164439,
-1.0434867263587853,
1.2153787936463343,
-0.9841260925764858,
1.5878541492278757,
0.5868781859412673,
0.26283212361760866,
-0.9081790869869905,
1.200367927444946,
3.155405601054565,
0.19810365707195607,
0.6950762676177132,
0.9262562990824308],
[0.11310194099017501,
0.35533489757839665,
0.7274669277129748,
0.7931766663622793,
0.9215124586161831,
1.5868010098342087,
1.6938861718192633,
1.6938861718192633,
2.4953100977078537,
2.4953100977078537,
2.8524614686988468,
2.852561718095858,
3.5986528286299406,
4.15163213852048,
4.658101059037985,
4.662433860401005,
5.170840359657884,
5.170840359657884,
7.389891885603243,
7.408023950817849,
7.472384801069389,
7.816163992050368,
8.61557017602429,
9.097989314327426,
9.523961629484033,
9.523961629484033,
9.983169099900275,
10.18670132329559,
10.652310618690768,
11.298231074807717,
0.0018299791732057594,
-2.098323349499344,
-0.32648794289911637,
-0.9015207614699141,
0.16889281860224642,
-0.23801018256690964,
-0.2710900692754434,
-0.354821626365823,
0.2834027319981757,
0.5370764326800048,
0.3290793485287995,
-1.7004803192621687,
-0.8267928250682555,

0.7226363685371004,
-0.23961420559359078,
-0.9892421071296497,
-0.3062192663980587,
-1.2857004554953664,
-0.00570902688287085,
-0.7828701256611971,
0.036023661097245405,
-1.405764493898646,
0.09523715910714706,
1.0678085339341385,
-0.3477609497961018,
-1.3254201123902754,
-0.5891224027277412,
-0.14272029692701035,
0.10490796904880517,
2.930201181567992,
0.04738429480019876,
0.0461816378739841,
-0.48875992312198635,
-0.1220291693117149,
-0.4148582659190601,
0.17851659149817373,
0.4130064273610762,
-0.06315869176871276,
0.3800783279561489,
-0.02544638146603332,
2.215655886557845,
0.11710086424953603,
0.9518430371086406,
-0.12584714752332027,
0.018420823357896687,
1.2484195950702188,
0.13344161240602143,
0.09153656466168367,
0.7487460411896285,
0.4894289612390041,
0.03364631653773711,
-0.40583141994665783,
-1.508449266642364,
-0.0876621026521168,
0.16953459800120027,
0.7644948380070223,
1.6192126500413289,
-0.10749735247146877,
2.0545430524982202,
0.024857404521674214,
-0.048992600661546246,
1.151942209932795,
-0.5651424971501683,
-0.3291333996292886,
1.3277495361297842,
0.3468681089352445,
0.25355521459481833,
-1.43151363095255,
-0.6791478024501341,
-0.11073517248868558,
0.6626722725413648,
-1.1931469775385106,
-0.5976012599191521,
-0.6367586148577103,
-0.2241173972369649,
-1.0874289140947961,
-0.4768780848197268,
1.2286899896766026,
1.7976782920817624,
0.10434792845100443,
-0.31719050779977503,
0.7620589119443072,
-0.6377447181256041,
0.484158469702177,
-0.5854682569359358,
0.7650979634948629,
-0.6510076440579313,
1.4650942622453875,

```

1.0359738562191985,
0.7750682527878552,
-0.37331893064057897,
-0.1914404490007497,
0.25187298257054047,
0.370887355356332,
0.9916451969775287,
2.255013915351184,
0.564469362172986,
-1.1669103509730057,
0.42926615022397796,
0.43095576457077756,
0.7007537480246202,
3.3633287927079483,
-0.5433314326657547,
0.03986376903117299,
-1.913610430050787,
-1.183549627581873,
-1.5712622883507832,
-1.282300011205569,
-0.31197479792826754,
-2.6880651613874877,
0.33078250437153595,
0.07570451779538533,
0.1724953212076054,
-1.3944682324956643,
-0.31604217181433353,
1.821638064235333,
-0.4612503790165566,
0.19201221884571829,
-2.5536752551832254,
-3.5133198239284997]],
'run_time_sec': 34029.58519411087}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,

```

6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,

```

-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},

```

False,


```

        'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                    'status': False,
                    'type': 'png'},
          'file_name_prefix':

'automate_ising_noise_free_time_evo_dephasing_ric3_sample#8'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},

'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
          'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                    '+ 1 Z 0 + '
                                                    '1 X 1 X 2 '
                                                    '+ 1 Z 1 + '
                                                    '1 X 2 X 3 '
                                                    '+ 1 Z 2 + '
                                                    '1 X 3 X 4 '
                                                    '+ 1 Z 3 + '
                                                    '1 X 4 X 5 '
                                                    '+ 1 Z 4 + '
                                                    '1 X 5 X 6 '
                                                    '+ 1 Z 5 + '
                                                    '1 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -6.029866062982579],
                             [12, 6, 0, 0, -5.108515955211577],
                             [20, 10, 0, 0, -4.320587490415481]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

```

```
0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4]],
'run_time_sec': 10.637134313583374}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.5483166258326411,
0.5788252343888002,
0.5803330366161238,
1.2169503585394688,
1.4292236971359287,
1.4293923120692815,
1.4365169928726365,
1.4365169928726365,
2.510196103669615,
2.510196103669615,
2.5544154452534937,
2.5544154452534937,
4.029649065019888,
4.067883071594656,
4.076808377533052,
4.1651466476623735,
4.656187563066811,
4.656187563066811,
5.18228186954163,
5.18228186954163,
6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
```

-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,

[illegible]

```
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 7.923394728339895e-17,
0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
5.879155312872552,
5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
```

-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,

False,

```
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.275631329374743],
[12, 6, 0, 0, -5.287235063864105],
[20, 10, 0, 0, -4.454788748195675]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_prob': [
```



```

0.001,
0.001,
0.0,
0.0],

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 9.310113906860352}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
'init_param': { 'value': [ 4.68500097130382e-19,
    0.00019188574128314593,
    0.4999939126216869,
    0.571475986197529,
    0.6379337879222081,
    1.2223970983691403,
    1.405127109337375,
    1.405127109337375,
    1.405127109337375,
    3.9334167957034354,
    3.9801359916712626,
    3.9801359916712626,
    4.868547987503254,
    4.868547987503254,
    5.067947873787899,
    5.231153270915478,
    5.2311532709154775,
    6.116966983963522,

```

6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,

```

-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [ 0.001,
                                     0.001,
                                     0.0,
                                     0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,

```

False,

```

0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                    'status': False,
                    'type': 'png'},
            'file_name_prefix':

'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#3'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
                             [12, 6, 0, 0, -5.679026952355478],
                             [20, 10, 0, 0, -4.762614433311011]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4},

```

```

        { 'gates_num': [20, 5, 1, 0],
          'identity_factors': [2, 2, 0, 0],
          'noise_level': [20, 10, 0, 0],
          'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':
'dephasing'},

                                'odd_wires': 4}},
          'run_time_sec': 9.617511510848999}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
  'init_param': { 'value': [ 0.0921501071458524,
                              1.5113409130757622,
                              1.5171500885968463,
                              1.9257567103877564,
                              2.103027702732541,
                              2.106266184153511,
                              2.3067654466695937,
                              2.7972204473176414,
                              2.7972204473176414,
                              2.9760063319143315,
                              2.9760063319143315,
                              2.976006331914331,
                              4.725403617162805,
                              4.725403617162805,
                              4.725403617162805,
                              5.3458155973379915,
                              5.3458155973379915,
                              5.3458155973379915,
                              5.3458155973379915,
                              5.511978345293384,
                              5.511978345293384,
                              7.852859529848597,
                              7.8664501138991945,
                              8.306657385522128,
                              8.306657385522128,
                              8.306657385522128,
                              8.77694772653372,
                              8.77694772653372,
                              8.77694772653372,
                              10.210889411786125,
                              -0.025196159437353705,
                              0.17256782673282392,
                              0.08925866669000004,
                              -0.2640737309697103,
                              0.08393384639489809,
                              0.818808421272339,
                              0.21848095983081994,
                              0.18119625098082034,
                              -0.011865447267727605,
                              1.6758619362933431,
                              0.20111234151604201,
```

0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,

```

0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,

```

```
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1
'+ 1 Z 0 +
'1 X 1 X 2
'+ 1 Z 1 +
'1 X 2 X 3
'+ 1 Z 2 +
'1 X 3 X 4
'+ 1 Z 3 +
'1 X 4 X 5
'+ 1 Z 4 +
'1 X 5 X 6
'+ 1 Z 5 +
'1 Z 6']}},
'output': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
[12, 6, 0, 0, -4.800941579254223],
[20, 10, 0, 0, -4.03450493736607]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4}]}],
'run_time_sec': 10.198833465576172}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
```



```
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.3675336714235261,
0.40575472279006464,
0.6952672462769884,
2.137221581425111,
2.137221581425111,
2.4653237039513405,
2.4653237039513414,
2.671087921572092,
2.6710879215720915,
3.889448133547601,
3.889448133547601,
5.229066948311018,
5.229066948311018,
5.736515631630257,
5.736515631630257,
5.736515631630257,
8.147692740111799,
8.179590109004724,
8.179590109004724,
8.179590109004724,
8.21310311779852,
8.21310311779852,
8.752241426452148,
8.752241426452148,
8.803045108086177,
9.928410928444814,
9.928410928444814,
9.928410928444814,
9.969030315653287,
14.058841247147779,
-0.593395889833616,
1.0797089435996057,
-0.18386516671058062,
0.9172540646403935,
0.5920567467507187,
0.5030949392578172,
-0.19817311203429638,
0.8732878271906367,
1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
```

-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,

False,

```
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.27430922876383],
[12, 6, 0, 0, -6.102630979708389],
[20, 10, 0, 0, -5.139058519967983]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 10.52130126953125}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
'init_param': { 'value': [ 0.23860472738944535,
    0.24738626223792576,
    0.368610904156845,
    0.9146619524924428,
    0.9146619524924426,
    1.6697383257267104,
    1.6697383257267104,
    2.054253047546413,
    2.3726208960540474,
    2.9146286026908905,
    2.9146286026908905,
    2.9146286026908905,

```

2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,

0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0},

'noise_prob': [0.001,
0.001,
0.0,
0.0],

'status': True,
'type': 'dephasing'},

'nqubits': 7,

'observable': { 'coefficients': { 'bn': [1.0,
1.0,

False,

```

        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
                0.5,
                0.5,
                0.5,
                0.5,
                0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
                          'status': False,
                          'type': 'png'},
               'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#10'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                          [1, 1, 0, 0],
                                          [2, 2, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
             'optimization': { 'algorithm': 'SLSQP',
                               'constraint': True,
                               'status': False}},
    'zne': { 'data_points': None,
            'degree': 0,
            'method': 'richardson',
            'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                     'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                     '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                     'X 5 X 6 + 1 Z 5 + 1 Z 6',
              'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -6.823046671583829],
                                 [12, 6, 0, 0, -5.597458502232875],
                                 [20, 10, 0, 0, -4.598788267087871]],
              'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                   'identity_factors': [0, 0, 0, 0],
                                   'noise_level': [4, 2, 0, 0],
                                   'noise_profile': {
'noise_on_init_param': { 'status': False,
                           'value': 0},

                                0.001,
                                0.001,
                                0.0,
                                0.0],

                                'noise_prob': [

                                'status': True,
                                'type':

                                'dephasing'},

                                'odd_wires': 4},
                    { 'gates_num': [12, 3, 1, 0],
                      'identity_factors': [1, 1, 0, 0],
                      'noise_level': [12, 6, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                           'value': 0},

                                0.001,
                                0.001,
```

```

0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4}},
'run_time_sec': 10.297536134719849}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,

```


0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,

```

-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',

```

False,

```

'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                    '+ 1 Z 0 + '
                                                    '1 X 1 X 2 '
                                                    '+ 1 Z 1 + '
                                                    '1 X 2 X 3 '
                                                    '+ 1 Z 2 + '
                                                    '1 X 3 X 4 '
                                                    '+ 1 Z 3 + '
                                                    '1 X 4 X 5 '
                                                    '+ 1 Z 4 + '
                                                    '1 X 5 X 6 '
                                                    '+ 1 Z 5 + '
                                                    '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.536883635090364],
                             [12, 6, 0, 0, -5.4912521012226865],
                             [20, 10, 0, 0, -4.624115876403395]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}],
'run_time_sec': 9.742278099060059}},
{ 'config': { 'ansatz': { 'gateset': 1,

```

```
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
```

-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,

False,

```
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '

```

```

'output': { 'data_points': [ [4, 2, 0, 0, -6.376486414275041],
                              [12, 6, 0, 0, -5.250000083007176],
                              [20, 10, 0, 0, -4.32383475666671]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                      'identity_factors': [0, 0, 0, 0],
                      'noise_level': [4, 2, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'dephasing'},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'dephasing'},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'dephasing'},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'odd_wires': 4}],
'run_time_sec': 9.482933282852173}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'},
'init_param': { 'value': [ 0.17096748946205537,
                           0.24182068788319971,
                           0.5294082221811705,
                           0.6436439076685059,
                           0.6752523962811601,
                           1.4661557963348293,

```

1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,

0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0},

'noise_prob': [0.001,
0.001,
0.0,

False,

```

0.0],
    'status': True,
    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.139980799817413],
[12, 6, 0, 0, -5.986699722057315],
[20, 10, 0, 0, -5.03269185703209]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_profile': {
    'noise_prob': [
        0.001,
        0.001,
        0.0,
        0.0],

        'status': True,
        'type':

            'odd_wires': 4},
            { 'gates_num': [20, 5, 1, 0],
              'identity_factors': [2, 2, 0, 0],
              'noise_level': [20, 10, 0, 0],
              'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

            'odd_wires': 4}},
            'run_time_sec': 10.040735960006714}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
            'r': 0},
            'time': {'max': 10.0, 'min': 0.0},
            'type': 'ising'}}},
    'init_param': {'value': 'random'},
    'noise_profile': { 'noise_on_init_param': { 'status': False,
        'value': 0},
        'noise_prob': [0.001, 0.001, 0.0, 0.0],
        'status': True,
        'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
            'r': 1},
            'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': True,
        'type': 'png'},
        'file_name_prefix': 'ising_noisefree_time_evo'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 1, 1]]},
    'run': 'vqe',

```

```

'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + '
                                '1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + '
                                '1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
                                '1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z '
                                                    '0 + 1 X 1 X 2 + '
                                                    '1 Z 1 + 1 X 2 X '
                                                    '3 + 1 Z 2 + 1 X '
                                                    '3 X 4 + 1 Z 3 + '
                                                    '1 X 4 X 5 + 1 Z '
                                                    '4 + 1 X 5 X 6 + '
                                                    '1 Z 5 + 1 Z 6']}],
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.358359699677247,
                                     4.598584386614463,
                                     5.205211991476004,
                                     2.498239515374607,
                                     3.620239544509655,
                                     3.5202907273108752,
                                     3.874294382863561,
                                     2.6240487828355272,
                                     3.466109018012848,
                                     4.215104097495168],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': {
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
                'status': True,
                'type': 'dephasing'},
            'odd_wires': 4},
            'optimized_minimum_cost': [ -7.274309228763741,
                                       -7.1399807998173275,
                                       -6.7841170852099,
                                       -6.206238059807503,
                                       -6.275631329374697,
                                       -5.70418119478636,
                                       -6.53688363509032,
                                       -6.029866062982587,
                                       -6.376486414275039,
                                       -6.823046671583788],
            'optimized_parameters': [ [ 0.3675336714235261,
                                       0.40575472279006464,
                                       0.6952672462769884,
                                       2.137221581425111,
                                       2.137221581425111,
                                       2.4653237039513405,
                                       2.4653237039513414,
                                       2.671087921572092,
                                       2.6710879215720915,
                                       3.889448133547601,
                                       3.889448133547601,
                                       5.229066948311018,
                                       5.229066948311018,
                                       5.736515631630257,
                                       5.736515631630257,
                                       5.736515631630257,
                                       8.147692740111799,
                                       8.179590109004724,
                                       8.179590109004724,
                                       8.179590109004724,

```

```

'status': False,
'value': 0},

```

8.21310311779852,
8.21310311779852,
8.752241426452148,
8.752241426452148,
8.803045108086177,
9.928410928444814,
9.928410928444814,
9.928410928444814,
9.969030315653287,
14.058841247147779,
-0.593395889833616,
1.0797089435996057,
-0.18386516671058062,
0.9172540646403935,
0.5920567467507187,
0.5030949392578172,
-0.19817311203429638,
0.8732878271906367,
1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,

-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804],
[0.17096748946205537,
0.24182068788319971,
0.5294082221811705,
0.6436439076685059,
0.6752523962811601,
1.4661557963348293,
1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,

6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,

-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993],
[4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,

7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,

0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138],
[0.5483166258326411,
0.5788252343888002,
0.5803330366161238,
1.2169503585394688,
1.4292236971359287,
1.4293923120692815,
1.4365169928726365,
1.4365169928726365,
2.510196103669615,
2.510196103669615,
2.5544154452534937,
2.5544154452534937,
4.029649065019888,
4.067883071594656,
4.076808377533052,
4.1651466476623735,
4.656187563066811,
4.656187563066811,
5.18228186954163,
5.18228186954163,

6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,

1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301],
[7.923394728339895e-17,
0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
5.879155312872552,

5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,

2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084],
[0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,

5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.08925866669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,

-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674],
[0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,

7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,

-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603],
[0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,

4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,

1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171],
[0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,

6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,

1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893],
[0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,

5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,

```

0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'run_time_sec': 46646.78074049955}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.0921501071458524,

```


1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.0892586669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,

0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674]],

'noise_profile': { 'noise_on_init_param': { 'status': False,

```

        'value': 0},
        'noise_prob': [0.001, 0.001, 0.0, 0.0],
        'status': True,
        'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
        [12, 6, 0, 0, -4.800941579254223],
        [20, 10, 0, 0, -4.03450493736607]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
        'run_time_sec': 0.0003864765167236328,
        'zne_values': { 'degree': 0,
        'extrapolated_value': -6.207102117668988,
        'others': { 'beta_coeffieicients': [ 1.875,
        -1.25,
        0.375],
        'cost_richardson_zne': 5.21875,
        'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254223,
-4.03450493736607],

        'sorted_noise': [6, 18, 30]}},
        'sampled data': [ [ 4,
        2,
        0,
        0,
        -5.704181194786395],
        [ 12,
        6,
        0,
        0,
        -4.800941579254223],
        [ 20,
        10,
        0,
        0,
        -4.03450493736607]],
        'sampling': 'default'}}},
    { 'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'r': 1},
        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
        [12, 6, 0, 0, -4.800941579254223],
        [20, 10, 0, 0, -4.03450493736607]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
        'run_time_sec': 0.0003864765167236328,
        'zne_values': { 'degree': 0,
        'extrapolated_value': -6.207102117668988,
        'others': { 'beta_coeffieicients': [ 1.875,
        -1.25,
        0.375],
        'cost_richardson_zne': 5.21875,
        'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254223,
-4.03450493736607],

        'sorted_noise': [6, 18, 30]}},
        'sampled data': [ [ 4,
        2,
        0,
        0,
        -5.704181194786395],
        [ 12,
        6,
        0,
        0,
        -4.800941579254223],
        [ 20,
        10,
        0,
        0,
        -4.03450493736607]],
        'sampling': 'default'}}},
        'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'r': 1},
        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
        [12, 6, 0, 0, -4.800941579254223],
        [20, 10, 0, 0, -4.03450493736607]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
        'run_time_sec': 0.0003864765167236328,
        'zne_values': { 'degree': 0,
        'extrapolated_value': -6.207102117668988,
        'others': { 'beta_coeffieicients': [ 1.875,
        -1.25,
        0.375],
        'cost_richardson_zne': 5.21875,
        'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254223,
-4.03450493736607],

        'sorted_noise': [6, 18, 30]}},
        'sampled data': [ [ 4,
        2,
        0,
        0,
        -5.704181194786395],
        [ 12,
        6,
        0,
        0,
        -4.800941579254223],
        [ 20,
        10,
        0,
        0,
        -4.03450493736607]],
        'sampling': 'default'}}},
        'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'r': 1},
        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
        [12, 6, 0, 0, -4.800941579254223],
        [20, 10, 0, 0, -4.03450493736607]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
        'run_time_sec': 0.0003864765167236328,
        'zne_values': { 'degree': 0,
        'extrapolated_value': -6.207102117668988,
        'others': { 'beta_coeffieicients': [ 1.875,
        -1.25,
        0.375],
        'cost_richardson_zne': 5.21875,
        'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254223,
-4.03450493736607],

        'sorted_noise': [6, 18, 30]}},
        'sampled data': [ [ 4,
        2,
        0,
        0,
        -5.704181194786395],
        [ 12,
        6,
        0,
        0,
        -4.800941579254223],
        [ 20,
        10,
        0,
        0,
        -4.03450493736607]],
        'sampling': 'default'}}},
        'config': { 'ansatz': { 'gateset': 1,
        'layer': 30,
        'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0,
        0,
        0,
        0,
        0,
        0],
        'r': 1},
        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#6'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
        [12, 6, 0, 0, -4.800941579254223],
        [20, 10, 0, 0, -4.03450493736607]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
        'run_time_sec': 0.0003864765167236328,
        'zne_values': { 'degree': 0,
        'extrapolated_value': -6.207102117668988,
        'others': { 'beta_coeffieicients': [ 1.875,
        -1.25,
        0.375],
        'cost_richardson_zne': 5.21875,
        'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254223,
-4.03450493736607],

        'sorted_noise': [6, 18, 30]}},
        'sampled data': [ [ 4,
        2,
        0,
        0,
        -5.704181194786395],
        [ 12,

```

```
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
```

-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,

```

        2.288296821099581,
        0.3797180776898864,
        0.43119211143988634,
        0.11587678539460887,
        -0.20259754629417875,
        -1.3297836588453886,
        -0.27310434987665144,
        1.0101242784162827,
        1.2218251312173358,
        0.4833587270164354,
        -1.4028471613931468,
        -0.7051086788097171]],
    'noise_profile': { 'noise_on_init_param': { 'status': False,
                                                'value': 0},
                      'noise_prob': [0.001, 0.001, 0.0, 0.0],
                      'status': True,
                      'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
                                              1.0,
                                              1.0,
                                              1.0,
                                              1.0,
                                              1.0],
                                      'cn': [ 0.5,
                                              0.5,
                                              0.5,
                                              0.5,
                                              0.5,
                                              0.5],
                                      'r': 1}},
                  'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
                          'status': False,
                          'type': 'png'},
               'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#8'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                         [1, 1, 0, 0],
                                         [2, 2, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
             'optimization': { 'algorithm': 'SLSQP',
                              'constraint': True,
                              'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -6.029866062982579],
                             [12, 6, 0, 0, -5.108515955211577],
                             [20, 10, 0, 0, -4.320587490415481]],
            'degree': 0,
            'method': 'richardson',
            'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
                'run_time_sec': 4.029273986816406e-05,
                'zne_values': { 'degree': 0,
                               'extrapolated_value': -6.540574232983671,
                               'others': { 'beta_coeffieicients': [ 1.875,
                                                                    -1.25,
                                                                    0.375],
                                           'cost_richardson_zne': 5.21875,
                                           'sorted_expectation_vals': [
-6.029866062982579,
-5.108515955211577,
-4.320587490415481],
                                           'sorted_noise': [6, 18, 30]},
                               'sampled_data': [ [ 4,
                                                    2,
                                                    0,
                                                    0,
                                                    -6.029866062982579],
                                                  [ 12,
                                                    6,
                                                    0,
                                                    0,
```

```

-5.108515955211577],
[ 20,
 10,
 0,
 0,
-4.320587490415481]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                        'cn': [ 0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5,
                                0.5],
                        'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'}},
'init_param': { 'value': [ 7.923394728339895e-17,
                            0.44053533716557003,
                            0.44053533716557,
                            0.8633875163783187,
                            0.8807523602927853,
                            0.8807523602927853,
                            1.0224141131682343,
                            1.599513745187798,
                            2.8464451213500714,
                            2.922472045892935,
                            2.922472045892935,
                            2.9239478183172296,
                            4.400338604169817,
                            4.721305166667179,
                            4.721305166667179,
                            4.721305166667178,
                            5.011876780494166,
                            5.011876780494166,
                            5.011876780494166,
                            5.879155312872552,
                            5.879155312872551,
                            6.750159190843359,
                            6.75015919084336,
                            8.012324492482197,
                            8.012324492482197,
                            8.7704267977754,
                            9.114848260128039,
                            9.114848260128039,
                            9.87613180402372,
                            10.23973172788205,
                            0.09774272849121603,
                            0.0018601482237552494,
                            -0.09414894191588159,
                            0.03531507271404531,
                            0.01801632266215566,
                            -0.031446214752801294,
                            0.04762079993218455,
                            0.16430279282954635,
                            -1.4212045547488918,
                            -0.051616216342739524,
                            -1.2094395115841,
                            1.961457452158631,
                            -0.09882210453707145,
                            1.4772087948778743,
                            -1.07174135216323,
                            -0.1881882042247284,
                            0.9429310525529949,
                            0.06018460007044777,
                            -2.137106479060228,
                            -0.9614924638003328,

```

0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,

-6.275631329374743,
-5.287235063864105,
-4.454788748195675],

```
        'sorted_noise': [6, 18, 30]],  
    'sampled data': [ [ 4,  
                        2,  
                        0,  
                        0,  
                        -6.275631329374743],  
                      [ 12,  
                        6,  
                        0,  
                        0,  
                        -5.287235063864105],  
                      [ 20,  
                        10,  
                        0,  
                        0,  
                        -4.454788748195675]],  
    'sampling': 'default']},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0],  
                                                              'cn': [ 0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5],  
                                                              'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'ising'}},  
    'init_param': { 'value': [ 0.47544493830108536,  
                               0.47591537288460184,  
                               0.4759153728846019,  
                               0.5985896638985994,  
                               1.3939903966429974,  
                               1.3939903966429974,  
                               3.2587118733130036,  
                               3.258711873313004,  
                               4.312142629456884,  
                               4.312142629456884,  
                               4.857905543812315,  
                               4.971569214842323,  
                               5.142019393710199,  
                               5.286850957206092,  
                               5.286850957206092,  
                               5.286850957206092,  
                               5.286850957206092,  
                               5.286850957206092,  
                               7.438635744061485,  
                               7.438635744061485,  
                               7.5101721136480535,  
                               7.510172113648054,  
                               8.498151007141688,  
                               8.498151007141688,  
                               8.89695633933963,  
                               8.89695633933963,  
                               9.949067520176195,  
                               10.53028906819762,  
                               10.53028906819762,  
                               -0.7640988398723291,  
                               -0.2609761344365046,  
                               0.3714650363100771,  
                               1.14227988754955,  
                               -0.5586008021142339,  
                               0.3232165993259411,  
                               -0.808074054347843,
```

0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,

[illegible]

-6.536883635090364,
-5.4912521012226865,
-4.624115876403395],

```
[20, 10, 0, 0, -4.624115876403395]],  
    'degree': 0,  
    'method': 'richardson',  
    'sampling': 'default'}},  
'output': { 'exact_sol': -8.566772233505624,  
    'run_time_sec': 5.698204040527344e-05,  
    'zne_values': { 'degree': 0,  
        'extrapolated_value': -7.126635142917348,  
        'others': { 'beta_coefficients': [ 1.875,  
            -1.25,  
            0.375],  
        'cost_richardson_zne': 5.21875,  
        'sorted_expectation_vals': [  
  
            'sorted_noise': [6, 18, 30]],  
    'sampled data': [ [ 4,  
        2,  
        0,  
        0,  
        -6.536883635090364],  
        [ 12,  
        6,  
        0,  
        0,  
        -5.4912521012226865],  
        [ 20,  
        10,  
        0,  
        0,  
        -4.624115876403395]],  
    'sampling': 'default'}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
    'layer': 30,  
    'ugate': { 'coefficients': { 'bn': [ 0,  
        0,  
        0,  
        0,  
        0,  
        0,  
        0],  
        'cn': [ 0.5,  
        0.5,  
        0.5,  
        0.5,  
        0.5,  
        0.5],  
        'r': 0},  
    'time': {'max': 10.0, 'min': 0.0},  
    'type': 'ising'}},  
    'init_param': { 'value': [ 0.3675336714235261,  
        0.40575472279006464,  
        0.6952672462769884,  
        2.137221581425111,  
        2.137221581425111,  
        2.4653237039513405,  
        2.4653237039513414,  
        2.671087921572092,  
        2.6710879215720915,  
        3.889448133547601,  
        3.889448133547601,  
        5.229066948311018,  
        5.229066948311018,  
        5.736515631630257,  
        5.736515631630257,  
        5.736515631630257,  
        8.147692740111799,  
        8.179590109004724,  
        8.179590109004724,  
        8.179590109004724,  
        8.21310311779852,  
        8.21310311779852,  
        8.752241426452148,  
        8.752241426452148,
```

8.803045108086177,
9.928410928444814,
9.928410928444814,
9.928410928444814,
9.969030315653287,
14.058841247147779,
-0.593395889833616,
1.0797089435996057,
-0.18386516671058062,
0.9172540646403935,
0.5920567467507187,
0.5030949392578172,
-0.19817311203429638,
0.8732878271906367,
1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,

```

0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},

```

```

'file_name_prefix':
'AUTOMATE_ising_noise_free_time_evo_dephasing_ric3_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.27430922876383],
                          [12, 6, 0, 0, -6.102630979708389],
                          [20, 10, 0, 0, -5.139058519967983]],

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.076957702636719e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.93818802428469,
                            'others': { 'beta_coefficients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                            'cost_richardson_zne': 5.21875,
                            'sorted_expectation_vals': [
-7.27430922876383,
-6.102630979708389,
-5.139058519967983],

                                'sorted_noise': [6, 18, 30]}},
            'sampled_data': [ [ 4,
                                2,
                                0,
                                0,
                                -7.27430922876383],
                              [ 12,
                                6,
                                0,
                                0,
                                -6.102630979708389],
                              [ 20,
                                10,
                                0,
                                0,
                                -5.139058519967983]],
            'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}}},
  'init_param': { 'value': [ 0.23860472738944535,
                             0.24738626223792576,
                             0.368610904156845,
                             0.9146619524924428,
                             0.9146619524924426,
                             1.6697383257267104,
                             1.6697383257267104,
                             2.054253047546413,
                             2.3726208960540474,
                             2.9146286026908905,
                             2.9146286026908905,

```


2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,

```
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                              'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
```

```

1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.823046671583829],
[12, 6, 0, 0, -5.597458502232875],
[20, 10, 0, 0, -4.598788267087871]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.315376281738281e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.520934981586538,
'others': { 'beta_coefficients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [
-6.823046671583829,
-5.597458502232875,
-4.598788267087871],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-6.823046671583829],
[ 12,
6,
0,
0,
-5.597458502232875],
[ 20,
10,
0,
0,
-4.598788267087871]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},

```

```
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.647503533031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
```

-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,

```

2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.376486414275041],
                          [12, 6, 0, 0, -5.250000083007176],
                          [20, 10, 0, 0, -4.32383475666671]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.482269287109375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.014849956756748,
                           'others': { 'beta_coefficients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-6.376486414275041,
-5.250000083007176,
-4.32383475666671],

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.376486414275041],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.250000083007176],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.32383475666671]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,

```

```
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
```

1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,


```

0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
                          [12, 6, 0, 0, -5.679026952355478],
                          [20, 10, 0, 0, -4.762614433311011]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.553794860839844e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -7.4074162568157735,
                            'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                      'cost_richardson_zne': 5.21875,
                                      'sorted_expectation_vals': [
-6.7841170852098625,
-5.679026952355478,
-4.762614433311011],

'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.7841170852098625],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.679026952355478],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.762614433311011]]]

```

```

6,
0,
0,
-5.679026952355478],
[ 20,
10,
0,
0,
-4.762614433311011]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.17096748946205537,
0.24182068788319971,
0.5294082221811705,
0.6436439076685059,
0.6752523962811601,
1.4661557963348293,
1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,

```

0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,

[illegible]

-7.139980799817413,
-5.986699722057315,
-5.03269185703209],

```
0.375],  
'cost_richardson_zne': 5.21875,  
'sorted_expectation_vals': [  
  
    'sorted_noise': [6, 18, 30]],  
'sampled data': [ [ 4,  
                    2,  
                    0,  
                    0,  
                    -7.139980799817413],  
                  [ 12,  
                    6,  
                    0,  
                    0,  
                    -5.986699722057315],  
                  [ 20,  
                    10,  
                    0,  
                    0,  
                    -5.03269185703209]],  
    'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                            0,  
                                                            0,  
                                                            0,  
                                                            0,  
                                                            0],  
                                'cn': [ 0.5,  
                                         0.5,  
                                         0.5,  
                                         0.5,  
                                         0.5,  
                                         0.5],  
                                'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'ising'}},  
  'init_param': { 'value': [ 0.5483166258326411,  
                             0.5788252343888002,  
                             0.5803330366161238,  
                             1.2169503585394688,  
                             1.4292236971359287,  
                             1.4293923120692815,  
                             1.4365169928726365,  
                             1.4365169928726365,  
                             2.510196103669615,  
                             2.510196103669615,  
                             2.5544154452534937,  
                             2.5544154452534937,  
                             4.029649065019888,  
                             4.067883071594656,  
                             4.076808377533052,  
                             4.1651466476623735,  
                             4.656187563066811,  
                             4.656187563066811,  
                             5.18228186954163,  
                             5.18228186954163,  
                             6.808184896226372,  
                             6.808184896226372,  
                             6.960163700827979,  
                             6.960163700827979,  
                             6.960163700827979,  
                             8.49759837936073,  
                             8.49759837936073,  
                             8.526772837307739,  
                             9.456275223201612,  
                             9.799125697936137,  
                             -1.1324033485399458,  
                             0.3066519452505097,  
                             0.43856397893505916,  
                             2.603476267806093,
```

0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,

```

-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric3_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```

```

        'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.2062380598075135],
                          [12, 6, 0, 0, -5.187681916745086],
                          [20, 10, 0, 0, -4.3460110500437334]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.53131103515625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.78184810997413,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                           'cost_richardson_zne': 5.21875,
                           'sorted_expectation_vals': [

```

```

-6.2062380598075135,
-5.187681916745086,
-4.3460110500437334],

```

```

        'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.2062380598075135],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.187681916745086],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.3460110500437334]],
'sampling': 'default'}}]],
'ising-ric4-uniform-scaling': { 'noiseoff': [ { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [ 0,
                                                                    0,
                                                                    0,
                                                                    0,
                                                                    0,
                                                                    0,
                                                                    0],
                              'cn': [ 0.5,
                                      0.5,
                                      0.5,
                                      0.5,
                                      0.5,
                                      0.5],
                              'r': 0},
                              'time': {'max': 10.0, 'min': 0.0},
                              'type': 'ising'}},
                              'init_param': {'value': 'random'},
                              'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value': 0},
                              'noise_prob': [0.001, 0.001, 0.0, 0.0],
                              'status': False,
                              'type': 'dephasing'},
                              'nqubits': 7,
                              'observable': { 'coefficients': { 'bn': [ 1.0,
                                                                        1.0,
                                                                        1.0,
                                                                        1.0,
                                                                        1.0,
                                                                        1.0,
                                                                        1.0],
                              'cn': [ 0.5,
                                      0.5,
                                      0.5,
                                      0.5,
                                      0.5,
                                      0.5],

```

False,


```

        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix': 'ising_noise_off'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 1, 1]]},
    'run': 'vqe',
    'state': 'dmatrix',
    'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': True}},
    'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
        'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
        '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
        'X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + '
            '1 Z 0 + 1 X '
            '1 X 2 + 1 Z '
            '1 + 1 X 2 X '
            '3 + 1 Z 2 + '
            '1 X 3 X 4 + '
            '1 Z 3 + 1 X '
            '4 X 5 + 1 Z '
            '4 + 1 X 5 X '
            '6 + 1 Z 5 + '
            '1 Z 6']},
    'output': { 'exact_sol': -8.56677223350563,
        'initial_cost_history': [ 4.217815007292055,
            3.394484740359831,
            3.3211044240247634,
            3.65155648872765,
            4.031781430865161,
            2.5147460401592765,
            2.635251011361358,
            2.6229212283850565,
            3.6175377967590236,
            3.125720978907869],
        'noise_details': { 'gates_num': [4, 1, 1, 0],
            'identity_factors': [0, 0, 0, 0],
            'noise_level': [0, 0, 0, 0],
            'noise_profile': {
                'noise_prob': [
                    0.001,
                    0.001,
                    0.0],
                    'status': False,
                    'type':
                        'dephasing'}},
            'odd_wires': 4},
        'optimized_minimum_cost': [ -6.945492229929243,
            -7.1165547898808885,
            -7.2842760350438445,
            -6.724195761448022,
            -7.306479477026572,
            -7.415156948711891,
            -7.202776954905004,
            -6.836396584821816,
            -7.611342393541609,
            -7.203866848572784],
        'optimized_parameters': [ [ 0.33159768767931425,
            0.3415216906329597,
            0.34152169063295984,
            0.5593316391966133,
            0.5593316391966133,
            2.351554445259766,

```

2.9331738085905035,
3.4748030853426815,
3.4748030853426815,
3.616855036152045,
3.616855036152045,
3.8672141033893017,
4.423346341593822,
4.423346341593821,
4.845338698326811,
4.845338698326812,
7.020561389723981,
7.020561389723981,
8.059188148673007,
8.486998646842132,
8.48889562974299,
8.48889562974299,
8.48889562974299,
8.852856222269512,
8.852856222269512,
8.85285622226951,
9.91447562836983,
9.968000692942592,
10.549940074823725,
10.549940074823727,
0.8537391604836984,
0.9168200846380319,
0.2559354991965516,
1.0210726229552618,
1.2632715564780685,
-0.3480423461281932,
1.7009717288666621,
0.8945988480638971,
0.0342878706217857,
1.5867242856094939,
-0.46323092808777266,
0.9197703726655558,
0.7442882527479103,
0.9265871236439417,
1.007434201350355,
0.029736173082484843,
0.47331341236562946,
0.4861485561257538,
1.0394118763980624,
-2.9809358180453613,
-0.2596551057591169,
0.274814845028427,
-1.4049453096263822,
-1.0753350027554103,
-1.11413441253153,
0.9507335703621095,
0.8064528289725565,
-0.4246412780854718,
0.4472568123237856,
1.634610917427341,
0.8557019227225735,
-0.10158119911628401,
-0.31211728480663814,
-0.9970023252564638,
0.25431186671884126,
0.7248953017537308,
-0.4073131817183126,
1.6792046485134604,
0.5505815433075344,
-0.8093860553948707,
-0.6172486758497164,
-0.12685386183371894,
0.21223631022564798,
0.8168562896383363,
-0.17647756432116687,
-0.5731604235592278,
0.4333036783698612,
-0.7682262923606079,
1.6429671181235093,
0.6337476059862421,
1.67916133236013,

-0.5926217976812627,
-0.298257419735069,
-0.6583760201171601,
-0.889601924248653,
-1.3248621012262398,
0.013854140862500447,
-0.7662338030087398,
-0.6194525061041963,
0.6815602243054589,
0.08161257575833167,
-0.3072072388036838,
0.42225747459757357,
0.06754004785449946,
0.4843941525302115,
-0.1684623955114203,
0.8138047240050311,
0.052754329692318934,
-0.03499144746108355,
0.04466125788054358,
-0.028799258738440928,
0.7127927371021741,
0.7421378220215715,
-0.6858695512918895,
-0.11720668569532972,
-1.3579343673989164,
-0.9607782638271641,
2.0876158669999754,
0.030744163991020267,
-0.6062544285001135,
-0.08046454707386355,
-0.9589458574514981,
0.6218575881236439,
0.008264172331609584,
0.1779434874985719,
-0.2072304227788729,
0.9938506593352908,
-1.0471692270924964,
1.0898429645649914,
-0.22066548691854398,
0.8076838524861312,
1.7135973843396541,
0.23198771018109937,
-0.14790366150484735,
-0.7779724161623084,
-0.5740177045120107,
-0.8356765454865094,
-0.12758989923704314,
-0.3192845855088301,
0.9992686200806561,
0.18910999621207844,
-0.3626298108536852,
0.5832805029120673,
-0.8863720585918194,
-1.6851050525249511,
-0.3551820269447921,
-0.4184387520471605,
1.05736378101237,
2.031993671638238,
0.20520618587592238,
1.6363439075818225,
0.14342936783484309,
-0.6043776905728043,
0.3922986049495649,
1.4322290738810806,
1.585876902523968,
-0.2738910870161085,
1.7714953537072202,
-0.29354248742932953,
-1.2273457901372085],
[2.0562096130838231e-16,
0.03538692433604476,
0.5493527934980842,
0.5695965809115139,
1.2739811069641827,
1.2739811069641827,

1.5440077454822854,
2.6187560157598857,
2.6193651223262795,
4.47656169572244,
4.47656169572244,
4.900258790927604,
4.903975205691072,
4.965015364809818,
4.971477584375831,
4.9714775843758305,
6.285851088616883,
6.285851088616883,
6.5727281068015415,
6.914906570174498,
6.914906570174498,
7.5825149096464,
7.5825149096464,
7.5825149096464,
8.266106449462097,
8.772625056171776,
8.772625056171774,
8.772625056171774,
9.888607991704035,
10.650763302381714,
-0.25736569540817156,
-0.13749177250114622,
0.6913360970917151,
-1.0251451053141543,
0.04952583576033172,
0.6839090084427677,
1.2841797475889756,
-1.4498903904619547,
-0.6609983487872932,
-0.9195300213913112,
1.0882692755947088,
2.28444907625875,
0.5013640158162996,
1.3195089116901169,
-0.44969609005592753,
-0.19760723927349785,
0.047416784300074,
-0.2592183302467101,
-0.95267369637524,
0.2510197484008247,
0.12431427724764418,
-1.9455191869747381,
0.0834449432923055,
0.6181240781879018,
0.06318897794461549,
0.7469995321450943,
-0.13869412828035013,
1.7789335199415868,
-0.2133263169614118,
-1.747801096143808,
1.7374950822862127,
-0.1617072567646695,
-0.13415848991929505,
2.0599871163251184,
1.2617098025896833,
0.7421863291532844,
0.6134289282076183,
-0.010288112985795142,
-0.164325473683837,
-0.19747193793554596,
-0.1095664145547814,
0.3448002170188953,
-0.008329131684078827,
-0.27021098586965386,
0.5642984785825904,
0.26951720880242647,
0.9416408813291534,
0.18894282686740613,
0.6744670741290004,
0.15073677601160052,
0.7600850898072902,

0.23785895970346504,
-0.3096016709123609,
-0.29939543701383164,
-1.2773708023878871,
0.012610842713240655,
-2.001134648031791,
0.5569056601340245,
-0.022823668895979605,
-0.44069878780800564,
0.9238325880974837,
0.627132735311806,
0.833823816751152,
1.174260792549426,
-0.8145978291018492,
-1.8911045685123373,
0.03189540714483889,
-0.6114896265404774,
0.18855141943106907,
-0.5415449375701122,
0.3556432019461298,
-0.02940037290701983,
0.7847914443788936,
-0.8204017405579882,
-0.22117427936396428,
0.9668207238887914,
0.6982235282720571,
1.1161787173582038,
-1.4967922582645339,
0.6955597513963624,
-0.08409555407567708,
-0.7515653427189505,
-0.13809041364831362,
0.694805471024674,
-0.16980356694456364,
-0.7378676064228189,
-0.5510006148273209,
-0.4664642602089831,
0.3131774610690113,
0.25515679468716435,
0.01472701586112837,
0.9612508062048253,
0.3876700512851733,
0.34932481898103945,
-0.0017068592723796239,
-0.9400752962906423,
0.20680236697496335,
1.2239109612714394,
1.5295424358244767,
-0.16313500004842318,
-0.6283905359663017,
0.6007919237891753,
1.4275642137136517,
0.8373669928162908,
-0.3460569718594813,
0.30478283310349635,
0.7741593665645724,
-1.49133186355475,
-0.40261318631906845,
-0.6780231584993508,
0.2796292694824153,
0.8154646298435348,
0.6705233696972283,
0.8515977010836673,
1.38835715204434,
1.5513313401022284,
-1.9456573322270103,
0.6825855385724343,
0.3411942765245547,
-1.8463135511065893],
[-1.394844222866792e-17,
0.2294538829927996,
0.7521460514270448,
1.524448292944527,
1.699326062163185,
1.699326062163185,

1.699326062163185,
1.6993260621631845,
1.6993260621631843,
2.04427958156354,
2.044279581563538,
3.760855608085718,
3.7608556080857176,
4.422362610147448,
4.422388771925709,
4.423247474933146,
5.088788131744124,
5.518902595462681,
5.518902595462682,
5.518902595462682,
5.518902595462682,
6.713884925045287,
6.713884925045287,
7.229900049876175,
7.23519041379673,
7.5137384820073985,
9.896584707205738,
9.89724831030544,
11.177528016222873,
11.333986770578415,
-0.5874618645254188,
-0.38498635294065175,
-0.14869891390826667,
-1.385574110751756,
0.3783147753212974,
0.1998192518416486,
1.8873134139140266,
-1.0608247810156872,
0.2365140987973328,
-0.9738696834833499,
1.5453125784784605,
1.8657042183850436,
1.5293157207302228,
-0.7517266227054149,
-0.8871771295479706,
-1.6950047018057137,
-0.5885280426519628,
0.04826899994478067,
-0.4330318552574016,
-1.388710608286563,
-0.08211595300211308,
-0.5283548125054784,
0.3763483175118206,
0.335884831266236,
-0.5070752646015597,
-0.2910412053619489,
0.2354568453295162,
-1.4336146483959564,
0.228779351175754,
-0.06520272691838713,
-0.8209928725245101,
0.7001970509858747,
0.5785044584879039,
-0.49932901566271715,
0.45729974286308867,
0.46052818546886504,
1.4986129468586522,
-1.4491938093059253,
0.58130192556905,
0.052769119433839204,
-0.00640289944448277,
-0.3687606063433655,
-0.006145485686113623,
-0.024597955898482345,
-1.411768598869635,
0.08097884701091772,
-0.9927225428945597,
-0.0075215312387787595,
0.010579195204485638,
-0.42659138073244207,
0.002225130694705108,

-0.17841021982057653,
0.9375108947888807,
-2.7757789566035065,
-0.010127033509228812,
-0.5146645773967682,
-0.3567686303606847,
1.4398753496798034,
-0.5703995887616833,
-0.05677408229384658,
0.957894065877575,
1.9071290282089846,
-0.6507219310887598,
-1.6162530141223683,
1.1818941119762436,
1.6868779692850424,
0.98800409304883,
-1.0587465299812662,
-1.094775277351337,
-0.2877600991535592,
1.8986470088901148,
0.8207273276904813,
1.2066175393079017,
-0.22168365465550993,
0.17258447386478676,
-1.8715762567403773,
-0.48240059727904117,
0.7145730346214052,
1.341149548855606,
1.3388694132947494,
1.1029357907173007,
1.7761192703736626,
0.389134329678578,
0.6998749042461945,
-0.4466329063075881,
0.9939411509507131,
0.5933993643264636,
-0.2238849148647213,
0.16806395587982337,
0.40153989063889134,
-0.3072506770355845,
-0.3084801740315303,
0.35020486394553546,
-0.6893875836471265,
1.1621768900261087,
-0.005422106750069378,
-0.6392006402141877,
-2.6960821965791135,
1.2350659365734071,
0.02344533464890316,
1.6191412587681537,
-0.624233381931643,
1.164388497878687,
2.2414892885807953,
2.3459185940732037,
-0.18816057823941634,
1.0703302771481322,
-0.6022027174173271,
1.5454586171163143,
0.49083778015320845,
-0.5414313403110282,
0.5943360939664153,
0.8531518824889626,
-2.0712921555535835,
-0.9507228787104378,
0.1663695492007906,
-0.4829435842745145,
-0.9852459132453085,
0.6444213968977472,
1.9304389405092637],
[0.47195138158384564,
0.4745413934062829,
0.6416788974326727,
0.6416788974326727,
1.3767805975406497,
1.3774084633016226,

1.3850075193774707,
1.385007519377471,
1.961329521314093,
1.961329521314093,
2.6299528219180073,
2.6299528219180073,
2.6462286434741538,
4.376679341839624,
5.443648648286045,
5.443648648286045,
5.443648648286044,
6.427045191271688,
6.427045191271688,
6.737104333282549,
7.345768854864493,
7.345768854864493,
7.523575375204561,
8.254985561732491,
8.914602409288278,
8.914602409288278,
8.914602409288278,
10.045438587538639,
10.045438587538639,
10.045438587538639,
-0.014294444089673569,
0.8050246736150516,
-0.4603650249158636,
3.8028784519891046,
-0.40535003035622474,
-0.20345609256189173,
0.6921728480649123,
1.2635655003066155,
-0.5816868768558133,
0.8004330576026809,
-1.2247112488961414,
-0.7226464348007927,
0.021600305535157273,
0.2422993600121551,
-0.2813766982328762,
-1.8376799098437777,
0.1842235451250743,
0.08033551390998013,
1.2509807369145618,
-0.43993928060300874,
0.02749412053256657,
0.9446032887644298,
0.5876918496487636,
-1.2193113596650038,
-0.30025121886784994,
0.40718701943488855,
-7.055881324256668e-05,
-0.3230027373880063,
-1.3595946794844804,
0.4025901861096578,
-0.331697269297153,
0.8588883241978998,
1.7499317067572346,
0.2974332602533821,
0.36978773801612186,
-0.1369546347590247,
0.17848126558694688,
0.5078793586065516,
-0.026587668618036284,
-0.32676172114586377,
0.043331505672591836,
0.7587982826624224,
0.5755592246710275,
0.5753466292739289,
0.6580990661684004,
-1.8654933062390762,
1.4954582293028102,
1.1206423350708015,
-0.6308795940680783,
-1.1220353310878195,
0.3113810383877842,

1.1285056041045296,
0.8014038814434817,
-0.3123476059423272,
-0.46649205373142577,
-1.0739566662407007,
1.7799335725034942,
0.3271320302071495,
-0.4957408509507477,
-0.8965339685216933,
0.9394687605669856,
-1.0123362977633281,
-0.8493226829357736,
1.9016506939491882,
0.39163185359556635,
1.2450374039748047,
0.6637318474595636,
1.4487854848879562,
0.8556001420669627,
0.2844708394448605,
1.19751608671342,
-0.2783699910645836,
0.2731694463477748,
-0.8573920204053973,
0.4298526210883498,
0.1620542969403055,
0.7385715014812971,
0.10161961925998468,
0.08514730828974114,
1.5393767966723602,
-0.7517708453085385,
-1.0667182547896288,
0.7653330843147801,
0.2770240276337771,
0.44649448354109117,
1.839692443661341,
0.6074846807891079,
0.5576661480775236,
-0.5735679701859718,
-0.5213243179488682,
-0.13640843594155702,
1.4993013794616976,
0.959504645257717,
-0.8869155163266706,
-1.0471786039389914,
0.6169717474797428,
0.010330090116839902,
-0.36051081047743266,
0.08240138993527565,
-1.19819214697301,
-1.2973765177480467,
-0.8056172495711347,
-0.04435763741623314,
0.14581600625591132,
0.26060016989542056,
0.07248105118801107,
0.5605125602873733,
0.7950998674453089,
1.1145979627095777,
0.00625222978811041,
-0.3867814373266647,
-0.1324550289145508,
-1.2907509838890496,
-2.272496441870613,
-0.4069714152055066,
-0.09281234013047389,
-0.02440219940061265,
-1.4477733222094868,
1.1238797279606656,
0.08808709541843764],
[0.4802107213676794,
0.5872443280976499,
1.2030235571335923,
1.6244635367917757,
1.6244635367917757,
3.26071227407772,

3.26071227407772,
3.8336805164974805,
3.8336805164974805,
4.826555535372041,
4.895860683399619,
4.953829011998897,
5.427903354045862,
5.904661066786138,
5.904661066786139,
6.1444777924663665,
6.1444777924663665,
8.501726489902856,
8.560790735911917,
8.56230393956988,
8.56230393956988,
8.56230393956988,
9.349516793328615,
9.383607475180085,
10.075280307758542,
10.10947387193881,
10.10947387193881,
10.676123926986733,
10.676123926986731,
10.676123926986731,
1.4057383265279821,
0.12082364332553383,
-0.26633169071888585,
0.8440095288231939,
-1.6788257521351209,
0.5654604704832584,
-0.8067769238497999,
-3.0754933205615647,
0.08918153353934634,
-0.44144180897560614,
-0.12233030448793467,
0.7466185674598658,
2.029541324404434,
2.6182216828178912,
-1.132391894978203,
-0.24681552984026395,
0.016701788041529435,
2.056331221347139,
0.4687565935163574,
-0.814652769544358,
1.3493029022207552,
-0.0979175823892474,
1.1060724549152727,
0.10932839153376225,
-0.021236171452369973,
-0.10240895200106154,
0.0678743319848748,
0.31105500906993255,
-1.3775189120014106,
0.1972259082837923,
0.5685166045457986,
-0.04805822649685886,
0.05656484641847995,
0.28963771198353216,
-0.0823457664175828,
0.10970044186707124,
0.4470565049690292,
0.3481580101245137,
1.429670826498759,
0.9929735218605864,
-0.118080682882002,
-0.24768077486638254,
0.14000119823560106,
-0.0033695140848767277,
0.28167924146812856,
0.33054680125209646,
0.25446310854586207,
0.4874199962063169,
-0.4292630164890844,
-0.8493517422041822,
1.10338617740252,

1.008639770575481,
-1.333715224046274,
1.5531304809990258,
-0.9099800622014602,
0.3709414533710614,
0.013272332284884182,
0.2649430125577457,
0.2281562083367391,
-0.8531201835868235,
0.3516874446864623,
-0.20156681030383022,
-0.20235738350440882,
-0.042569687456911996,
0.15952312088272655,
-1.5291183017296281,
-0.14843597874824083,
0.29841129212111156,
-1.0541286856947945,
0.1912222645636818,
-0.6584533547304796,
-0.8290997819567264,
-0.8585614987938113,
-0.925446343244433,
-0.31902871745414296,
0.059023427516986,
1.0543033158567225,
0.06361350569662501,
-0.24874942116320348,
0.14393135270630777,
-0.6226920500261081,
-0.9314119807859277,
0.08452985204639987,
-1.009402214771574,
-1.0062750028527756,
-0.28841331398033154,
0.5886312985955309,
0.25138239542645724,
0.4840684579382569,
1.66880370795529,
0.4947479443765874,
-0.5723605201777653,
-0.16848739586132902,
-1.2344957342488987,
-0.21521798449848678,
-0.061827721816023445,
-0.2248741094525013,
-0.14743116036125348,
1.3439891420398065,
0.26093100927833396,
2.340498096953527,
0.48577707139825455,
1.3703195203028469,
-0.03611016151365655,
-0.39327539415785684,
0.19418290589939505,
-1.2131007544515844,
-0.2043806706538651,
0.10177491262713155,
-0.04073135722774902,
-0.6647700988785717,
1.8155909796605325,
-0.9294726054396647,
0.06362652018359347,
-0.3885510819409685,
0.2259437203319192,
-0.29241696657905075,
1.5301954112147191,
1.6475949707589932,
-0.20840788688387885],
[0.42838040299935903,
0.43437695063037796,
0.7671204362948421,
1.4217493914297048,
1.532465319753789,
1.532465319753789,

2.7699209108087253,
2.7699209108087257,
4.156218307623238,
4.2237366972759505,
5.0429088489710905,
5.613108479440035,
5.957592523811918,
5.957592523811918,
7.200159227676843,
7.200159227676843,
7.511063978540512,
7.511063978540512,
7.5293149335701015,
8.184114862205785,
8.59759634646416,
8.59759634646416,
8.967813742088222,
8.967813742088222,
8.967813742088222,
9.731896873309175,
9.95178485601347,
10.151200227727777,
10.572249489684078,
11.21122434413465,
0.6103673192422585,
-2.2449370868204332,
-0.5531680672940967,
1.6213620933184167,
0.9025434887524114,
0.7267620669083164,
0.688558576899684,
-0.6961599873617196,
0.1412349369195377,
0.7459812495914973,
0.4704664842082417,
-0.38033435666108695,
0.35864815337369327,
-0.9120464289406482,
-0.39019902865511685,
0.8340346919183055,
-2.0544585551745573,
1.25105004616506,
-2.352820292981314,
0.4238462318210131,
0.07915902405687532,
0.20643459967058658,
0.23897101893275827,
-0.8664160098114893,
-1.685166937074754,
-0.06832039714759808,
-0.9791405090065629,
0.025203133044215387,
0.0013271421621207855,
-1.0715671968015688,
-0.008370532310604396,
0.13052855552000728,
0.497310552248094,
0.30798451910916236,
-0.6467356358635334,
-0.411683867969946,
-0.10065713602082733,
0.5062131798374145,
-0.10337024350765668,
-0.9070679783587652,
-0.6555610326122162,
2.2255239458177773,
1.727993426652873,
-1.440768657668252,
-1.3828846188724404,
-0.4461523368168654,
1.0838317458165323,
0.38330860036255543,
-1.499700854710827,
-1.3408221959756346,
0.7419236696397365,

-0.26719837498867116,
-0.2377148775462395,
2.1193168382343814,
-0.3237137921938176,
0.30565535651577447,
-0.3512550922173362,
-0.12043790779624768,
-0.7473597606659746,
0.04801624913587568,
0.0018824715948976833,
-1.0616614621946072,
0.0026415486583119922,
-0.3446629765669456,
1.7936621861690503,
0.6484630040536189,
0.911272728030746,
1.1140004969268402,
-0.7180242734781173,
0.5255116360573389,
0.21141077992540375,
-0.45008152208030955,
0.27750531007906354,
0.32764881051843897,
-0.018511194189945894,
0.3875950700305173,
-0.3325176658862957,
-1.8796967870400432,
-0.6725696299533431,
-0.016521571218157557,
-0.08350862144921116,
-0.5447867614937103,
0.09550333474849668,
0.008862454446484716,
0.22308949354372234,
0.43503110895758224,
0.08385997823794214,
0.8516706560998085,
0.37101021681963386,
-1.5090995065898887,
0.3045648040580248,
0.6581099200957676,
0.35742409273461206,
1.7397310471751102,
0.19044279487017676,
0.9986798590605289,
0.31257833320056017,
0.19093877798180586,
-0.12038962867457878,
-0.4069447684098612,
-1.0520519597232647,
-0.45448340350076705,
-0.09417833232522027,
1.3653492736312018,
0.6606235759628448,
-0.1314405098844259,
1.082951248594517,
0.2015466917595484,
0.16965874881834048,
0.2649109746606924,
0.8389747424450928,
1.4332963880188982,
0.288517036279826,
-0.13031950550882798,
0.31918733601921234,
1.3571234506254648,
1.4386307786107537,
0.4541349276361828,
-1.1895164340492679,
-0.27888567947798426],
[0.403034597497222,
0.4502729934454179,
0.5700423320933159,
0.5700423320933157,
0.6781241992806778,
1.3786879425414615,

1.4501147159778993,
2.4844307076867937,
2.4844307076867937,
3.0256317298652835,
3.0256317298652835,
4.15344004944777,
4.15344004944777,
4.2038017426165855,
4.6259251670404895,
4.6259251670404895,
4.797805401698216,
4.797805401698216,
5.52869212939408,
5.565345896572568,
7.178724808111429,
7.362080567239832,
9.32786374549076,
9.32786374549076,
9.69844688689066,
9.970706574916212,
9.970706574916212,
10.264471030354098,
10.264471030354098,
10.987148477608002,
-0.928231690167026,
0.8057374905304977,
-0.09124110656415482,
-1.2764053392323713,
-0.1699102369047808,
0.2854283542912807,
-0.38934147484457715,
1.9562479348306092,
-0.08583239283592208,
0.1375503575288643,
-0.3029413642384467,
0.3573494966906954,
0.4106808877553357,
1.0168255078067534,
1.020920504974372,
1.182081306796335,
-0.381751450929397,
0.7872502675264477,
0.11650702005295199,
-0.7041962676262727,
-0.9664718849607234,
0.0017002173560816299,
1.2143682588418563,
0.27506501793685123,
0.20402751900538993,
-0.18948284766781337,
0.003995331337594065,
1.1960049920193474,
0.5655953653806649,
0.045530409523104075,
-0.6946141334022584,
-0.014434864812455215,
0.011912663267278172,
0.3387818931591513,
0.03142887465994337,
-0.6463763503344852,
0.17750599329640465,
0.014482690293971139,
1.8640640992205775,
-0.048924475364900355,
0.07047222436652016,
0.5836409098942513,
-0.033227003016975516,
0.5638116666909924,
0.6159645449318359,
-0.28815587863252157,
1.0812127139774732,
0.379778365443053,
1.5589007157572847,
0.42162649727394796,
0.05475856745796003,

0.7192262097201122,
0.3263213716214118,
-1.3661280753618785,
-1.292135594114921,
-0.8807331583336575,
0.4025768854927813,
0.5295243646894815,
-0.8002135124103957,
0.1750760012142075,
-0.054032551513248435,
-0.40188384379722514,
0.07562722660439603,
0.9832576708267071,
-0.2899613446815228,
-0.06710629496184976,
0.7055866137794157,
-0.007144529560920351,
0.12285567358407816,
1.143563250691634,
0.15056960823466894,
1.2837012011259124,
-0.6023181038676345,
0.045174795264068736,
-0.16273344906061393,
0.06659839307873978,
-0.018202395096242784,
-0.39240523180078335,
-0.08377351665795224,
0.97309387436951,
2.220420875428843,
0.26956954819495804,
-0.9599252243572389,
-0.804945736082422,
-0.11945317743340336,
-0.8427551316765194,
0.30866551832235317,
2.4872766722318986,
-0.6424953924634126,
1.3858530207015087,
-0.28083276964757203,
0.4192227932599254,
-0.823908670962852,
-1.1126529481475467,
0.21496241595348659,
-0.08934899247793467,
0.1474411947098728,
0.5744211770388311,
-0.04170488097810379,
-0.22724449437110839,
1.9972780099267948,
-0.8591254315731605,
1.1685333425381264,
0.35810812429508854,
0.4121168481727771,
-0.15141964973712507,
-0.4596563680482823,
0.2311510501006746,
-1.1273861464736383,
0.5254398836685342,
0.8562906940888275,
0.2485294131097614,
0.5338805724034291,
0.4862154010986529,
-0.6197058235436752,
0.20248976894253998,
-0.8370526377715465,
-0.2508663863331003,
0.895805892021248,
0.966286773500402],
[0.45986682004385615,
0.587884042281194,
0.587884042281194,
0.8634874478193177,
0.8634874478193176,
2.1638996627094382,

2.6675296177894383,
2.7047736270183154,
4.061300455462818,
5.103553089959934,
5.103553089959934,
6.751801791048447,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.020369336633852,
7.781371497069936,
7.781371497069936,
7.781371497069936,
8.420463205941301,
8.420463205941301,
8.420463205941301,
8.420463205941301,
9.122294773242167,
9.202160901745463,
9.814044623063202,
10.300168251850302,
10.300979681078617,
10.504460883512992,
11.02433376042467,
0.6573588347385189,
0.5828935548145123,
-0.574462871025295,
-1.546656732274899,
-0.27660055256222155,
1.273681854258797,
1.2384634856925236,
0.2379867607060117,
-0.9322977855669335,
-0.7820979498963491,
2.372805485180463,
-0.032501906954750276,
-0.7800293806349019,
0.4387330183441131,
-0.7149138216831045,
-1.1824362472270316,
0.5918930779181975,
0.829172686537006,
0.977795961853665,
-1.6128990556187608,
0.19086262573792945,
-1.3821924448334508,
-0.37777709213522187,
-0.7880705929271598,
-0.17099996546962573,
0.7254760017497719,
0.9819125877821568,
0.3047932425929136,
-0.02742999568379597,
0.5485981155871978,
0.11485583255370253,
-0.32472329363004176,
-0.6331630162839701,
1.9724401417574067,
1.7698339388439421,
-1.1254891310675255,
-1.1920329227643298,
-0.11899515772905239,
0.6149649547843763,
0.08915119991626204,
0.4168497404293414,
1.795330514128595,
0.13783861826417038,
-0.9065003531161022,
-0.3985523491151137,
-0.36816430284390833,
-1.1942463088521762,
-1.129677747629269,
0.6710845981341038,
-0.6149988458361377,
0.4665365160069161,

-0.19209648586871406,
0.013763508723981719,
-0.8135432145712183,
-0.4057592495812904,
-0.6414147049863589,
0.8876536381187553,
-0.7757241021147325,
0.40873541229116084,
0.514258938404948,
0.3722392877827026,
1.8009017308404462,
0.116225182388376,
0.2191004740219447,
0.8255374993004068,
0.8886674208207993,
-0.579134397544579,
0.06903106340432566,
-0.7849541624527424,
1.2193465265846974,
-0.5822657541432643,
-0.5730195760225829,
-0.08026565436629246,
-0.5615010238556278,
-0.24965155606323408,
0.9668325985251349,
0.32484880948626715,
-0.419910520268625,
0.9941102073040009,
0.290737643308207,
-0.05558286792755536,
-0.07023052502159025,
0.6542319702891408,
-0.7125513821501609,
0.3724489707231321,
0.6482228950077775,
0.6813557635502394,
0.5904217885770654,
0.3344272908394536,
0.9892271354161819,
-0.6202922067549016,
0.12947475126138008,
0.35715623223344434,
-2.26745180129,
-0.20006687506113488,
-0.6290028667838425,
0.08022472893529967,
-0.961248450361056,
-0.20142658286918888,
-0.26748808365437937,
0.3448273909474204,
0.6331413989953146,
-0.5916282783618806,
-0.951916970919588,
-1.150804089444678,
-0.8692217930277405,
0.2236835205843831,
-0.10881486654903144,
-0.004228302858090612,
0.16762641757787602,
1.4232995099688168,
0.17348586714896594,
1.0253598045594392,
0.3864684766177986,
1.0340175270140362,
1.1184694851102563,
1.066055569095364,
0.07092789808311199,
-0.8620685771604671,
0.46490022331063124],
[0.49845241666821427,
0.5080489367341008,
0.9738886018422865,
1.569493854319001,
1.5867336495888638,
2.22018578038962,

2.222975784710255,
2.7355212499054673,
3.167907147345212,
3.912280465424413,
3.9324465712853787,
4.601639034821639,
4.6023794814303285,
4.622186895887535,
4.629761229768061,
5.8119318913753775,
5.81529780322803,
5.862601007224966,
5.868327135385091,
7.931887080538795,
8.400559524273383,
8.400626671117248,
8.400626709821431,
9.338300284642187,
9.338300284937413,
10.032003053056542,
10.032003053750085,
10.151775184539204,
10.15177520525735,
12.777878105202333,
-1.0652764150946168,
0.1030911905785532,
2.1938566419651506,
-1.687039127173871,
-1.2892167937928738,
0.454901922887811,
-0.10701158278386386,
-0.27610154265055814,
0.3864936366984189,
-2.1008557969724673,
0.03526686701549039,
1.2075362869846002,
0.9729996942576583,
0.011602974506752167,
-1.7901344125733514,
-0.5312859895436812,
-0.6764114285522254,
-0.568464725902531,
0.7761280206550648,
0.10820918045329357,
-1.191918699884087,
0.6710933645796882,
-0.998882902161297,
0.9668399094583318,
-0.5526879367651495,
-0.3448144391622963,
0.3546231755031622,
0.36267379522525617,
-1.3188129954125047,
0.7986540408445592,
0.5385879858279273,
1.3987311647526153,
0.17032526930657013,
-0.2765839794470922,
-0.4561318195988584,
0.7779672645593669,
-1.6618874628312856,
0.7145344738782802,
-0.8703397140730336,
0.3664383221285875,
0.09488224223418325,
0.17319381149242297,
-0.15027776082229982,
-1.4690811524212324,
-1.4342980641635596,
-0.0435375212610062,
-1.1192667984833862,
1.489571420955581,
-1.385268501544282,
0.07500036651414835,
-1.0373356270085057,

1.6055364169931783,
-1.1908916083661096,
0.47394571447103884,
-0.23948832756930136,
1.6865746730435722,
-0.07643328425683309,
-0.26016480364478584,
0.1230869402232525,
0.24188831995971777,
-0.12202515710835343,
-1.3273176630150638,
-0.8685353737050294,
0.16431995234498184,
0.5641710812415553,
-0.1851122077039956,
-0.3773492143584495,
-1.5621470223099339,
-1.4505536127106367,
0.571773409102061,
0.20129944645039902,
1.1233832799408359,
0.09021785654141615,
-0.09203212520614017,
1.4656739277180684,
1.5116863426084852,
-1.4548391160838867,
-0.8665597553970019,
-0.8311755953666669,
-1.2962978849547035,
2.7863849971356167,
-0.026842981867963016,
-0.5147225252861769,
1.0767631952186236,
0.23432587613057304,
-0.3849014581870786,
-0.5492566246326465,
-1.590706022355079,
0.2729988606074437,
-1.0566347351025576,
0.3490824852559465,
-0.08255987668621423,
2.389615689206544,
-0.5033499561524312,
0.8858613199859888,
0.002820627497544732,
0.19486994113578365,
-0.12438999158785295,
-0.2658298270496061,
-0.2763678550214429,
0.4048526015479744,
-0.06841768991626676,
-0.32117880607439586,
1.6019482252187531,
0.41890698857400605,
-0.8510923614851694,
0.16411947273918887,
0.3096085421164439,
-1.0434867263587853,
1.2153787936463343,
-0.9841260925764858,
1.5878541492278757,
0.5868781859412673,
0.26283212361760866,
-0.9081790869869905,
1.200367927444946,
3.155405601054565,
0.19810365707195607,
0.6950762676177132,
0.9262562990824308],
[0.11310194099017501,
0.35533489757839665,
0.7274669277129748,
0.7931766663622793,
0.9215124586161831,
1.5868010098342087,

1.6938861718192633,
1.6938861718192633,
2.4953100977078537,
2.4953100977078537,
2.8524614686988468,
2.852561718095858,
3.5986528286299406,
4.15163213852048,
4.658101059037985,
4.662433860401005,
5.170840359657884,
5.170840359657884,
7.389891885603243,
7.408023950817849,
7.472384801069389,
7.816163992050368,
8.61557017602429,
9.097989314327426,
9.523961629484033,
9.523961629484033,
9.983169099900275,
10.18670132329559,
10.652310618690768,
11.298231074807717,
0.0018299791732057594,
-2.098323349499344,
-0.32648794289911637,
-0.9015207614699141,
0.16889281860224642,
-0.23801018256690964,
-0.2710900692754434,
-0.354821626365823,
0.2834027319981757,
0.5370764326800048,
0.3290793485287995,
-1.7004803192621687,
-0.8267928250682555,
0.7226363685371004,
-0.23961420559359078,
-0.9892421071296497,
-0.3062192663980587,
-1.2857004554953664,
-0.00570902688287085,
-0.7828701256611971,
0.036023661097245405,
-1.405764493898646,
0.09523715910714706,
1.0678085339341385,
-0.3477609497961018,
-1.3254201123902754,
-0.5891224027277412,
-0.14272029692701035,
0.10490796904880517,
2.930201181567992,
0.04738429480019876,
0.0461816378739841,
-0.48875992312198635,
-0.1220291693117149,
-0.4148582659190601,
0.17851659149817373,
0.4130064273610762,
-0.06315869176871276,
0.3800783279561489,
-0.02544638146603332,
2.215655886557845,
0.11710086424953603,
0.9518430371086406,
-0.12584714752332027,
0.018420823357896687,
1.2484195950702188,
0.13344161240602143,
0.09153656466168367,
0.7487460411896285,
0.4894289612390041,
0.03364631653773711,

[illegible]

```
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
```

-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,

False,

```
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                                    'value': 0},
                    'noise_prob': [ 0.001,
                                     0.001,
                                     0.0,
                                     0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.82304667158383],
```


[illegible]

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,
0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
```

1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,

```

False,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.029866062982578],
[12, 6, 0, 0, -5.108515955211577],
[20, 10, 0, 0, -4.320587490415481],
[28, 14, 0, 0, -3.6439280425186893]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],

```

```
'noise_on_init_param': { 'status': False,  
    'value': 0},  
  
    0.001,  
    0.001,  
    0.0,  
    0.0],  
  
    'dephasing'},  
  
'noise_on_init_param': { 'status': False,  
    'value': 0},  
  
    0.001,  
    0.001,  
    0.0,  
    0.0],  
  
    'dephasing'},  
  
'noise_on_init_param': { 'status': False,  
    'value': 0},  
  
    0.001,  
    0.001,  
    0.0,  
    0.0],  
  
    'dephasing'},  
  
'noise_on_init_param': { 'status': False,  
    'value': 0},  
  
    0.001,  
    0.001,  
    0.0,  
    0.0],  
  
    'dephasing'},  
  
    'odd_wires': 4}},  
{'run_time_sec': 12.810645580291748}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
        'layer': 30,  
        'ugate': { 'coefficients': { 'bn': [ 0,  
            0,  
            0,  
            0,  
            0,  
            0],  
            'cn': [ 0.5,  
                0.5,  
                0.5,  
                0.5,  
                0.5,  
                0.5],  
            'r': 0},  
            'time': {'max': 10.0, 'min': 0.0}
```

```
'type': 'ising'}}},
'init_param': { 'value': [ 0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.0892586669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
```

-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,

```
False,
```

```
'value': 0},
    'noise_prob': [ 0.001,
                    0.001,
                    0.0,
                    0.0],
    'status': True,
    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
          'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                      '+ 1 Z 0 + '
                                                      '1 X 1 X 2 '
                                                      '+ 1 Z 1 + '
                                                      '1 X 2 X 3 '
                                                      '+ 1 Z 2 + '
                                                      '1 X 3 X 4 '
                                                      '+ 1 Z 3 + '
                                                      '1 X 4 X 5 '
                                                      '+ 1 Z 4 + '
                                                      '1 X 5 X 6 '
                                                      '+ 1 Z 5 + '
                                                      '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
                             [12, 6, 0, 0, -4.800941579254224],
                             [20, 10, 0, 0, -4.034504937366069],
                             [28, 14, 0, 0, -3.3806822209161007]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
'noise_prob': [
```



```

0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4}},
'run_time_sec': 13.264791011810303}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,

```

1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,

-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603]],

'noise_profile': { 'noise_on_init_param': { 'status':

'value': 0}},

'noise_prob': [0.001,
0.001,

False,

```

0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -6.536883635090364],
[12, 6, 0, 0, -5.4912521012226865],
[20, 10, 0, 0, -4.624115876403395],
[28, 14, 0, 0, -3.8988670742113007]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

      'noise_prob': [

          'status': True,
          'type':

              'odd_wires': 4},
              { 'gates_num': [28, 7, 1, 0],
                'identity_factors': [3, 3, 0, 0],
                'noise_level': [28, 14, 0, 0],
                'noise_profile': {

                    'noise_prob': [

                        'status': True,
                        'type':

                            'odd_wires': 4}},
                    'run_time_sec': 13.687187194824219}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
            'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'ising'}}},
'init_param': { 'value': [ 4.68500097130382e-19,
    0.00019188574128314593,
    0.4999939126216869,
    0.571475986197529,
    0.6379337879222081,
    1.2223970983691403,
    1.405127109337375,
    1.405127109337375,
    1.405127109337375,
    3.9334167957034354,
    3.9801359916712626,

```

3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,

```

-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```

False,

```

1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
'4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
'X 5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
'+ 1 Z 0 + '
'1 X 1 X 2 '
'+ 1 Z 1 + '
'1 X 2 X 3 '
'+ 1 Z 2 + '
'1 X 3 X 4 '
'+ 1 Z 3 + '
'1 X 4 X 5 '
'+ 1 Z 4 + '
'1 X 5 X 6 '
'+ 1 Z 5 + '
'1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
[12, 6, 0, 0, -5.679026952355479],
[20, 10, 0, 0, -4.762614433311011],
[28, 14, 0, 0, -3.9973832617084413]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},

```



```

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4}},
  'run_time_sec': 12.833034753799438}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                         'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                         'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
  'init_param': { 'value': [ 0.5483166258326411,
                              0.5788252343888002,
                              0.5803330366161238,
                              1.2169503585394688,
                              1.4292236971359287,
                              1.4293923120692815,
                              1.4365169928726365,
                              1.4365169928726365,
                              2.510196103669615,
                              2.510196103669615,
                              2.5544154452534937,
                              2.5544154452534937,
                              4.029649065019888,
                              4.067883071594656,
                              4.076808377533052,
                              4.1651466476623735,
                              4.656187563066811,

```

4.656187563066811,
5.18228186954163,
5.18228186954163,
6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,
1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,

```

0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],

```

False,

```

'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':

'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                         '+ 1 Z 0 + '
                                                         '1 X 1 X 2 '
                                                         '+ 1 Z 1 + '
                                                         '1 X 2 X 3 '
                                                         '+ 1 Z 2 + '
                                                         '1 X 3 X 4 '
                                                         '+ 1 Z 3 + '
                                                         '1 X 4 X 5 '
                                                         '+ 1 Z 4 + '
                                                         '1 X 5 X 6 '
                                                         '+ 1 Z 5 + '
                                                         '1 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -6.206238059807513],
                              [12, 6, 0, 0, -5.187681916745086],
                              [20, 10, 0, 0, -4.3460110500437334],
                              [28, 14, 0, 0, -3.6444352165959075]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

'noise_prob': [

0.001,
0.001,
0.0,
0.0],

'status': True,

```

```
'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

'dephasing'},

'odd_wires': 4}},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {

'noise_prob': [

'status': True,
'type':

'odd_wires': 4}}],
    'run_time_sec': 12.855448246002197}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
      'init_param': { 'value': [ 0.17096748946205537,
                                  0.24182068788319971,
                                  0.5294082221811705,
                                  0.6436439076685059,
                                  0.6752523962811601,
                                  1.4661557963348293,
                                  1.466155796334829,
                                  1.5116530105229151,
                                  3.783489526832599,
                                  3.7842138074378933,
                                  3.8804047790072995,
                                  3.8804047790072995,
                                  4.154125602724648,
                                  4.692532374514593,
                                  4.692532374514593,
                                  5.064132124282894,
                                  5.064132124282894,
                                  5.064132124282894,
                                  6.382147767045899,
                                  6.751768360743096,
                                  6.751768360743096,
                                  6.751768360743096,
                                  7.743234563929715,
```

8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,

```

-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],

```

False,

```
'r': 1},
    'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
                          'status': False,
                          'type': 'png'},
                'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#2'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                         [1, 1, 0, 0],
                                         [2, 2, 0, 0],
                                         [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': None,
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                     'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                     '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                     'X 5 X 6 + 1 Z 5 + 1 Z 6',
                 'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                             '+ 1 Z 0 + '
                                                             '1 X 1 X 2 '
                                                             '+ 1 Z 1 + '
                                                             '1 X 2 X 3 '
                                                             '+ 1 Z 2 + '
                                                             '1 X 3 X 4 '
                                                             '+ 1 Z 3 + '
                                                             '1 X 4 X 5 '
                                                             '+ 1 Z 4 + '
                                                             '1 X 5 X 6 '
                                                             '+ 1 Z 5 + '
                                                             '1 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.139980799817412],
                                  [12, 6, 0, 0, -5.986699722057316],
                                  [20, 10, 0, 0, -5.03269185703209],
                                  [28, 14, 0, 0, -4.237780524148987]],
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                         'identity_factors': [0, 0, 0, 0],
                         'noise_level': [4, 2, 0, 0],
                         'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0,
0.0],
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
```



```

'noise_on_init_param': { 'status': False,
    'value': 0},

    'noise_prob': [

0.001,
0.001,
0.0,
0.0],

    'status': True,
    'type':

'dephasing'},

    'odd_wires': 4},
    { 'gates_num': [28, 7, 1, 0],
      'identity_factors': [3, 3, 0, 0],
      'noise_level': [28, 14, 0, 0],
      'noise_profile': {

'noise_on_init_param': { 'status': False,
    'value': 0},

    'noise_prob': [

0.001,
0.001,
0.0,
0.0],

    'status': True,
    'type':

'dephasing'},

    'odd_wires': 4}},
    { 'run_time_sec': 14.22638750076294}},
    { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'ising'}}},
    'init_param': { 'value': [ 7.923394728339895e-17,
    0.44053533716557003,
    0.44053533716557,
    0.8633875163783187,
    0.8807523602927853,
    0.8807523602927853,
    1.0224141131682343,
    1.599513745187798,
    2.8464451213500714,
    2.922472045892935,
    2.922472045892935,
    2.9239478183172296,
    4.400338604169817,
    4.721305166667179,
    4.721305166667179,
    4.721305166667178,
    5.011876780494166,
    5.011876780494166,
    5.011876780494166,
    5.879155312872552,
    5.879155312872551,
    6.750159190843359,
    6.75015919084336,
    8.012324492482197,
    8.012324492482197,
    8.7704267977754,
    9.114848260128039,
    9.114848260128039,
    9.87613180402372,

```

10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,

```

0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084]],
'noise_profile': { 'noise_on_init_param': { 'status':
'value': 0},
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':

```

False,

```

'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#5'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                          [1, 1, 0, 0],
                                          [2, 2, 0, 0],
                                          [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
             'optimization': { 'algorithm': 'SLSQP',
                               'constraint': True,
                               'status': False}},
    'zne': { 'data_points': None,
             'degree': 0,
             'method': 'richardson',
             'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                     'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                     '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                     'X 5 X 6 + 1 Z 5 + 1 Z 6',
               'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                           '+ 1 Z 0 + '
                                                           '1 X 1 X 2 '
                                                           '+ 1 Z 1 + '
                                                           '1 X 2 X 3 '
                                                           '+ 1 Z 2 + '
                                                           '1 X 3 X 4 '
                                                           '+ 1 Z 3 + '
                                                           '1 X 4 X 5 '
                                                           '+ 1 Z 4 + '
                                                           '1 X 5 X 6 '
                                                           '+ 1 Z 5 + '
                                                           '1 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -6.275631329374743],
                                  [12, 6, 0, 0, -5.287235063864106],
                                  [20, 10, 0, 0, -4.454788748195676],
                                  [28, 14, 0, 0, -3.749549420096481]],
               'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                    'identity_factors': [0, 0, 0, 0],
                                    'noise_level': [4, 2, 0, 0],
                                    'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'noise_prob': [
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'noise_prob': [
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},
0.001,
0.001,
0.0,
0.0],
'noise_prob': [

```

```

0.0,
0.0],

'status': True,
'type':

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4}},
'run_time_sec': 14.052101850509644}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,

```

0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,
1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,

```

-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': { 'status':
'noise_prob': [ 0.001,
0.001,
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'redundant',

```

False,

```

'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
                                'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
                                '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
                                'X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
                                                    '+ 1 Z 0 + '
                                                    '1 X 1 X 2 '
                                                    '+ 1 Z 1 + '
                                                    '1 X 2 X 3 '
                                                    '+ 1 Z 2 + '
                                                    '1 X 3 X 4 '
                                                    '+ 1 Z 3 + '
                                                    '1 X 4 X 5 '
                                                    '+ 1 Z 4 + '
                                                    '1 X 5 X 6 '
                                                    '+ 1 Z 5 + '
                                                    '1 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -6.37648641427504],
                             [12, 6, 0, 0, -5.250000083007176],
                             [20, 10, 0, 0, -4.3238347566667095],
                             [28, 14, 0, 0, -3.556658050073089]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':
'dephasing'},

                                'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':
'dephasing'},

                                'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':
'dephasing'},

                                'odd_wires': 4},

```



```

        { 'gates_num': [28, 7, 1, 0],
          'identity_factors': [3, 3, 0, 0],
          'noise_level': [28, 14, 0, 0],
          'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

0.001,
0.001,
0.0,
0.0],

                                'noise_prob': [

                                'status': True,
                                'type':

'depasing'},

                                'odd_wires': 4]],
                                'run_time_sec': 14.338054656982422}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'ising'}}},
'init_param': { 'value': [ 0.3675336714235261,
                           0.40575472279006464,
                           0.6952672462769884,
                           2.137221581425111,
                           2.137221581425111,
                           2.4653237039513405,
                           2.4653237039513414,
                           2.671087921572092,
                           2.6710879215720915,
                           3.889448133547601,
                           3.889448133547601,
                           5.229066948311018,
                           5.229066948311018,
                           5.736515631630257,
                           5.736515631630257,
                           5.736515631630257,
                           8.147692740111799,
                           8.179590109004724,
                           8.179590109004724,
                           8.179590109004724,
                           8.21310311779852,
                           8.21310311779852,
                           8.752241426452148,
                           8.752241426452148,
                           8.803045108086177,
                           9.928410928444814,
                           9.928410928444814,
                           9.928410928444814,
                           9.969030315653287,
                           14.058841247147779,
                           -0.593395889833616,
                           1.0797089435996057,
                           -0.18386516671058062,
                           0.9172540646403935,
                           0.5920567467507187,
                           0.5030949392578172,
                           -0.19817311203429638,
                           0.8732878271906367,
                           1.3773497870778637,
                           -0.2091442862578227,
                           0.9912881446359345,

```

2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,

```

1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804]],
'noise_profile': { 'noise_on_init_param': { 'status':
                                'value': 0},
'noise_prob': [ 0.001,
                0.001,
                0.0,
                0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,

```

False,

```

        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 '
        'Z 1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X '
        '4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 '
        'X 5 X 6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 '
        '+ 1 Z 0 + '
        '1 X 1 X 2 '
        '+ 1 Z 1 + '
        '1 X 2 X 3 '
        '+ 1 Z 2 + '
        '1 X 3 X 4 '
        '+ 1 Z 3 + '
        '1 X 4 X 5 '
        '+ 1 Z 4 + '
        '1 X 5 X 6 '
        '+ 1 Z 5 + '
        '1 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.274309228763829],
        [12, 6, 0, 0, -6.102630979708389],
        [20, 10, 0, 0, -5.139058519967983],
        [28, 14, 0, 0, -4.341721712357192]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': {

'noise_on_init_param': { 'status': False,
        'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
        'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                'identity_factors': [2, 2, 0, 0],
                                'noise_level': [20, 10, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
        'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0,
0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [28, 7, 1, 0],
                                'identity_factors': [3, 3, 0, 0],
                                'noise_level': [28, 14, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
        'value': 0},

```

```

0.001,
0.001,
0.0,
0.0],

'noise_prob': [

'status': True,
'type':

'dephasing'},

'odd_wires': 4}},
'run_time_sec': 13.112362623214722}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
1.0,
1.0,
1.0,
1.0,
1.0,
1.0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': True,
'type': 'png'},
'file_name_prefix': 'ising_noisefree_time_evo'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': True}},
'zne': { 'data_points': None,
'degree': 1,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + '
'1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + '
'1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + '
'1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '1 X 0 X 1 + 1 Z '
'0 + 1 X 1 X 2 + '
'1 Z 1 + 1 X 2 X '
'3 + 1 Z 2 + 1 X '
'3 X 4 + 1 Z 3 + '

```

```

'1 X 4 X 5 + 1 Z '
'4 + 1 X 5 X 6 + '
'1 Z 5 + 1 Z 6']],
'output': { 'exact_sol': -8.56677223350563,
  'initial_cost_history': [ 4.358359699677247,
    4.598584386614463,
    5.205211991476004,
    2.498239515374607,
    3.620239544509655,
    3.5202907273108752,
    3.874294382863561,
    2.6240487828355272,
    3.466109018012848,
    4.215104097495168],
  'noise_details': { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [4, 2, 0, 0],
    'noise_profile': { 'noise_on_init_param': {
      'noise_prob': [ 0.001,
        0.001,
        0.0,
        0.0],
      'status': True,
      'type': 'dephasing'},
    'odd_wires': 4},
  'optimized_minimum_cost': [ -7.274309228763741,
    -7.1399807998173275,
    -6.7841170852099,
    -6.206238059807503,
    -6.275631329374697,
    -5.70418119478636,
    -6.53688363509032,
    -6.029866062982587,
    -6.376486414275039,
    -6.823046671583788],
  'optimized_parameters': [ [ 0.3675336714235261,
    0.40575472279006464,
    0.6952672462769884,
    2.137221581425111,
    2.137221581425111,
    2.4653237039513405,
    2.4653237039513414,
    2.671087921572092,
    2.6710879215720915,
    3.889448133547601,
    3.889448133547601,
    5.229066948311018,
    5.229066948311018,
    5.736515631630257,
    5.736515631630257,
    5.736515631630257,
    8.147692740111799,
    8.179590109004724,
    8.179590109004724,
    8.179590109004724,
    8.21310311779852,
    8.21310311779852,
    8.752241426452148,
    8.752241426452148,
    8.803045108086177,
    9.928410928444814,
    9.928410928444814,
    9.928410928444814,
    9.969030315653287,
    14.058841247147779,
    -0.593395889833616,
    1.0797089435996057,
    -0.18386516671058062,
    0.9172540646403935,
    0.5920567467507187,
    0.5030949392578172,
    -0.19817311203429638,
    0.8732878271906367,

```

1.3773497870778637,
-0.2091442862578227,
0.9912881446359345,
2.1471684900095735,
1.8011951688988466,
0.08640841047975001,
0.12889510785919772,
0.05509015903577425,
0.056646924667161334,
-0.24683198530133132,
0.10458007099731792,
1.1863156573315168,
-2.4620703709527216,
0.7027036608466238,
1.7463688042058148,
-0.22769098390292125,
0.021713799462876483,
-0.0317703225281112,
0.35077834879841635,
0.06872812315186226,
-0.39611130359042956,
-0.7384605336871046,
0.876054711487725,
-0.15325999071538723,
0.12119402352536833,
0.20800082622423624,
-0.15476782660464156,
-0.5287189417532797,
-0.03286635866013837,
-0.049418991474155675,
0.9668813506694777,
-0.07420876651127355,
0.09295283805923779,
-0.08326516159803334,
-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,

0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,
-3.1432344127842007,
-0.46456701138365963,
-0.18850231543614804],
[0.17096748946205537,
0.24182068788319971,
0.5294082221811705,
0.6436439076685059,
0.6752523962811601,
1.4661557963348293,
1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,

0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,

1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993],
[4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,

-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,

1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,
1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138],
[0.5483166258326411,
0.5788252343888002,
0.5803330366161238,
1.2169503585394688,
1.4292236971359287,
1.4293923120692815,
1.4365169928726365,
1.4365169928726365,
2.510196103669615,
2.510196103669615,
2.5544154452534937,
2.5544154452534937,
4.029649065019888,
4.067883071594656,
4.076808377533052,
4.1651466476623735,
4.656187563066811,
4.656187563066811,
5.18228186954163,
5.18228186954163,
6.808184896226372,
6.808184896226372,
6.960163700827979,
6.960163700827979,
6.960163700827979,
8.49759837936073,
8.49759837936073,
8.526772837307739,
9.456275223201612,
9.799125697936137,
-1.1324033485399458,
0.3066519452505097,
0.43856397893505916,
2.603476267806093,
0.0006914105351412576,
1.1085344664632693,
0.4735183530419211,
1.0969942437601548,

1.7819039437129416,
1.2220922008004673,
-0.516581792367389,
-1.8223032952581883,
-0.1454480817152758,
-1.2131062816976166,
-0.02593999163205417,
0.5898254930964741,
0.7377016495064371,
-0.18729526225026624,
-0.24589424731168963,
0.8550874647467951,
-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,

1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,
1.492284082755186,
0.5913401617935748,
-1.0815960254278196,
0.39120930996344166,
-0.47318263710851194,
0.3405336256152792,
-0.004335634039844681,
0.9981842015674961,
0.03844807043591727,
1.2262083751618813,
-0.8292110587477036,
0.0538674203763304,
1.8573594984180253,
0.5926260677957834,
2.195238147282677,
-1.4749752374650615,
2.6643724981663564,
1.1623882981123574,
0.3480159661399594,
2.1792323100927695,
1.2551578493616056,
-0.35423101063456375,
1.9204695063595076,
-1.1085689776690866,
-1.1607555456918301],
[7.923394728339895e-17,
0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
5.879155312872552,
5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,

-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,
0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,

-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084],
[0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,
8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.0892586669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,

-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,
-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,

-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674],
[0.47544493830108536,
0.47591537288460184,
0.4759153728846019,
0.5985896638985994,
1.3939903966429974,
1.3939903966429974,
3.2587118733130036,
3.258711873313004,
4.312142629456884,
4.312142629456884,
4.857905543812315,
4.971569214842323,
5.142019393710199,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,

0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,
-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,

-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603],
[0.18025020416614318,
0.18025020416614315,
0.4610778566518408,
0.6219220463806964,
0.7278326366057072,
0.8201944581570828,
1.2788460286146952,
1.4054470533314678,
1.4054470533314678,
2.282706833033757,
2.2962450119862856,
2.3028314628270254,
4.069670148635338,
4.069670148635338,
4.605143078506114,
4.605143078506114,
4.605143078506114,
4.712341711691788,
4.712341711691788,
4.71827378027349,
4.71827378027349,
5.585132855182541,
5.585132855182542,
6.936688988043727,
6.996547924306228,
7.327044121023376,
7.847751295142583,
7.8562202008953275,
8.679046525727268,
9.452714795876016,
0.01979341787597725,
-2.1649513485863423,
-0.3469207112923858,
-0.19054577342605242,
0.08943279174694625,
-0.40023385514982557,
-0.3378513267335698,
-0.32184385507584645,

0.17281592706447121,
0.5432446490193347,
-0.2468193649067236,
-1.491986060579654,
-0.6929963404149894,
0.030224915087063667,
-0.08162385956356048,
-1.6881840196674878,
-0.5438179745034216,
0.2710568887289234,
-0.3546480276303292,
0.4606888067368054,
0.19895915418638402,
-1.5118185786211353,
-0.022782253138705504,
-0.3090460309752015,
-0.3518945333066163,
0.9690796168990217,
-0.010538359546097177,
-0.5968380191117454,
-1.3356264059285607,
-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,

0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,
-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171],
[0.22237247586897502,
0.3434883293456931,
0.600978934518952,
0.6193810427253835,
1.6475035333031534,
2.204130997587115,
2.4096412381325085,
2.515509113107616,
2.5155322756928205,
2.9680516150838403,
2.9680516150838403,
3.0576851370343725,
3.8016986766973213,
4.264365678603219,
4.264365678603219,
5.4290969626181775,
5.4290969626181775,
5.453519119574046,
6.855500326085536,
6.858194453958984,
6.956907552727685,
7.899693187867781,
8.578442687794452,
8.60387860142029,
9.193247925689713,
9.205803142575261,
9.205803142575261,
9.815391824330385,
9.815391824330385,
10.97297451152796,
-0.4325514790797943,
-1.0860755794816048,
-0.34917301406803075,
0.9554400980633908,
-0.3272724546684993,
0.13698615504934158,
-0.3120459104790063,
-0.5324325213761384,

1.6476833500899999,
2.184163963641692,
1.0643894246063128,
-2.7436302898700338,
-0.9203607373406255,
0.2960792725949677,
1.646667339143857,
-0.34154797106217816,
0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,

-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,
-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893],
[0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,
0.5345472321438205,

0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.2018222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,
-0.10883760977815361,

```
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'run_time_sec': 46646.7807404995}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}}},
'init_param': { 'value': [ 7.923394728339895e-17,
0.44053533716557003,
0.44053533716557,
0.8633875163783187,
0.8807523602927853,
0.8807523602927853,
1.0224141131682343,
1.599513745187798,
2.8464451213500714,
2.922472045892935,
2.922472045892935,
2.9239478183172296,
4.400338604169817,
4.721305166667179,
4.721305166667179,
4.721305166667178,
5.011876780494166,
5.011876780494166,
5.011876780494166,
```

5.879155312872552,
5.879155312872551,
6.750159190843359,
6.75015919084336,
8.012324492482197,
8.012324492482197,
8.7704267977754,
9.114848260128039,
9.114848260128039,
9.87613180402372,
10.23973172788205,
0.09774272849121603,
0.0018601482237552494,
-0.09414894191588159,
0.03531507271404531,
0.01801632266215566,
-0.031446214752801294,
0.04762079993218455,
0.16430279282954635,
-1.4212045547488918,
-0.051616216342739524,
-1.2094395115841,
1.961457452158631,
-0.09882210453707145,
1.4772087948778743,
-1.07174135216323,
-0.1881882042247284,
0.9429310525529949,
0.06018460007044777,
-2.137106479060228,
-0.9614924638003328,
0.6761415275532413,
-0.2012038247290061,
0.45087642211311835,
-0.023367740958275957,
0.8166584231413961,
0.20437280688240314,
-0.2844425853304819,
1.1888019453741512,
1.5075763398120259,
-0.2932965681072964,
-1.1651486090429806,
0.8516736950903361,
0.5562774324383903,
0.08880625225982887,
-0.20510729519447557,
0.07958047309161287,
1.150148720793205,
0.752058017629903,
1.4826233092486047,
-0.8085401459238525,
-0.21105856720357505,
0.021601053395518256,
0.005125366751275638,
-1.1933032602708362,
-1.5268625421370876,
-0.8973979375162666,
-0.43714085354227866,
-0.6923079565655138,
-0.7819644917507689,
0.2464309258627461,
0.9752292851604389,
1.705366721310855,
2.670086120293194,
-0.10159593932583255,
-0.6106112146039476,
-1.1296882955679428,
1.0064992616215405,
0.40385713042582505,
-2.0943184858836723,
1.7647548623477693,
0.3967863740889611,
-0.5786939645933067,
0.35645911000135366,
0.09854741962650256,

```

0.029750480285362434,
2.7402600540547537,
0.7814340906834192,
-0.11244124912070508,
-0.12898049179814935,
0.21936143575795827,
1.8005854889670254,
1.0042932350117184,
1.1984508988919396,
0.5935372296951723,
0.13212966566398052,
0.6023186740029555,
-0.21274807564009723,
-0.1033977056759172,
-0.9467553063617766,
-0.10557581411347433,
-0.0698721995740029,
1.2622344420372125,
-0.06633441601507087,
-0.03888422959879465,
0.31835507851505285,
-0.031743979466302155,
0.8149831955043099,
0.06247398644885955,
-0.03215998926308989,
-0.5395956587277304,
0.04913536676265748,
1.0701144892582446,
-0.34252580263801796,
0.014005130614426067,
-1.377473860355396,
-0.11545427586689089,
-0.13711392533889225,
0.04860684499271863,
0.19873936309866688,
-0.4369649987147877,
-2.0430772807931668,
0.04994916401386025,
0.807678373786241,
0.12290360689510754,
-0.11366092566160511,
0.3816234685943951,
-0.8533208797465992,
-0.6618437681078548,
-0.37099628864670037,
-1.4955777077514916,
0.25868529123713524,
-0.2229684260155789,
-1.6341415907725487,
-1.0518371238906572,
0.533023275284551,
1.231775507694775,
0.9994647567984135,
1.5668333358779631,
-1.809645687899931,
-1.1171127998349084]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],

```

```

        'r': 1},
        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
                              'status': False,
                              'type': 'png'},
                    'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#5'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0],
                                              [3, 3, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -6.275631329374743],
                                   [12, 6, 0, 0, -5.287235063864106],
                                   [20, 10, 0, 0, -4.454788748195676],
                                   [28, 14, 0, 0, -3.749549420096481]],
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
                     'run_time_sec': 5.364418029785156e-05,
                     'zne_values': { 'degree': 0,
                                      'extrapolated_value': -6.837292869031195,
                                      'others': { 'beta_coeffieicients': [ 2.1875,
                                                                           -2.1875,
                                                                           1.3125,
                                                                           -0.3125],
                                                  'cost_richardson_zne': 11.390625,
                                                  'sorted_expectation_vals': [
-6.275631329374743,
-5.287235063864106,
-4.454788748195676,
-3.749549420096481],

                                     'sorted_noise': [6, 18, 30, 42]}},
                     'sampled data': [ [ 4,
                                           2,
                                           0,
                                           0,
                                           -6.275631329374743],
                                         [ 12,
                                           6,
                                           0,
                                           0,
                                           -5.287235063864106],
                                         [ 20,
                                           10,
                                           0,
                                           0,
                                           -4.454788748195676],
                                         [ 28,
                                           14,
                                           0,
                                           0,
                                           -3.749549420096481]],
                     'sampling': 'default'}}},
        { 'config': { 'ansatz': { 'gateset': 1,
                                   'layer': 30,
                                   'ugate': { 'coefficients': { 'bn': [ 0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    }
                                   }
        }

```

```
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'ising'}},
'init_param': { 'value': [ 0.3675336714235261,
                             0.40575472279006464,
                             0.6952672462769884,
                             2.137221581425111,
                             2.137221581425111,
                             2.4653237039513405,
                             2.4653237039513414,
                             2.671087921572092,
                             2.6710879215720915,
                             3.889448133547601,
                             3.889448133547601,
                             5.229066948311018,
                             5.229066948311018,
                             5.736515631630257,
                             5.736515631630257,
                             5.736515631630257,
                             8.147692740111799,
                             8.179590109004724,
                             8.179590109004724,
                             8.179590109004724,
                             8.21310311779852,
                             8.21310311779852,
                             8.752241426452148,
                             8.752241426452148,
                             8.803045108086177,
                             9.928410928444814,
                             9.928410928444814,
                             9.928410928444814,
                             9.969030315653287,
                             14.058841247147779,
                             -0.593395889833616,
                             1.0797089435996057,
                             -0.18386516671058062,
                             0.9172540646403935,
                             0.5920567467507187,
                             0.5030949392578172,
                             -0.19817311203429638,
                             0.8732878271906367,
                             1.3773497870778637,
                             -0.2091442862578227,
                             0.9912881446359345,
                             2.1471684900095735,
                             1.8011951688988466,
                             0.08640841047975001,
                             0.12889510785919772,
                             0.05509015903577425,
                             0.056646924667161334,
                             -0.24683198530133132,
                             0.10458007099731792,
                             1.1863156573315168,
                             -2.4620703709527216,
                             0.7027036608466238,
                             1.7463688042058148,
                             -0.22769098390292125,
                             0.021713799462876483,
                             -0.0317703225281112,
                             0.35077834879841635,
                             0.06872812315186226,
                             -0.39611130359042956,
                             -0.7384605336871046,
                             0.876054711487725,
                             -0.15325999071538723,
                             0.12119402352536833,
                             0.20800082622423624,
                             -0.15476782660464156,
                             -0.5287189417532797,
                             -0.03286635866013837,
                             -0.049418991474155675,
                             0.9668813506694777,
                             -0.07420876651127355,
                             0.09295283805923779,
                             -0.08326516159803334,
```

-0.0427482361921042,
1.342369647984463,
0.34072690643450376,
0.5933444311387184,
0.31417941006555294,
1.6304762150342247,
1.891349621534731,
0.899218544502096,
0.48577075731530633,
-2.2306225778710727,
0.8866764919462848,
0.2762222157368045,
-1.0925021661494219,
-0.1494913660340356,
0.8289891017783336,
-2.0678071859966467,
0.717418966449541,
1.0294808193164726,
0.13580094343779148,
1.2154221461203674,
-0.072393033465795,
0.5936030701283823,
0.6414219801572755,
-0.5026874400746322,
0.5048570237991846,
0.4837080454239775,
-0.41713967597961793,
0.16774379184496263,
0.12299750952594474,
-0.453572084558526,
0.8026213332271597,
-1.3045370621813255,
-0.7515694999188218,
0.22628589276852773,
1.8030093044263258,
-0.09378305233784993,
-0.5023529855902581,
1.391117839405254,
0.5371312888835978,
-0.04231396987827893,
0.003390151575822153,
0.4591655267057695,
0.5544793368338241,
-0.18222740098302695,
1.0345797235555083,
-0.9427667898653518,
0.7051815648216593,
-1.691304548797814,
-0.4922082934466421,
-0.5924038004988397,
-1.246401400099831,
-0.08030391458868152,
-1.3647885430537856,
-1.6331668457314883,
-0.9002918947913646,
0.2613439182073027,
0.956367170861211,
-0.7033177975360739,
-0.06022273848082935,
1.5178321229807306,
0.4402028677909301,
0.4872881773773794,
1.5034707343822726,
-0.8692844176300923,
0.024725220715673252,
1.4212144932422452,
-1.5211626552940618,
-0.13019129994104797,
1.6367648887242354,
-0.8381394798024493,
2.1088433598418392,
-1.654966911713182,
1.577409737852239,
2.6372197738973933,
-0.09882697532942787,

```
-3.1432344127842007,  
-0.46456701138365963,  
-0.18850231543614804]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
      'value': 0},  
    'noise_prob': [0.001, 0.001, 0.0, 0.0],  
    'status': True,  
    'type': 'dephasing'},  
'nqubits': 7,  
'observable': { 'coefficients': { 'bn': [ 1.0,  
      1.0,  
      1.0,  
      1.0,  
      1.0,  
      1.0],  
    'cn': [ 0.5,  
      0.5,  
      0.5,  
      0.5,  
      0.5,  
      0.5],  
    'r': 1},  
    'def': 'ising'},  
'output': { 'draw': { 'fig_dpi': 100,  
    'status': False,  
    'type': 'png'},  
    'file_name_prefix':  
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#1'},  
'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
    [1, 1, 0, 0],  
    [2, 2, 0, 0],  
    [3, 3, 0, 0]]},  
'run': 'zne',  
'state': 'dmatrix',  
'vqe': { 'iteration': 1,  
    'optimization': { 'algorithm': 'SLSQP',  
    'constraint': True,  
    'status': False}},  
'zne': { 'data_points': [ [4, 2, 0, 0, -7.274309228763829],  
    [12, 6, 0, 0, -6.102630979708389],  
    [20, 10, 0, 0, -5.139058519967983],  
    [28, 14, 0, 0, -4.341721712357192]],  
    'degree': 0,  
    'method': 'richardson',  
    'sampling': 'default'}},  
'output': { 'exact_sol': -8.566772233505624,  
    'run_time_sec': 4.7206878662109375e-05,  
    'zne_values': { 'degree': 0,  
    'extrapolated_value': -7.9512724421551315,  
    'others': { 'beta_coeffieicients': [ 2.1875,  
      -2.1875,  
      1.3125,  
      -0.3125],  
    'cost_richardson_zne': 11.390625,  
    'sorted_expectation_vals': [  
  
-7.274309228763829,  
-6.102630979708389,  
-5.139058519967983,  
-4.341721712357192],  
  
    'sorted_noise': [6, 18, 30, 42]},  
    'sampled data': [ [ 4,  
      2,  
      0,  
      0,  
      -7.274309228763829],  
    [ 12,  
      6,  
      0,  
      0,  
      -6.102630979708389],  
    [ 20,  
      10,  
      0,  
      0,
```



```

-5.139058519967983],
[ 28,
 14,
 0,
 0,
-4.341721712357192]],
'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}},
  'init_param': { 'value': [ 0.5483166258326411,
                              0.5788252343888002,
                              0.5803330366161238,
                              1.2169503585394688,
                              1.4292236971359287,
                              1.4293923120692815,
                              1.4365169928726365,
                              1.4365169928726365,
                              2.510196103669615,
                              2.510196103669615,
                              2.5544154452534937,
                              2.5544154452534937,
                              4.029649065019888,
                              4.067883071594656,
                              4.076808377533052,
                              4.1651466476623735,
                              4.656187563066811,
                              4.656187563066811,
                              5.18228186954163,
                              5.18228186954163,
                              6.808184896226372,
                              6.808184896226372,
                              6.960163700827979,
                              6.960163700827979,
                              6.960163700827979,
                              8.49759837936073,
                              8.49759837936073,
                              8.526772837307739,
                              9.456275223201612,
                              9.799125697936137,
                              -1.1324033485399458,
                              0.3066519452505097,
                              0.43856397893505916,
                              2.603476267806093,
                              0.0006914105351412576,
                              1.1085344664632693,
                              0.4735183530419211,
                              1.0969942437601548,
                              1.7819039437129416,
                              1.2220922008004673,
                              -0.516581792367389,
                              -1.8223032952581883,
                              -0.1454480817152758,
                              -1.2131062816976166,
                              -0.02593999163205417,
                              0.5898254930964741,
                              0.7377016495064371,
                              -0.18729526225026624,
                              -0.24589424731168963,
                              0.8550874647467951,

```

-0.3224945411184162,
-0.36832376208491285,
-0.881745818749188,
-0.4517353913610538,
-2.6507955691692096,
1.4089040411711087,
-1.98647805300985,
0.09207178440352033,
0.1176545491584157,
-0.4036806074380365,
0.28240629583972154,
-0.5198549065378171,
1.966487693291819,
0.12100547054089811,
-0.9655075075590218,
0.060623052899729045,
-0.04466432031915775,
-0.012043574647936208,
-0.07368488370701011,
-0.3912984657978762,
0.3138735215497361,
-0.029528339465838396,
-0.5194819171696577,
-0.01406438454151942,
0.06673368769020717,
0.0013821621856672335,
0.05802396168464121,
-0.2296532226147227,
-0.7250509399845758,
1.1403585484853793,
-0.7425421891989091,
0.4698659760148211,
-0.46030905741208916,
-0.837579768242913,
0.34096381957080085,
-1.4754011527557147,
0.06613174433317287,
0.11886218195100834,
-0.13737271588733974,
0.8709143480388293,
0.5849694357909463,
-0.007686748415400163,
0.7194710241447149,
-1.798020970709118,
1.4074770089484554,
1.2469256446499142,
1.8355324378456597,
0.5054387836842794,
-0.0700868560931335,
-0.5230414063813488,
-0.014861433288994705,
-1.4400446529278175,
0.8775230818925047,
0.22796149766495682,
0.9448222091815641,
0.23129451823823294,
0.052565790394436165,
-0.08048294755479887,
0.012280358203632009,
-0.6913975392794202,
-1.929287003081602,
-0.5241849295015967,
-1.1103968265168844,
1.066674647946539,
-0.3094206762856894,
0.24061064720772918,
0.5933116395873674,
-0.7615900433566246,
1.5438732810865121,
-0.8507280711774536,
0.04498895715050362,
1.0586212232587777,
0.718069845727537,
2.1595911429392896,
-1.2227632347997481,

[illegible]

-6.206238059807513,
-5.187681916745086,
-4.3460110500437334,
-3.6444352165959075],

-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [[4,
2,
0,
0,
-6.206238059807513],
[12,
6,
0,
0,
-5.187681916745086],
[20,
10,
0,
0,
-4.3460110500437334],
[28,
14,
0,
0,
-3.6444352165959075]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0,
0,
0,
0,
0,
0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [0.0921501071458524,
1.5113409130757622,
1.5171500885968463,
1.9257567103877564,
2.103027702732541,
2.106266184153511,
2.3067654466695937,
2.7972204473176414,
2.7972204473176414,
2.9760063319143315,
2.9760063319143315,
2.976006331914331,
4.725403617162805,
4.725403617162805,
4.725403617162805,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.3458155973379915,
5.511978345293384,
5.511978345293384,
7.852859529848597,
7.8664501138991945,
8.306657385522128,
8.306657385522128,
8.306657385522128,
8.77694772653372,
8.77694772653372,

8.77694772653372,
10.210889411786125,
-0.025196159437353705,
0.17256782673282392,
0.08925866669000004,
-0.2640737309697103,
0.08393384639489809,
0.818808421272339,
0.21848095983081994,
0.18119625098082034,
-0.011865447267727605,
1.6758619362933431,
0.20111234151604201,
0.17176612144416795,
-0.1795592352150498,
-1.2018318923099678,
0.08170248187102172,
1.409214819126438,
0.5648299128933032,
0.054353065507126555,
0.08392111936409213,
1.16950265151543,
0.07477897396775408,
-0.008829449454164306,
0.10880476724845219,
-0.5783743284523922,
-0.11357835811377229,
0.6354312612952143,
0.021708821949977686,
0.5385210917583648,
0.5834796179427235,
-1.1874307484947892,
0.7150667707424543,
0.9396080412103254,
1.1135753498523808,
-1.5522802709475478,
-0.11308565226407502,
-0.902328599922437,
0.6105061273776329,
-1.3024815918849448,
0.6409780710433735,
0.7784151779199293,
-2.056218886007638,
-0.06787511914459264,
1.4619100440682296,
-1.3881564722971613,
-0.09934457547262787,
0.4291845959725414,
0.013413399120346666,
0.7707917768117696,
-0.49515018167266756,
-0.05680542820434917,
0.09888186726019127,
-0.13266874357516228,
0.5579796398209254,
-0.36536355362662937,
1.550247183456112,
0.4215244697474259,
1.483841509854635,
1.497286463884731,
-0.45255762081826245,
1.6344877637268267,
-1.2325840660540837,
-0.7228757279646764,
1.6755710458344988,
-0.9658932498706531,
0.3687148201845115,
-1.0361645288732706,
0.6714735411027907,
-1.8607345659746353,
-0.32849943810405213,
0.4104159551336501,
0.043359551765467436,
-0.35832842782405305,
0.07584315690008855,

```

-0.40564547005989215,
-0.022573498410409683,
1.3899131661322317,
-0.2963541740239415,
0.6150504501864379,
0.53617597781661,
-2.041141744280377,
-0.05637704635582033,
-0.8231020660745568,
-0.08041961726151937,
-1.1911113266202114,
-0.05193542833783212,
0.901593969250458,
0.15361083432968986,
-1.0377872297998838,
-0.13207592600043197,
-0.23526927736020792,
0.55396369886821,
0.16839817958859243,
0.5421272687474321,
-0.06420574173480442,
-0.24879762058904856,
-0.4683402667266821,
-0.15969454258349336,
1.546951248667097,
-0.139836175289928,
-1.1248147648654399,
-0.13845807313151898,
0.6340168818617303,
0.2697324606427981,
1.2528349670264023,
-0.6436895920276484,
-1.0149182288490899,
0.4272419055563505,
-1.0099536441419146,
-1.073322301534239,
-0.8044844231302818,
1.2626718925874854,
1.6101762356912184,
0.05905275705907849,
1.4976490655687835,
-0.07924121082687685,
-0.08423147815827382,
-1.223171666189706,
-0.31170725162430535,
1.6458525776311825,
1.3191825149633674]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],

```

```

[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -5.704181194786395],
                          [12, 6, 0, 0, -4.800941579254224],
                          [20, 10, 0, 0, -4.034504937366069],
                          [28, 14, 0, 0, -3.3806822209161007]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.220008850097656e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -6.214661195233308,
                          'others': { 'beta_coefficients': [ 2.1875,
                                                            -2.1875,
                                                            1.3125,
                                                            -0.3125],
                          'cost_richardson_zne': 11.390625,
                          'sorted_expectation_vals': [
-5.704181194786395,
-4.800941579254224,
-4.034504937366069,
-3.3806822209161007],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -5.704181194786395],
                  [ 12,
                    6,
                    0,
                    0,
                    -4.800941579254224],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.034504937366069],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.3806822209161007]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                        'cn': [ 0.5,
                              0.5,
                              0.5,
                              0.5,
                              0.5],
                        'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'ising'}},
'init_param': { 'value': [ 0.17096748946205537,
                          0.24182068788319971,
                          0.5294082221811705,
                          0.6436439076685059,
                          0.6752523962811601,
                          1.4661557963348293,

```

1.466155796334829,
1.5116530105229151,
3.783489526832599,
3.7842138074378933,
3.8804047790072995,
3.8804047790072995,
4.154125602724648,
4.692532374514593,
4.692532374514593,
5.064132124282894,
5.064132124282894,
5.064132124282894,
6.382147767045899,
6.751768360743096,
6.751768360743096,
6.751768360743096,
7.743234563929715,
8.1520396126177,
8.1520396126177,
8.546274908334533,
8.589779496078958,
8.589779496078958,
9.859248780822462,
13.3058320214528,
0.25900326422191966,
0.7885699864164276,
0.41320213613295365,
-1.1189869522952365,
0.26027865496517016,
0.9396191194055923,
-0.036590936599100087,
0.4526959896459153,
0.23545371285521352,
-0.08089020453927935,
-0.04037254840266629,
1.8055011929897393,
1.0437583322662911,
0.8301172091681238,
1.4721458579301823,
1.0565790032678872,
-0.0620204872484031,
0.7778737999221846,
-0.07568060381342757,
-0.4832550478503789,
1.561646348208689,
1.0276654568346077,
0.6741324750771829,
0.5058216800397286,
-1.0558943326715462,
-0.14283396260407313,
-0.12020755874532785,
0.37547144626073276,
-0.3549607031100976,
-0.10410956462681178,
1.5153634922337307,
1.9495981063680297,
0.31118583706033537,
-0.7235645457595358,
0.17173795551894178,
-0.33392096172962504,
0.2993290942776598,
-0.07002988913014692,
0.10352604443544179,
0.5331508884476422,
-0.49057208375166744,
2.0651610484381075,
-1.109792891712479,
0.06126285218750582,
-0.509814145327961,
-1.6155787633490326,
0.3541018130871613,
0.0659140468444337,
0.03819797934790045,
0.06882428190974979,
0.35856824659128844,


```
0.0650751753706436,
0.6837216421306276,
-0.884828447980924,
0.9361885296400346,
0.3256709539855342,
0.5012252138218954,
3.103575831496634,
-0.9690138803920368,
0.45458362305325123,
-0.5797429988309931,
0.316601123478116,
-1.065017325075906,
-0.9047966767153034,
0.3932903597036206,
-0.9114113366629168,
-0.17114920824853386,
1.4258993796813948,
-0.07964065351532917,
0.1994846715071765,
0.12923396176016305,
-0.5223531844359014,
0.841012534727648,
-0.3282172658303134,
0.0933235612390595,
-1.02439069873068,
0.2893856117659334,
0.11058803685263456,
-0.9777267211094056,
-1.1176856724072852,
0.1085498190459602,
-0.9880748918428673,
1.11986650737539,
1.1314622575020645,
-0.21476359665028094,
1.5188657965163455,
0.4484622062828435,
0.8945514888585444,
0.2870188784367138,
-0.015550768886605199,
-0.896475749571893,
-1.014435869983,
-1.9768898365892673,
-0.3672926690545215,
0.1914933122291532,
-0.5244934480882791,
-0.3536397670021093,
1.3749243130690434,
0.24384412635951458,
0.6792216927248427,
-1.2133197112413483,
0.6880710371414687,
0.2810854885236179,
0.6678202051241985,
-0.22746971306552316,
-1.0728265220925242,
0.9226641349031418,
0.6521511693216135,
-0.06400174728010423,
1.3813950780275313,
0.12045063559874317,
-1.330667821680903,
1.1126517740274287,
1.8183845077029084,
-0.3680979251526694,
1.5797361276787631,
0.02967118776857895,
3.3215372828432215,
0.2632835778696766,
0.25781524624649993]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
```

```

        'observable': { 'coefficients': { 'bn': [ 1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0],
                                            'cn': [ 0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5],
                                            'r': 1}},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
                          'status': False,
                          'type': 'png'},
                'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#2'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                          [1, 1, 0, 0],
                                          [2, 2, 0, 0],
                                          [3, 3, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -7.139980799817412],
                              [12, 6, 0, 0, -5.986699722057316],
                              [20, 10, 0, 0, -5.03269185703209],
                              [28, 14, 0, 0, -4.237780524148987]],
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
                'run_time_sec': 5.221366882324219e-05,
                'zne_values': { 'degree': 0,
                                'extrapolated_value': -7.8039040061582705,
                                'others': { 'beta_coefficients': [ 2.1875,
                                                                    -2.1875,
                                                                    1.3125,
                                                                    -0.3125],
                                'cost_richardson_zne': 11.390625,
                                'sorted_expectation_vals': [
-7.139980799817412,
-5.986699722057316,
-5.03269185703209,
-4.237780524148987],

                                'sorted_noise': [6, 18, 30, 42]},
                                'sampled data': [ [ 4,
                                                       2,
                                                       0,
                                                       0,
                                                       -7.139980799817412],
                                                    [ 12,
                                                       6,
                                                       0,
                                                       0,
                                                       -5.986699722057316],
                                                    [ 20,
                                                       10,
                                                       0,
                                                       0,
                                                       -5.03269185703209],
                                                    [ 28,
                                                       14,
                                                       0,
                                                       0,
                                                       -4.237780524148987]],
                                'sampling': 'default'}}}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,

```

```
'ugate': { 'coefficients': { 'bn': [ 0,
                                     0,
                                     0,
                                     0,
                                     0,
                                     0,
                                     0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 0},
          'time': {'max': 10.0, 'min': 0.0},
          'type': 'ising'}},
'init_param': { 'value': [ 0.18025020416614318,
                           0.18025020416614315,
                           0.4610778566518408,
                           0.6219220463806964,
                           0.7278326366057072,
                           0.8201944581570828,
                           1.2788460286146952,
                           1.4054470533314678,
                           1.4054470533314678,
                           2.282706833033757,
                           2.2962450119862856,
                           2.3028314628270254,
                           4.069670148635338,
                           4.069670148635338,
                           4.605143078506114,
                           4.605143078506114,
                           4.605143078506114,
                           4.712341711691788,
                           4.712341711691788,
                           4.71827378027349,
                           4.71827378027349,
                           5.585132855182541,
                           5.585132855182542,
                           6.936688988043727,
                           6.996547924306228,
                           7.327044121023376,
                           7.847751295142583,
                           7.8562202008953275,
                           8.679046525727268,
                           9.452714795876016,
                           0.01979341787597725,
                           -2.1649513485863423,
                           -0.3469207112923858,
                           -0.19054577342605242,
                           0.08943279174694625,
                           -0.40023385514982557,
                           -0.3378513267335698,
                           -0.32184385507584645,
                           0.17281592706447121,
                           0.5432446490193347,
                           -0.2468193649067236,
                           -1.491986060579654,
                           -0.6929963404149894,
                           0.030224915087063667,
                           -0.08162385956356048,
                           -1.6881840196674878,
                           -0.5438179745034216,
                           0.2710568887289234,
                           -0.3546480276303292,
                           0.4606888067368054,
                           0.19895915418638402,
                           -1.5118185786211353,
                           -0.022782253138705504,
                           -0.3090460309752015,
                           -0.3518945333066163,
                           0.9690796168990217,
                           -0.010538359546097177,
                           -0.5968380191117454,
                           -1.3356264059285607,
```

-0.912996133699909,
-2.02529425814526,
-0.3406423605421284,
0.3136438177132789,
-0.42139037173019495,
0.23198446622909977,
0.13901623117318299,
0.7004860770740983,
0.7653757135038268,
1.4117144880057488,
1.1918513262099462,
0.39926108972604557,
-1.3581477835087472,
2.1078396880727714,
0.9239545795858499,
0.06743094136295699,
-2.0545138482102785,
0.0549672030604437,
0.7029276558677147,
0.37787637288188336,
-0.08668853138413642,
-1.1015951353466904,
0.06267964588455946,
-0.03642619393844789,
0.34216202779535876,
0.07342907652461796,
-1.2254969917464302,
-1.210778593655016,
-0.6369271911659602,
0.12209934303399678,
-1.431543513983085,
0.25504854629883444,
0.016147036189938364,
0.14060301402789038,
1.4194518936414688,
-0.24219812172079702,
1.1798186446582548,
-0.11061811278437873,
-0.10696841273552876,
0.317456635319965,
-1.0710732932698277,
-0.6160878210144879,
0.25517233968164993,
0.22085959752648107,
-0.4329813535491156,
-0.013993117466929031,
1.3430648546414072,
-1.658483972566601,
0.7135818554895808,
-0.5029938030490397,
-0.5201602573093934,
-0.06316551993691735,
0.6230390756472122,
-0.2747250957433143,
0.2704418935383965,
-0.028423887623808652,
-0.21971614582631055,
-0.1849598653755513,
-0.21638988776145818,
-0.17584760033744762,
-0.20098638279684936,
-0.009830605169654295,
-0.3731393105434215,
0.016652737908834807,
1.5293419437677958,
-0.5586654763342961,
0.3143750300974426,
-0.8135231056303758,
0.4488655619968851,
0.12184849651506084,
-0.3514328367886149,
-1.0730523191169916,
0.20999659473927512,
0.4198496313207449,
-0.2949126664376062,

```

-0.7531087300641465,
0.4016553127140967,
1.7780699757893914,
2.338109295497774,
2.288296821099581,
0.3797180776898864,
0.43119211143988634,
0.11587678539460887,
-0.20259754629417875,
-1.3297836588453886,
-0.27310434987665144,
1.0101242784162827,
1.2218251312173358,
0.4833587270164354,
-1.4028471613931468,
-0.7051086788097171]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.029866062982578],
                          [12, 6, 0, 0, -5.108515955211577],
                          [20, 10, 0, 0, -4.320587490415481],
                          [28, 14, 0, 0, -3.6439280425186893]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.1961669921875e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -6.547496928632296,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                         'cost_richardson_zne': 11.390625,
                                         'sorted_expectation_vals': [
-6.029866062982578,
-5.108515955211577,
-4.320587490415481,
-3.6439280425186893],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,

```

```

2,
0,
0,
-6.029866062982578],
[ 12,
6,
0,
0,
-5.108515955211577],
[ 20,
10,
0,
0,
-4.320587490415481],
[ 28,
14,
0,
0,
-3.6439280425186893]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 0.23860472738944535,
0.24738626223792576,
0.368610904156845,
0.9146619524924428,
0.9146619524924426,
1.6697383257267104,
1.6697383257267104,
2.054253047546413,
2.3726208960540474,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
2.9146286026908905,
3.4924659345227917,
4.165338212258259,
4.165338212258259,
4.795197187789762,
5.022734448096858,
5.707107792388194,
5.707107792388194,
8.323917907910845,
8.967030236986108,
9.27289747465919,
9.27289747465919,
9.651472654450101,
9.651472654450103,
10.4607661524399,
10.4607661524399,
13.149200610156198,
0.08980902398510449,
0.6419619591442681,
0.3976886140600265,
-0.1339261955535598,
-0.07178891234334085,
0.20844549930040607,
-0.6039755260047369,

```

0.5345472321438205,
0.4885623179827016,
0.9007848202255657,
0.29262188635397585,
1.0311974603475478,
-0.7987894429612461,
-0.6323817753587336,
-0.8923011457984407,
-0.37936698482152614,
0.06585808205756116,
0.5122477185376288,
-0.2246894945378454,
-0.8796883548310565,
0.4023829077269032,
-0.6752839410958273,
0.20069365692379137,
0.5033852680281804,
0.24960186092606218,
0.058252930735255344,
0.2566722934175659,
-0.1810188232578722,
-1.0819601868914308,
0.6548412011669319,
-0.14141206607793577,
1.592915324309131,
-0.8278164458051894,
-0.3812348095673508,
-0.9060991568034324,
0.5864630340426413,
-0.2612726529884887,
-0.23456784875555411,
0.3993109852728946,
-1.6100891847686745,
-0.025086921642605235,
0.17965009471210514,
-0.5472865017332094,
1.4757076309755235,
-0.20182222210036213,
-0.4881953537109863,
-0.47862039265515355,
1.3275479824310896,
1.5474312317405958,
-0.365003055970495,
0.4312357294916266,
-0.18299603845951454,
0.706673586721164,
-0.7147228256515984,
0.21227947796353308,
0.0588792409702735,
0.37104688015833304,
0.5624160446643004,
-0.9359341524655871,
1.3022223571582927,
-0.08087685811880535,
-0.9026141041038307,
0.16238954726297725,
1.5451829819142204,
0.7129468042556484,
0.7230095383104325,
0.17583341711081185,
0.8889071634718573,
1.3752518501200566,
0.0640253150209717,
1.2836939598332835,
0.3344321550301345,
0.3035357788500751,
0.049516329509299586,
0.012257236048235042,
1.5823783745435407,
-0.09537942982333185,
0.5341486888465604,
-4.067358953041359,
-0.2870183941358629,
0.4906845201891505,
0.6709676538790386,

```

-0.10883760977815361,
-0.9015075659614459,
0.5057755407774712,
0.2603749012295441,
0.8480384556641941,
0.3777251280717097,
-0.7517099448725303,
0.5824300904263875,
0.42464435669423206,
-0.9724368356834788,
-1.035580562026805,
-0.008606395290722258,
2.0040454226391233,
-0.326939809664038,
0.0009364740955449761,
1.6853561871618157,
0.2235885662175078,
1.2569352560498372,
-0.35590863169612874,
-0.40928456866831275,
-0.8730825626925035,
-0.2434340162934524,
0.014655642366467374,
-0.07506183248897615,
0.12344014428925512,
0.4451771135631272,
-1.1817851818398777,
-0.17749779140411484,
0.48159794071814915,
-0.12344850129759947,
0.0485840790320397,
-0.8467407381542291,
0.027429753657626434,
0.7224917290233174,
-0.9085522245342863,
-0.08852023391973605,
-3.0071097182531115,
-1.466370256379572]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 0, 0],
                                      [2, 2, 0, 0],
                                      [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.82304667158383],

```


5.286850957206092,
5.286850957206092,
5.286850957206092,
5.286850957206092,
7.438635744061485,
7.438635744061485,
7.5101721136480535,
7.510172113648054,
8.498151007141688,
8.498151007141688,
8.89695633933963,
8.89695633933963,
9.949067520176195,
10.53028906819762,
10.53028906819762,
-0.7640988398723291,
-0.2609761344365046,
0.3714650363100771,
1.14227988754955,
-0.5586008021142339,
0.3232165993259411,
-0.808074054347843,
0.8156349361921491,
0.5302830682191738,
1.2319021875526037,
0.7713987684014778,
0.9552417745832319,
-1.2597420027575141,
1.0859901786608446,
1.3293719083173852,
1.9257516083881598,
0.36155074069433935,
0.15691696817317585,
-1.9831252740130665,
3.1611300866181464,
0.11297659540190035,
-0.8383054978325493,
-0.9072138705135362,
1.1291432376637918,
0.9684633525808153,
-0.0915018961421912,
1.3220745686094677,
-0.08267898444549507,
-0.0327735825742729,
0.35325234586292803,
0.008207563720112603,
0.6513781432301641,
0.7029739077472655,
-0.30374084774584015,
0.2705932145389305,
0.4299328079102274,
-0.25194341915550755,
0.9555661315796072,
0.006863497152706496,
-0.7799531748854542,
-0.9930641289246434,
0.029544121182944284,
-0.4089593191007625,
-0.9011793012188939,
0.24123797575417072,
0.7293698333067726,
0.2522025803398678,
-0.3662781893779444,
-0.040548739311108306,
-1.1705851767004056,
-0.37343950340573,
-0.6338759916409307,
-1.0085221962652005,
-0.08535366899581824,
-0.28965931149413715,
0.6099806260244993,
0.08931779318894412,
-0.043967227459777145,
-1.5650626910124423,
0.08045184558426326,

```

-1.1210462921934545,
-0.15581574650646343,
0.15100421645590653,
0.16901536922727223,
0.32421335800103107,
-0.05835809947146797,
-0.37767124143713443,
-0.5071581711302545,
0.26136135560376417,
-1.430365725839871,
0.19826339752764172,
0.38748357107642595,
-0.016413511028994576,
1.079152349973569,
-0.0926766024920056,
-1.4325837619063462,
0.4781305719489115,
1.5750151057909785,
-0.9642636149798988,
-0.2651388761233712,
-0.34395940038347866,
-0.571420352449216,
-0.1898017063159329,
-0.029075471666140197,
1.2134703230092165,
-1.526574130631626,
0.9149281390831964,
-0.16334350156219624,
0.0018380876570853188,
-0.27074661169481956,
0.01901564596467799,
0.17724273634666915,
-1.142032047382486,
-1.6992780971386738,
0.4921981051800609,
-1.4231374674303492,
1.279359294401491,
0.8796211692249704,
-0.6047896272897909,
-0.7770358631085403,
0.23322523970444167,
-0.7024356181264217,
-1.670552786717462,
-0.8176247209127478,
0.94419234756782,
0.5802716769380636,
0.27231979684377106,
0.0184909631791703,
1.2296098255675691,
-0.1467151618098401,
-0.9946525590855883,
-1.04257063761292,
1.4403642773133565,
-1.3523803858041898,
0.498491237471849,
1.015725079101807,
0.3090226723240105,
1.8267891711304018,
1.0609400619024847,
0.6031333925595603]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,

```

```

0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.536883635090364],
[12, 6, 0, 0, -5.4912521012226865],
[20, 10, 0, 0, -4.624115876403395],
[28, 14, 0, 0, -3.8988670742113007]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 0.00017976760864257812,
'zne_values': { 'degree': 0,
'extrapolated_value': -7.138075107423967,
'others': { 'beta_coefficients': [ 2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [
-6.536883635090364,
-5.4912521012226865,
-4.624115876403395,
-3.8988670742113007],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
2,
0,
0,
-6.536883635090364],
[ 12,
6,
0,
0,
-5.4912521012226865],
[ 20,
10,
0,
0,
-4.624115876403395],
[ 28,
14,
0,
0,
-3.8988670742113007]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,

```

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'ising'}},
'init_param': { 'value': [ 4.68500097130382e-19,
0.00019188574128314593,
0.4999939126216869,
0.571475986197529,
0.6379337879222081,
1.2223970983691403,
1.405127109337375,
1.405127109337375,
1.405127109337375,
3.9334167957034354,
3.9801359916712626,
3.9801359916712626,
4.868547987503254,
4.868547987503254,
5.067947873787899,
5.231153270915478,
5.2311532709154775,
6.116966983963522,
6.116966983963522,
7.100862078116925,
7.100862078116925,
7.100862078116925,
8.658171837297239,
8.670337232190553,
8.670337232190553,
8.670337232190553,
10.099540897359793,
10.10264280920201,
10.102675042499124,
11.255727095089838,
-0.004954852195250444,
0.49334633432738895,
0.4083998920567888,
0.7005599603945909,
-0.2726710135013912,
1.7017720535726606,
0.31242824153108295,
0.18213977913198362,
-0.768972357244276,
-1.2675908133090252,
0.24851075547279064,
0.758632579424417,
0.15831380308868334,
-0.40713384248795287,
0.5550964450005433,
0.49124548320800016,
0.12908691722609594,
0.7495294852763479,
0.2183951382187659,
0.3869596991035125,
-0.6237223499220805,
-0.8588440479719404,
0.3249743723844063,
1.1202240010341376,
-0.9665348984507536,
0.014548405539254808,
-0.9303555965408071,
-0.1828885288132902,
-0.03149136941083403,
1.028843591032342,
1.716975329829541,
2.1417393440093173,
-0.3249810914100684,
-0.29580699897963963,
0.03474070302461121,
-0.047604331969969146,
0.47339919478545306,
1.1366650605651316,
```

0.20275438481263153,
0.23966710331644123,
-1.3523342039629156,
-0.9754463664850115,
-0.2001650111968273,
-1.5422757838665726,
-0.0789634310315002,
-0.4791620247951226,
0.3726938787973833,
1.5571588855871121,
2.9076766591454613,
0.0805159751714424,
1.2397608690078437,
-0.01725082954866091,
-0.08478544671108038,
0.5035755545702888,
0.0483150970198388,
0.39318121771041326,
-0.22904257662027977,
0.5457943597837928,
1.626832845335358,
1.740382733640172,
-1.083884361853876,
1.4904397154767375,
0.15453140539699517,
-0.3408650280808243,
-0.14903551647542937,
0.6286230044280887,
-0.11628788661254054,
0.10414683520207971,
-2.1088223185456503,
-0.1212852764853123,
-0.5775366030665824,
-0.17694325541481232,
-0.037210874584927306,
-0.20680178616321185,
-0.03652613740982194,
-0.08764043729389985,
-0.7677620465647192,
1.5711877406264763,
-0.7996083277851588,
-0.797598947346768,
0.14961771067626958,
-0.01602963805135088,
0.5562356172649764,
1.7058411163304328,
0.08841953300538762,
1.6796692259198813,
-0.0683168196107479,
0.412631445931469,
0.7690268328461398,
-1.1965988622765378,
-1.634850410955949,
-0.3585108711773816,
1.4515485245761623,
-0.5118812625179237,
-0.33796173975350213,
0.9046803820753733,
1.919749746146965,
0.8291464157254846,
-0.1350562984990471,
0.07957078412968303,
-0.8960128872012803,
0.7674820757881975,
-1.9940192829172003,
0.7589351716787923,
0.45973502216902334,
0.040526985048879915,
0.17528949190523055,
-1.2894028841444158,
-0.011475752292035499,
2.0713928129783454,
0.20597398926497368,
0.9592210916748978,
0.42837585835406544,

```

1.3426423381373416,
-1.9221135042950956,
-1.997216137032599,
-1.1075910870533463,
-0.24932031450285055,
1.3098010190882858,
1.0657612241892138]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.7841170852098625],
                          [12, 6, 0, 0, -5.679026952355479],
                          [20, 10, 0, 0, -4.762614433311011],
                          [28, 14, 0, 0, -3.9973832617084413]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 0.00017833709716796875,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -7.419133840055779,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-6.7841170852098625,
-5.679026952355479,
-4.762614433311011,
-3.9973832617084413],

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -6.7841170852098625],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.679026952355479],

```

```

[ 20,
 10,
 0,
 0,
-4.762614433311011],
[ 28,
 14,
 0,
 0,
-3.9973832617084413]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'ising'}},
  'init_param': { 'value': [ 0.22237247586897502,
                              0.3434883293456931,
                              0.600978934518952,
                              0.6193810427253835,
                              1.6475035333031534,
                              2.204130997587115,
                              2.4096412381325085,
                              2.515509113107616,
                              2.5155322756928205,
                              2.9680516150838403,
                              2.9680516150838403,
                              3.0576851370343725,
                              3.8016986766973213,
                              4.264365678603219,
                              4.264365678603219,
                              5.4290969626181775,
                              5.4290969626181775,
                              5.453519119574046,
                              6.855500326085536,
                              6.858194453958984,
                              6.956907552727685,
                              7.899693187867781,
                              8.578442687794452,
                              8.60387860142029,
                              9.193247925689713,
                              9.205803142575261,
                              9.205803142575261,
                              9.815391824330385,
                              9.815391824330385,
                              10.97297451152796,
                              -0.4325514790797943,
                              -1.0860755794816048,
                              -0.34917301406803075,
                              0.9554400980633908,
                              -0.3272724546684993,
                              0.13698615504934158,
                              -0.3120459104790063,
                              -0.5324325213761384,
                              1.6476833500899999,
                              2.184163963641692,
                              1.0643894246063128,
                              -2.7436302898700338,
                              -0.9203607373406255,
                              0.2960792725949677,
                              1.646667339143857,
                              -0.34154797106217816,

```


0.5971775478384601,
-0.8758117372527006,
-0.5328437036469449,
1.0093796179148329,
-0.3390771908975529,
-1.0516631129606762,
-1.899848962378923,
1.0171271290916915,
1.938798354471919,
-0.9913918052370154,
-0.5882123471692551,
-0.2591906358136659,
-0.16051793033532072,
-1.00404705451914,
-0.6166532433204007,
-0.36188450836273767,
0.050500866856171264,
-1.9050925803490544,
0.44677704942019497,
-1.1753602972946917,
2.7447289966589064,
0.6045447447119641,
0.8998543123679479,
-0.6953603830902315,
0.6417236531456995,
-0.5066995361238301,
-0.6557671363535594,
-1.0249921742780712,
-0.7707842488201082,
-0.7447403034964661,
0.02205807195035654,
1.422919970075721,
0.21691798134603987,
0.019101707085961515,
-1.5459948748132235,
-0.8336804070854993,
-1.1349385342158511,
-1.763233663517991,
0.9698040529008946,
-0.44077663287617147,
-0.037314749661943425,
1.6452841075012843,
0.06530823237303056,
-0.5191753532394425,
-3.3927849916233557,
-0.9466255969188528,
1.1621532796642329,
2.728652480060894,
-1.4097883504698188,
1.6597459198542186,
-1.6022660999296543,
1.1929284141495398,
0.2936047355153142,
-0.039596579367738485,
-0.40612830245611553,
0.27386094225966673,
0.619794935541383,
-0.06440450102281042,
0.2885024090042384,
-0.4666286809510815,
-1.0437342022538576,
0.6208113670044477,
-2.311602134698998,
0.5891252871010197,
-1.9541222671932912,
1.910301252167151,
-1.5078706659624532,
-1.3581199892036264,
-1.4501597709616996,
-0.1395100013007674,
0.4345418228024135,
-0.04062033627254925,
0.12975822420449573,
0.7006910131419305,
-0.4577672118378531,

```

-0.8504301982960907,
-0.6206071827471396,
-1.2689654888714474,
-0.6028452314331775,
0.14098909118247002,
0.2582405914138898,
1.1153023482335702,
0.28953043606843626,
-0.34737036965609414,
-0.18623907101172993,
-0.7264540299251561,
1.491275503355475,
0.9818216668009369,
-0.06763614743578832,
0.45977626052089293,
0.4489021740623341,
0.4542312767253669,
-0.3771631048098388,
-0.48778145745612106,
-0.7778747545839428,
0.7808334581652677,
-0.25809112185919825,
2.0032973367261864,
-0.3387989838818156,
-0.2225214088051193,
-0.9736339495299617,
0.15018800357199702,
2.1788568254522622,
2.092251568848893]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix':
'AUTOMATE_ising_noisefree_time_evo_dephasing_ric4_sample#9'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -6.37648641427504],
                          [12, 6, 0, 0, -5.250000083007176],
                          [20, 10, 0, 0, -4.3238347566667095],
                          [28, 14, 0, 0, -3.556658050073089]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.695487976074219e-05,
            'zne_values': { 'degree': 0,

```

[illegible]

```

'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                               'identity_factors': [0, 0, 0, 0],
                               'noise_level': [0, 0, 0, 0],
                               'noise_profile': { 'noise_on_init_param': { 'status':
                                                                              'value': 0},
                                                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                                                  'status': False,
                                                  'type': 'dephasing'},
                               'odd_wires': 4},
            'optimized_minimum_cost': [ -8.213984832683868,
                                       -8.467734328099501,
                                       -8.447208232910596,
                                       -8.461814133720251,
                                       -8.483392807973491,
                                       -8.44723907770434,
                                       -8.362559674056595,
                                       -8.366604609651855,
                                       -8.163432600733032,
                                       -8.478925857587775],
            'optimized_parameters': [ [ 0.5489081084990874,
                                       0.6882459728440448,
                                       0.7729468967125317,
                                       0.8347489553606998,
                                       1.0946318761773612,
                                       1.9534300183374533,
                                       2.2829395447313403,
                                       3.232524496500775,
                                       3.2325244965209894,
                                       3.852765644418336,
                                       3.8527656444207725,
                                       3.8527673900689967,
                                       3.8860323667767283,
                                       4.588361079261242,
                                       4.600166894773711,
                                       5.527977091209439,
                                       5.556427027819143,
                                       5.803787289292802,
                                       6.218904325824693,
                                       6.239985329573426,
                                       6.538211674407394,
                                       6.925082879783563,
                                       7.640591161405884,
                                       7.75471653665469,
                                       7.896761387122529,
                                       8.932525392602521,
                                       9.128535074996934,
                                       9.925693656125251,
                                       9.925693656687802,
                                       11.037683520466203,
                                       -0.10948884769018298,
                                       0.012576058394267018,
                                       0.9054362612114637,
                                       -0.5027178523290218,
                                       1.9293813165204678,
                                       0.12560680736570712,
                                       0.40022822712854333,
                                       -2.471918351448183,
                                       0.5257346934244156,
                                       1.1130547436006022,
                                       1.0771121143603033,
                                       -0.1314162667650016,

```

False,

0.02294938233855663,
0.36225076948659746,
-1.3206473358532678,
-0.2449807752862754,
0.29890651347484976,
1.0095017101029446,
0.6150854645533661,
0.2815584721561816,
0.3601395589466358,
-0.5388375319509757,
0.9923143444796582,
0.8473755438313292,
-0.4646214972755992,
0.8527825907455423,
-0.6098121229644526,
0.12413023074624807,
-0.06044402625536853,
-0.29258159286196256,
1.3610999143074116,
0.09999631572477011,
-0.18981827210653637,
0.2696177153747519,
0.4272428992844361,
-0.26705761267067063,
-0.37988713130471985,
0.75645724651605,
0.8109920260438386,
-1.028544143258335,
0.44106157371844235,
-1.6090649394170073,
-0.44600474997528494,
-0.2579853712948814,
0.3063055059682419,
0.9497745366960584,
-0.2701835817775974,
0.9807489498405011,
-0.507582861688708,
-1.25201940633301,
0.5281948255512322,
0.3440155511307017,
-0.6791241507669356,
-0.7498601012695852,
-0.6946373774642369,
-0.6177642266777679,
-0.2856729357201208,
0.4851947472201811,
0.09726903107154261,
-0.4045609051365335,
0.18389954360123525,
-0.2034787076783578,
-0.2260359270986582,
0.8035576635033466,
-0.7931715620009508,
-0.9362607227109441,
0.5578839793416537,
0.15959477202733782,
-0.454821222216028,
-0.545593252337095,
0.2205461704673383,
-0.35716661949785156,
-0.6934665190405042,
0.0830933564238442,
0.17268207109916753,
-0.8013515313723008,
-0.20341695383964103,
1.4288863566434409,
-0.8458196045603313,
0.3893336367327371,
-0.34888961728294265,
0.7608524424568026,
0.06661544595138999,
2.0595559407890174,
0.6490263079624171,
-0.4572869614004236,
-1.1373095831864677,

-0.3649376822063403,
-0.22071540101499584,
-0.6587338793306329,
-2.059884237950364,
-0.08358300020061853,
1.0319197724415174,
-0.2751839419251212,
1.3088618827914082,
0.7764572920081257,
0.8767331962684795,
0.955714857858676,
-0.5966677196420035,
-0.2715939735155754,
0.12847285565670696,
0.4974981016237345,
0.11157624833218974,
2.280970649154798,
0.7801190316815027,
-0.311773417229458,
1.158014603168253,
0.5949520544831722,
1.6123203236495465,
0.09856141876014059,
-2.5192133286473677,
1.414165495054559,
0.568818345966238,
-0.7689353156130335,
-0.016254732770980017,
-0.044305575074125,
0.05295776050958731,
1.0200318030747035,
1.4632071591524118,
0.7778847321553402],
[7.194344081667924e-18,
0.22666307415943535,
1.9342085859558782,
1.944505522513897,
2.976191287506401,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.469325640936666,
3.4873843407422243,
4.118956189268777,
4.118956189268777,
5.942584534769547,
5.9611960725237205,
6.340111863711309,
6.460202466469716,
6.7852294872778645,
6.979402607245081,
7.168797029298139,
7.227045817029271,
7.302238353497494,
7.404556177953146,
8.659014511382493,
8.681357101968592,
10.220537951103626,
10.220537951103626,
10.990885177615443,
-0.4696397519192671,
0.27796231748191075,
-0.5848844010182828,
-0.9399496817226014,
-0.6791144953934183,
1.7599237567532016,
-0.5153866079362172,
-0.9884960511226192,
0.5991380415167059,
0.03526597855649831,
-0.4928202951963407,
-0.17823604247966235,

-0.5598665649984788,
-0.7477193118385453,
1.3145784058485683,
-0.01658289220788679,
1.3749987665160877,
1.5410101089396069,
0.5381593361770547,
-1.5686214512244299,
0.5414340679559935,
0.17960267336650448,
1.002730580710403,
-0.5079352878091554,
0.27127261026651706,
0.8523739943313656,
-0.22490410891672527,
-0.5177684122022047,
-0.7335136739541019,
0.9537821935011014,
0.9299309014905508,
0.027634097755273066,
0.11678727620864918,
1.329875652999114,
0.22154652145214473,
-0.5149535741083672,
-0.0935144974812166,
0.8101347453355321,
1.002535906635114,
-0.21669230492415195,
0.25505956278191033,
0.8322911879517776,
0.24105173903016894,
0.3010018335234525,
1.0269077594015688,
1.1653005827455005,
-0.7139789414025479,
0.1802432624729846,
-0.43621452412416506,
0.10805015233497241,
-0.22099264006117136,
-0.06821093567938447,
0.4991913033327633,
-0.09623960812205685,
1.7645880679165642,
-0.057916644058093394,
-1.1151840835769227,
-0.7145527671981541,
2.0743689798531384,
-0.04887843441837349,
1.5069809195108836,
-0.3256203951764359,
-0.8875710119086927,
-0.20072291205396614,
-0.8322399466388898,
0.46293426681558253,
1.4946138988374058,
0.09519796067520077,
-0.7389501244350067,
0.7163596590753283,
-0.9779834532530196,
0.24121899590668236,
0.5362525857496123,
0.7178873526537328,
-0.09720605867652783,
-0.8844816300920153,
0.6879494830422364,
-0.556207799173706,
-2.085706465222243,
-0.6990668132647135,
-1.060060203430475,
0.5108382979368807,
-1.195038624073349,
-0.4693850277739393,
0.4997731574718765,
-0.7981866110564311,
-1.3492284551615505,

-0.8150525999866463,
-0.023113205223160647,
-0.1007660060236893,
0.5860867369253601,
0.15047431993488672,
0.8082527357931191,
0.6508889280029133,
-1.3215333882547653,
0.13580576430947058,
1.0768338700600115,
0.09630564743985112,
0.5304856979004331,
-0.2470170488957273,
-0.6273296913896085,
-0.012309334672496116,
0.8219063430566509,
-2.522381990553264,
-0.5959392155078911,
-0.9015453006612434,
0.25028124043210703,
-0.035229743179394535,
0.46028190301395183,
0.030645450102450806,
0.9152462362128074,
-0.09021729137847402,
0.014451484462572361,
-2.194500168454895,
0.0026823747896166745,
-0.021868499065043765,
0.25783762601108967,
1.5371718314814482,
1.9085420695857682,
1.5522802210039741],
[-7.180915735189581e-18,
0.2741144583980765,
1.4514670623449988,
1.4514670623449988,
1.6489164272497665,
1.6489164272497667,
1.73149137801893,
1.830467910520106,
1.830467910520106,
2.708217872437681,
2.708217872437681,
4.495539722625299,
4.495539722625299,
4.495539722625299,
4.570233660709978,
5.002505460470289,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.812797709933179,
5.812797709933179,
6.677130390626635,
6.677130390626635,
6.861335344270694,
8.772688787903297,
8.772688787903297,
10.006873752440399,
10.006873752440399,
10.006873752440399,
1.5494767629045267,
1.0888430758932834,
0.24316211822093795,
0.5394610508612123,
1.6831866222617424,
-0.15121133640766382,
0.37446291095683387,
-0.39557168857427305,
1.895734214431956,
-0.2769106697905878,
0.4098744736876165,
-0.8550979914567345,

0.497698744019723,
1.1661969870744597,
0.4884385698850034,
0.1585303339769962,
-0.04038005447014292,
0.20331204987112447,
1.2717970037539772,
-0.024603186152083865,
-0.06561207027284033,
-0.17578386607253835,
0.4554538169128916,
-0.3458478405442004,
0.003014344905820474,
0.04935807070525889,
0.7414733867293872,
2.191797855158184,
0.7288491023643235,
0.6487285676710397,
0.0823080431430117,
0.23749626587684475,
0.1076729676697226,
-1.5131488602297158,
1.1994751078587316,
1.7910688286856344,
0.0059408425852493195,
-0.0215678421858246,
0.5920431102259514,
1.1797184458204304,
-0.4858383686169137,
1.1738772137731162,
1.917509405697279,
0.04253149516337592,
0.5485055496810046,
-0.35206499609814007,
0.4164896121092416,
-0.5778229043728815,
0.7927744863095695,
-0.40276124854461287,
-1.4077641380820347,
0.5847373551160024,
0.7110370445087761,
-0.272001728150977,
0.6801732294133973,
1.5986611291818411,
-1.7123595531876228,
0.9030593081535582,
0.5579581015351607,
-0.12955463976185633,
0.17158208553128768,
-0.4284115544697507,
0.41166090312183995,
-0.05704879108721846,
0.6260803344487521,
-0.1444379656414329,
-0.03323955152691923,
-0.5463524110157282,
1.9483453113378342,
0.24554581836345657,
-0.3912953380733674,
-0.2551086789875916,
1.7941408116066884,
0.5297614111369608,
-1.055305153377236,
0.5851894984947826,
-0.6277708161900404,
0.9023397588106277,
0.15473896858281969,
-0.5045582428336748,
0.35128474752318256,
0.27447050561374786,
0.2453530023237907,
-0.881260065779903,
1.2893822185874217,
0.034404783051078594,
-0.8093290700723595,

0.020829229564375124,
-0.5496174311783282,
1.5855246509676788,
-1.9597925195753434,
-0.35765423396713947,
0.0645095765054153,
1.2837142304502733,
0.006624180460715028,
0.35085061348827623,
-1.8681642835419228,
0.1423820815596781,
0.5399990382746707,
1.7033903387378666,
-0.1065388317224254,
-0.05044042000338169,
0.3207599083179648,
1.1530017150539378,
1.5387803133146891,
0.9426514512946884,
1.654923602736178,
0.15672212072564318,
0.5579523293912716,
0.03301127288721064,
0.2831128671935505,
-0.8296250007412259,
0.3295014726214846,
0.8099539632201395,
0.6731359481535688,
0.07726614260275587,
0.2823379588881529,
-1.6102046054887462,
1.3421086989005693,
1.506675537221725],
[0.5436487956056492,
1.0318754867837048,
1.031878809784634,
1.0354032636682196,
1.6023669004016947,
1.644001042836531,
2.7828327622495883,
2.987856277002514,
2.9878562849693426,
3.0896481585306774,
4.808772683377482,
4.9189322116630665,
5.053349250959963,
5.0705845742336875,
5.0876264564887625,
6.123589402685677,
6.123850407213233,
6.291601953687259,
6.291798238238624,
7.418249505998958,
7.418249506075792,
8.141624649997873,
8.163960362736292,
8.208498408727728,
8.208859361112212,
9.089278643760485,
9.180997185672972,
9.211346803231834,
9.871115455051829,
10.914592222542833,
-0.10714466270129372,
0.0414005813417363,
0.31017108200298504,
-1.9041640853184163,
-0.2149394741481629,
-0.954748928787851,
0.12809368425258344,
-1.068243921180568,
-0.5079865095515741,
-0.28605329513797556,
-0.19740513387232797,
-0.22325695706426568,

0.270216664817167,
-1.1670200531314157,
0.17357556599169296,
2.554477008278338,
0.2569346421962659,
-0.44255822443493054,
-1.0047996143631401,
1.3462842603333594,
1.1512175477641275,
-0.23524577874182742,
0.25862583919023807,
-0.03981155232548289,
0.12000346737958213,
-0.5231937940865384,
-0.07831458803809532,
-0.7352847060942096,
0.36694131393913326,
-0.2794757421840953,
0.8531564292966232,
0.11577732227315543,
-0.2692221140803255,
-0.01593534730278469,
0.5023269794705598,
-1.0201597672225167,
-1.217555742220479,
1.0019872098347335,
-0.9863383366879347,
-0.505076260127648,
0.44147961683678577,
0.17102749819972465,
-0.3872030295771854,
-1.0804196760959297,
-0.46011856071048735,
0.6404554735741855,
-0.1019842395221057,
-0.8338441616008496,
-0.1862543853067883,
0.7729351219902972,
0.1291698795888973,
1.84480481040653,
0.7902011519638132,
0.8035704757103046,
0.16307460637128957,
0.3879631381162318,
0.7553290974087437,
1.3544126344921927,
-0.6410152429142767,
0.30610263763196005,
-0.8028999721479468,
0.026104270638268326,
-0.7549249175514674,
0.3272210288981157,
-0.4017438577258813,
0.17639539137663426,
-0.0599886006454003,
-0.029000197801291885,
0.46434674317946,
-0.009657445164941145,
-1.4594578401188716,
1.0749651071987714,
-0.2858651376363263,
0.9236407241821395,
0.804989773126084,
-0.006131190880034288,
-0.41182615341479456,
0.0345636295388565,
-1.4528618699977163,
1.059940111163351,
-0.293288558562868,
-0.19261506463839523,
-0.013768907394512604,
-0.0058499255440563995,
0.9958119726697335,
1.6968974761241398,
-0.8295102319053347,

-0.12979840000582374,
0.1192301496450713,
0.36205695435395635,
0.18145573379217902,
-0.5008220240501698,
0.9951331553730755,
1.5809655477939468,
-0.44266547932504335,
0.5749472054038229,
-0.41342219281521114,
-0.6949087257678164,
-0.9951368874890167,
0.2815866277793241,
-0.14025691028810777,
0.36907964729939857,
0.10131186103604999,
-1.7520150960815608,
1.9938164611139795,
0.8620398797652439,
-1.2922896821720897,
0.28060229891238847,
2.8035209903051896,
1.568376396433029,
0.595816726425649,
-1.2842096282862903,
-0.6778028363202987,
-1.5573241486707392,
-0.7609920700723998,
-1.5773563548919856,
1.324267114980915,
1.5509238379824284,
-0.3029567022174245,
-1.5980321700442846],
[0.04778381839807009,
0.052328729194328676,
0.05232872919432868,
2.5402034202141324,
2.5402034202141324,
2.540203420214133,
2.9078994845512853,
2.9078994845512853,
4.626405021642842,
4.657483894791151,
4.657483894791151,
5.1543302051209485,
5.168582880456237,
5.168582880456237,
5.193083562661445,
6.29476745947969,
6.321713228151687,
6.406123865320823,
6.406123865320823,
6.415662129910562,
6.415662129910562,
6.4885710573049025,
8.891108534137606,
8.923842890216621,
8.941256389861133,
10.14144676971709,
10.14144676971709,
10.14144676971709,
10.173407474763442,
10.173407474763442,
0.9049919196609981,
-0.10571917073418317,
0.984067433195123,
-0.5046397239329845,
0.826983938894292,
-0.300506222622187,
0.6691119536121557,
-0.6359242615836268,
-0.9075291794736813,
0.7311597085342896,
0.9472124240542854,
0.9252859784089732,

0.2654320249732376,
-0.19570931474329503,
-0.23156782982391674,
0.15992044574971995,
-0.9344783087557579,
0.574058594523507,
-0.5900698195003219,
-0.015276834175791246,
0.8277476347193486,
1.701428746099737,
0.29486306492791453,
-1.5727896129747672,
1.2542378717585188,
-0.029060964767507632,
-2.410312077661991,
0.7794852804981598,
0.10382543392432386,
-0.27522638598589344,
-0.7737995873864438,
-0.17484621194736175,
-0.8869497632214399,
0.0739036632402581,
-0.11675807314923795,
-0.7813933016619123,
1.2237386371188925,
0.254266955819111,
0.8344221074025743,
-0.05418013806720967,
-0.35023932132414914,
1.5635792749140942,
2.0001489123678207,
-1.469548386716051,
0.6018811590660905,
1.0932588013795888,
0.1626638394237113,
-0.4071815610867397,
-0.09033843760896655,
1.5066284256317157,
-1.0091863719510874,
-0.5055220266722595,
0.8659824216894423,
1.5341961411404397,
-0.10472552531482807,
-0.9747080994118024,
-0.15896534579332847,
0.4283545141751367,
0.48555121925603906,
-1.0541413967651159,
0.7844732422535036,
1.231237717810785,
0.2304560895908817,
0.8964687802289353,
0.3608400337996031,
0.20964144290191944,
-0.3812502250633237,
-0.05244827792509459,
0.7363786261301344,
-0.5373684827470173,
-0.041135046809633154,
0.022494455398660802,
-0.1427510282638101,
0.15254373676169164,
-0.3538441331700385,
-0.6013550669628682,
1.177355155310087,
-0.25553253513169044,
0.24230205662919607,
-0.856739664506789,
-0.24258353085003115,
1.0855460718544134,
-0.2303523457340619,
-0.4228181709882648,
0.6300184916809138,
0.7448691834325954,
0.8306231103131142,

0.8921818612650112,
-1.2232038552468776,
1.1518310980776325,
-0.10925895973168652,
-1.5546501465626024,
0.25882124214269925,
0.2964831640226424,
0.20056526647993123,
0.6466051276015505,
0.2958855303900712,
1.574298633668074,
-0.2963531534087623,
-0.5079019143621002,
-0.9468892366026295,
1.2752258488383597,
0.7403271380521786,
0.2806459845384513,
0.00045412531964782697,
-0.30343305939372256,
-0.2826391347063689,
-0.24868728346285462,
0.6190941948455959,
0.22356779638775762,
1.0235473251809233,
0.10726603124753692,
0.2726476947387006,
0.34464280290355076,
0.1533269771575057,
-0.19391561509244587,
0.4350499735853799,
0.07688310057706789,
1.3928303170166854,
1.2150647846497966],
[-1.5612511283791264e-17,
0.37090300553176103,
0.7507019437657014,
0.7507019437657013,
0.8504313269395809,
0.8504313269395809,
1.0288453745501207,
1.0288453745501207,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.6746234572416636,
3.6746234572416636,
5.070304872258586,
5.070304872258586,
5.070304872258586,
6.244278301066357,
6.244278301066357,
7.550988432333311,
7.550988432333311,
7.550988432333311,
8.583669958606022,
8.583669958606022,
8.761663968184166,
8.762824899576403,
9.78739432508435,
9.78739432508435,
9.78739432508435,
-1.0196497144708154,
0.30674985458989995,
-0.4164372967959927,
0.3795733998494057,
-0.3589945609302255,
1.1740756105856336,
-0.881009363192087,
0.6233231770354686,
0.014641892239634163,
1.5178326228084182,
1.0177798404109606,
0.04930799061014925,

0.1259706076860049,
0.05495189820733549,
0.008740541357319203,
1.0293402971653869,
0.6712877763592521,
1.3470276988715923,
0.5204896552003178,
0.5839104657425647,
-0.09863982249226241,
0.44754254744872723,
1.0699570996956442,
0.052847134004663066,
0.4378537260228811,
0.8337645558299386,
0.26799172757326156,
-0.2370989308824484,
0.4268946106495037,
0.10394671906247668,
1.0368119557194113,
0.8780368078369323,
1.391762381585933,
-0.4887346750129456,
0.23129082798818065,
-1.185985414153505,
-0.1399486075576189,
0.624884164407104,
-0.394202372239471,
1.4744661957359235,
0.1333476321252126,
0.7992370886557384,
-0.032444927920587764,
0.2278570517496479,
-0.12707464851531813,
-1.1803807605079757,
0.12619040548444643,
0.8842719846976165,
-0.8876928774682807,
0.25819408346986983,
-0.15120492941863672,
-0.34186031332584377,
0.05717115426327923,
-0.1287980699517455,
-0.6384961775059668,
0.2028843608113393,
0.6968559028533619,
-1.0254791626625297,
0.04639881891449554,
-0.034568027823247464,
-0.01382953519480725,
-0.28736510780605545,
-0.27712541400117535,
1.7289913866690068,
0.3590804569357109,
0.5212797413877558,
0.3325950057479984,
0.31414178485416083,
-0.007677216540470154,
1.5276609022748393,
-0.19648884450771267,
1.7007124456248135,
-0.16165644607389268,
-0.19438002002138255,
0.42844370320601005,
1.0333499299480355,
0.8652144019307915,
-0.23530130050937098,
-0.07871384718095216,
-0.11451985500506619,
-0.9958204124079787,
2.0359256308075255,
0.9323993817814743,
1.5157870599134722,
-0.21153949564011645,
1.3854977335188041,
0.5637663837671051,

-2.0565965924917213,
0.4516254671982243,
1.5460814866423853,
0.3644961596633484,
1.429982344445419,
-0.6505068989688967,
-0.054141570934051676,
1.0286289924316303,
1.0001903806983123,
0.5202120156033159,
-0.4848333781219212,
1.2810399381016777,
-0.030825717336130737,
1.1138939193895847,
0.009665204926618747,
0.35242092505791633,
0.20878415377993897,
-0.4360318389272872,
0.12430046579509,
0.02631778400042499,
0.052498295792397884,
1.6355891166025114,
1.6103252652105555,
0.5534147808336787,
1.5586288893399751,
-1.292487508100558,
-0.008775390375330459,
1.149807410139418,
-0.5197517748175643,
-1.4661127557094134,
0.8953775577555926,
-0.7735285435355932,
-0.02192283351673928],
[-2.8132202298922583e-17,
0.28576499380662435,
0.5188712429506999,
0.5260718495211498,
1.6875510996286314,
1.769957047543949,
1.769957047543949,
2.1144412566849082,
2.2736787675443475,
2.2736787675443475,
3.8733552051866122,
3.8733552051866122,
4.144421713026456,
4.144421713026456,
5.356566841798323,
5.5388531097345926,
5.5388531097345926,
6.035393779471804,
6.303140775366703,
6.861779885684816,
6.944533564960904,
7.13357233256276,
7.13357233256276,
7.277389780428127,
7.277389780428127,
8.82462477385959,
8.845795504558339,
9.548198337876569,
9.548198337876569,
9.548198337876569,
0.25174303748303517,
0.025975082425550772,
-0.40984308440331035,
-0.38070585583844174,
1.799681239992893,
0.5594462945854607,
-0.4636478431530039,
-0.2780917996856129,
1.2810553386613224,
1.1310784748475073,
0.45914878725036795,
-1.1856954723964535,

0.7961435885090882,
0.57941026174794,
-0.13815313828722545,
-0.7574725065912581,
1.604820017143439,
-0.3603689506520919,
0.27196300072613677,
0.2722313257155289,
2.990969455340513,
1.62020725835794,
0.13735249860534363,
0.519616354535985,
0.46064754704679234,
-0.14807050848061099,
0.6713871000380094,
-0.2458402564805999,
0.9463308813112179,
0.2567843451270063,
0.4629107250573343,
-0.6212938478257929,
-0.22704009998639554,
0.3296464835751119,
2.1654921123752615,
0.6636756285031978,
0.9936965640543358,
-0.6185742210548419,
1.749534825479387,
0.24548213508704153,
-1.3019305728065236,
0.056631533618999905,
1.0932304270217696,
-0.5530783180440034,
0.019231105171754005,
1.3923960145800571,
-0.5916513325234946,
-0.003848263362819536,
0.03781527162426783,
0.5864098422960301,
-0.8251793549162847,
0.11206477827525356,
-0.09225750098811783,
-0.4770216140176493,
0.13695039906816245,
-0.08154474181557379,
1.3791196308847455,
-0.8857160703287424,
-1.129732421053176,
-0.44824804793710993,
1.565274483618525,
-0.47775652201553226,
0.06810484542081006,
0.17151342519304494,
-1.2660646869274135,
1.3724437443522657,
0.455376664537531,
-0.4968402042883,
1.636176060280602,
1.8168167124913772,
-0.6591502510366976,
1.3845331149359692,
-0.13680523979595852,
-0.6670867326095455,
-0.25188954569138783,
0.36341441995068174,
0.27861221206959597,
2.253753293233874,
-0.44572473607203233,
0.08052163176334037,
-0.3801650824413073,
0.7561740409138518,
0.4179895401049083,
0.7668389261341573,
2.5895839574180672,
2.3057077789471467,
0.009170869870097192,

-0.48391820047143247,
-0.1788385119964603,
0.08989915668495536,
0.36995891095455813,
-0.22641283869292964,
0.6178226857138855,
-0.025557409724796323,
-0.5624997309051616,
1.978118628762505,
-0.7161151168610898,
1.889934912715883,
-0.7926498270401485,
0.8062612160869115,
-1.7535002724636062,
-0.12121642361966994,
1.2556359822665695,
1.4004947542019472,
0.4094181922934832,
-0.7237598159979259,
0.8095124829350078,
-0.07921500563810784,
0.7941695908203817,
1.5790924811449014,
1.2151033673114386,
0.2760044806386616,
0.03215735026840476,
1.3335719235125942,
-0.01527696316380547,
1.667307482924314,
-0.5852882148598142,
-0.2634738849840575,
0.4308024302775489,
1.564240027869587],
[0.3981028774646537,
1.3970863863849614,
1.4091876724203942,
1.4095196690866725,
1.435309832652522,
1.6159440641430487,
1.6590822800214955,
2.5041994971471193,
2.5046214861953655,
2.504687307075914,
2.6463903945527543,
2.646392107563143,
3.5650194444109125,
3.568763188733049,
4.893404038177111,
4.8993251206087916,
5.047882242480286,
5.059087496587666,
5.734377927903264,
5.7343992879002315,
5.734611661260137,
6.672744415088379,
6.716855027417073,
7.532922291325236,
7.76042742188154,
7.760528016690849,
8.741894091408147,
9.173605790810564,
9.476175665329622,
10.771165370152604,
0.07585840339400536,
-0.32559648608449443,
-0.3584877554976206,
0.24942288785390876,
0.8910963785962387,
0.41120182935281147,
0.06965634755603489,
0.20611943841300834,
-0.5108736052092635,
1.0351003083741233,
0.23303911085101622,
-0.8140715483421308,

1.3592937988371023,
1.1741125230549692,
-0.7984232202856042,
1.1431263661345903,
-0.42271729328296237,
1.1685998262673312,
-0.8900747330133342,
0.16620411037713437,
-0.06441490232378988,
0.5721569257130531,
-0.6405877965847887,
0.1435721627905685,
-1.0023253187840875,
-0.8834734269597262,
0.873011521015721,
-0.5294133334283674,
0.42491694402044683,
-0.24819133021277387,
-0.006858394744271055,
0.10239876199833975,
-1.0945255159067344,
-0.5209588569771533,
0.07684624481803015,
1.149902758246962,
0.6387502322543195,
0.43311665397553645,
0.2524037362289789,
0.5970991691948484,
-0.8659392186972171,
-0.2075467141416817,
-0.471169586924007,
0.3474995850944464,
0.8407912910032019,
-0.39456865353795484,
0.007483066013485144,
-0.3000031511517205,
-0.27926483264869584,
-0.02627276579859505,
-0.7790488347588548,
-0.6185889956791905,
-0.09477367060629449,
1.4923529367376538,
-0.10445531715928387,
0.12530829976044264,
-0.6395373064435654,
0.08277402847573397,
0.7939754550840359,
-0.45257909256973483,
-0.6794716273620853,
1.5141539142037763,
0.4312816489513316,
-0.1679469048132705,
-0.17124947545547786,
0.09347199374723385,
-0.1562849778095718,
1.3537610374278526,
0.9008040355239236,
-0.9700720508412178,
0.3256401758946771,
0.29535126871409256,
0.18912935978590353,
-1.3766878479597657,
-1.9912841200531366,
0.14684386063156718,
0.9485726251766791,
-1.1647041818090063,
0.47731661524457764,
-0.05220023510180973,
1.3678085559740152,
-0.9056449204835322,
-1.0124342233515295,
-0.9002321917007622,
0.25948869946042935,
0.6916196201790348,
-2.867057384272419,

0.10161679048335649,
-0.202200597369352,
-0.574167005177845,
-0.19403420003548572,
-0.47967113273306583,
0.5285998620423336,
0.1951966167292077,
-0.01680550493300095,
1.2402630343430938,
0.30327521875472907,
-0.21038560271353512,
-0.2563819703743369,
0.61553927096869,
0.8894393642077769,
-0.7281429405128931,
-0.5382166701178036,
-0.564888527870778,
0.9324104873196495,
1.3153762213142488,
1.6087794639717614,
-1.4809112056457674,
2.6089661079426194,
1.6134732700667287,
-1.1392791071004762,
-1.7031247999122032,
-0.12594303016721067,
-1.552828751326619,
0.7484033373249843,
-1.627916556868976,
-1.4613739383316358,
1.6030230459146289,
-1.4549302634648913,
1.5145658855223556],
[0.00014920185539273834,
0.9238380862195603,
1.1653356841767,
1.167832939600511,
1.2191691700431773,
1.2210447233093105,
1.9314710919546132,
2.0244898268357567,
2.025055317728332,
2.025055317755535,
2.035108012828863,
2.0351100169574385,
2.250256137680943,
3.15082115258344,
3.246453048346403,
3.2464530484741454,
3.2616906588279586,
3.267255104894572,
4.694356606797461,
4.6943575411597775,
4.694357544022533,
6.488320294504503,
6.488320350076187,
7.5482403718064734,
7.549174906557876,
7.567191669993812,
7.5932047752247485,
7.6539061864591025,
9.752212018804654,
9.752227380423289,
-1.324607236830295,
1.9077202588182536,
-0.8802070447756194,
-2.0149948315634574,
1.4533735387069473,
0.997568141525462,
-0.7074886190930807,
-0.2560389197368321,
0.4012267462832038,
1.558634946129638,
0.4929137086923578,
-1.110627649535145,

0.02522612974783881,
0.9667142992980035,
-0.5237885436922995,
0.4478493220686497,
1.4081655138312377,
-0.6604234389026553,
0.8156393803790498,
0.9162727880779341,
1.2376705325338233,
0.8905837956479322,
-0.7295152882741086,
0.24561292878662663,
1.7665604379212652,
0.17703116950798709,
0.4931954498011207,
1.0631695242763428,
0.5506819823950141,
0.49574385725394715,
-0.9067001814641317,
0.03244215799947241,
1.5112600834704966,
-0.08582031254909145,
0.04823393263399211,
-0.1471510424202064,
1.1677829257421761,
-0.4795697461798601,
-0.4872565354105511,
0.22168062822432655,
1.4072421499069951,
-0.811680819103198,
1.651055088666158,
-0.7313303670161658,
0.5404312773997829,
-0.3816094570928806,
0.5640067451570394,
0.6555156836542061,
-0.09410372652444776,
0.844617621120923,
1.4181988757356223,
0.6157327018229515,
-1.8426182279645893,
0.7475476733327415,
0.9297363986538332,
1.5219888966448256,
0.12784560574897466,
1.3383878758441214,
-0.3849331170513322,
-2.5759820228897086,
-0.44171402566963375,
-1.1481005382735594,
-0.11732003654377818,
1.4261012490056857,
0.37374285139811464,
1.3264125331467558,
-0.3101118867274816,
-0.5580221641216343,
-0.8733903562168412,
0.763630147907968,
-0.22407005303761457,
-0.4427553244179856,
0.25630540825178894,
-0.9744141539121856,
0.7035909537131968,
-1.1511448415135326,
-0.02706398350090099,
-1.4364455854835774,
1.760537605923817,
0.08770997893488061,
-0.4257007958221735,
0.8079796656424563,
0.0842113896113966,
0.43919545053894077,
1.4638006778586652,
-0.01014009539920246,
-1.8469337659391836,

-1.3494797221531116,
0.7904481242840491,
1.4234316624418009,
-1.4054528119805059,
-0.041414215577158846,
0.7865492524837541,
0.7547742563728446,
0.4892505595197172,
-0.7646468207281831,
-1.2841992554361892,
0.014780110538174546,
0.800978697085001,
0.938237883807168,
-0.43390682676117964,
-0.488905643723897,
-1.7892652616586333,
2.0824770359325453,
0.6465813428295442,
0.813778987942834,
-0.4618041833296182,
0.5940671128875871,
-0.49307011143397783,
0.652155769110408,
-2.662957327611224,
1.0258915732895864,
-0.8259876341171806,
0.023143897149311463,
1.3124040163708688,
2.6574917189601317,
-0.6227996400692593,
-0.539154106968482,
1.702015600845961,
0.03237978965554299],
[0.21389598228651574,
0.5393241017358461,
1.336052351787059,
1.4548884734431182,
1.4631515545587506,
2.2996776596727213,
2.303486102556458,
3.672114740986732,
3.738311693108453,
4.172103126491601,
4.185637203208781,
4.879786029287921,
4.879786029287921,
5.514641520665522,
6.187890324824944,
6.199874857261559,
6.337185103245036,
6.337673866716393,
8.339356224610325,
8.355342425303002,
8.376069528482942,
8.376069528482942,
8.376069528482942,
8.376069528482942,
8.405896742002076,
8.450718053610304,
8.463555902040309,
10.004417530570722,
10.004417530570722,
11.277411014627386,
-0.4550197171928084,
0.8726389677058429,
0.32380330098542204,
-0.9102552663335266,
-0.3566919533881965,
1.616361456517712,
0.17449327037699397,
-1.1936385630270505,
-0.3626977992670382,
0.6065214216445567,
-0.27266827511951447,
-0.008337250595426905,

-0.11442883582944025,
-0.2552879859027531,
0.8608200568301817,
0.9797863499294986,
-0.5454704270163543,
-0.12465088219091108,
1.0680711786011012,
0.008287966764571728,
0.8340806285400814,
-0.10467902667059636,
0.6944226087871278,
-1.1848583772718688,
-2.1289584360902833,
2.0815012423116896,
0.7888919711742388,
-0.12601348190113199,
0.5814024197737417,
-0.32142652665182836,
-1.0761996135691065,
1.1437448939093156,
-0.9665568900718753,
-0.751505855358832,
-0.8777599880692168,
-0.0665611067528402,
1.625406556377903,
0.14073065902471577,
-0.008887426812216265,
0.11314595138478645,
0.23282575925386234,
-1.3029158546977586,
0.39815363644872126,
-0.28006162282581953,
0.5981370714824289,
-0.08964177383705696,
-0.23973482016502778,
-1.1546366940781616,
-0.3335643458249379,
1.4313780024296703,
0.6615033850837366,
0.09540664540173596,
1.6701354164263156,
1.6037180066269185,
0.45366725021426957,
1.5306014722736203,
-0.7114276522073563,
-0.0387128408482294,
-0.40951463995726844,
-0.14821428534018466,
1.6120880297451547,
-0.1578213810367529,
-0.4551176093567465,
0.12643398042842974,
0.29421606580526033,
0.3517029135915254,
-0.21200478261515812,
1.503225888163223,
0.6867008326592696,
1.2990703939363029,
-1.2633831104757771,
-0.35212753263373886,
0.2558013627761136,
0.1557716944165696,
-0.2985274551715097,
0.3096340969306992,
1.1065921140021713,
0.5655703287990456,
0.29407543941401004,
1.87556201451638,
-0.005713899736395531,
-0.47654203677386653,
-0.33487139893703466,
0.28841760275861744,
0.881872547749141,
-0.2998307941218902,
1.3565987475579786,

```

0.39905124343442133,
0.37529991980433175,
0.4721044202054456,
-0.0837323332258721,
0.2128419559212174,
1.445893331967549,
-0.5917542945785007,
-0.33453435504213846,
-0.05724156447138151,
0.5825344314121998,
0.05851962203923921,
0.13841145669138022,
-0.07195386411014011,
1.546635282716052,
-0.3465571437275552,
0.9484886642506297,
0.41941222891459207,
0.0901808762528231,
-0.8871255799219674,
-0.4695342320219668,
0.6930505657922351,
1.1594181773830308,
0.042534590889293046,
0.45776921806648474,
0.8831536077453052,
-1.883349137185357,
2.293435211334673,
1.3066760169202165,
-0.040640374761399124,
-0.8716978424494161,
1.5544327319775963,
0.5977489456678642,
-1.5697728131838606]],
'run_time_sec': 35225.583796978}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
                                     'layer': 30,
                                     'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                                    'cn': [ 0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5,
                                                                    0.5],
                                                                    'r': 0},
                                     'time': {'max': 10.0, 'min': 0.0},
                                     'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.0780449993030362,
                           0.10783439075300198,
                           0.5845642021744096,
                           0.7814268466054133,
                           0.8443273792280553,
                           2.8661173620125435,
                           2.866459808493013,
                           2.866459808493013,
                           2.866459808493013,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.787331067871434,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           7.013227786440474,
                           7.016905064536492,
                           8.483022393696176,
                           8.483022393696176,
                           8.483022393696176,
                           8.483022393696176,
                           10.30807680489275,
                           10.30807680489275,

```


10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,

```

-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```


12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,

```

-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,
0.0475457555114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noise_free_time_evo_dephasing_ric2_sample#4'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```


12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,

```

-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```


10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,

```

0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```


12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,
-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,

[illegible]

```

'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.524810336842993],
                             [12, 6, 0, 0, -6.145141411469215]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.0,
                                                                                       0.0],
                                                                                       'status': True,
                                                                                       'type': 'dephasing'},
                              'odd_wires': 4},
                              { 'gates_num': [12, 3, 1, 0],
                              'identity_factors': [1, 1, 0, 0],
                              'noise_level': [12, 6, 0, 0],
                              'noise_profile': { 'noise_on_init_param': { 'status':
                                                                                       'value':
                                                                                       'noise_prob': [ 0.001,
                                                                                       0.001,
                                                                                       0.0,
                                                                                       0.0],
                                                                                       'status': True,
                                                                                       'type': 'dephasing'},
                              'odd_wires': 4}],
                              'run_time_sec': 5.684312105178833}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.1021834784589592,
                             0.9196384424899031,
                             0.9261694014921312,
                             0.983882928321485,
                             1.8913212439431486,
                             1.9426406276748993,
                             1.9426406276748993,
                             1.9426406276748993,
                             2.7354895381565307,
                             2.7475159984062567,
                             4.89472071800884,
                             4.89472071800884,
                             4.89472071800884,
                             4.951228136281907,
                             4.9774006951115535,
                             4.9774006951115535,
                             5.053403895048273,
                             5.053403895048273,
                             5.101252532365191,
                             7.988961801418946,
                             7.988961801418945,
                             7.988961801418946,
                             7.988961801418946,
                             8.978507898154866,
                             8.993364745936525,
                             9.010251752506093,
                             9.010251752506093,
                             9.808175975385067,
                             9.808175975385067,

```

10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,

[illegible]

```

'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.588052513021028],
                             [12, 6, 0, 0, -6.143369929824417]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [0, 0, 0, 0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                            'status': True,
                                                            'type': 'dephasing'},
                       'odd_wires': 4},
                       { 'gates_num': [12, 3, 1, 0],
                         'identity_factors': [1, 1, 0, 0],
                         'noise_level': [12, 6, 0, 0],
                         'noise_profile': { 'noise_on_init_param': { 'status':
                                                                      'value':
                                                                      'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                            'status': True,
                                                            'type': 'dephasing'},
                         'odd_wires': 4}],
                       'run_time_sec': 7.301640748977661}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.009858867127598406,
                             0.6371504690778502,
                             1.0666846377193393,
                             1.104415373659906,
                             1.464838116920491,
                             1.4648381169204907,
                             1.9080243917751187,
                             2.1981829040528402,
                             2.6297267537087063,
                             2.658272884291024,
                             3.752650276988564,
                             3.752985814574736,
                             4.407595284592648,
                             4.407595284592648,
                             5.031509128956631,
                             5.031509128956631,
                             5.031509128956631,
                             5.033233798634394,
                             5.033233798634395,
                             5.036830700020435,
                             5.036830700020434,
                             7.933675292188165,
                             7.950623161195741,
                             7.950659720589115,
                             8.023568123130133,
                             8.309873389450049,
                             8.310972029308223,
                             8.959490059358307,
                             9.904746541347446,

```

9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,

```

-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```


10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,

[illegible]

```

'0.5 X 3 X 4 + 0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + 0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],
                             [12, 6, 0, 0, -6.098488474706269]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [0, 0, 0, 0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                            'status': True,
                                                            'type': 'dephasing'},
                       'odd_wires': 4},
                       { 'gates_num': [12, 3, 1, 0],
                         'identity_factors': [1, 1, 0, 0],
                         'noise_level': [12, 6, 0, 0],
                         'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                            'status': True,
                                                            'type': 'dephasing'},
                         'odd_wires': 4}],
                       'run_time_sec': 7.421908140182495}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.08803864939944098,
                             0.2533857257734638,
                             0.4067196832055002,
                             0.4411063797089926,
                             1.0492461473989636,
                             1.0543168722856693,
                             2.21020815530381,
                             2.218381560687012,
                             2.367234970755274,
                             2.658389587322544,
                             3.2990226425967433,
                             3.5549209440787726,
                             4.0112101619321985,
                             4.138810393646779,
                             4.424339957187965,
                             6.02518256832302,
                             6.064174786863594,
                             6.069208430366886,
                             6.069208430366886,
                             6.941565801297405,
                             7.124864639242438,
                             7.129421087787242,
                             7.45818921038985,
                             7.468062221399231,
                             8.693928169186727,
                             8.75723930487613,
                             8.77623189944677,
                             8.812275470753457,
                             8.81486926281639,

```


11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,

```

-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': None,
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + '
'1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X '
'5 X 6 + 1 Z 5 + 1 Z 6',
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + '
'0.5 X 1 X 2 + 0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + 0.5 Z 2 Z 3 + '

```

$\emptyset\},$

11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,

[illegible]

```

'output': { 'data_points': [ [4, 2, 0, 0, -7.597273562260126],
                             [12, 6, 0, 0, -6.262099522076846]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                       'identity_factors': [0, 0, 0, 0],
                       'noise_level': [4, 2, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                                    'status': True,
                                                                    'type': 'dephasing'},
                       'odd_wires': 4},
                     { 'gates_num': [12, 3, 1, 0],
                       'identity_factors': [1, 1, 0, 0],
                       'noise_level': [12, 6, 0, 0],
                       'noise_profile': { 'noise_on_init_param': { 'status':
                                                                    'value':
                                                                    'noise_prob': [ 0.001,
                                                            0.001,
                                                            0.0,
                                                            0.0],
                                                                    'status': True,
                                                                    'type': 'dephasing'},
                       'odd_wires': 4}],
  'run_time_sec': 7.239971399307251}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                   'layer': 30,
                                   'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 0},
                                   'time': {'max': 10.0, 'min': 0.0},
                                   'type': 'xy-iss'},
                                   'init_param': {'value': 'random'},
                                   'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                                                     'noise_prob': [0.001, 0.001, 0.0, 0.0],
                                                     'status': True,
                                                     'type': 'dephasing'},
                                   'nqubits': 7,
                                   'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                                                'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                                'r': 1},
                                   'def': 'ising'},
                                   'output': { 'draw': {'fig_dpi': 100, 'status': True, 'type': 'png'},
                                               'file_name_prefix': 'xy_noise_free_time_evo'},
                                   'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
                                   'run': 'vqe',
                                   'state': 'dmatrix',
                                   'vqe': { 'iteration': 10,
                                           'optimization': { 'algorithm': 'SLSQP',
                                                             'constraint': True,
                                                             'status': True}},
                                   'zne': { 'data_points': None,
                                           'degree': 1,
                                           'method': 'richardson',
                                           'sampling': 'default'}},
  'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 X 2 X 3 + 1 Z 2 '
                                   '+ 1 X 3 X 4 + 1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z '
                                   '5 + 1 Z 6',
  'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z 0 Z 1 + 0.5 X 1 '
                                              'X 2 + 0.5 Z 1 Z 2 + 0.5 X 2 X 3 + '
                                              '0.5 Z 2 Z 3 + 0.5 X 3 X 4 + 0.5 Z 3 '
                                              'Z 4 + 0.5 X 4 X 5 + 0.5 Z 4 Z 5 + '
                                              '0.5 X 5 X 6 + 0.5 Z 5 Z 6']]],
'output': { 'exact_sol': -8.56677223350563,
  'initial_cost_history': [ 2.490874640234538,
                           3.123425159431485,

```

```

3.646429251689861,
2.764268013304591,
3.17022836931859,
3.7735896776807554,
3.622216535636285,
2.438144835449103,
3.477758051251201,
4.254627215438117],
'noise_details': { 'gates_num': [4, 1, 1, 0],
'identity_factors': [0, 0, 0, 0],
'noise_level': [4, 2, 0, 0],
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -7.597273562260196,
-7.556050750105993,
-7.517540187131898,
-7.665909699744024,
-7.524810336843006,
-7.40701382554743,
-7.601751527683981,
-7.561979391804892,
-7.59036798324113,
-7.588052513021082],
'optimized_parameters': [ [ 0.00487815438629808,
0.3836991336608517,
0.3836991336608516,
1.1027183519022283,
1.430822963194853,
1.7336849694382164,
1.9079110557306378,
1.9079110557306376,
2.407942958541018,
2.4384441996231097,
2.7198331200698176,
2.7363529901218957,
3.8707803400773457,
3.8707803400773457,
4.283774948106429,
4.283774948106428,
4.283774948106429,
4.283774948106429,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,

```

-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,

1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559],
[0.0780449993030362,
0.10783439075300198,
0.5845642021744096,
0.7814268466054133,
0.8443273792280553,
2.8661173620125435,
2.866459808493013,
2.866459808493013,
2.866459808493013,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.787331067871434,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
7.013227786440474,
7.016905064536492,
8.483022393696176,
8.483022393696176,
8.483022393696176,
8.483022393696176,
10.30807680489275,
10.30807680489275,
10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,

0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,

-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892],
[2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,
8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,

-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,

0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109],
[0.8953934585022342,
1.0441540460805852,
1.2122882144067701,
1.3902546362390118,
2.4452392605971704,
2.4476376114298906,
3.59167707372567,
3.592168933497119,
4.752173032506677,
4.770483368888809,
5.226089600509016,
5.227272839220446,
5.277877376458118,
6.286147534970271,
6.330458303415314,
7.343951335722083,
7.407616509271585,
7.5311385683234535,
7.532344729415073,
7.683350056306,
7.684095349251904,
8.372928051556965,
8.450880672852195,
8.893521328096941,
8.933105958948394,
10.009729102620259,
10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.4073492906335524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,

1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,

-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.4018894448333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684],
[1.0344655484192336e-17,
0.36119418156962224,
0.3611941815696223,
0.36119418156962224,
0.8440759344131337,
1.1727045787919286,
1.5692279974524073,
1.570888864611923,
1.592214192172409,
1.7598342563277258,
3.059056443368036,
3.0591149843524144,
4.3676699123871625,
4.3676699123871625,
5.510828301667605,
5.51914482396091,
6.113847931320903,
6.113847931320903,
6.533178927747682,
6.533178927747682,
6.755940863606281,
6.837142696961756,
7.694830693654466,
7.694830693654467,
7.694830693654466,
7.873525558910739,
7.873525558910739,
7.922034998277628,
7.922034998277628,
12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,

-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,

1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,
0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,
-0.2901105978964291,
-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,
1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593],
[0.009858867127598406,
0.6371504690778502,
1.0666846377193393,
1.104415373659906,
1.464838116920491,
1.4648381169204907,
1.9080243917751187,
2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,

-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,

0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012],
[0.08803864939944098,
0.2533857257734638,
0.4067196832055002,
0.4411063797089926,
1.0492461473989636,
1.0543168722856693,
2.21020815530381,
2.218381560687012,
2.367234970755274,
2.658389587322544,
3.2990226425967433,
3.5549209440787726,
4.0112101619321985,
4.138810393646779,
4.424339957187965,
6.02518256832302,
6.064174786863594,
6.069208430366886,
6.069208430366886,
6.941565801297405,
7.124864639242438,
7.129421087787242,
7.45818921038985,
7.468062221399231,
8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,

0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,

1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548],
[0.029287487040924362,
0.6764270956551205,
1.6620807976810363,
1.7056653351603042,
1.8485187197716686,
1.9397654142515774,
1.9426163313124116,
1.9760071204347747,
2.0652508647096677,
3.338212066261205,
3.3382549495561102,
3.507308503786301,
3.668597772988933,
3.6686655840136484,
3.7105300968471533,
3.737148430160667,
5.361211694679849,
5.40430867517309,
5.45325014669751,
5.461795374069968,
5.465563084002408,
5.492346359399902,
7.18245326721486,
7.245488992311329,
7.248698529808956,
8.18726919118503,
8.197326336426727,
8.213328813441153,
8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,

-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,

-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623],
[0.25473970784448047,
1.2025511746035562,
1.5481560170840756,
1.8324395977107744,
1.8439662803769572,
2.9830338898217192,
2.9830338898217192,
3.241693555484232,
3.634822278674692,
3.7522709686014135,
3.7522709686014135,
4.94705318367592,
4.955555446952993,
5.343973550433924,
5.3577622539956655,
6.390003257794682,
6.419263094824833,
8.039405153723413,
8.039804264859526,
8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,

-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,

1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956],
[0.1021834784589592,
0.9196384424899031,
0.9261694014921312,
0.983882928321485,
1.8913212439431486,
1.9426406276748993,
1.9426406276748993,
1.9426406276748993,
2.7354895381565307,
2.7475159984062567,
4.89472071800884,
4.89472071800884,
4.89472071800884,
4.951228136281907,
4.9774006951115535,
4.9774006951115535,
5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,

-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,

```

0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,
0.028627573515451674,
1.9279053503307635,
0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'run_time_sec': 42239.92721557617}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
'init_param': { 'value': [ 0.1021834784589592,
0.9196384424899031,
0.9261694014921312,
0.983882928321485,
1.8913212439431486,
1.9426406276748993,
1.9426406276748993,
1.9426406276748993,
2.7354895381565307,
2.7475159984062567,
4.89472071800884,
4.89472071800884,
4.89472071800884,
4.951228136281907,
4.9774006951115535,
4.9774006951115535,
5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,

```

0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,

```

1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,
0.028627573515451674,
1.9279053503307635,
0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#10'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.588052513021028],
                          [12, 6, 0, 0, -6.143369929824417]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.6716461181640625e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.310393804619334,
                           'others': { 'beta_coeffieicients': [1.5, -0.5],
                                        'cost_richardson_zne': 2.5,
                                        'sorted_expectation_vals': [ -7.588052513021028,
                                                                    -6.143369929824417],
                                        'sorted_noise': [6, 18]},
                           'sampled data': [ [4, 2, 0, 0, -7.588052513021028],
                                              [12, 6, 0, 0, -6.143369929824417]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],

```

```
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0780449993030362,
    0.10783439075300198,
    0.5845642021744096,
    0.7814268466054133,
    0.8443273792280553,
    2.8661173620125435,
    2.866459808493013,
    2.866459808493013,
    2.866459808493013,
    4.786930144034059,
    4.786930144034059,
    4.786930144034059,
    4.786930144034059,
    4.786930144034059,
    4.786930144034059,
    4.787331067871434,
    6.292785833628836,
    6.292785833628836,
    6.292785833628836,
    6.292785833628836,
    6.292785833628836,
    7.013227786440474,
    7.016905064536492,
    8.483022393696176,
    8.483022393696176,
    8.483022393696176,
    8.483022393696176,
    10.30807680489275,
    10.30807680489275,
    10.987173174448158,
    -0.09982429964200479,
    -0.1011988902651152,
    -0.11924685050397593,
    0.11938936982148393,
    0.26480809900884805,
    0.9860195530322333,
    0.32479711291812374,
    -0.24554677857208176,
    -0.049000425450729294,
    2.064341791548283,
    -0.10644000592090332,
    1.3304809249936382,
    0.46278363868645017,
    0.14042287528230984,
    0.2133289095055383,
    -0.6423967247784159,
    -0.42996733188660946,
    -0.3318304058008573,
    0.6204697000624867,
    -0.5183204366214588,
    0.6244473440442605,
    0.0183714264553683,
    -0.3522016603760132,
    -0.9892342164188427,
    -1.502610568052502,
    0.3280599415277172,
    -0.48553815029820807,
    -0.010371641322887242,
    1.573878039413477,
    -0.020575326946359336,
    -0.14841518088300737,
    0.26016829146348547,
    0.15593168727347026,
    -0.5856941876551979,
    0.12654079580108715,
    -0.007993830462140899,
    0.5218497886375917,
    1.4495850566219788,
    -1.1997007030306313,
    0.005867405286106471,
    -1.158661564042903,
    0.7480115816132973,
```

0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,

```

1.5813078296578018,
0.45463950449288687,
1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                  'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                  'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#2'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.556050750105972],
                          [12, 6, 0, 0, -6.124528497687962]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 2.7179718017578125e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.271811876314977,
                          'others': { 'beta_coeffieicients': [1.5, -0.5],
                                      'cost_richardson_zne': 2.5,
                                      'sorted_expectation_vals': [ -7.556050750105972,
                                                                    -6.124528497687962],
                                      'sorted_noise': [6, 18]},
                          'sampled data': [ [4, 2, 0, 0, -7.556050750105972],
                                              [12, 6, 0, 0, -6.124528497687962]],
                          'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                    'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                    'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.009858867127598406,
                             0.6371504690778502,
                             1.0666846377193393,
                             1.104415373659906,
                             1.464838116920491,
                             1.4648381169204907,
                             1.9080243917751187,
                             2.1981829040528402,
                             2.6297267537087063,
                             2.658272884291024,
                             3.752650276988564,
                             3.752985814574736,
                             4.407595284592648,
                             4.407595284592648,
                             5.031509128956631,
                             5.031509128956631,
                             5.031509128956631,
                             5.033233798634394,
                             5.033233798634395,
                             5.036830700020435,
                             5.036830700020434,
                             7.933675292188165,
                             7.950623161195741,
                             7.950659720589115,
                             8.023568123130133,
                             8.309873389450049,
                             8.310972029308223,
                             8.959490059358307,
                             9.904746541347446,

```


9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,

```

-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#6'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.407013825547437],
                          [12, 6, 0, 0, -6.114813208156221]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 0.00011587142944335938,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.053114134243044,
                           'others': { 'beta_coeffieicients': [1.5, -0.5],

```

```

        'cost_richardson_zne': 2.5,
        'sorted_expectation_vals': [ -7.407013825547437,
                                      -6.114813208156221],
        'sorted_noise': [6, 18]],
    'sampled data': [ [4, 2, 0, 0, -7.407013825547437],
                      [12, 6, 0, 0, -6.114813208156221]],
    'sampling': 'default']},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                      'time': {'max': 10.0, 'min': 0.0},
                                      'type': 'xy-iss'}},
  'init_param': { 'value': [ 1.0344655484192336e-17,
                             0.36119418156962224,
                             0.3611941815696223,
                             0.36119418156962224,
                             0.8440759344131337,
                             1.1727045787919286,
                             1.5692279974524073,
                             1.570888864611923,
                             1.592214192172409,
                             1.7598342563277258,
                             3.059056443368036,
                             3.0591149843524144,
                             4.3676699123871625,
                             4.3676699123871625,
                             5.510828301667605,
                             5.51914482396091,
                             6.113847931320903,
                             6.113847931320903,
                             6.533178927747682,
                             6.533178927747682,
                             6.755940863606281,
                             6.837142696961756,
                             7.694830693654466,
                             7.694830693654467,
                             7.694830693654466,
                             7.873525558910739,
                             7.873525558910739,
                             7.922034998277628,
                             7.922034998277628,
                             12.310276979046629,
                             -0.6972922025249428,
                             -0.0508578639309505,
                             0.03511585194544882,
                             -0.05290285417260422,
                             0.6885525491076263,
                             0.23164759975262209,
                             0.06930481892124933,
                             -0.05051704727786026,
                             0.2745998381310896,
                             2.8540854853592244,
                             0.1762956340846527,
                             -0.46176266603345323,
                             0.4487337371763968,
                             1.96119949254268,
                             -0.028430732456242037,
                             1.020183549895116,
                             0.09052188212177692,
                             0.6283703459809612,
                             -0.2008669437139055,
                             -0.333213682394554,
                             0.49298533108167947,
                             -0.073860646808665,
                             -0.13593481376514907,
                             -0.597217576988091,
                             0.38422303935103047,
                             -0.1405570093523461,
                             -0.17552659439894247,
                             0.802391437181995,
                             0.41745069672968155,
                             -0.3651927864609047,
                             0.40673923582512855,

```

1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,
0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,
-0.2901105978964291,

```

-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,
1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#5'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.524810336842993],
[12, 6, 0, 0, -6.145141411469215]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 3.910064697265625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.214644799529882,
'others': { 'beta_coeffieicients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.524810336842993,
-6.145141411469215],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.524810336842993],
[12, 6, 0, 0, -6.145141411469215]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00487815438629808,
0.3836991336608517,
0.3836991336608516,
1.1027183519022283,
1.430822963194853,
1.7336849694382164,
1.9079110557306378,
1.9079110557306376,
2.407942958541018,
2.4384441996231097,
2.7198331200698176,
2.7363529901218957,
3.8707803400773457,
3.8707803400773457,
4.283774948106429,
4.283774948106428,
4.283774948106429,
4.283774948106429,
4.283774948106429,

```

5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,

```

0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#1'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```


0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,

```

-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
'noise_prob': [0.001, 0.001, 0.0, 0.0],
'status': True,
'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#3'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.517540187131846],
[12, 6, 0, 0, -5.995430553311076]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.9591064453125e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.27859500404223,
'others': { 'beta_coefficients': [1.5, -0.5],
'cost_richardson_zne': 2.5,
'sorted_expectation_vals': [ -7.517540187131846,
-5.995430553311076],
'sorted_noise': [6, 18]},
'sampled data': [ [4, 2, 0, 0, -7.517540187131846],
[12, 6, 0, 0, -5.995430553311076]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.8953934585022342,
1.0441540460805852,
1.2122882144067701,
1.3902546362390118,
2.4452392605971704,
2.4476376114298906,
3.59167707372567,

```

3.592168933497119,
4.752173032506677,
4.770483368888809,
5.226089600509016,
5.227272839220446,
5.277877376458118,
6.286147534970271,
6.330458303415314,
7.343951335722083,
7.407616509271585,
7.5311385683234535,
7.532344729415073,
7.683350056306,
7.684095349251904,
8.372928051556965,
8.450880672852195,
8.893521328096941,
8.933105958948394,
10.009729102620259,
10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,

```

1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],

```

```

        'r': 1},
        'def': 'ising'},
    'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
                'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#4'},
    'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
    'run': 'zne',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
              'optimization': { 'algorithm': 'SLSQP',
                                'constraint': True,
                                'status': False}},
    'zne': { 'data_points': [ [4, 2, 0, 0, -7.665909699744034],
                              [12, 6, 0, 0, -6.335722541155832]],
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
    'output': { 'exact_sol': -8.566772233505624,
                'run_time_sec': 3.218650817871094e-05,
                'zne_values': { 'degree': 0,
                                'extrapolated_value': -8.331003279038136,
                                'others': { 'beta_coefficients': [1.5, -0.5],
                                              'cost_richardson_zne': 2.5,
                                              'sorted_expectation_vals': [ -7.665909699744034,
                                                                           -6.335722541155832],
                                              'sorted_noise': [6, 18]},
                                'sampled data': [ [4, 2, 0, 0, -7.665909699744034],
                                                  [12, 6, 0, 0, -6.335722541155832]],
                                'sampling': 'default'}}},
    { 'config': { 'ansatz': { 'gateset': 1,
                              'layer': 30,
                              'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0, 0],
                                                            'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                            'r': 0},
                                          'time': {'max': 10.0, 'min': 0.0},
                                          'type': 'xy-iss'}}},
      'init_param': { 'value': [ 0.08803864939944098,
                                0.2533857257734638,
                                0.4067196832055002,
                                0.4411063797089926,
                                1.0492461473989636,
                                1.0543168722856693,
                                2.21020815530381,
                                2.218381560687012,
                                2.367234970755274,
                                2.658389587322544,
                                3.2990226425967433,
                                3.5549209440787726,
                                4.0112101619321985,
                                4.138810393646779,
                                4.424339957187965,
                                6.02518256832302,
                                6.064174786863594,
                                6.069208430366886,
                                6.069208430366886,
                                6.941565801297405,
                                7.124864639242438,
                                7.129421087787242,
                                7.45818921038985,
                                7.468062221399231,
                                8.693928169186727,
                                8.75723930487613,
                                8.77623189944677,
                                8.812275470753457,
                                8.81486926281639,
                                11.019616289520474,
                                -0.611476740771433,
                                0.05158007641927867,
                                0.011569665901033028,
                                -0.054784977759570794,
                                0.634383931417704,
                                0.04908784681904958,
                                0.14144925064556785,
                                0.11185153756258273,
                                -0.11524892510531,

```

1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,

```

-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#7'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.6017515276839065],
                          [12, 6, 0, 0, -6.187268174258091]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.933906555175781e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.308993204396815,
                           'others': { 'beta_coefficients': [1.5, -0.5],
                                       'cost_richardson_zne': 2.5,
                                       'sorted_expectation_vals': [ -7.6017515276839065,
                                                                    -6.187268174258091],
                                       'sorted_noise': [6, 18]},
                           'sampled data': [ [4, 2, 0, 0, -7.6017515276839065],
                                              [12, 6, 0, 0, -6.187268174258091]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],

```

```
        'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
        'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.25473970784448047,
    1.2025511746035562,
    1.5481560170840756,
    1.8324395977107744,
    1.8439662803769572,
    2.9830338898217192,
    2.9830338898217192,
    3.241693555484232,
    3.634822278674692,
    3.7522709686014135,
    3.7522709686014135,
    4.94705318367592,
    4.955555446952993,
    5.343973550433924,
    5.3577622539956655,
    6.390003257794682,
    6.419263094824833,
    8.039405153723413,
    8.039804264859526,
    8.039804264859526,
    8.097462604506834,
    8.099750106093039,
    8.685653778392616,
    8.685653778392616,
    8.685653778392616,
    8.686674341764832,
    8.842213669739554,
    10.173486136133235,
    10.173486136133235,
    12.284287827852737,
    -0.5275758730734342,
    0.9066548612174815,
    0.21500588132446097,
    -1.0687833608508335,
    0.6112018055607131,
    1.4602447412527104,
    0.5461149477957931,
    0.7262216027427579,
    0.29207423930862947,
    1.0780988617567295,
    0.030019770603227844,
    0.6811845529236524,
    0.16899250416157743,
    0.3817265788697317,
    0.8504511882214915,
    0.2943749516217996,
    2.5678685545322057,
    -0.09707287225998511,
    -0.05255145791734958,
    0.2596628656770196,
    1.4709186207405707,
    0.36806659135242686,
    1.1820332803522335,
    -1.8688297897861523,
    -0.023758258884169974,
    -0.1539594509982686,
    0.8699600311365924,
    0.026062392574156816,
    2.0579809580202704,
    0.36704866549095816,
    -0.42819223169421283,
    -0.5093566943293824,
    -1.9075487684328258,
    -1.3810384294539255,
    -1.5948684983369932,
    0.9548136506191117,
    0.48748532474959005,
    -0.002608217350405503,
    -0.6436639802673768,
    -0.33250098360488384,
    -2.1988529165637916,
```


-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.895038777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,

```

1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#9'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.590367983241113],
                          [12, 6, 0, 0, -6.1449349379730975]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 5.173683166503906e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.313084505875121,
                           'others': { 'beta_coefficients': [1.5, -0.5],
                                       'cost_richardson_zne': 2.5,
                                       'sorted_expectation_vals': [ -7.590367983241113,
                                                                    -6.1449349379730975],
                                       'sorted_noise': [6, 18]},
                           'sampled data': [ [4, 2, 0, 0, -7.590367983241113],
                                              [12, 6, 0, 0, -6.1449349379730975]],
                           'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [0, 0, 0, 0, 0, 0, 0],
                                                         'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                                         'r': 0},
                                    'time': {'max': 10.0, 'min': 0.0},
                                    'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.029287487040924362,
                             0.6764270956551205,
                             1.6620807976810363,
                             1.7056653351603042,
                             1.8485187197716686,
                             1.9397654142515774,
                             1.9426163313124116,
                             1.9760071204347747,
                             2.0652508647096677,
                             3.338212066261205,
                             3.3382549495561102,
                             3.507308503786301,
                             3.668597772988933,
                             3.6686655840136484,
                             3.7105300968471533,
                             3.737148430160667,
                             5.361211694679849,
                             5.40430867517309,
                             5.45325014669751,
                             5.461795374069968,
                             5.465563084002408,
                             5.492346359399902,
                             7.18245326721486,
                             7.245488992311329,
                             7.248698529808956,
                             8.18726919118503,
                             8.197326336426727,
                             8.213328813441153,

```

8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,

```

-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623]],
'noise_profile': { 'noise_on_init_param': {'status': False, 'value': 0},
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
                    'status': True,
                    'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0],
                                   'cn': [0.5, 0.5, 0.5, 0.5, 0.5, 0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': {'fig_dpi': 100, 'status': False, 'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric2_sample#8'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],
                          [12, 6, 0, 0, -6.098488474706269]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.127357482910156e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.293724850354089,

```

[illegible]

```

'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']],
'output': { 'exact_sol': -8.56677223350563,
  'initial_cost_history': [ 4.032700960953626,
    2.876037164175563,
    3.024801381527762,
    2.6026187891538575,
    2.558855794921029,
    3.233596797235869,
    4.0713546948778445,
    4.434976242432684,
    2.895971800514176,
    2.6461223700347904],
  'noise_details': { 'gates_num': [4, 1, 1, 0],
    'identity_factors': [0, 0, 0, 0],
    'noise_level': [0, 0, 0, 0],
    'noise_profile': { 'noise_on_init_param':

'noise_prob': [ 0.001,
  0.001,
  0.0,
  0.0],
  'status': False,
  'type': 'dephasing'},
  'odd_wires': 4},
  'optimized_minimum_cost': [ -8.213984832683868,
    -8.467734328099501,
    -8.447208232910596,
    -8.461814133720251,
    -8.483392807973491,
    -8.44723907770434,
    -8.362559674056595,
    -8.366604609651855,
    -8.163432600733032,
    -8.478925857587775],
  'optimized_parameters': [ [ 0.5489081084990874,
    0.6882459728440448,
    0.7729468967125317,
    0.8347489553606998,
    1.0946318761773612,
    1.9534300183374533,
    2.2829395447313403,
    3.232524496500775,
    3.2325244965209894,
    3.852765644418336,
    3.8527656444207725,
    3.8527673900689967,
    3.8860323667767283,
    4.588361079261242,
    4.600166894773711,
    5.527977091209439,
    5.556427027819143,
    5.803787289292802,
    6.218904325824693,
    6.239985329573426,
    6.538211674407394,
    6.925082879783563,
    7.640591161405884,
    7.75471653665469,
    7.896761387122529,
    8.932525392602521,
    9.128535074996934,
    9.925693656125251,
    9.925693656687802,
    11.037683520466203,
    -0.10948884769018298,
    0.012576058394267018,
    0.9054362612114637,
    -0.5027178523290218,
    1.9293813165204678,
    0.12560680736570712,

```

0.40022822712854333,
-2.471918351448183,
0.5257346934244156,
1.1130547436006022,
1.0771121143603033,
-0.1314162667650016,
0.02294938233855663,
0.36225076948659746,
-1.3206473358532678,
-0.2449807752862754,
0.29890651347484976,
1.0095017101029446,
0.6150854645533661,
0.2815584721561816,
0.3601395589466358,
-0.5388375319509757,
0.9923143444796582,
0.8473755438313292,
-0.4646214972755992,
0.8527825907455423,
-0.6098121229644526,
0.12413023074624807,
-0.06044402625536853,
-0.29258159286196256,
1.3610999143074116,
0.09999631572477011,
-0.18981827210653637,
0.2696177153747519,
0.4272428992844361,
-0.26705761267067063,
-0.37988713130471985,
0.75645724651605,
0.8109920260438386,
-1.028544143258335,
0.44106157371844235,
-1.6090649394170073,
-0.44600474997528494,
-0.2579853712948814,
0.3063055059682419,
0.9497745366960584,
-0.2701835817775974,
0.9807489498405011,
-0.507582861688708,
-1.25201940633301,
0.5281948255512322,
0.3440155511307017,
-0.6791241507669356,
-0.7498601012695852,
-0.6946373774642369,
-0.6177642266777679,
-0.2856729357201208,
0.4851947472201811,
0.09726903107154261,
-0.4045609051365335,
0.18389954360123525,
-0.2034787076783578,
-0.2260359270986582,
0.8035576635033466,
-0.7931715620009508,
-0.9362607227109441,
0.5578839793416537,
0.15959477202733782,
-0.454821222216028,
-0.545593252337095,
0.2205461704673383,
-0.35716661949785156,
-0.6934665190405042,
0.0830933564238442,
0.17268207109916753,
-0.8013515313723008,
-0.20341695383964103,
1.4288863566434409,
-0.8458196045603313,
0.3893336367327371,
-0.34888961728294265,

0.7608524424568026,
0.06661544595138999,
2.0595559407890174,
0.6490263079624171,
-0.4572869614004236,
-1.1373095831864677,
-0.3649376822063403,
-0.22071540101499584,
-0.6587338793306329,
-2.059884237950364,
-0.08358300020061853,
1.0319197724415174,
-0.2751839419251212,
1.3088618827914082,
0.7764572920081257,
0.8767331962684795,
0.955714857858676,
-0.5966677196420035,
-0.2715939735155754,
0.12847285565670696,
0.4974981016237345,
0.11157624833218974,
2.280970649154798,
0.7801190316815027,
-0.311773417229458,
1.158014603168253,
0.5949520544831722,
1.6123203236495465,
0.09856141876014059,
-2.5192133286473677,
1.414165495054559,
0.568818345966238,
-0.7689353156130335,
-0.016254732770980017,
-0.044305575074125,
0.05295776050958731,
1.0200318030747035,
1.4632071591524118,
0.7778847321553402],
[7.194344081667924e-18,
0.22666307415943535,
1.9342085859558782,
1.944505522513897,
2.976191287506401,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.469325640936666,
3.4873843407422243,
4.118956189268777,
4.118956189268777,
5.942584534769547,
5.9611960725237205,
6.340111863711309,
6.460202466469716,
6.7852294872778645,
6.979402607245081,
7.168797029298139,
7.227045817029271,
7.302238353497494,
7.404556177953146,
8.659014511382493,
8.681357101968592,
10.220537951103626,
10.220537951103626,
10.990885177615443,
-0.4696397519192671,
0.27796231748191075,
-0.5848844010182828,
-0.9399496817226014,
-0.6791144953934183,
1.7599237567532016,

-0.5153866079362172,
-0.9884960511226192,
0.5991380415167059,
0.03526597855649831,
-0.4928202951963407,
-0.17823604247966235,
-0.5598665649984788,
-0.7477193118385453,
1.3145784058485683,
-0.01658289220788679,
1.3749987665160877,
1.5410101089396069,
0.5381593361770547,
-1.5686214512244299,
0.5414340679559935,
0.17960267336650448,
1.002730580710403,
-0.5079352878091554,
0.27127261026651706,
0.8523739943313656,
-0.22490410891672527,
-0.5177684122022047,
-0.7335136739541019,
0.9537821935011014,
0.9299309014905508,
0.027634097755273066,
0.11678727620864918,
1.329875652999114,
0.22154652145214473,
-0.5149535741083672,
-0.0935144974812166,
0.8101347453355321,
1.002535906635114,
-0.21669230492415195,
0.25505956278191033,
0.8322911879517776,
0.24105173903016894,
0.3010018335234525,
1.0269077594015688,
1.1653005827455005,
-0.7139789414025479,
0.1802432624729846,
-0.43621452412416506,
0.10805015233497241,
-0.22099264006117136,
-0.06821093567938447,
0.4991913033327633,
-0.09623960812205685,
1.7645880679165642,
-0.057916644058093394,
-1.1151840835769227,
-0.7145527671981541,
2.0743689798531384,
-0.04887843441837349,
1.5069809195108836,
-0.3256203951764359,
-0.8875710119086927,
-0.20072291205396614,
-0.8322399466388898,
0.46293426681558253,
1.4946138988374058,
0.09519796067520077,
-0.7389501244350067,
0.7163596590753283,
-0.9779834532530196,
0.24121899590668236,
0.5362525857496123,
0.7178873526537328,
-0.09720605867652783,
-0.8844816300920153,
0.6879494830422364,
-0.556207799173706,
-2.085706465222243,
-0.6990668132647135,
-1.060060203430475,

0.5108382979368807,
-1.195038624073349,
-0.4693850277739393,
0.4997731574718765,
-0.7981866110564311,
-1.3492284551615505,
-0.8150525999866463,
-0.023113205223160647,
-0.1007660060236893,
0.5860867369253601,
0.15047431993488672,
0.8082527357931191,
0.6508889280029133,
-1.3215333882547653,
0.13580576430947058,
1.0768338700600115,
0.09630564743985112,
0.5304856979004331,
-0.2470170488957273,
-0.6273296913896085,
-0.012309334672496116,
0.8219063430566509,
-2.522381990553264,
-0.5959392155078911,
-0.9015453006612434,
0.25028124043210703,
-0.035229743179394535,
0.46028190301395183,
0.030645450102450806,
0.9152462362128074,
-0.09021729137847402,
0.014451484462572361,
-2.194500168454895,
0.0026823747896166745,
-0.021868499065043765,
0.25783762601108967,
1.5371718314814482,
1.9085420695857682,
1.5522802210039741],
[-7.180915735189581e-18,
0.2741144583980765,
1.4514670623449988,
1.4514670623449988,
1.6489164272497665,
1.6489164272497667,
1.73149137801893,
1.830467910520106,
1.830467910520106,
2.708217872437681,
2.708217872437681,
4.495539722625299,
4.495539722625299,
4.495539722625299,
4.570233660709978,
5.002505460470289,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.812797709933179,
5.812797709933179,
6.677130390626635,
6.677130390626635,
6.861335344270694,
8.772688787903297,
8.772688787903297,
10.006873752440399,
10.006873752440399,
10.006873752440399,
1.5494767629045267,
1.0888430758932834,
0.24316211822093795,
0.5394610508612123,
1.6831866222617424,
-0.15121133640766382,

0.37446291095683387,
-0.39557168857427305,
1.895734214431956,
-0.2769106697905878,
0.4098744736876165,
-0.8550979914567345,
0.497698744019723,
1.1661969870744597,
0.4884385698850034,
0.1585303339769962,
-0.04038005447014292,
0.20331204987112447,
1.2717970037539772,
-0.024603186152083865,
-0.06561207027284033,
-0.17578386607253835,
0.4554538169128916,
-0.3458478405442004,
0.003014344905820474,
0.04935807070525889,
0.7414733867293872,
2.191797855158184,
0.7288491023643235,
0.6487285676710397,
0.0823080431430117,
0.23749626587684475,
0.1076729676697226,
-1.5131488602297158,
1.1994751078587316,
1.7910688286856344,
0.0059408425852493195,
-0.0215678421858246,
0.5920431102259514,
1.1797184458204304,
-0.4858383686169137,
1.1738772137731162,
1.917509405697279,
0.04253149516337592,
0.5485055496810046,
-0.35206499609814007,
0.4164896121092416,
-0.5778229043728815,
0.7927744863095695,
-0.40276124854461287,
-1.4077641380820347,
0.5847373551160024,
0.7110370445087761,
-0.272001728150977,
0.6801732294133973,
1.5986611291818411,
-1.7123595531876228,
0.9030593081535582,
0.5579581015351607,
-0.12955463976185633,
0.17158208553128768,
-0.4284115544697507,
0.41166090312183995,
-0.05704879108721846,
0.6260803344487521,
-0.1444379656414329,
-0.03323955152691923,
-0.5463524110157282,
1.9483453113378342,
0.24554581836345657,
-0.3912953380733674,
-0.2551086789875916,
1.7941408116066884,
0.5297614111369608,
-1.055305153377236,
0.5851894984947826,
-0.6277708161900404,
0.9023397588106277,
0.15473896858281969,
-0.5045582428336748,
0.35128474752318256,

0.27447050561374786,
0.2453530023237907,
-0.881260065779903,
1.2893822185874217,
0.034404783051078594,
-0.8093290700723595,
0.020829229564375124,
-0.5496174311783282,
1.5855246509676788,
-1.9597925195753434,
-0.35765423396713947,
0.0645095765054153,
1.2837142304502733,
0.006624180460715028,
0.35085061348827623,
-1.8681642835419228,
0.1423820815596781,
0.5399990382746707,
1.7033903387378666,
-0.1065388317224254,
-0.05044042000338169,
0.3207599083179648,
1.1530017150539378,
1.5387803133146891,
0.9426514512946884,
1.654923602736178,
0.15672212072564318,
0.5579523293912716,
0.03301127288721064,
0.2831128671935505,
-0.8296250007412259,
0.3295014726214846,
0.8099539632201395,
0.6731359481535688,
0.07726614260275587,
0.2823379588881529,
-1.6102046054887462,
1.3421086989005693,
1.506675537221725],
[0.5436487956056492,
1.0318754867837048,
1.031878809784634,
1.0354032636682196,
1.6023669004016947,
1.644001042836531,
2.7828327622495883,
2.987856277002514,
2.9878562849693426,
3.0896481585306774,
4.808772683377482,
4.9189322116630665,
5.053349250959963,
5.0705845742336875,
5.0876264564887625,
6.123589402685677,
6.123850407213233,
6.291601953687259,
6.291798238238624,
7.418249505998958,
7.418249506075792,
8.141624649997873,
8.163960362736292,
8.208498408727728,
8.208859361112212,
9.089278643760485,
9.180997185672972,
9.211346803231834,
9.871115455051829,
10.914592222542833,
-0.10714466270129372,
0.0414005813417363,
0.31017108200298504,
-1.9041640853184163,
-0.2149394741481629,
-0.954748928787851,

0.12809368425258344,
-1.068243921180568,
-0.5079865095515741,
-0.28605329513797556,
-0.19740513387232797,
-0.22325695706426568,
0.270216664817167,
-1.1670200531314157,
0.17357556599169296,
2.554477008278338,
0.2569346421962659,
-0.44255822443493054,
-1.0047996143631401,
1.3462842603333594,
1.1512175477641275,
-0.23524577874182742,
0.25862583919023807,
-0.03981155232548289,
0.12000346737958213,
-0.5231937940865384,
-0.07831458803809532,
-0.7352847060942096,
0.36694131393913326,
-0.2794757421840953,
0.8531564292966232,
0.11577732227315543,
-0.2692221140803255,
-0.01593534730278469,
0.5023269794705598,
-1.0201597672225167,
-1.217555742220479,
1.0019872098347335,
-0.9863383366879347,
-0.505076260127648,
0.44147961683678577,
0.17102749819972465,
-0.3872030295771854,
-1.0804196760959297,
-0.46011856071048735,
0.6404554735741855,
-0.1019842395221057,
-0.8338441616008496,
-0.1862543853067883,
0.7729351219902972,
0.1291698795888973,
1.84480481040653,
0.7902011519638132,
0.8035704757103046,
0.16307460637128957,
0.3879631381162318,
0.7553290974087437,
1.3544126344921927,
-0.6410152429142767,
0.30610263763196005,
-0.8028999721479468,
0.026104270638268326,
-0.7549249175514674,
0.3272210288981157,
-0.4017438577258813,
0.17639539137663426,
-0.0599886006454003,
-0.029000197801291885,
0.46434674317946,
-0.009657445164941145,
-1.4594578401188716,
1.0749651071987714,
-0.2858651376363263,
0.9236407241821395,
0.804989773126084,
-0.006131190880034288,
-0.41182615341479456,
0.0345636295388565,
-1.4528618699977163,
1.059940111163351,
-0.293288558562868,

-0.19261506463839523,
-0.013768907394512604,
-0.0058499255440563995,
0.9958119726697335,
1.6968974761241398,
-0.8295102319053347,
-0.12979840000582374,
0.1192301496450713,
0.36205695435395635,
0.18145573379217902,
-0.5008220240501698,
0.9951331553730755,
1.5809655477939468,
-0.44266547932504335,
0.5749472054038229,
-0.41342219281521114,
-0.6949087257678164,
-0.9951368874890167,
0.2815866277793241,
-0.14025691028810777,
0.36907964729939857,
0.10131186103604999,
-1.7520150960815608,
1.9938164611139795,
0.8620398797652439,
-1.2922896821720897,
0.28060229891238847,
2.8035209903051896,
1.568376396433029,
0.595816726425649,
-1.2842096282862903,
-0.6778028363202987,
-1.5573241486707392,
-0.7609920700723998,
-1.5773563548919856,
1.324267114980915,
1.5509238379824284,
-0.3029567022174245,
-1.5980321700442846],
[0.04778381839807009,
0.052328729194328676,
0.05232872919432868,
2.5402034202141324,
2.5402034202141324,
2.540203420214133,
2.9078994845512853,
2.9078994845512853,
4.626405021642842,
4.657483894791151,
4.657483894791151,
5.1543302051209485,
5.168582880456237,
5.168582880456237,
5.193083562661445,
6.29476745947969,
6.321713228151687,
6.406123865320823,
6.406123865320823,
6.415662129910562,
6.415662129910562,
6.4885710573049025,
8.891108534137606,
8.923842890216621,
8.941256389861133,
10.14144676971709,
10.14144676971709,
10.14144676971709,
10.173407474763442,
10.173407474763442,
0.9049919196609981,
-0.10571917073418317,
0.984067433195123,
-0.5046397239329845,
0.826983938894292,
-0.300506222622187,

0.6691119536121557,
-0.6359242615836268,
-0.9075291794736813,
0.7311597085342896,
0.9472124240542854,
0.9252859784089732,
0.2654320249732376,
-0.19570931474329503,
-0.23156782982391674,
0.15992044574971995,
-0.9344783087557579,
0.574058594523507,
-0.5900698195003219,
-0.015276834175791246,
0.8277476347193486,
1.701428746099737,
0.29486306492791453,
-1.5727896129747672,
1.2542378717585188,
-0.029060964767507632,
-2.410312077661991,
0.7794852804981598,
0.10382543392432386,
-0.27522638598589344,
-0.7737995873864438,
-0.17484621194736175,
-0.8869497632214399,
0.0739036632402581,
-0.11675807314923795,
-0.7813933016619123,
1.2237386371188925,
0.254266955819111,
0.8344221074025743,
-0.05418013806720967,
-0.35023932132414914,
1.5635792749140942,
2.0001489123678207,
-1.469548386716051,
0.6018811590660905,
1.0932588013795888,
0.1626638394237113,
-0.4071815610867397,
-0.09033843760896655,
1.5066284256317157,
-1.0091863719510874,
-0.5055220266722595,
0.8659824216894423,
1.5341961411404397,
-0.10472552531482807,
-0.9747080994118024,
-0.15896534579332847,
0.4283545141751367,
0.48555121925603906,
-1.0541413967651159,
0.7844732422535036,
1.231237717810785,
0.2304560895908817,
0.8964687802289353,
0.3608400337996031,
0.20964144290191944,
-0.3812502250633237,
-0.05244827792509459,
0.7363786261301344,
-0.5373684827470173,
-0.041135046809633154,
0.022494455398660802,
-0.1427510282638101,
0.15254373676169164,
-0.3538441331700385,
-0.6013550669628682,
1.177355155310087,
-0.25553253513169044,
0.24230205662919607,
-0.856739664506789,
-0.24258353085003115,

1.0855460718544134,
-0.2303523457340619,
-0.4228181709882648,
0.6300184916809138,
0.7448691834325954,
0.8306231103131142,
0.8921818612650112,
-1.2232038552468776,
1.1518310980776325,
-0.10925895973168652,
-1.5546501465626024,
0.25882124214269925,
0.2964831640226424,
0.20056526647993123,
0.6466051276015505,
0.2958855303900712,
1.574298633668074,
-0.2963531534087623,
-0.5079019143621002,
-0.9468892366026295,
1.2752258488383597,
0.7403271380521786,
0.2806459845384513,
0.00045412531964782697,
-0.30343305939372256,
-0.2826391347063689,
-0.24868728346285462,
0.6190941948455959,
0.22356779638775762,
1.0235473251809233,
0.10726603124753692,
0.2726476947387006,
0.34464280290355076,
0.1533269771575057,
-0.19391561509244587,
0.4350499735853799,
0.07688310057706789,
1.3928303170166854,
1.2150647846497966],
[-1.5612511283791264e-17,
0.37090300553176103,
0.7507019437657014,
0.7507019437657013,
0.8504313269395809,
0.8504313269395809,
1.0288453745501207,
1.0288453745501207,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.6746234572416636,
3.6746234572416636,
5.070304872258586,
5.070304872258586,
5.070304872258586,
6.244278301066357,
6.244278301066357,
7.550988432333311,
7.550988432333311,
7.550988432333311,
8.583669958606022,
8.583669958606022,
8.761663968184166,
8.762824899576403,
9.78739432508435,
9.78739432508435,
9.78739432508435,
-1.0196497144708154,
0.30674985458989995,
-0.4164372967959927,
0.3795733998494057,
-0.3589945609302255,
1.1740756105856336,

-0.881009363192087,
0.6233231770354686,
0.014641892239634163,
1.5178326228084182,
1.0177798404109606,
0.04930799061014925,
0.1259706076860049,
0.05495189820733549,
0.008740541357319203,
1.0293402971653869,
0.6712877763592521,
1.3470276988715923,
0.5204896552003178,
0.5839104657425647,
-0.09863982249226241,
0.44754254744872723,
1.0699570996956442,
0.052847134004663066,
0.4378537260228811,
0.8337645558299386,
0.26799172757326156,
-0.2370989308824484,
0.4268946106495037,
0.10394671906247668,
1.0368119557194113,
0.8780368078369323,
1.391762381585933,
-0.4887346750129456,
0.23129082798818065,
-1.185985414153505,
-0.1399486075576189,
0.624884164407104,
-0.394202372239471,
1.4744661957359235,
0.1333476321252126,
0.7992370886557384,
-0.032444927920587764,
0.2278570517496479,
-0.12707464851531813,
-1.1803807605079757,
0.12619040548444643,
0.8842719846976165,
-0.8876928774682807,
0.25819408346986983,
-0.15120492941863672,
-0.34186031332584377,
0.05717115426327923,
-0.1287980699517455,
-0.6384961775059668,
0.2028843608113393,
0.6968559028533619,
-1.0254791626625297,
0.04639881891449554,
-0.034568027823247464,
-0.01382953519480725,
-0.28736510780605545,
-0.27712541400117535,
1.7289913866690068,
0.3590804569357109,
0.5212797413877558,
0.3325950057479984,
0.31414178485416083,
-0.007677216540470154,
1.5276609022748393,
-0.19648884450771267,
1.7007124456248135,
-0.16165644607389268,
-0.19438002002138255,
0.42844370320601005,
1.0333499299480355,
0.8652144019307915,
-0.23530130050937098,
-0.07871384718095216,
-0.11451985500506619,
-0.9958204124079787,

2.0359256308075255,
0.9323993817814743,
1.5157870599134722,
-0.21153949564011645,
1.3854977335188041,
0.5637663837671051,
-2.0565965924917213,
0.4516254671982243,
1.5460814866423853,
0.3644961596633484,
1.429982344445419,
-0.6505068989688967,
-0.054141570934051676,
1.0286289924316303,
1.0001903806983123,
0.5202120156033159,
-0.4848333781219212,
1.2810399381016777,
-0.030825717336130737,
1.1138939193895847,
0.009665204926618747,
0.35242092505791633,
0.20878415377993897,
-0.4360318389272872,
0.12430046579509,
0.02631778400042499,
0.052498295792397884,
1.6355891166025114,
1.6103252652105555,
0.5534147808336787,
1.5586288893399751,
-1.292487508100558,
-0.008775390375330459,
1.149807410139418,
-0.5197517748175643,
-1.4661127557094134,
0.8953775577555926,
-0.7735285435355932,
-0.02192283351673928],
[-2.8132202298922583e-17,
0.28576499380662435,
0.5188712429506999,
0.5260718495211498,
1.6875510996286314,
1.769957047543949,
1.769957047543949,
2.1144412566849082,
2.2736787675443475,
2.2736787675443475,
3.8733552051866122,
3.8733552051866122,
4.144421713026456,
4.144421713026456,
5.356566841798323,
5.5388531097345926,
5.5388531097345926,
6.035393779471804,
6.303140775366703,
6.861779885684816,
6.944533564960904,
7.13357233256276,
7.13357233256276,
7.277389780428127,
7.277389780428127,
8.82462477385959,
8.845795504558339,
9.548198337876569,
9.548198337876569,
9.548198337876569,
0.25174303748303517,
0.025975082425550772,
-0.40984308440331035,
-0.38070585583844174,
1.799681239992893,
0.5594462945854607,

-0.4636478431530039,
-0.2780917996856129,
1.2810553386613224,
1.1310784748475073,
0.45914878725036795,
-1.1856954723964535,
0.7961435885090882,
0.57941026174794,
-0.13815313828722545,
-0.7574725065912581,
1.604820017143439,
-0.3603689506520919,
0.27196300072613677,
0.2722313257155289,
2.990969455340513,
1.62020725835794,
0.13735249860534363,
0.519616354535985,
0.46064754704679234,
-0.14807050848061099,
0.6713871000380094,
-0.2458402564805999,
0.9463308813112179,
0.2567843451270063,
0.4629107250573343,
-0.6212938478257929,
-0.22704009998639554,
0.3296464835751119,
2.1654921123752615,
0.6636756285031978,
0.9936965640543358,
-0.6185742210548419,
1.749534825479387,
0.24548213508704153,
-1.3019305728065236,
0.056631533618999905,
1.0932304270217696,
-0.5530783180440034,
0.019231105171754005,
1.3923960145800571,
-0.5916513325234946,
-0.003848263362819536,
0.03781527162426783,
0.5864098422960301,
-0.8251793549162847,
0.11206477827525356,
-0.09225750098811783,
-0.4770216140176493,
0.13695039906816245,
-0.08154474181557379,
1.3791196308847455,
-0.8857160703287424,
-1.129732421053176,
-0.44824804793710993,
1.565274483618525,
-0.47775652201553226,
0.06810484542081006,
0.17151342519304494,
-1.2660646869274135,
1.3724437443522657,
0.455376664537531,
-0.4968402042883,
1.636176060280602,
1.8168167124913772,
-0.6591502510366976,
1.3845331149359692,
-0.13680523979595852,
-0.6670867326095455,
-0.25188954569138783,
0.36341441995068174,
0.27861221206959597,
2.253753293233874,
-0.44572473607203233,
0.08052163176334037,
-0.3801650824413073,

0.7561740409138518,
0.4179895401049083,
0.7668389261341573,
2.5895839574180672,
2.3057077789471467,
0.009170869870097192,
-0.48391820047143247,
-0.1788385119964603,
0.08989915668495536,
0.36995891095455813,
-0.22641283869292964,
0.6178226857138855,
-0.025557409724796323,
-0.5624997309051616,
1.978118628762505,
-0.7161151168610898,
1.889934912715883,
-0.7926498270401485,
0.8062612160869115,
-1.7535002724636062,
-0.12121642361966994,
1.2556359822665695,
1.4004947542019472,
0.4094181922934832,
-0.7237598159979259,
0.8095124829350078,
-0.07921500563810784,
0.7941695908203817,
1.5790924811449014,
1.2151033673114386,
0.2760044806386616,
0.03215735026840476,
1.3335719235125942,
-0.01527696316380547,
1.667307482924314,
-0.5852882148598142,
-0.2634738849840575,
0.4308024302775489,
1.564240027869587],
[0.3981028774646537,
1.3970863863849614,
1.4091876724203942,
1.4095196690866725,
1.435309832652522,
1.6159440641430487,
1.6590822800214955,
2.5041994971471193,
2.5046214861953655,
2.504687307075914,
2.6463903945527543,
2.646392107563143,
3.5650194444109125,
3.568763188733049,
4.893404038177111,
4.8993251206087916,
5.047882242480286,
5.059087496587666,
5.734377927903264,
5.7343992879002315,
5.734611661260137,
6.672744415088379,
6.716855027417073,
7.532922291325236,
7.76042742188154,
7.760528016690849,
8.741894091408147,
9.173605790810564,
9.476175665329622,
10.771165370152604,
0.07585840339400536,
-0.32559648608449443,
-0.3584877554976206,
0.24942288785390876,
0.8910963785962387,
0.41120182935281147,

0.06965634755603489,
0.20611943841300834,
-0.5108736052092635,
1.0351003083741233,
0.23303911085101622,
-0.8140715483421308,
1.3592937988371023,
1.1741125230549692,
-0.7984232202856042,
1.1431263661345903,
-0.42271729328296237,
1.1685998262673312,
-0.8900747330133342,
0.16620411037713437,
-0.06441490232378988,
0.5721569257130531,
-0.6405877965847887,
0.1435721627905685,
-1.0023253187840875,
-0.8834734269597262,
0.873011521015721,
-0.5294133334283674,
0.42491694402044683,
-0.24819133021277387,
-0.006858394744271055,
0.10239876199833975,
-1.0945255159067344,
-0.5209588569771533,
0.07684624481803015,
1.149902758246962,
0.6387502322543195,
0.43311665397553645,
0.2524037362289789,
0.5970991691948484,
-0.8659392186972171,
-0.2075467141416817,
-0.471169586924007,
0.3474995850944464,
0.8407912910032019,
-0.39456865353795484,
0.007483066013485144,
-0.3000031511517205,
-0.27926483264869584,
-0.02627276579859505,
-0.7790488347588548,
-0.6185889956791905,
-0.09477367060629449,
1.4923529367376538,
-0.10445531715928387,
0.12530829976044264,
-0.6395373064435654,
0.08277402847573397,
0.7939754550840359,
-0.45257909256973483,
-0.6794716273620853,
1.5141539142037763,
0.4312816489513316,
-0.1679469048132705,
-0.17124947545547786,
0.09347199374723385,
-0.1562849778095718,
1.3537610374278526,
0.9008040355239236,
-0.9700720508412178,
0.3256401758946771,
0.29535126871409256,
0.18912935978590353,
-1.3766878479597657,
-1.9912841200531366,
0.14684386063156718,
0.9485726251766791,
-1.1647041818090063,
0.47731661524457764,
-0.05220023510180973,
1.3678085559740152,

-0.9056449204835322,
-1.0124342233515295,
-0.9002321917007622,
0.25948869946042935,
0.6916196201790348,
-2.867057384272419,
0.10161679048335649,
-0.202200597369352,
-0.574167005177845,
-0.19403420003548572,
-0.47967113273306583,
0.5285998620423336,
0.1951966167292077,
-0.01680550493300095,
1.2402630343430938,
0.30327521875472907,
-0.21038560271353512,
-0.2563819703743369,
0.61553927096869,
0.8894393642077769,
-0.7281429405128931,
-0.5382166701178036,
-0.564888527870778,
0.9324104873196495,
1.3153762213142488,
1.6087794639717614,
-1.4809112056457674,
2.6089661079426194,
1.6134732700667287,
-1.1392791071004762,
-1.7031247999122032,
-0.12594303016721067,
-1.552828751326619,
0.7484033373249843,
-1.627916556868976,
-1.4613739383316358,
1.6030230459146289,
-1.4549302634648913,
1.5145658855223556],
[0.00014920185539273834,
0.9238380862195603,
1.1653356841767,
1.167832939600511,
1.2191691700431773,
1.2210447233093105,
1.9314710919546132,
2.0244898268357567,
2.025055317728332,
2.025055317755535,
2.035108012828863,
2.0351100169574385,
2.250256137680943,
3.15082115258344,
3.246453048346403,
3.2464530484741454,
3.2616906588279586,
3.267255104894572,
4.694356606797461,
4.6943575411597775,
4.694357544022533,
6.488320294504503,
6.488320350076187,
7.5482403718064734,
7.549174906557876,
7.567191669993812,
7.5932047752247485,
7.6539061864591025,
9.752212018804654,
9.752227380423289,
-1.324607236830295,
1.9077202588182536,
-0.8802070447756194,
-2.0149948315634574,
1.4533735387069473,
0.997568141525462,

-0.7074886190930807,
-0.2560389197368321,
0.4012267462832038,
1.558634946129638,
0.4929137086923578,
-1.110627649535145,
0.02522612974783881,
0.9667142992980035,
-0.5237885436922995,
0.4478493220686497,
1.4081655138312377,
-0.6604234389026553,
0.8156393803790498,
0.9162727880779341,
1.2376705325338233,
0.8905837956479322,
-0.7295152882741086,
0.24561292878662663,
1.7665604379212652,
0.17703116950798709,
0.4931954498011207,
1.0631695242763428,
0.5506819823950141,
0.49574385725394715,
-0.9067001814641317,
0.03244215799947241,
1.5112600834704966,
-0.08582031254909145,
0.04823393263399211,
-0.1471510424202064,
1.1677829257421761,
-0.4795697461798601,
-0.4872565354105511,
0.22168062822432655,
1.4072421499069951,
-0.811680819103198,
1.651055088666158,
-0.7313303670161658,
0.5404312773997829,
-0.3816094570928806,
0.5640067451570394,
0.6555156836542061,
-0.09410372652444776,
0.844617621120923,
1.4181988757356223,
0.6157327018229515,
-1.8426182279645893,
0.7475476733327415,
0.9297363986538332,
1.5219888966448256,
0.12784560574897466,
1.3383878758441214,
-0.3849331170513322,
-2.5759820228897086,
-0.44171402566963375,
-1.1481005382735594,
-0.11732003654377818,
1.4261012490056857,
0.37374285139811464,
1.3264125331467558,
-0.3101118867274816,
-0.5580221641216343,
-0.8733903562168412,
0.763630147907968,
-0.22407005303761457,
-0.4427553244179856,
0.25630540825178894,
-0.9744141539121856,
0.7035909537131968,
-1.1511448415135326,
-0.02706398350090099,
-1.4364455854835774,
1.760537605923817,
0.08770997893488061,
-0.4257007958221735,

0.8079796656424563,
0.0842113896113966,
0.43919545053894077,
1.4638006778586652,
-0.01014009539920246,
-1.8469337659391836,
-1.3494797221531116,
0.7904481242840491,
1.4234316624418009,
-1.4054528119805059,
-0.041414215577158846,
0.7865492524837541,
0.7547742563728446,
0.4892505595197172,
-0.7646468207281831,
-1.2841992554361892,
0.014780110538174546,
0.800978697085001,
0.938237883807168,
-0.43390682676117964,
-0.488905643723897,
-1.7892652616586333,
2.0824770359325453,
0.6465813428295442,
0.813778987942834,
-0.4618041833296182,
0.5940671128875871,
-0.49307011143397783,
0.652155769110408,
-2.662957327611224,
1.0258915732895864,
-0.8259876341171806,
0.023143897149311463,
1.3124040163708688,
2.6574917189601317,
-0.6227996400692593,
-0.539154106968482,
1.702015600845961,
0.03237978965554299],
[0.21389598228651574,
0.5393241017358461,
1.336052351787059,
1.4548884734431182,
1.4631515545587506,
2.2996776596727213,
2.303486102556458,
3.672114740986732,
3.738311693108453,
4.172103126491601,
4.185637203208781,
4.879786029287921,
4.879786029287921,
5.514641520665522,
6.187890324824944,
6.199874857261559,
6.337185103245036,
6.337673866716393,
8.339356224610325,
8.355342425303002,
8.376069528482942,
8.376069528482942,
8.376069528482942,
8.405896742002076,
8.450718053610304,
8.463555902040309,
10.004417530570722,
10.004417530570722,
11.277411014627386,
-0.4550197171928084,
0.8726389677058429,
0.32380330098542204,
-0.9102552663335266,
-0.3566919533881965,
1.616361456517712,

0.17449327037699397,
-1.1936385630270505,
-0.3626977992670382,
0.6065214216445567,
-0.27266827511951447,
-0.008337250595426905,
-0.11442883582944025,
-0.2552879859027531,
0.8608200568301817,
0.9797863499294986,
-0.5454704270163543,
-0.12465088219091108,
1.0680711786011012,
0.008287966764571728,
0.8340806285400814,
-0.10467902667059636,
0.6944226087871278,
-1.1848583772718688,
-2.1289584360902833,
2.0815012423116896,
0.7888919711742388,
-0.12601348190113199,
0.5814024197737417,
-0.32142652665182836,
-1.0761996135691065,
1.1437448939093156,
-0.9665568900718753,
-0.751505855358832,
-0.8777599880692168,
-0.0665611067528402,
1.625406556377903,
0.14073065902471577,
-0.008887426812216265,
0.11314595138478645,
0.23282575925386234,
-1.3029158546977586,
0.39815363644872126,
-0.28006162282581953,
0.5981370714824289,
-0.08964177383705696,
-0.23973482016502778,
-1.1546366940781616,
-0.3335643458249379,
1.4313780024296703,
0.6615033850837366,
0.09540664540173596,
1.6701354164263156,
1.6037180066269185,
0.45366725021426957,
1.5306014722736203,
-0.7114276522073563,
-0.0387128408482294,
-0.40951463995726844,
-0.14821428534018466,
1.6120880297451547,
-0.1578213810367529,
-0.4551176093567465,
0.12643398042842974,
0.29421606580526033,
0.3517029135915254,
-0.21200478261515812,
1.503225888163223,
0.6867008326592696,
1.2990703939363029,
-1.2633831104757771,
-0.35212753263373886,
0.2558013627761136,
0.1557716944165696,
-0.2985274551715097,
0.3096340969306992,
1.1065921140021713,
0.5655703287990456,
0.29407543941401004,
1.87556201451638,
-0.005713899736395531,

```

-0.47654203677386653,
-0.33487139893703466,
0.28841760275861744,
0.881872547749141,
-0.2998307941218902,
1.3565987475579786,
0.39905124343442133,
0.37529991980433175,
0.4721044202054456,
-0.0837323332258721,
0.2128419559212174,
1.445893331967549,
-0.5917542945785007,
-0.33453435504213846,
-0.05724156447138151,
0.5825344314121998,
0.05851962203923921,
0.13841145669138022,
-0.07195386411014011,
1.546635282716052,
-0.3465571437275552,
0.9484886642506297,
0.41941222891459207,
0.0901808762528231,
-0.8871255799219674,
-0.4695342320219668,
0.6930505657922351,
1.1594181773830308,
0.042534590889293046,
0.45776921806648474,
0.8831536077453052,
-1.883349137185357,
2.293435211334673,
1.3066760169202165,
-0.040640374761399124,
-0.8716978424494161,
1.5544327319775963,
0.5977489456678642,
-1.5697728131838606]],
    'run_time_sec': 35225.583796978}}],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
                                     'layer': 30,
                                     'ugate': { 'coefficients': { 'bn': [ 0,
                                                                           0,
                                                                           0,
                                                                           0,
                                                                           0,
                                                                           0,
                                                                           0],
                                                                           'cn': [ 0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5,
                                                                           0.5],
                                                                           'r': 0},
                                     'time': {'max': 10.0, 'min': 0.0},
                                     'type': 'xy-iss'}}},
    'init_param': { 'value': [ 0.25473970784448047,
                                1.2025511746035562,
                                1.5481560170840756,
                                1.8324395977107744,
                                1.8439662803769572,
                                2.9830338898217192,
                                2.9830338898217192,
                                3.241693555484232,
                                3.634822278674692,
                                3.7522709686014135,
                                3.7522709686014135,
                                4.94705318367592,
                                4.955555446952993,
                                5.343973550433924,
                                5.3577622539956655,
                                6.390003257794682,
                                6.419263094824833,

```

8.039405153723413,
8.039804264859526,
8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,

```
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
```

```

iss_noisefree_time_evo_dephasing_ric3_sample#9'},
    'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix': 'AUTOMATE_xy-
0.5,
0.5],
    'r': 1},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
        '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
        '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
        '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
            '0.5 Z 0 Z 1 + '
            '0.5 X 1 X 2 + '
            '0.5 Z 1 Z 2 + '
            '0.5 X 2 X 3 + '
            '0.5 Z 2 Z 3 + '
            '0.5 X 3 X 4 + '
            '0.5 Z 3 Z 4 + '
            '0.5 X 4 X 5 + '
            '0.5 Z 4 Z 5 + '
            '0.5 X 5 X 6 + '
            '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.590367983241113],
        [12, 6, 0, 0, -6.1449349379730975],
        [20, 10, 0, 0, -5.022599820662391]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
            'identity_factors': [0, 0, 0, 0],
            'noise_level': [4, 2, 0, 0],
            'noise_profile': {
'noise_on_init_param': { 'status': False,
    'value': 0},
0.001,
0.001,
0.0],
'dephasing'},
'noise_prob': [
0.0,
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
    'value': 0},
0.001,
0.001,
0.0],
'dephasing'},
'noise_prob': [
0.0,
'status': True,
'type':
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

```

```
'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

0.0],

'noise_prob': [

0.0,

'status': True,
'type':

'oddd_wires': 4]],
'run_time_sec': 10.862261056900024}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
'init_param': { 'value': [ 0.1021834784589592,
    0.9196384424899031,
    0.9261694014921312,
    0.983882928321485,
    1.8913212439431486,
    1.9426406276748993,
    1.9426406276748993,
    1.9426406276748993,
    2.7354895381565307,
    2.7475159984062567,
    4.89472071800884,
    4.89472071800884,
    4.89472071800884,
    4.951228136281907,
    4.9774006951115535,
    4.9774006951115535,
    5.053403895048273,
    5.053403895048273,
    5.101252532365191,
    7.988961801418946,
    7.988961801418945,
    7.988961801418946,
    7.988961801418946,
    8.978507898154866,
    8.993364745936525,
    9.010251752506093,
    9.010251752506093,
    9.808175975385067,
    9.808175975385067,
    10.987620898296862,
    -0.17474125502348634,
    0.15377532976778396,
    0.10530559022128577,
    -0.002734731613738592,
    0.18566397098861895,
    -0.21248706612806187,
    0.00824191786492345,
    -0.17814151074061357,
    0.01796252087548974,
    0.34324522632459664,
    0.0074801226415391366,
    0.7599717323032058,
    -0.10515906185801696,
    1.0828859909757265,
    0.414437884618902,
```

1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,

[illegible]


```
'0.5 X 1 X 2 + '  
'0.5 Z 1 Z 2 + '  
'0.5 X 2 X 3 + '  
'0.5 Z 2 Z 3 + '  
'0.5 X 3 X 4 + '  
'0.5 Z 3 Z 4 + '  
'0.5 X 4 X 5 + '  
'0.5 Z 4 Z 5 + '  
'0.5 X 5 X 6 + '  
'0.5 Z 5 Z 6']]],  
    'output': { 'data_points': [ [4, 2, 0, 0, -7.5880525130210295],  
                                  [12, 6, 0, 0, -6.143369929824418],  
                                  [20, 10, 0, 0, -4.998571859003289]],  
              'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                                   'identity_factors': [0, 0, 0, 0],  
                                   'noise_level': [4, 2, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.001,  
                                         0.001,  
  
                                         0.0],  
  
                                     'status': True,  
                                     'type':  
  
                                     'dephasing'},  
  
                                   'odd_wires': 4},  
                                { 'gates_num': [12, 3, 1, 0],  
                                   'identity_factors': [1, 1, 0, 0],  
                                   'noise_level': [12, 6, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.001,  
                                         0.001,  
  
                                         0.0],  
  
                                     'status': True,  
                                     'type':  
  
                                     'dephasing'},  
  
                                   'odd_wires': 4},  
                                { 'gates_num': [20, 5, 1, 0],  
                                   'identity_factors': [2, 2, 0, 0],  
                                   'noise_level': [20, 10, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.001,  
                                         0.001,  
  
                                         0.0],  
  
                                     'status': True,  
                                     'type':  
  
                                     'dephasing'},  
  
                                   'odd_wires': 4}],  
                              'run_time_sec': 10.673752307891846}},  
    { 'config': { 'ansatz': { 'gateset': 1,  
                             'layer': 30,  
                             'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0,  
                                                                0],  
                                           'cn': [ 0.5,  
                                                    0.5,  
                                                    0.5,  
                                                    0.5,  
                                                    0.5],  
                                           'r': 0},
```

```
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0780449993030362,
0.10783439075300198,
0.5845642021744096,
0.7814268466054133,
0.8443273792280553,
2.8661173620125435,
2.866459808493013,
2.866459808493013,
2.866459808493013,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.787331067871434,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
7.013227786440474,
7.016905064536492,
8.483022393696176,
8.483022393696176,
8.483022393696176,
8.483022393696176,
10.30807680489275,
10.30807680489275,
10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
```

0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,

```

0.45463950449288687,
1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-

iss_noisefree_time_evo_dephasing_ric3_sample#2'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.556050750105971],
                             [12, 6, 0, 0, -6.124528497687961],
                             [20, 10, 0, 0, -5.005182740239412]],
          'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0],

                                'status': True,
                                'type':
'dephasing'},

                                'odd_wires': 4},

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

      'noise_prob': [

          0.0,

          'status': True,
          'type':

              'odd_wires': 4},
          { 'gates_num': [20, 5, 1, 0],
            'identity_factors': [2, 2, 0, 0],
            'noise_level': [20, 10, 0, 0],
            'noise_profile': {

                'noise_prob': [

                    0.0,

                    'status': True,
                    'type':

                        'odd_wires': 4]],
                'run_time_sec': 9.584286212921143}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
                                          0,
                                          0,
                                          0,
                                          0,
                                          0,
                                          0],
                                'cn': [ 0.5,
                                          0.5,
                                          0.5,
                                          0.5,
                                          0.5,
                                          0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.8953934585022342,
                                1.0441540460805852,
                                1.2122882144067701,
                                1.3902546362390118,
                                2.4452392605971704,
                                2.4476376114298906,
                                3.59167707372567,
                                3.592168933497119,
                                4.752173032506677,
                                4.770483368888809,
                                5.226089600509016,
                                5.227272839220446,
                                5.277877376458118,
                                6.286147534970271,
                                6.330458303415314,
                                7.343951335722083,
                                7.407616509271585,
                                7.5311385683234535,
                                7.532344729415073,
                                7.683350056306,
                                7.684095349251904,
                                8.372928051556965,
                                8.450880672852195,
                                8.893521328096941,
                                8.933105958948394,
                                10.009729102620259,

```

10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,

```

0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#4'},

```

```

        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0]]},

        'run': 'redundant',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},

        'zne': { 'data_points': None,
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                     '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                     '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                     '6 + 1 Z 5 + 1 Z 6',
                  'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                                '0.5 Z 0 Z 1 + '
                                                                '0.5 X 1 X 2 + '
                                                                '0.5 Z 1 Z 2 + '
                                                                '0.5 X 2 X 3 + '
                                                                '0.5 Z 2 Z 3 + '
                                                                '0.5 X 3 X 4 + '
                                                                '0.5 Z 3 Z 4 + '
                                                                '0.5 X 4 X 5 + '
                                                                '0.5 Z 4 Z 5 + '
                                                                '0.5 X 5 X 6 + '
                                                                '0.5 Z 5 Z 6']},

    'output': { 'data_points': [ [4, 2, 0, 0, -7.665909699744034],
                                  [12, 6, 0, 0, -6.335722541155832],
                                  [20, 10, 0, 0, -5.278211380427869]],
                'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                      'identity_factors': [0, 0, 0, 0],
                                      'noise_level': [4, 2, 0, 0],
                                      'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0],

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0],

                                'status': True,
                                'type':

```


'dephasing'},

```
                                'odd_wires': 4]],
                                'run_time_sec': 8.782463073730469}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 2.067283500702386e-18,
                              0.00305317943967044,
                              0.6707334318791758,
                              0.6819425533370418,
                              2.0804317671049177,
                              2.0811966167088074,
                              3.3186336689124984,
                              3.3186336689124984,
                              4.0962902377799315,
                              4.0962902377799315,
                              4.0962902377799315,
                              4.235998445520904,
                              4.662355223605685,
                              4.662355223605685,
                              4.801118955358666,
                              4.801118955358666,
                              5.157394229481825,
                              6.5093436750150495,
                              6.5217715636056575,
                              7.126394528770992,
                              7.126394528770992,
                              7.5433952316895745,
                              7.5433952316895745,
                              8.461929873684097,
                              8.461929873684097,
                              8.516782749532563,
                              8.558798125131,
                              8.770317023984102,
                              8.869770466030348,
                              10.891182667941633,
                              -0.17551436837428686,
                              -0.09405623650981064,
                              1.4401112083844008,
                              -0.21984615875458338,
                              -1.2243534933440952,
                              -0.21317591292102225,
                              1.1730311609183601,
                              0.18285933627515502,
                              -0.16374038568220986,
                              0.49758703398713866,
                              -0.11662974810194524,
                              -1.8302240243419805,
                              -0.8876433539714791,
                              -0.5042106139218809,
                              0.2510944840638433,
                              0.2904423394991934,
                              -0.15052766025181108,
                              0.0485323125786155,
                              -0.23225254624760458,
                              -1.4505258693023013,
                              0.29319886843755527,
                              0.6416318622903661,
                              0.6810789378910713,
                              -0.03223238490139622,
```

-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,

[illegible]

```

                                '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.517540187131846],
                                [12, 6, 0, 0, -5.995430553311077],
                                [20, 10, 0, 0, -4.811503569399131]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                        'identity_factors': [0, 0, 0, 0],
                        'noise_level': [4, 2, 0, 0],
                        'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0],

                                'status': True,
                                'type':
'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                'identity_factors': [1, 1, 0, 0],
                                'noise_level': [12, 6, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0],

                                'status': True,
                                'type':
'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                'identity_factors': [2, 2, 0, 0],
                                'noise_level': [20, 10, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [
0.001,
0.001,
0.0],

                                'status': True,
                                'type':
'odd_wires': 4}],
'run_time_sec': 10.744234323501587}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
'init_param': { 'value': [ 0.009858867127598406,
                            0.6371504690778502,
                            1.0666846377193393,
                            1.104415373659906,
                            1.464838116920491,
                            1.4648381169204907,
                            1.9080243917751187,

```

2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,

```

-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,

```

```
iss_noisefree_time_evo_dephasing_ric3_sample#6'},
    'cn': [0.5, 0.5, 0.5, 0.5, 0.5],
    'r': 1},
    'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#6'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
            'constraint': True,
            'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
        '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
        '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
        '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
            '0.5 Z 0 Z 1 + '
            '0.5 X 1 X 2 + '
            '0.5 Z 1 Z 2 + '
            '0.5 X 2 X 3 + '
            '0.5 Z 2 Z 3 + '
            '0.5 X 3 X 4 + '
            '0.5 Z 3 Z 4 + '
            '0.5 X 4 X 5 + '
            '0.5 Z 4 Z 5 + '
            '0.5 X 5 X 6 + '
            '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.407013825547438],
        [12, 6, 0, 0, -6.114813208156221],
        [20, 10, 0, 0, -5.082084674797013]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
            'identity_factors': [0, 0, 0, 0],
            'noise_level': [4, 2, 0, 0],
            'noise_profile': {
                'noise_prob': [
                    0.001,
                    0.001,
                    0.0],
                'status': True,
                'type':
                    'dephasing'},
            'odd_wires': 4},
        { 'gates_num': [12, 3, 1, 0],
            'identity_factors': [1, 1, 0, 0],
            'noise_level': [12, 6, 0, 0],
            'noise_profile': {
                'noise_prob': [
                    0.001,
                    0.001,
```

```

0.0],
0.0,
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_on_init_param': { 'status': False,
'value': 0},
'noise_prob': [
0.001,
0.001,
0.0,
'status': True,
'type':
'dephasing'},
'odd_wires': 4}},
'run_time_sec': 10.834871768951416}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.08803864939944098,
0.2533857257734638,
0.4067196832055002,
0.4411063797089926,
1.0492461473989636,
1.0543168722856693,
2.21020815530381,
2.218381560687012,
2.367234970755274,
2.658389587322544,
3.2990226425967433,
3.5549209440787726,
4.0112101619321985,
4.138810393646779,
4.424339957187965,
6.02518256832302,
6.064174786863594,
6.069208430366886,
6.069208430366886,
6.941565801297405,
7.124864639242438,
7.129421087787242,
7.45818921038985,
7.468062221399231,
8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,

```


0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.010069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,

```

0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

```

[illegible]

```
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.00487815438629808,
0.3836991336608517,
0.3836991336608516,
1.1027183519022283,
1.430822963194853,
1.7336849694382164,
1.9079110557306378,
1.9079110557306376,
2.407942958541018,
2.4384441996231097,
2.7198331200698176,
2.7363529901218957,
3.8707803400773457,
3.8707803400773457,
4.283774948106429,
4.283774948106428,
4.283774948106429,
4.283774948106429,
4.283774948106429,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
```

0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,

```

0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'data_points': [ [4, 2, 0, 0, -7.597273562260126],
                             [12, 6, 0, 0, -6.262099522076846],
                             [20, 10, 0, 0, -5.200592923966883]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                     'identity_factors': [0, 0, 0, 0],
                     'noise_level': [4, 2, 0, 0],
                     'noise_profile': {
'noise_on_init_param': { 'status': False,

```

```

'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4}},
'run_time_sec': 9.490588665008545}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 1.0344655484192336e-17,
0.36119418156962224,
0.3611941815696223,
0.36119418156962224,
0.8440759344131337,
1.1727045787919286,
1.5692279974524073,
1.570888864611923,
1.592214192172409,
1.7598342563277258,
3.059056443368036,
3.0591149843524144,
4.3676699123871625,
4.3676699123871625,
5.510828301667605,
5.51914482396091,

```

6.113847931320903,
6.113847931320903,
6.533178927747682,
6.533178927747682,
6.755940863606281,
6.837142696961756,
7.694830693654466,
7.694830693654467,
7.694830693654466,
7.873525558910739,
7.873525558910739,
7.922034998277628,
7.922034998277628,
12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,
-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,


```
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,
0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,
-0.2901105978964291,
-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,
1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
```

```

                                0.5,
                                0.5,
                                0.5],
                                'r': 1},
                                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                        'status': False,
                        'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-

iss_noisefree_time_evo_dephasing_ric3_sample#5'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']],
            'output': { 'data_points': [ [4, 2, 0, 0, -7.524810336842993],
                                          [12, 6, 0, 0, -6.145141411469215],
                                          [20, 10, 0, 0, -5.055735320587113]],
                        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                              'identity_factors': [0, 0, 0, 0],
                                              'noise_level': [4, 2, 0, 0],
                                              'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                  'identity_factors': [1, 1, 0, 0],
                                  'noise_level': [12, 6, 0, 0],
                                  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'dephasing'},

                                'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                  'identity_factors': [2, 2, 0, 0],
                                  'noise_level': [20, 10, 0, 0],
                                  'noise_profile': {

```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

    0.001,
    0.001,

    0.0],

    'status': True,
    'type':

'odd_wires': 4]],
    'run_time_sec': 10.338340997695923}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
            0.5,
            0.5,
            0.5,
            0.5,
            0.5],
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.029287487040924362,
        0.6764270956551205,
        1.6620807976810363,
        1.7056653351603042,
        1.8485187197716686,
        1.9397654142515774,
        1.9426163313124116,
        1.9760071204347747,
        2.0652508647096677,
        3.338212066261205,
        3.3382549495561102,
        3.507308503786301,
        3.668597772988933,
        3.6686655840136484,
        3.7105300968471533,
        3.737148430160667,
        5.361211694679849,
        5.40430867517309,
        5.45325014669751,
        5.461795374069968,
        5.465563084002408,
        5.492346359399902,
        7.18245326721486,
        7.245488992311329,
        7.248698529808956,
        8.18726919118503,
        8.197326336426727,
        8.213328813441153,
        8.251719559085661,
        10.791930409527371,
        -0.48557603875244904,
        -0.06546231446775812,
        -1.4810159853023965,
        -0.6439509610243448,
        -0.4889611776104492,
        0.10983589949376514,
        -1.2155949975003408,
        1.1338047187824702,
        0.18557769650103798,
        -0.36145572824608646,
        -0.7821715666301856,
        0.8512861656756193,
        0.30536456746903484,
        -0.30807040011779185,

```

0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,

```

-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '

```

```
'0.5 Z 0 Z 1 + '  
'0.5 X 1 X 2 + '  
'0.5 Z 1 Z 2 + '  
'0.5 X 2 X 3 + '  
'0.5 Z 2 Z 3 + '  
'0.5 X 3 X 4 + '  
'0.5 Z 3 Z 4 + '  
'0.5 X 4 X 5 + '  
'0.5 Z 4 Z 5 + '  
'0.5 X 5 X 6 + '  
'0.5 Z 5 Z 6']},  
    'output': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],  
                                  [12, 6, 0, 0, -6.098488474706267],  
                                  [20, 10, 0, 0, -4.939317874668221]],  
              'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                                   'identity_factors': [0, 0, 0, 0],  
                                   'noise_level': [4, 2, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.0,  
  
                                     'status': True,  
                                     'type':  
  
                                     'odd_wires': 4},  
                                   { 'gates_num': [12, 3, 1, 0],  
                                   'identity_factors': [1, 1, 0, 0],  
                                   'noise_level': [12, 6, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.0,  
  
                                     'status': True,  
                                     'type':  
  
                                     'odd_wires': 4},  
                                   { 'gates_num': [20, 5, 1, 0],  
                                   'identity_factors': [2, 2, 0, 0],  
                                   'noise_level': [20, 10, 0, 0],  
                                   'noise_profile': {  
  
                                     'noise_prob': [  
  
                                         0.0,  
  
                                     'status': True,  
                                     'type':  
  
                                     'odd_wires': 4}},  
                                'run_time_sec': 9.619635820388794}}],  
    'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,  
                                       'layer': 30,  
                                       'ugate': { 'coefficients': { 'bn': [ 0,  
                                                                           0,  
                                                                           0,  
                                                                           0,  
                                                                           0,  
                                                                           0],  
                               'cn': [ 0.5,  
                                        0.5,  
                                        0.5,  
                                        0.5,  
                                        0.5],
```

```

        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': True,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': True,
                      'type': 'png'},
            'file_name_prefix': 'xy_noisefree_time_evo'},
'redundant': {'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 '
                                'X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z '
                                                         '0 Z 1 + 0.5 X 1 X 2 '
                                                         '+ 0.5 Z 1 Z 2 + 0.5 '
                                                         'X 2 X 3 + 0.5 Z 2 Z '
                                                         '3 + 0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + 0.5 X '
                                                         '4 X 5 + 0.5 Z 4 Z 5 '
                                                         '+ 0.5 X 5 X 6 + 0.5 '
                                                         'Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 2.490874640234538,
                                     3.123425159431485,
                                     3.646429251689861,
                                     2.764268013304591,
                                     3.17022836931859,
                                     3.7735896776807554,
                                     3.622216535636285,
                                     2.438144835449103,
                                     3.477758051251201,
                                     4.254627215438117],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': {
'status': False,
'value': 0},
'noise_prob': [ 0.001,
                 0.001,
                 0.0,
                 0.0],
'status': True,
'type': 'dephasing'},

```

```
        'odd_wires': 4},
'optimized_minimum_cost': [ -7.597273562260196,
                             -7.556050750105993,
                             -7.517540187131898,
                             -7.665909699744024,
                             -7.524810336843006,
                             -7.40701382554743,
                             -7.601751527683981,
                             -7.561979391804892,
                             -7.59036798324113,
                             -7.588052513021082],
'optimized_parameters': [ [ 0.00487815438629808,
                             0.3836991336608517,
                             0.3836991336608516,
                             1.1027183519022283,
                             1.430822963194853,
                             1.7336849694382164,
                             1.9079110557306378,
                             1.9079110557306376,
                             2.407942958541018,
                             2.4384441996231097,
                             2.7198331200698176,
                             2.7363529901218957,
                             3.8707803400773457,
                             3.8707803400773457,
                             4.283774948106429,
                             4.283774948106428,
                             4.283774948106429,
                             4.283774948106429,
                             5.706222745811317,
                             5.706222745811317,
                             5.706222745811317,
                             5.706222745811317,
                             5.706222745811317,
                             5.706222745811317,
                             5.706222745811317,
                             6.386180553174607,
                             6.421245480123581,
                             8.31501274226813,
                             8.345179306265157,
                             11.444382336838434,
                             0.4142599738840607,
                             -0.0179112834331289,
                             -1.916401049394031,
                             0.14120666199524137,
                             0.7700670528228858,
                             0.4673833851581437,
                             -1.0389421964094037,
                             0.4744345335688107,
                             -0.02892536105987987,
                             0.7177135899887972,
                             0.03424375267071561,
                             -0.8966767104461824,
                             -0.2754786671261514,
                             -0.38073768598802404,
                             -0.05488081631876704,
                             1.3786960897007534,
                             0.49169988282833965,
                             -0.0017740881267394302,
                             -0.0730349088569547,
                             1.022071163826922,
                             0.35709372961339875,
                             -0.25522274095888026,
                             0.1780496791518154,
                             1.1037611427142897,
                             0.41779504446715204,
                             0.5262749279658233,
                             0.06490819244365402,
                             -1.4552807970719621,
                             -0.3245263390561476,
                             1.0554193966046508,
                             0.022339750983574838,
                             -0.3669003487168035,
                             2.663679114021979,
                             0.10849122591523966,
```


-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,

0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559],
[0.0780449993030362,
0.10783439075300198,
0.5845642021744096,
0.7814268466054133,
0.8443273792280553,
2.8661173620125435,
2.866459808493013,
2.866459808493013,
2.866459808493013,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.787331067871434,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
7.013227786440474,
7.016905064536492,
8.483022393696176,
8.483022393696176,
8.483022393696176,
8.483022393696176,
10.30807680489275,
10.30807680489275,
10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,

0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,

-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892],
[2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,
8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,

-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,

0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109],
[0.8953934585022342,
1.0441540460805852,
1.2122882144067701,
1.3902546362390118,
2.4452392605971704,
2.4476376114298906,
3.59167707372567,
3.592168933497119,
4.752173032506677,
4.770483368888809,
5.226089600509016,
5.227272839220446,
5.277877376458118,
6.286147534970271,
6.330458303415314,
7.343951335722083,
7.407616509271585,
7.5311385683234535,
7.532344729415073,
7.683350056306,
7.684095349251904,
8.372928051556965,
8.450880672852195,
8.893521328096941,
8.933105958948394,
10.009729102620259,
10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,

-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,

0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684],
[1.0344655484192336e-17,
0.36119418156962224,
0.3611941815696223,
0.36119418156962224,
0.8440759344131337,
1.1727045787919286,
1.5692279974524073,
1.570888864611923,
1.592214192172409,
1.7598342563277258,
3.059056443368036,
3.0591149843524144,
4.3676699123871625,
4.3676699123871625,
5.510828301667605,
5.51914482396091,
6.113847931320903,
6.113847931320903,
6.533178927747682,
6.533178927747682,
6.755940863606281,
6.837142696961756,
7.694830693654466,
7.694830693654467,
7.694830693654466,
7.873525558910739,
7.873525558910739,
7.922034998277628,
7.922034998277628,
12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,
-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,

-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,
0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,
-0.2901105978964291,
-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,

1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593],
[0.009858867127598406,
0.6371504690778502,
1.0666846377193393,
1.104415373659906,
1.464838116920491,
1.4648381169204907,
1.9080243917751187,
2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,

-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,

0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012],
[0.08803864939944098,
0.2533857257734638,
0.4067196832055002,
0.4411063797089926,
1.0492461473989636,
1.0543168722856693,
2.21020815530381,
2.218381560687012,
2.367234970755274,
2.658389587322544,
3.2990226425967433,
3.5549209440787726,
4.0112101619321985,
4.138810393646779,
4.424339957187965,
6.02518256832302,
6.064174786863594,
6.069208430366886,
6.069208430366886,
6.941565801297405,
7.124864639242438,
7.129421087787242,
7.45818921038985,
7.468062221399231,
8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.627735568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,

-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,

0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548],
[0.029287487040924362,
0.6764270956551205,
1.6620807976810363,
1.7056653351603042,
1.8485187197716686,
1.9397654142515774,
1.9426163313124116,
1.9760071204347747,
2.0652508647096677,
3.338212066261205,
3.3382549495561102,
3.507308503786301,
3.668597772988933,
3.6686655840136484,
3.7105300968471533,
3.737148430160667,
5.361211694679849,
5.40430867517309,
5.45325014669751,
5.461795374069968,
5.465563084002408,
5.492346359399902,
7.18245326721486,
7.245488992311329,
7.248698529808956,
8.18726919118503,
8.197326336426727,
8.213328813441153,
8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,

0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,

1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623],
[0.25473970784448047,
1.2025511746035562,
1.5481560170840756,
1.8324395977107744,
1.8439662803769572,
2.9830338898217192,
2.9830338898217192,
3.241693555484232,
3.634822278674692,
3.7522709686014135,
3.7522709686014135,
4.94705318367592,
4.955555446952993,
5.343973550433924,
5.3577622539956655,
6.390003257794682,
6.419263094824833,
8.039405153723413,
8.039804264859526,
8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,

-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,

-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956],
[0.1021834784589592,
0.9196384424899031,
0.9261694014921312,
0.983882928321485,
1.8913212439431486,
1.9426406276748993,
1.9426406276748993,
1.9426406276748993,
2.7354895381565307,
2.7475159984062567,
4.89472071800884,
4.89472071800884,
4.89472071800884,
4.951228136281907,
4.9774006951115535,
4.9774006951115535,
5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,

0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,

```
0.028627573515451674,
1.9279053503307635,
0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'run_time_sec': 42239.92721557617}}],
'zne': [ { 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.08803864939944098,
0.2533857257734638,
0.4067196832055002,
0.4411063797089926,
1.0492461473989636,
1.0543168722856693,
2.21020815530381,
2.218381560687012,
2.367234970755274,
2.658389587322544,
3.2990226425967433,
3.5549209440787726,
4.0112101619321985,
4.138810393646779,
4.424339957187965,
6.02518256832302,
6.064174786863594,
6.069208430366886,
6.069208430366886,
6.941565801297405,
7.124864639242438,
7.129421087787242,
7.45818921038985,
7.468062221399231,
8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
```

-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,

```

-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.6017515276839065],
                          [12, 6, 0, 0, -6.187268174258091],
                          [20, 10, 0, 0, -5.079662736732887]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.553794860839844e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.424072422859544,

```

[illegible]

-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,


```

1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noise_free_time_evo_dephasing_ric3_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

```

      'optimization': { 'algorithm': 'SLSQP',
                        'constraint': True,
                        'status': False}},
      'zne': { 'data_points': [ [4, 2, 0, 0, -7.556050750105971],
                                [12, 6, 0, 0, -6.124528497687961],
                                [20, 10, 0, 0, -5.005182740239412]],
              'degree': 0,
              'method': 'richardson',
              'sampling': 'default'}},
      'output': { 'exact_sol': -8.566772233505624,
                  'run_time_sec': 4.839897155761719e-05,
                  'zne_values': { 'degree': 0,
                                  'extrapolated_value': -8.388878061928525,
                                  'others': { 'beta_coefiecients': [ 1.875,
                                                                    -1.25,
                                                                    0.375],
                                  'cost_richardson_zne': 5.21875,
                                  'sorted_expectation_vals': [

```

```

-7.556050750105971,
-6.124528497687961,
-5.005182740239412],

```

```

                                'sorted_noise': [6, 18, 30]]},
      'sampled data': [ [ 4,
                          2,
                          0,
                          0,
                          -7.556050750105971],
                        [ 12,
                          6,
                          0,
                          0,
                          -6.124528497687961],
                        [ 20,
                          10,
                          0,
                          0,
                          -5.005182740239412]],
      'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                            'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                            'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}}},
  'init_param': { 'value': [ 0.25473970784448047,
                              1.2025511746035562,
                              1.5481560170840756,
                              1.8324395977107744,
                              1.8439662803769572,
                              2.9830338898217192,
                              2.9830338898217192,
                              3.241693555484232,
                              3.634822278674692,
                              3.7522709686014135,
                              3.7522709686014135,
                              4.94705318367592,
                              4.955555446952993,
                              5.343973550433924,
                              5.3577622539956655,
                              6.390003257794682,
                              6.419263094824833,
                              8.039405153723413,
                              8.039804264859526,

```

8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,

```

0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],

```

```
iss_noisefree_time_evo_dephasing_ric3_sample#9'},  
    'def': 'ising'},  
    'output': { 'draw': { 'fig_dpi': 100,  
        'status': False,  
        'type': 'png'},  
        'file_name_prefix': 'AUTOMATE_xy-'  
    },  
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],  
        [1, 1, 0, 0],  
        [2, 2, 0, 0]]},  
    'run': 'zne',  
    'state': 'dmatrix',  
    'vqe': { 'iteration': 1,  
        'optimization': { 'algorithm': 'SLSQP',  
            'constraint': True,  
            'status': False}},  
    'zne': { 'data_points': [ [4, 2, 0, 0, -7.590367983241113],  
        [12, 6, 0, 0, -6.1449349379730975],  
        [20, 10, 0, 0, -5.022599820662391]],  
        'degree': 0,  
        'method': 'richardson',  
        'sampling': 'default'}},  
    'output': { 'exact_sol': -8.566772233505624,  
        'run_time_sec': 4.506111145019531e-05,  
        'zne_values': { 'degree': 0,  
            'extrapolated_value': -8.434246228859113,  
            'others': { 'beta_coefficients': [ 1.875,  
                -1.25,  
                0.375],  
            'cost_richardson_zne': 5.21875,  
            'sorted_expectation_vals': [  
  
                'sorted_noise': [6, 18, 30]},  
        'sampled data': [ [ 4,  
            2,  
            0,  
            0,  
            -7.590367983241113],  
            [ 12,  
            6,  
            0,  
            0,  
            -6.1449349379730975],  
            [ 20,  
            10,  
            0,  
            0,  
            -5.022599820662391]],  
        'sampling': 'default'}}},  
    { 'config': { 'ansatz': { 'gateset': 1,  
        'layer': 30,  
        'ugate': { 'coefficients': { 'bn': [ 0,  
            0,  
            0,  
            0,  
            0,  
            0],  
            'cn': [ 0.5,  
                0.5,  
                0.5,  
                0.5,  
                0.5,  
                0.5],  
            'r': 0},  
        'time': {'max': 10.0, 'min': 0.0},  
        'type': 'xy-iss'}},  
    'init_param': { 'value': [ 0.00487815438629808,  
        0.3836991336608517,  
        0.3836991336608516,  
        1.1027183519022283,  
        1.430822963194853,  
        1.7336849694382164,
```

1.9079110557306378,
1.9079110557306376,
2.407942958541018,
2.4384441996231097,
2.7198331200698176,
2.7363529901218957,
3.8707803400773457,
3.8707803400773457,
4.283774948106429,
4.283774948106428,
4.283774948106429,
4.283774948106429,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,


```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.029287487040924362,
0.6764270956551205,
1.6620807976810363,
1.7056653351603042,
1.8485187197716686,
1.9397654142515774,
1.9426163313124116,
1.9760071204347747,
2.0652508647096677,
3.338212066261205,
3.3382549495561102,
3.507308503786301,
3.668597772988933,
3.6686655840136484,
3.7105300968471533,
3.737148430160667,
5.361211694679849,
5.40430867517309,
5.45325014669751,
5.461795374069968,
5.465563084002408,
5.492346359399902,
7.18245326721486,
7.245488992311329,
7.248698529808956,
8.18726919118503,
8.197326336426727,
8.213328813441153,
8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
```

-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,

```

0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],
                          [12, 6, 0, 0, -6.098488474706267],
                          [20, 10, 0, 0, -4.939317874668221]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.601478576660156e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.407844969251778,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-7.561979391804815,
-6.098488474706267,
-4.939317874668221],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.561979391804815],
                  [ 12,
                    6,
                    0,
                    0,
                    -6.098488474706267],
                  [ 20,
                    10,
                    0,
                    0,
                    0,

```

```

-4.939317874668221]],
    'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}}},
  'init_param': { 'value': [ 1.0344655484192336e-17,
                              0.36119418156962224,
                              0.3611941815696223,
                              0.36119418156962224,
                              0.8440759344131337,
                              1.1727045787919286,
                              1.5692279974524073,
                              1.570888864611923,
                              1.592214192172409,
                              1.7598342563277258,
                              3.059056443368036,
                              3.0591149843524144,
                              4.3676699123871625,
                              4.3676699123871625,
                              5.510828301667605,
                              5.51914482396091,
                              6.113847931320903,
                              6.113847931320903,
                              6.533178927747682,
                              6.533178927747682,
                              6.755940863606281,
                              6.837142696961756,
                              7.694830693654466,
                              7.694830693654467,
                              7.694830693654466,
                              7.873525558910739,
                              7.873525558910739,
                              7.922034998277628,
                              7.922034998277628,
                              12.310276979046629,
                              -0.6972922025249428,
                              -0.0508578639309505,
                              0.03511585194544882,
                              -0.05290285417260422,
                              0.6885525491076263,
                              0.23164759975262209,
                              0.06930481892124933,
                              -0.05051704727786026,
                              0.2745998381310896,
                              2.8540854853592244,
                              0.1762956340846527,
                              -0.46176266603345323,
                              0.4487337371763968,
                              1.96119949254268,
                              -0.028430732456242037,
                              1.020183549895116,
                              0.09052188212177692,
                              0.6283703459809612,
                              -0.2008669437139055,
                              -0.333213682394554,
                              0.49298533108167947,
                              -0.073860646808665,
                              -0.13593481376514907,
                              -0.597217576988091,
                              0.38422303935103047,

```

-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,

```

0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,
-0.2901105978964291,
-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,
1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#5'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.524810336842993],
                          [12, 6, 0, 0, -6.145141411469215],
                          [20, 10, 0, 0, -5.055735320587113]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.482269287109375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.32349336246426,
                           'others': { 'beta_coeffieicients': [ 1.875,
                                                                -1.25,
                                                                0.375],
                                       'cost_richardson_zne': 5.21875,
                                       'sorted_expectation_vals': [
-7.524810336842993,
-6.145141411469215,
-5.055735320587113],
'sorted_noise': [6, 18, 30]],
'sampled data': [ [ 4,

```

```

2,
0,
0,
-7.524810336842993],
[ 12,
6,
0,
0,
-6.145141411469215],
[ 20,
10,
0,
0,
-5.055735320587113]],
'sampling': 'default']]],
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,
8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,

```

-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,


```

-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noise_free_time_evo_dephasing_ric3_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.517540187131846],
                          [12, 6, 0, 0, -5.995430553311077],
                          [20, 10, 0, 0, -4.811503569399131]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,

```

-7.517540187131846,
-5.995430553311077,
-4.811503569399131],

```
'run_time_sec': 4.00543212890625e-05,  
'zne_values': { 'degree': 0,  
                'extrapolated_value': -8.40541349775804,  
                'others': { 'beta_coeffieicients': [ 1.875,  
                                                    -1.25,  
                                                    0.375],  
                'cost_richardson_zne': 5.21875,  
                'sorted_expectation_vals': [  
  
                'sorted_noise': [6, 18, 30]],  
        'sampled data': [ [ 4,  
                            2,  
                            0,  
                            0,  
                            -7.517540187131846],  
                          [ 12,  
                            6,  
                            0,  
                            0,  
                            -5.995430553311077],  
                          [ 20,  
                            10,  
                            0,  
                            0,  
                            -4.811503569399131]],  
        'sampling': 'default'}}},  
{ 'config': { 'ansatz': { 'gateset': 1,  
                          'layer': 30,  
                          'ugate': { 'coefficients': { 'bn': [ 0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0,  
                                                              0],  
                                                              'cn': [ 0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5,  
                                                              0.5],  
                                                              'r': 0},  
                          'time': {'max': 10.0, 'min': 0.0},  
                          'type': 'xy-iss'}},  
  'init_param': { 'value': [ 0.8953934585022342,  
                             1.0441540460805852,  
                             1.2122882144067701,  
                             1.3902546362390118,  
                             2.4452392605971704,  
                             2.4476376114298906,  
                             3.59167707372567,  
                             3.592168933497119,  
                             4.752173032506677,  
                             4.770483368888809,  
                             5.226089600509016,  
                             5.227272839220446,  
                             5.277877376458118,  
                             6.286147534970271,  
                             6.330458303415314,  
                             7.343951335722083,  
                             7.407616509271585,  
                             7.5311385683234535,  
                             7.532344729415073,  
                             7.683350056306,  
                             7.684095349251904,  
                             8.372928051556965,  
                             8.450880672852195,  
                             8.893521328096941,  
                             8.933105958948394,  
                             10.009729102620259,  
                             10.02381465220654,  
                             10.426876716424587,  
                             10.426876716424587,
```

12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,

```

-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5,
                                         0.5],
                                'r': 1},
               'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0]]},

```

-7.665909699744034,
-6.335722541155832,
-5.278211380427869],

-7.665909699744034,
-6.335722541155832,
-5.278211380427869],

5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,

```
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,
0.028627573515451674,
1.9279053503307635,
0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,
                                         0.5,
```

```

0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric3_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.5880525130210295],
[12, 6, 0, 0, -6.143369929824418],
[20, 10, 0, 0, -4.998571859003289]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.506111145019531e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.422850496760141,
'others': { 'beta_coeffieicients': [ 1.875,
-1.25,
0.375],
'cost_richardson_zne': 5.21875,
'sorted_expectation_vals': [
-7.5880525130210295,
-6.143369929824418,
-4.998571859003289],
'sorted_noise': [6, 18, 30]},
'sampled data': [ [ 4,
2,
0,
0,
-7.5880525130210295],
[ 12,
6,
0,
0,
-6.143369929824418],
[ 20,
10,
0,
0,
-4.998571859003289]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.009858867127598406,
0.6371504690778502,
1.0666846377193393,

```


1.104415373659906,
1.464838116920491,
1.4648381169204907,
1.9080243917751187,
2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,

```
-0.5652644350797837,  
-0.04661359801412543,  
-0.8211476277064503,  
0.2688028104686384,  
-0.12556704820535647,  
-0.023840075766175,  
-0.18959105165277507,  
-0.026652605847716424,  
-0.0368848436080881,  
-0.17033806404624055,  
-0.24172965402850155,  
1.8740805145949957,  
3.3954947367399346,  
0.685069809753845,  
0.18877298628713768,  
-0.8116148922898279,  
0.3642216955021626,  
0.776870193177977,  
-0.2915626595335885,  
0.48528749335602656,  
0.3052797687185898,  
0.8094152496869139,  
-0.8933636925729215,  
-0.927850020272857,  
0.26040310486864426,  
-0.02621475103327714,  
-0.3044278549592545,  
-0.45445142071133066,  
-0.660404261852709,  
1.1611135994497148,  
-0.08594794341260105,  
0.7467695488502767,  
-0.5557725464437727,  
-0.16927866426422344,  
-0.043602634805518736,  
0.9335958048537285,  
-0.6399661361094481,  
0.23806458661693586,  
1.596804769065565,  
0.05733925195491384,  
0.7574169294466017,  
0.040475318628828375,  
-0.8594601463323506,  
-0.09619413741349449,  
-0.21197930687835795,  
0.2757964313499001,  
0.06286971719145724,  
1.1856485993065717,  
-0.08746041433688896,  
-0.3598130921460406,  
-0.15364860008047934,  
1.2858349919806815,  
0.18001757541551167,  
-0.5562723277178536,  
0.3666171831013428,  
1.6125643575569677,  
-1.3673007848855094,  
0.09427109776859108,  
0.5265559951107008,  
-1.7052545510096164,  
2.291596524870159,  
0.046069640183881794,  
0.6559544004774626,  
-0.5991904786957847,  
-0.6867617223561892,  
-0.1637234892612084,  
0.30313993491542573,  
1.548639749817063,  
-0.9376256579293304,  
1.0052059218812999,  
0.9493196821469583,  
0.13892828355847012]],  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                             'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],
```

[illegible]

```

                                0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                           'value': 0},
                  'noise_prob': [0.001, 0.001, 0.0, 0.0],
                  'status': False,
                  'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                     'status': False,
                     'type': 'png'},
            'file_name_prefix': 'xy_noise_off'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': True}},
'zne': { 'data_points': None,
        'degree': 1,
        'method': 'richardson',
        'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 '
                                '+ 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z '
                                '+ 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 '
                                'Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost': 4.032700960953626,
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.895971800514176,
                                     2.6461223700347904],
            'initial_cost_history': [ 4.032700960953626,
                                     2.876037164175563,
                                     3.024801381527762,
                                     2.6026187891538575,
                                     2.558855794921029,
                                     3.233596797235869,
                                     4.0713546948778445,
                                     4.434976242432684,
                                     2.8959718005
```

```

{ 'status': False,
  'value': 0},

'noise_details': { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [0, 0, 0, 0],
                    'noise_profile': { 'noise_on_init_param':

                                'noise_prob': [ 0.001,
                                                0.001,
                                                0.0,
                                                0.0],
                                'status': False,
                                'type': 'dephasing'},

                    'odd_wires': 4},
'optimized_minimum_cost': [ -8.213984832683868,
                             -8.467734328099501,
                             -8.447208232910596,
                             -8.461814133720251,
                             -8.483392807973491,
                             -8.44723907770434,
                             -8.362559674056595,
                             -8.366604609651855,
                             -8.163432600733032,
                             -8.478925857587775],
'optimized_parameters': [ [ 0.5489081084990874,
                             0.6882459728440448,
                             0.7729468967125317,
                             0.8347489553606998,
                             1.0946318761773612,
                             1.9534300183374533,
                             2.2829395447313403,
                             3.232524496500775,
                             3.2325244965209894,
                             3.852765644418336,
                             3.8527656444207725,
                             3.8527673900689967,
                             3.8860323667767283,
                             4.588361079261242,
                             4.600166894773711,
                             5.527977091209439,
                             5.556427027819143,
                             5.803787289292802,
                             6.218904325824693,
                             6.239985329573426,
                             6.538211674407394,
                             6.925082879783563,
                             7.640591161405884,
                             7.75471653665469,
                             7.896761387122529,
                             8.932525392602521,
                             9.128535074996934,
                             9.925693656125251,
                             9.925693656687802,
                             11.037683520466203,
                             -0.10948884769018298,
                             0.012576058394267018,
                             0.9054362612114637,
                             -0.5027178523290218,
                             1.9293813165204678,
                             0.12560680736570712,
                             0.40022822712854333,
                             -2.471918351448183,
                             0.5257346934244156,
                             1.1130547436006022,
                             1.0771121143603033,
                             -0.1314162667650016,
                             0.02294938233855663,
                             0.36225076948659746,
                             -1.3206473358532678,
                             -0.2449807752862754,
                             0.29890651347484976,
                             1.0095017101029446,
                             0.6150854645533661,
                             0.2815584721561816,
                             0.3601395589466358,
                             -0.5388375319509757,
```

0.9923143444796582,
0.8473755438313292,
-0.4646214972755992,
0.8527825907455423,
-0.6098121229644526,
0.12413023074624807,
-0.06044402625536853,
-0.29258159286196256,
1.3610999143074116,
0.09999631572477011,
-0.18981827210653637,
0.2696177153747519,
0.4272428992844361,
-0.26705761267067063,
-0.37988713130471985,
0.75645724651605,
0.8109920260438386,
-1.028544143258335,
0.44106157371844235,
-1.6090649394170073,
-0.44600474997528494,
-0.2579853712948814,
0.3063055059682419,
0.9497745366960584,
-0.2701835817775974,
0.9807489498405011,
-0.507582861688708,
-1.25201940633301,
0.5281948255512322,
0.3440155511307017,
-0.6791241507669356,
-0.7498601012695852,
-0.6946373774642369,
-0.6177642266777679,
-0.2856729357201208,
0.4851947472201811,
0.09726903107154261,
-0.4045609051365335,
0.18389954360123525,
-0.2034787076783578,
-0.2260359270986582,
0.8035576635033466,
-0.7931715620009508,
-0.9362607227109441,
0.5578839793416537,
0.15959477202733782,
-0.454821222216028,
-0.545593252337095,
0.2205461704673383,
-0.35716661949785156,
-0.6934665190405042,
0.0830933564238442,
0.17268207109916753,
-0.8013515313723008,
-0.20341695383964103,
1.4288863566434409,
-0.8458196045603313,
0.3893336367327371,
-0.34888961728294265,
0.7608524424568026,
0.06661544595138999,
2.0595559407890174,
0.6490263079624171,
-0.4572869614004236,
-1.1373095831864677,
-0.3649376822063403,
-0.22071540101499584,
-0.6587338793306329,
-2.059884237950364,
-0.08358300020061853,
1.0319197724415174,
-0.2751839419251212,
1.3088618827914082,
0.7764572920081257,
0.8767331962684795,

0.955714857858676,
-0.5966677196420035,
-0.2715939735155754,
0.12847285565670696,
0.4974981016237345,
0.11157624833218974,
2.280970649154798,
0.7801190316815027,
-0.311773417229458,
1.158014603168253,
0.5949520544831722,
1.6123203236495465,
0.09856141876014059,
-2.5192133286473677,
1.414165495054559,
0.568818345966238,
-0.7689353156130335,
-0.016254732770980017,
-0.044305575074125,
0.05295776050958731,
1.0200318030747035,
1.4632071591524118,
0.7778847321553402],
[7.194344081667924e-18,
0.22666307415943535,
1.9342085859558782,
1.944505522513897,
2.976191287506401,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.4411909222312294,
3.469325640936666,
3.4873843407422243,
4.118956189268777,
4.118956189268777,
5.942584534769547,
5.9611960725237205,
6.340111863711309,
6.460202466469716,
6.7852294872778645,
6.979402607245081,
7.168797029298139,
7.227045817029271,
7.302238353497494,
7.404556177953146,
8.659014511382493,
8.681357101968592,
10.220537951103626,
10.220537951103626,
10.990885177615443,
-0.4696397519192671,
0.27796231748191075,
-0.5848844010182828,
-0.9399496817226014,
-0.6791144953934183,
1.7599237567532016,
-0.5153866079362172,
-0.9884960511226192,
0.5991380415167059,
0.03526597855649831,
-0.4928202951963407,
-0.17823604247966235,
-0.5598665649984788,
-0.7477193118385453,
1.3145784058485683,
-0.01658289220788679,
1.3749987665160877,
1.5410101089396069,
0.5381593361770547,
-1.5686214512244299,
0.5414340679559935,
0.17960267336650448,

1.002730580710403,
-0.5079352878091554,
0.27127261026651706,
0.8523739943313656,
-0.22490410891672527,
-0.5177684122022047,
-0.7335136739541019,
0.9537821935011014,
0.9299309014905508,
0.027634097755273066,
0.11678727620864918,
1.329875652999114,
0.22154652145214473,
-0.5149535741083672,
-0.0935144974812166,
0.8101347453355321,
1.002535906635114,
-0.21669230492415195,
0.25505956278191033,
0.8322911879517776,
0.24105173903016894,
0.3010018335234525,
1.0269077594015688,
1.1653005827455005,
-0.7139789414025479,
0.1802432624729846,
-0.43621452412416506,
0.10805015233497241,
-0.22099264006117136,
-0.06821093567938447,
0.4991913033327633,
-0.09623960812205685,
1.7645880679165642,
-0.057916644058093394,
-1.1151840835769227,
-0.7145527671981541,
2.0743689798531384,
-0.04887843441837349,
1.5069809195108836,
-0.3256203951764359,
-0.8875710119086927,
-0.20072291205396614,
-0.8322399466388898,
0.46293426681558253,
1.4946138988374058,
0.09519796067520077,
-0.7389501244350067,
0.7163596590753283,
-0.9779834532530196,
0.24121899590668236,
0.5362525857496123,
0.7178873526537328,
-0.09720605867652783,
-0.8844816300920153,
0.6879494830422364,
-0.556207799173706,
-2.085706465222243,
-0.6990668132647135,
-1.060060203430475,
0.5108382979368807,
-1.195038624073349,
-0.4693850277739393,
0.4997731574718765,
-0.7981866110564311,
-1.3492284551615505,
-0.8150525999866463,
-0.023113205223160647,
-0.1007660060236893,
0.5860867369253601,
0.15047431993488672,
0.8082527357931191,
0.6508889280029133,
-1.3215333882547653,
0.13580576430947058,
1.0768338700600115,

0.09630564743985112,
0.5304856979004331,
-0.2470170488957273,
-0.6273296913896085,
-0.012309334672496116,
0.8219063430566509,
-2.522381990553264,
-0.5959392155078911,
-0.9015453006612434,
0.25028124043210703,
-0.035229743179394535,
0.46028190301395183,
0.030645450102450806,
0.9152462362128074,
-0.09021729137847402,
0.014451484462572361,
-2.194500168454895,
0.0026823747896166745,
-0.021868499065043765,
0.25783762601108967,
1.5371718314814482,
1.9085420695857682,
1.5522802210039741],
[-7.180915735189581e-18,
0.2741144583980765,
1.4514670623449988,
1.4514670623449988,
1.6489164272497665,
1.6489164272497667,
1.73149137801893,
1.830467910520106,
1.830467910520106,
2.708217872437681,
2.708217872437681,
4.495539722625299,
4.495539722625299,
4.495539722625299,
4.570233660709978,
5.002505460470289,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.380500339711581,
5.812797709933179,
5.812797709933179,
6.677130390626635,
6.677130390626635,
6.861335344270694,
8.772688787903297,
8.772688787903297,
10.006873752440399,
10.006873752440399,
10.006873752440399,
1.5494767629045267,
1.0888430758932834,
0.24316211822093795,
0.5394610508612123,
1.6831866222617424,
-0.15121133640766382,
0.37446291095683387,
-0.39557168857427305,
1.895734214431956,
-0.2769106697905878,
0.4098744736876165,
-0.8550979914567345,
0.497698744019723,
1.1661969870744597,
0.4884385698850034,
0.1585303339769962,
-0.04038005447014292,
0.20331204987112447,
1.2717970037539772,
-0.024603186152083865,
-0.06561207027284033,
-0.17578386607253835,

0.4554538169128916,
-0.3458478405442004,
0.003014344905820474,
0.04935807070525889,
0.7414733867293872,
2.191797855158184,
0.7288491023643235,
0.6487285676710397,
0.0823080431430117,
0.23749626587684475,
0.1076729676697226,
-1.5131488602297158,
1.1994751078587316,
1.7910688286856344,
0.0059408425852493195,
-0.0215678421858246,
0.5920431102259514,
1.1797184458204304,
-0.4858383686169137,
1.1738772137731162,
1.917509405697279,
0.04253149516337592,
0.5485055496810046,
-0.35206499609814007,
0.4164896121092416,
-0.5778229043728815,
0.7927744863095695,
-0.40276124854461287,
-1.4077641380820347,
0.5847373551160024,
0.7110370445087761,
-0.272001728150977,
0.6801732294133973,
1.5986611291818411,
-1.7123595531876228,
0.9030593081535582,
0.5579581015351607,
-0.12955463976185633,
0.17158208553128768,
-0.4284115544697507,
0.41166090312183995,
-0.05704879108721846,
0.6260803344487521,
-0.1444379656414329,
-0.03323955152691923,
-0.5463524110157282,
1.9483453113378342,
0.24554581836345657,
-0.3912953380733674,
-0.2551086789875916,
1.7941408116066884,
0.5297614111369608,
-1.055305153377236,
0.5851894984947826,
-0.6277708161900404,
0.9023397588106277,
0.15473896858281969,
-0.5045582428336748,
0.35128474752318256,
0.27447050561374786,
0.2453530023237907,
-0.881260065779903,
1.2893822185874217,
0.034404783051078594,
-0.8093290700723595,
0.020829229564375124,
-0.5496174311783282,
1.5855246509676788,
-1.9597925195753434,
-0.35765423396713947,
0.0645095765054153,
1.2837142304502733,
0.006624180460715028,
0.35085061348827623,
-1.8681642835419228,

0.1423820815596781,
0.5399990382746707,
1.7033903387378666,
-0.1065388317224254,
-0.05044042000338169,
0.3207599083179648,
1.1530017150539378,
1.5387803133146891,
0.9426514512946884,
1.654923602736178,
0.15672212072564318,
0.5579523293912716,
0.03301127288721064,
0.2831128671935505,
-0.8296250007412259,
0.3295014726214846,
0.8099539632201395,
0.6731359481535688,
0.07726614260275587,
0.2823379588881529,
-1.6102046054887462,
1.3421086989005693,
1.506675537221725],
[0.5436487956056492,
1.0318754867837048,
1.031878809784634,
1.0354032636682196,
1.6023669004016947,
1.644001042836531,
2.7828327622495883,
2.987856277002514,
2.9878562849693426,
3.0896481585306774,
4.808772683377482,
4.9189322116630665,
5.053349250959963,
5.0705845742336875,
5.0876264564887625,
6.123589402685677,
6.123850407213233,
6.291601953687259,
6.291798238238624,
7.41824950598958,
7.418249506075792,
8.141624649997873,
8.163960362736292,
8.208498408727728,
8.208859361112212,
9.089278643760485,
9.180997185672972,
9.211346803231834,
9.871115455051829,
10.914592222542833,
-0.10714466270129372,
0.0414005813417363,
0.31017108200298504,
-1.9041640853184163,
-0.2149394741481629,
-0.954748928787851,
0.12809368425258344,
-1.068243921180568,
-0.5079865095515741,
-0.28605329513797556,
-0.19740513387232797,
-0.22325695706426568,
0.270216664817167,
-1.1670200531314157,
0.17357556599169296,
2.554477008278338,
0.2569346421962659,
-0.44255822443493054,
-1.0047996143631401,
1.3462842603333594,
1.1512175477641275,
-0.23524577874182742,

0.25862583919023807,
-0.03981155232548289,
0.12000346737958213,
-0.5231937940865384,
-0.07831458803809532,
-0.7352847060942096,
0.36694131393913326,
-0.2794757421840953,
0.8531564292966232,
0.11577732227315543,
-0.2692221140803255,
-0.01593534730278469,
0.5023269794705598,
-1.0201597672225167,
-1.217555742220479,
1.0019872098347335,
-0.9863383366879347,
-0.505076260127648,
0.44147961683678577,
0.17102749819972465,
-0.3872030295771854,
-1.0804196760959297,
-0.46011856071048735,
0.6404554735741855,
-0.1019842395221057,
-0.8338441616008496,
-0.1862543853067883,
0.7729351219902972,
0.1291698795888973,
1.84480481040653,
0.7902011519638132,
0.8035704757103046,
0.16307460637128957,
0.3879631381162318,
0.7553290974087437,
1.3544126344921927,
-0.6410152429142767,
0.30610263763196005,
-0.8028999721479468,
0.026104270638268326,
-0.7549249175514674,
0.3272210288981157,
-0.4017438577258813,
0.17639539137663426,
-0.0599886006454003,
-0.029000197801291885,
0.46434674317946,
-0.009657445164941145,
-1.4594578401188716,
1.0749651071987714,
-0.2858651376363263,
0.9236407241821395,
0.804989773126084,
-0.006131190880034288,
-0.41182615341479456,
0.0345636295388565,
-1.4528618699977163,
1.059940111163351,
-0.293288558562868,
-0.19261506463839523,
-0.013768907394512604,
-0.0058499255440563995,
0.9958119726697335,
1.6968974761241398,
-0.8295102319053347,
-0.12979840000582374,
0.1192301496450713,
0.36205695435395635,
0.18145573379217902,
-0.5008220240501698,
0.9951331553730755,
1.5809655477939468,
-0.44266547932504335,
0.5749472054038229,
-0.41342219281521114,

-0.6949087257678164,
-0.9951368874890167,
0.2815866277793241,
-0.14025691028810777,
0.36907964729939857,
0.10131186103604999,
-1.7520150960815608,
1.9938164611139795,
0.8620398797652439,
-1.2922896821720897,
0.28060229891238847,
2.8035209903051896,
1.568376396433029,
0.595816726425649,
-1.2842096282862903,
-0.6778028363202987,
-1.5573241486707392,
-0.7609920700723998,
-1.5773563548919856,
1.324267114980915,
1.5509238379824284,
-0.3029567022174245,
-1.5980321700442846],
[0.04778381839807009,
0.052328729194328676,
0.05232872919432868,
2.5402034202141324,
2.5402034202141324,
2.540203420214133,
2.9078994845512853,
2.9078994845512853,
4.626405021642842,
4.657483894791151,
4.657483894791151,
5.1543302051209485,
5.168582880456237,
5.168582880456237,
5.193083562661445,
6.29476745947969,
6.321713228151687,
6.406123865320823,
6.406123865320823,
6.415662129910562,
6.415662129910562,
6.4885710573049025,
8.891108534137606,
8.923842890216621,
8.941256389861133,
10.14144676971709,
10.14144676971709,
10.14144676971709,
10.173407474763442,
10.173407474763442,
0.9049919196609981,
-0.10571917073418317,
0.984067433195123,
-0.5046397239329845,
0.826983938894292,
-0.300506222622187,
0.6691119536121557,
-0.6359242615836268,
-0.9075291794736813,
0.7311597085342896,
0.9472124240542854,
0.9252859784089732,
0.2654320249732376,
-0.19570931474329503,
-0.23156782982391674,
0.15992044574971995,
-0.9344783087557579,
0.574058594523507,
-0.5900698195003219,
-0.015276834175791246,
0.8277476347193486,
1.701428746099737,

0.29486306492791453,
-1.5727896129747672,
1.2542378717585188,
-0.029060964767507632,
-2.410312077661991,
0.7794852804981598,
0.10382543392432386,
-0.27522638598589344,
-0.7737995873864438,
-0.17484621194736175,
-0.8869497632214399,
0.0739036632402581,
-0.11675807314923795,
-0.7813933016619123,
1.2237386371188925,
0.254266955819111,
0.8344221074025743,
-0.05418013806720967,
-0.35023932132414914,
1.5635792749140942,
2.0001489123678207,
-1.469548386716051,
0.6018811590660905,
1.0932588013795888,
0.1626638394237113,
-0.4071815610867397,
-0.09033843760896655,
1.5066284256317157,
-1.0091863719510874,
-0.5055220266722595,
0.8659824216894423,
1.5341961411404397,
-0.10472552531482807,
-0.9747080994118024,
-0.15896534579332847,
0.4283545141751367,
0.48555121925603906,
-1.0541413967651159,
0.7844732422535036,
1.231237717810785,
0.2304560895908817,
0.8964687802289353,
0.3608400337996031,
0.20964144290191944,
-0.3812502250633237,
-0.05244827792509459,
0.7363786261301344,
-0.5373684827470173,
-0.041135046809633154,
0.022494455398660802,
-0.1427510282638101,
0.15254373676169164,
-0.3538441331700385,
-0.6013550669628682,
1.177355155310087,
-0.25553253513169044,
0.24230205662919607,
-0.856739664506789,
-0.24258353085003115,
1.0855460718544134,
-0.2303523457340619,
-0.4228181709882648,
0.6300184916809138,
0.7448691834325954,
0.8306231103131142,
0.8921818612650112,
-1.2232038552468776,
1.1518310980776325,
-0.10925895973168652,
-1.5546501465626024,
0.25882124214269925,
0.2964831640226424,
0.20056526647993123,
0.6466051276015505,
0.2958855303900712,

1.574298633668074,
-0.2963531534087623,
-0.5079019143621002,
-0.9468892366026295,
1.2752258488383597,
0.7403271380521786,
0.2806459845384513,
0.00045412531964782697,
-0.30343305939372256,
-0.2826391347063689,
-0.24868728346285462,
0.6190941948455959,
0.22356779638775762,
1.0235473251809233,
0.10726603124753692,
0.2726476947387006,
0.34464280290355076,
0.1533269771575057,
-0.19391561509244587,
0.4350499735853799,
0.07688310057706789,
1.3928303170166854,
1.2150647846497966],
[-1.5612511283791264e-17,
0.37090300553176103,
0.7507019437657014,
0.7507019437657013,
0.8504313269395809,
0.8504313269395809,
1.0288453745501207,
1.0288453745501207,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.220429414477101,
3.6746234572416636,
3.6746234572416636,
5.070304872258586,
5.070304872258586,
5.070304872258586,
5.070304872258586,
6.244278301066357,
6.244278301066357,
7.550988432333311,
7.550988432333311,
7.550988432333311,
8.583669958606022,
8.583669958606022,
8.761663968184166,
8.762824899576403,
9.78739432508435,
9.78739432508435,
9.78739432508435,
-1.0196497144708154,
0.30674985458989995,
-0.4164372967959927,
0.3795733998494057,
-0.3589945609302255,
1.1740756105856336,
-0.881009363192087,
0.6233231770354686,
0.014641892239634163,
1.5178326228084182,
1.0177798404109606,
0.04930799061014925,
0.1259706076860049,
0.05495189820733549,
0.008740541357319203,
1.0293402971653869,
0.6712877763592521,
1.3470276988715923,
0.5204896552003178,
0.5839104657425647,
-0.09863982249226241,
0.44754254744872723,

1.0699570996956442,
0.052847134004663066,
0.4378537260228811,
0.8337645558299386,
0.26799172757326156,
-0.2370989308824484,
0.4268946106495037,
0.10394671906247668,
1.0368119557194113,
0.8780368078369323,
1.391762381585933,
-0.4887346750129456,
0.23129082798818065,
-1.185985414153505,
-0.1399486075576189,
0.624884164407104,
-0.394202372239471,
1.4744661957359235,
0.1333476321252126,
0.7992370886557384,
-0.032444927920587764,
0.2278570517496479,
-0.12707464851531813,
-1.1803807605079757,
0.12619040548444643,
0.8842719846976165,
-0.8876928774682807,
0.25819408346986983,
-0.15120492941863672,
-0.34186031332584377,
0.05717115426327923,
-0.1287980699517455,
-0.6384961775059668,
0.2028843608113393,
0.6968559028533619,
-1.0254791626625297,
0.04639881891449554,
-0.034568027823247464,
-0.01382953519480725,
-0.28736510780605545,
-0.27712541400117535,
1.7289913866690068,
0.3590804569357109,
0.5212797413877558,
0.3325950057479984,
0.31414178485416083,
-0.007677216540470154,
1.5276609022748393,
-0.19648884450771267,
1.7007124456248135,
-0.16165644607389268,
-0.19438002002138255,
0.42844370320601005,
1.0333499299480355,
0.8652144019307915,
-0.23530130050937098,
-0.07871384718095216,
-0.11451985500506619,
-0.9958204124079787,
2.0359256308075255,
0.9323993817814743,
1.5157870599134722,
-0.21153949564011645,
1.3854977335188041,
0.5637663837671051,
-2.0565965924917213,
0.4516254671982243,
1.5460814866423853,
0.3644961596633484,
1.429982344445419,
-0.6505068989688967,
-0.054141570934051676,
1.0286289924316303,
1.0001903806983123,
0.5202120156033159,

-0.4848333781219212,
1.2810399381016777,
-0.030825717336130737,
1.1138939193895847,
0.009665204926618747,
0.35242092505791633,
0.20878415377993897,
-0.4360318389272872,
0.12430046579509,
0.02631778400042499,
0.052498295792397884,
1.6355891166025114,
1.6103252652105555,
0.5534147808336787,
1.5586288893399751,
-1.292487508100558,
-0.008775390375330459,
1.149807410139418,
-0.5197517748175643,
-1.4661127557094134,
0.8953775577555926,
-0.7735285435355932,
-0.02192283351673928],
[-2.8132202298922583e-17,
0.28576499380662435,
0.5188712429506999,
0.5260718495211498,
1.6875510996286314,
1.769957047543949,
1.769957047543949,
2.1144412566849082,
2.2736787675443475,
2.2736787675443475,
3.8733552051866122,
3.8733552051866122,
4.144421713026456,
4.144421713026456,
5.356566841798323,
5.5388531097345926,
5.5388531097345926,
6.035393779471804,
6.303140775366703,
6.861779885684816,
6.944533564960904,
7.13357233256276,
7.13357233256276,
7.277389780428127,
7.277389780428127,
8.82462477385959,
8.845795504558339,
9.548198337876569,
9.548198337876569,
9.548198337876569,
0.25174303748303517,
0.025975082425550772,
-0.40984308440331035,
-0.38070585583844174,
1.799681239992893,
0.5594462945854607,
-0.4636478431530039,
-0.2780917996856129,
1.2810553386613224,
1.1310784748475073,
0.45914878725036795,
-1.1856954723964535,
0.7961435885090882,
0.57941026174794,
-0.13815313828722545,
-0.7574725065912581,
1.604820017143439,
-0.3603689506520919,
0.27196300072613677,
0.2722313257155289,
2.990969455340513,
1.62020725835794,

0.13735249860534363,
0.519616354535985,
0.46064754704679234,
-0.14807050848061099,
0.6713871000380094,
-0.2458402564805999,
0.9463308813112179,
0.2567843451270063,
0.4629107250573343,
-0.6212938478257929,
-0.22704009998639554,
0.3296464835751119,
2.1654921123752615,
0.6636756285031978,
0.9936965640543358,
-0.6185742210548419,
1.749534825479387,
0.24548213508704153,
-1.3019305728065236,
0.056631533618999905,
1.0932304270217696,
-0.5530783180440034,
0.019231105171754005,
1.3923960145800571,
-0.5916513325234946,
-0.003848263362819536,
0.03781527162426783,
0.5864098422960301,
-0.8251793549162847,
0.11206477827525356,
-0.09225750098811783,
-0.4770216140176493,
0.13695039906816245,
-0.08154474181557379,
1.3791196308847455,
-0.8857160703287424,
-1.129732421053176,
-0.44824804793710993,
1.565274483618525,
-0.47775652201553226,
0.06810484542081006,
0.17151342519304494,
-1.2660646869274135,
1.3724437443522657,
0.455376664537531,
-0.4968402042883,
1.636176060280602,
1.8168167124913772,
-0.6591502510366976,
1.3845331149359692,
-0.13680523979595852,
-0.6670867326095455,
-0.25188954569138783,
0.36341441995068174,
0.27861221206959597,
2.253753293233874,
-0.44572473607203233,
0.08052163176334037,
-0.3801650824413073,
0.7561740409138518,
0.4179895401049083,
0.7668389261341573,
2.5895839574180672,
2.3057077789471467,
0.009170869870097192,
-0.48391820047143247,
-0.1788385119964603,
0.08989915668495536,
0.36995891095455813,
-0.22641283869292964,
0.6178226857138855,
-0.025557409724796323,
-0.5624997309051616,
1.978118628762505,
-0.7161151168610898,

1.889934912715883,
-0.7926498270401485,
0.8062612160869115,
-1.7535002724636062,
-0.12121642361966994,
1.2556359822665695,
1.4004947542019472,
0.4094181922934832,
-0.7237598159979259,
0.8095124829350078,
-0.07921500563810784,
0.7941695908203817,
1.5790924811449014,
1.2151033673114386,
0.2760044806386616,
0.03215735026840476,
1.3335719235125942,
-0.01527696316380547,
1.667307482924314,
-0.5852882148598142,
-0.2634738849840575,
0.4308024302775489,
1.564240027869587],
[0.3981028774646537,
1.3970863863849614,
1.4091876724203942,
1.4095196690866725,
1.435309832652522,
1.6159440641430487,
1.6590822800214955,
2.5041994971471193,
2.5046214861953655,
2.504687307075914,
2.6463903945527543,
2.646392107563143,
3.5650194444109125,
3.568763188733049,
4.893404038177111,
4.8993251206087916,
5.047882242480286,
5.059087496587666,
5.734377927903264,
5.7343992879002315,
5.734611661260137,
6.672744415088379,
6.716855027417073,
7.532922291325236,
7.76042742188154,
7.760528016690849,
8.741894091408147,
9.173605790810564,
9.476175665329622,
10.771165370152604,
0.07585840339400536,
-0.32559648608449443,
-0.3584877554976206,
0.24942288785390876,
0.8910963785962387,
0.41120182935281147,
0.06965634755603489,
0.20611943841300834,
-0.5108736052092635,
1.0351003083741233,
0.23303911085101622,
-0.8140715483421308,
1.3592937988371023,
1.1741125230549692,
-0.7984232202856042,
1.1431263661345903,
-0.42271729328296237,
1.1685998262673312,
-0.8900747330133342,
0.16620411037713437,
-0.06441490232378988,
0.5721569257130531,

-0.6405877965847887,
0.1435721627905685,
-1.0023253187840875,
-0.8834734269597262,
0.873011521015721,
-0.5294133334283674,
0.42491694402044683,
-0.24819133021277387,
-0.006858394744271055,
0.10239876199833975,
-1.0945255159067344,
-0.5209588569771533,
0.07684624481803015,
1.149902758246962,
0.6387502322543195,
0.43311665397553645,
0.2524037362289789,
0.5970991691948484,
-0.8659392186972171,
-0.2075467141416817,
-0.471169586924007,
0.3474995850944464,
0.8407912910032019,
-0.39456865353795484,
0.007483066013485144,
-0.3000031511517205,
-0.27926483264869584,
-0.02627276579859505,
-0.7790488347588548,
-0.6185889956791905,
-0.09477367060629449,
1.4923529367376538,
-0.10445531715928387,
0.12530829976044264,
-0.6395373064435654,
0.08277402847573397,
0.7939754550840359,
-0.45257909256973483,
-0.6794716273620853,
1.5141539142037763,
0.4312816489513316,
-0.1679469048132705,
-0.17124947545547786,
0.09347199374723385,
-0.1562849778095718,
1.3537610374278526,
0.9008040355239236,
-0.9700720508412178,
0.3256401758946771,
0.29535126871409256,
0.18912935978590353,
-1.3766878479597657,
-1.9912841200531366,
0.14684386063156718,
0.9485726251766791,
-1.1647041818090063,
0.47731661524457764,
-0.05220023510180973,
1.3678085559740152,
-0.9056449204835322,
-1.0124342233515295,
-0.9002321917007622,
0.25948869946042935,
0.6916196201790348,
-2.867057384272419,
0.10161679048335649,
-0.202200597369352,
-0.574167005177845,
-0.19403420003548572,
-0.47967113273306583,
0.5285998620423336,
0.1951966167292077,
-0.01680550493300095,
1.2402630343430938,
0.30327521875472907,

-0.21038560271353512,
-0.2563819703743369,
0.61553927096869,
0.8894393642077769,
-0.7281429405128931,
-0.5382166701178036,
-0.564888527870778,
0.9324104873196495,
1.3153762213142488,
1.6087794639717614,
-1.4809112056457674,
2.6089661079426194,
1.6134732700667287,
-1.1392791071004762,
-1.7031247999122032,
-0.12594303016721067,
-1.552828751326619,
0.7484033373249843,
-1.627916556868976,
-1.4613739383316358,
1.6030230459146289,
-1.4549302634648913,
1.5145658855223556],
[0.00014920185539273834,
0.9238380862195603,
1.1653356841767,
1.167832939600511,
1.2191691700431773,
1.2210447233093105,
1.9314710919546132,
2.0244898268357567,
2.025055317728332,
2.025055317755535,
2.035108012828863,
2.0351100169574385,
2.250256137680943,
3.15082115258344,
3.246453048346403,
3.2464530484741454,
3.2616906588279586,
3.267255104894572,
4.694356606797461,
4.6943575411597775,
4.694357544022533,
6.488320294504503,
6.488320350076187,
7.5482403718064734,
7.549174906557876,
7.567191669993812,
7.5932047752247485,
7.6539061864591025,
9.752212018804654,
9.752227380423289,
-1.324607236830295,
1.9077202588182536,
-0.8802070447756194,
-2.0149948315634574,
1.4533735387069473,
0.997568141525462,
-0.7074886190930807,
-0.2560389197368321,
0.4012267462832038,
1.558634946129638,
0.4929137086923578,
-1.110627649535145,
0.02522612974783881,
0.9667142992980035,
-0.5237885436922995,
0.4478493220686497,
1.4081655138312377,
-0.6604234389026553,
0.8156393803790498,
0.9162727880779341,
1.2376705325338233,
0.8905837956479322,

-0.7295152882741086,
0.24561292878662663,
1.7665604379212652,
0.17703116950798709,
0.4931954498011207,
1.0631695242763428,
0.5506819823950141,
0.49574385725394715,
-0.9067001814641317,
0.03244215799947241,
1.5112600834704966,
-0.08582031254909145,
0.04823393263399211,
-0.1471510424202064,
1.1677829257421761,
-0.4795697461798601,
-0.4872565354105511,
0.22168062822432655,
1.4072421499069951,
-0.811680819103198,
1.651055088666158,
-0.7313303670161658,
0.5404312773997829,
-0.3816094570928806,
0.5640067451570394,
0.6555156836542061,
-0.09410372652444776,
0.844617621120923,
1.4181988757356223,
0.6157327018229515,
-1.8426182279645893,
0.7475476733327415,
0.9297363986538332,
1.5219888966448256,
0.12784560574897466,
1.3383878758441214,
-0.3849331170513322,
-2.5759820228897086,
-0.44171402566963375,
-1.1481005382735594,
-0.11732003654377818,
1.4261012490056857,
0.37374285139811464,
1.3264125331467558,
-0.3101118867274816,
-0.5580221641216343,
-0.8733903562168412,
0.763630147907968,
-0.22407005303761457,
-0.4427553244179856,
0.25630540825178894,
-0.9744141539121856,
0.7035909537131968,
-1.1511448415135326,
-0.02706398350090099,
-1.4364455854835774,
1.760537605923817,
0.08770997893488061,
-0.4257007958221735,
0.8079796656424563,
0.0842113896113966,
0.43919545053894077,
1.4638006778586652,
-0.01014009539920246,
-1.8469337659391836,
-1.3494797221531116,
0.7904481242840491,
1.4234316624418009,
-1.4054528119805059,
-0.041414215577158846,
0.7865492524837541,
0.7547742563728446,
0.4892505595197172,
-0.7646468207281831,
-1.2841992554361892,

0.014780110538174546,
0.800978697085001,
0.938237883807168,
-0.43390682676117964,
-0.488905643723897,
-1.7892652616586333,
2.0824770359325453,
0.6465813428295442,
0.813778987942834,
-0.4618041833296182,
0.5940671128875871,
-0.49307011143397783,
0.652155769110408,
-2.662957327611224,
1.0258915732895864,
-0.8259876341171806,
0.023143897149311463,
1.3124040163708688,
2.6574917189601317,
-0.6227996400692593,
-0.539154106968482,
1.702015600845961,
0.03237978965554299],
[0.21389598228651574,
0.5393241017358461,
1.336052351787059,
1.4548884734431182,
1.4631515545587506,
2.2996776596727213,
2.303486102556458,
3.672114740986732,
3.738311693108453,
4.172103126491601,
4.185637203208781,
4.879786029287921,
4.879786029287921,
5.514641520665522,
6.187890324824944,
6.199874857261559,
6.337185103245036,
6.337673866716393,
8.339356224610325,
8.355342425303002,
8.376069528482942,
8.376069528482942,
8.376069528482942,
8.376069528482942,
8.405896742002076,
8.450718053610304,
8.463555902040309,
10.004417530570722,
10.004417530570722,
11.277411014627386,
-0.4550197171928084,
0.8726389677058429,
0.32380330098542204,
-0.9102552663335266,
-0.3566919533881965,
1.616361456517712,
0.17449327037699397,
-1.1936385630270505,
-0.3626977992670382,
0.6065214216445567,
-0.27266827511951447,
-0.008337250595426905,
-0.11442883582944025,
-0.2552879859027531,
0.8608200568301817,
0.9797863499294986,
-0.5454704270163543,
-0.12465088219091108,
1.0680711786011012,
0.008287966764571728,
0.8340806285400814,
-0.10467902667059636,

0.6944226087871278,
-1.1848583772718688,
-2.1289584360902833,
2.0815012423116896,
0.7888919711742388,
-0.12601348190113199,
0.5814024197737417,
-0.32142652665182836,
-1.0761996135691065,
1.1437448939093156,
-0.9665568900718753,
-0.751505855358832,
-0.8777599880692168,
-0.0665611067528402,
1.625406556377903,
0.14073065902471577,
-0.008887426812216265,
0.11314595138478645,
0.23282575925386234,
-1.3029158546977586,
0.39815363644872126,
-0.28006162282581953,
0.5981370714824289,
-0.08964177383705696,
-0.23973482016502778,
-1.1546366940781616,
-0.3335643458249379,
1.4313780024296703,
0.6615033850837366,
0.09540664540173596,
1.6701354164263156,
1.6037180066269185,
0.45366725021426957,
1.5306014722736203,
-0.7114276522073563,
-0.0387128408482294,
-0.40951463995726844,
-0.14821428534018466,
1.6120880297451547,
-0.1578213810367529,
-0.4551176093567465,
0.12643398042842974,
0.29421606580526033,
0.3517029135915254,
-0.21200478261515812,
1.503225888163223,
0.6867008326592696,
1.2990703939363029,
-1.2633831104757771,
-0.35212753263373886,
0.2558013627761136,
0.1557716944165696,
-0.2985274551715097,
0.3096340969306992,
1.1065921140021713,
0.5655703287990456,
0.29407543941401004,
1.87556201451638,
-0.005713899736395531,
-0.47654203677386653,
-0.33487139893703466,
0.28841760275861744,
0.881872547749141,
-0.2998307941218902,
1.3565987475579786,
0.39905124343442133,
0.37529991980433175,
0.4721044202054456,
-0.0837323332258721,
0.2128419559212174,
1.445893331967549,
-0.5917542945785007,
-0.33453435504213846,
-0.05724156447138151,
0.5825344314121998,


```

0.05851962203923921,
0.13841145669138022,
-0.07195386411014011,
1.546635282716052,
-0.3465571437275552,
0.9484886642506297,
0.41941222891459207,
0.0901808762528231,
-0.8871255799219674,
-0.4695342320219668,
0.6930505657922351,
1.1594181773830308,
0.042534590889293046,
0.45776921806648474,
0.8831536077453052,
-1.883349137185357,
2.293435211334673,
1.3066760169202165,
-0.040640374761399124,
-0.8716978424494161,
1.5544327319775963,
0.5977489456678642,
-1.5697728131838606]],
    'run_time_sec': 35225.583796978}}]],
'redundant': [ { 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
    0,
    0,
    0,
    0,
    0],
    'cn': [ 0.5,
    0.5,
    0.5,
    0.5,
    0.5,
    0.5],
    'r': 0},
    'time': {'max': 10.0, 'min': 0.0},
    'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.00487815438629808,
    0.3836991336608517,
    0.3836991336608516,
    1.1027183519022283,
    1.430822963194853,
    1.7336849694382164,
    1.9079110557306378,
    1.9079110557306376,
    2.407942958541018,
    2.4384441996231097,
    2.7198331200698176,
    2.7363529901218957,
    3.8707803400773457,
    3.8707803400773457,
    4.283774948106429,
    4.283774948106428,
    4.283774948106429,
    4.283774948106429,
    5.706222745811317,
    5.706222745811317,
    5.706222745811317,
    5.706222745811317,
    5.706222745811317,
    5.706222745811317,
    5.706222745811317,
    6.386180553174607,
    6.421245480123581,
    8.31501274226813,
    8.345179306265157,
    11.444382336838434,
    0.4142599738840607,
    -0.0179112834331289,
    -1.916401049394031,

```

0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,

```

0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,

```

```
'optimization': { 'algorithm': 'SLSQP',
                  'constraint': True,
                  'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']]],
'output': { 'data_points': [ [4, 2, 0, 0, -7.597273562260126],
                             [12, 6, 0, 0, -6.262099522076846],
                             [20, 10, 0, 0, -5.200592923966883],
                             [28, 14, 0, 0, -4.350606799333773]]},
noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {
                        'noise_prob': [
                            0.001,
                            0.001,
                            0.0],
                        'status': True,
                        'type':
                            'dephasing'},
                    'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
      'noise_prob': [
          0.001,
          0.001,
          0.0],
      'status': True,
      'type':
          'dephasing'},
      'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
      'noise_prob': [
          0.001,
          0.001,
          0.0],
      'status': True,
      'type':
          'dephasing'},
      'odd_wires': 4}
]
```

```

'noise_on_init_param': { 'status': False,
    'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_profile': {

    'noise_prob': [

        0.0,

        'status': True,
        'type':

            'odd_wires': 4]],
            'run_time_sec': 14.164844751358032}},
{ 'config': { 'ansatz': { 'gateset': 1,
    'layer': 30,
    'ugate': { 'coefficients': { 'bn': [ 0,
        0,
        0,
        0,
        0,
        0,
        0],
        'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
    'init_param': { 'value': [ 0.029287487040924362,
        0.6764270956551205,
        1.6620807976810363,
        1.7056653351603042,
        1.8485187197716686,
        1.9397654142515774,
        1.9426163313124116,
        1.9760071204347747,
        2.0652508647096677,
        3.338212066261205,
        3.3382549495561102,
        3.507308503786301,
        3.668597772988933,
        3.6686655840136484,
        3.7105300968471533,
        3.737148430160667,
        5.361211694679849,
        5.40430867517309,
        5.45325014669751,
        5.461795374069968,
        5.465563084002408,
        5.492346359399902,
        7.18245326721486,
        7.245488992311329,
        7.248698529808956,
        8.18726919118503,
        8.197326336426727,
        8.213328813441153,
        8.251719559085661,
        10.791930409527371,
        -0.48557603875244904,
        -0.06546231446775812,
        -1.4810159853023965,
        -0.6439509610243448,
        -0.4889611776104492,
        0.10983589949376514,
        -1.2155949975003408,
        1.1338047187824702,
        0.18557769650103798,
        -0.36145572824608646,
        -0.7821715666301856,
        0.8512861656756193,
        0.30536456746903484,
        -0.30807040011779185,

```

0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,

```

-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noise_free_time_evo_dephasing_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',

```

```
'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '  
      '0.5 Z 0 Z 1 + '  
      '0.5 X 1 X 2 + '  
      '0.5 Z 1 Z 2 + '  
      '0.5 X 2 X 3 + '  
      '0.5 Z 2 Z 3 + '  
      '0.5 X 3 X 4 + '  
      '0.5 Z 3 Z 4 + '  
      '0.5 X 4 X 5 + '  
      '0.5 Z 4 Z 5 + '  
      '0.5 X 5 X 6 + '  
      '0.5 Z 5 Z 6']],  
  
  'output': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],  
                                [12, 6, 0, 0, -6.098488474706268],  
                                [20, 10, 0, 0, -4.939317874668222],  
                                [28, 14, 0, 0, -4.015452245150805]],  
    'noise_details': [ { 'gates_num': [4, 1, 1, 0],  
                          'identity_factors': [0, 0, 0, 0],  
                          'noise_level': [4, 2, 0, 0],  
                          'noise_profile': {  
  
                            'noise_prob': [  
  
                              0.0,  
  
                              'status': True,  
                              'type':  
  
                                'odd_wires': 4},  
                        { 'gates_num': [12, 3, 1, 0],  
                          'identity_factors': [1, 1, 0, 0],  
                          'noise_level': [12, 6, 0, 0],  
                          'noise_profile': {  
  
                            'noise_prob': [  
  
                              0.0,  
  
                              'status': True,  
                              'type':  
  
                                'odd_wires': 4},  
                        { 'gates_num': [20, 5, 1, 0],  
                          'identity_factors': [2, 2, 0, 0],  
                          'noise_level': [20, 10, 0, 0],  
                          'noise_profile': {  
  
                            'noise_prob': [  
  
                              0.0,  
  
                              'status': True,  
                              'type':  
  
                                'odd_wires': 4},  
                        { 'gates_num': [28, 7, 1, 0],  
                          'identity_factors': [3, 3, 0, 0],  
                          'noise_level': [28, 14, 0, 0],  
                          'noise_profile': {  
  
                            'noise_prob': [  
  
                              0.0,  
  
                              'status': True,  
                              'type':  
  
                                'odd_wires': 4},  
                        { 'gates_num': [28, 7, 1, 0],  
                          'identity_factors': [3, 3, 0, 0],  
                          'noise_level': [28, 14, 0, 0],  
                          'noise_profile': {  
  
                            'noise_prob': [  
  
                              0.0,
```



```

        'odd_wires': 4]],
        'run_time_sec': 13.985239267349243}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                            'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                            'r': 0}},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}}},
'init_param': { 'value': [ 1.0344655484192336e-17,
                           0.36119418156962224,
                           0.3611941815696223,
                           0.36119418156962224,
                           0.8440759344131337,
                           1.1727045787919286,
                           1.5692279974524073,
                           1.570888864611923,
                           1.592214192172409,
                           1.7598342563277258,
                           3.059056443368036,
                           3.0591149843524144,
                           4.3676699123871625,
                           4.3676699123871625,
                           5.510828301667605,
                           5.51914482396091,
                           6.113847931320903,
                           6.113847931320903,
                           6.533178927747682,
                           6.533178927747682,
                           6.755940863606281,
                           6.837142696961756,
                           7.694830693654466,
                           7.694830693654467,
                           7.694830693654466,
                           7.873525558910739,
                           7.873525558910739,
                           7.922034998277628,
                           7.922034998277628,
                           12.310276979046629,
                           -0.6972922025249428,
                           -0.0508578639309505,
                           0.03511585194544882,
                           -0.05290285417260422,
                           0.6885525491076263,
                           0.23164759975262209,
                           0.06930481892124933,
                           -0.05051704727786026,
                           0.2745998381310896,
                           2.8540854853592244,
                           0.1762956340846527,
                           -0.46176266603345323,
                           0.4487337371763968,
                           1.96119949254268,
                           -0.028430732456242037,
                           1.020183549895116,
                           0.09052188212177692,
                           0.6283703459809612,
                           -0.2008669437139055,
                           -0.333213682394554,
                           0.49298533108167947,
                           -0.073860646808665,
                           -0.13593481376514907,
                           -0.597217576988091,
                           0.38422303935103047,

```

-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,

[illegible]

[illegible]

```
                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 0},
                                'time': {'max': 10.0, 'min': 0.0},
                                'type': 'xy-iss'}},
'init_param': { 'value': [ 0.1021834784589592,
                             0.9196384424899031,
                             0.9261694014921312,
                             0.983882928321485,
                             1.8913212439431486,
                             1.9426406276748993,
                             1.9426406276748993,
                             1.9426406276748993,
                             2.7354895381565307,
                             2.7475159984062567,
                             4.89472071800884,
                             4.89472071800884,
                             4.89472071800884,
                             4.951228136281907,
                             4.9774006951115535,
                             4.9774006951115535,
                             5.053403895048273,
                             5.053403895048273,
                             5.101252532365191,
                             7.988961801418946,
                             7.988961801418945,
                             7.988961801418946,
                             7.988961801418946,
                             8.978507898154866,
                             8.993364745936525,
                             9.010251752506093,
                             9.010251752506093,
                             9.808175975385067,
                             9.808175975385067,
                             10.987620898296862,
                             -0.17474125502348634,
                             0.15377532976778396,
                             0.10530559022128577,
                             -0.002734731613738592,
                             0.18566397098861895,
                             -0.21248706612806187,
                             0.00824191786492345,
                             -0.17814151074061357,
                             0.01796252087548974,
                             0.34324522632459664,
                             0.0074801226415391366,
                             0.7599717323032058,
                             -0.10515906185801696,
                             1.0828859909757265,
                             0.414437884618902,
                             1.5228608411556923,
                             0.33559085686809753,
                             0.015737214192853767,
                             -0.5136276162084019,
                             -1.8563551585814209,
                             0.5712467952174463,
                             0.8929866894843974,
                             0.13811028968068004,
                             -0.2763065211700681,
                             -0.7071702673229635,
                             -0.18135433485295524,
                             -0.27765639095770533,
                             0.7217433560748705,
                             -0.8828879013995732,
                             -2.0374813859941323,
                             1.36473854657663,
                             0.034109249352092696,
                             -0.6149254344617949,
                             0.03372845053867827,
                             0.503938133787832,
                             -0.6883442402732416,
```

0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,
0.028627573515451674,
1.9279053503307635,

```

0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-

iss_noisefree_time_evo_dephasing_ric4_sample#10'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
{others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -7.588052513021028],
                              [12, 6, 0, 0, -6.143369929824418],
                              [20, 10, 0, 0, -4.998571859003288],
                              [28, 14, 0, 0, -4.085121357543176]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

```

```
'noise_prob': [
0.001,
0.001,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
'identity_factors': [1, 1, 0, 0],
'noise_level': [12, 6, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0],
'status': True,
'type':
'dephasing'},
'odd_wires': 4}],
'run_time_sec': 14.418097496032715}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.25473970784448047,
1.2025511746035562,
```


1.5481560170840756,
1.8324395977107744,
1.8439662803769572,
2.9830338898217192,
2.9830338898217192,
3.241693555484232,
3.634822278674692,
3.7522709686014135,
3.7522709686014135,
4.94705318367592,
4.955555446952993,
5.343973550433924,
5.3577622539956655,
6.390003257794682,
6.419263094824833,
8.039405153723413,
8.039804264859526,
8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,

0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
'value': 0 },

```

        'noise_prob': [0.001, 0.001, 0.0, 0.0],
        'status': True,
        'type': 'dephasing'},
    'nqubits': 7,
    'observable': { 'coefficients': { 'bn': [ 1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0,
        1.0],
        'cn': [ 0.5,
        0.5,
        0.5,
        0.5,
        0.5,
        0.5],
        'r': 1},
        'def': 'ising'},
    'output': { 'draw': { 'fig_dpi': 100,
        'status': False,
        'type': 'png'},
        'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#9'},
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
        [1, 1, 0, 0],
        [2, 2, 0, 0],
        [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
        'constraint': True,
        'status': False}},
    'zne': { 'data_points': None,
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
        '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
        '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
        '6 + 1 Z 5 + 1 Z 6',
        'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
        '0.5 Z 0 Z 1 + '
        '0.5 X 1 X 2 + '
        '0.5 Z 1 Z 2 + '
        '0.5 X 2 X 3 + '
        '0.5 Z 2 Z 3 + '
        '0.5 X 3 X 4 + '
        '0.5 Z 3 Z 4 + '
        '0.5 X 4 X 5 + '
        '0.5 Z 4 Z 5 + '
        '0.5 X 5 X 6 + '
        '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.590367983241113],
        [12, 6, 0, 0, -6.1449349379730975],
        [20, 10, 0, 0, -5.02259982066239],
        [28, 14, 0, 0, -4.142640637847531]],
        'noise_details': [ { 'gates_num': [4, 1, 1, 0],
        'identity_factors': [0, 0, 0, 0],
        'noise_level': [4, 2, 0, 0],
        'noise_profile': {
'noise_on_init_param': { 'status': False,
        'value': 0},
        'noise_prob': [
0.001,
0.001,
0.0],
        'status': True,
        'type':
'dephasing'},
        'odd_wires': 4},
        { 'gates_num': [12, 3, 1, 0],
        'identity_factors': [1, 1, 0, 0],

```

```

'noise_level': [12, 6, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,

0.0],

0.0,

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
'identity_factors': [2, 2, 0, 0],
'noise_level': [20, 10, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,

0.0],

0.0,

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {

'noise_prob': [

0.001,
0.001,

0.0],

0.0,

'status': True,
'type':

'odd_wires': 4}],
'run_time_sec': 13.20900011062622}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.8953934585022342,
1.0441540460805852,
1.2122882144067701,
1.3902546362390118,
2.4452392605971704,
2.4476376114298906,
3.59167707372567,
3.592168933497119,
4.752173032506677,
4.770483368888809,
5.226089600509016,
5.227272839220446,
5.277877376458118,

```

6.286147534970271,
6.330458303415314,
7.343951335722083,
7.407616509271585,
7.5311385683234535,
7.532344729415073,
7.683350056306,
7.684095349251904,
8.372928051556965,
8.450880672852195,
8.893521328096941,
8.933105958948394,
10.009729102620259,
10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,

[illegible]

```

                                'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                'r': 1},
                                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-

iss_noisefree_time_evo_dephasing_ric4_sample#4'},

'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},

'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},

'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']},

'output': { 'data_points': [ [4, 2, 0, 0, -7.665909699744035],
                             [12, 6, 0, 0, -6.335722541155832],
                             [20, 10, 0, 0, -5.2782113804278685],
                             [28, 14, 0, 0, -4.431307763135173]],
            'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                'identity_factors': [0, 0, 0, 0],
                                'noise_level': [4, 2, 0, 0],
                                'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'oddd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {

'noise_on_init_param': { 'status': False,
                          'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

```

'dephasing'},

'noise_on_init_param': { 'status': False,
 'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'noise_on_init_param': { 'status': False,
 'value': 0},

0.001,
0.001,

0.0],

'dephasing'},

'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
 'identity_factors': [2, 2, 0, 0],
 'noise_level': [20, 10, 0, 0],
 'noise_profile': {

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
 'identity_factors': [3, 3, 0, 0],
 'noise_level': [28, 14, 0, 0],
 'noise_profile': {

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4]],
 'run_time_sec': 13.768438577651978}},
{ 'config': { 'ansatz': { 'gateset': 1,
 'layer': 30,
 'ugate': { 'coefficients': { 'bn': [0,
 0,
 0,
 0,
 0,
 0,
 0],
 'cn': [0.5,
 0.5,
 0.5,
 0.5,
 0.5,
 0.5],
 'r': 0},
 'time': {'max': 10.0, 'min': 0.0},
 'type': 'xy-iss'}},
 'init_param': { 'value': [0.08803864939944098,
 0.2533857257734638,
 0.4067196832055002,
 0.4411063797089926,
 1.0492461473989636,
 1.0543168722856693,
 2.21020815530381,
 2.218381560687012,
 2.367234970755274,
 2.658389587322544,
 3.2990226425967433,
 3.5549209440787726,
 4.0112101619321985,
 4.138810393646779,
 4.424339957187965,
 6.02518256832302,
 6.064174786863594,
 6.069208430366886,
 6.069208430366886,
 6.941565801297405,
 7.124864639242438,
 7.129421087787242,
 7.45818921038985,
 7.468062221399231,

8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,

```

0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},

```

```

iss_noisefree_time_evo_dephasing_ric4_sample#7'},
    'file_name_prefix': 'AUTOMATE_xy-
    'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                           [1, 1, 0, 0],
                                           [2, 2, 0, 0],
                                           [3, 3, 0, 0]]},
    'run': 'redundant',
    'state': 'dmatrix',
    'vqe': { 'iteration': 1,
             'optimization': { 'algorithm': 'SLSQP',
                               'constraint': True,
                               'status': False}},
    'zne': { 'data_points': None,
             'degree': 0,
             'method': 'richardson',
             'sampling': 'default'}},
    'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                     '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                     '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                     '6 + 1 Z 5 + 1 Z 6',
                'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                            '0.5 Z 0 Z 1 + '
                                                            '0.5 X 1 X 2 + '
                                                            '0.5 Z 1 Z 2 + '
                                                            '0.5 X 2 X 3 + '
                                                            '0.5 Z 2 Z 3 + '
                                                            '0.5 X 3 X 4 + '
                                                            '0.5 Z 3 Z 4 + '
                                                            '0.5 X 4 X 5 + '
                                                            '0.5 Z 4 Z 5 + '
                                                            '0.5 X 5 X 6 + '
                                                            '0.5 Z 5 Z 6']},
    'output': { 'data_points': [ [4, 2, 0, 0, -7.6017515276839065],
                                  [12, 6, 0, 0, -6.187268174258091],
                                  [20, 10, 0, 0, -5.079662736732888],
                                  [28, 14, 0, 0, -4.206145082489222]],
              'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                                   'identity_factors': [0, 0, 0, 0],
                                   'noise_level': [4, 2, 0, 0],
                                   'noise_profile': {
                                     'noise_prob': [
                                         0.001,
                                         0.001,
                                         0.0],
                                     'status': True,
                                     'type':
                                         'dephasing'},
                                   'odd_wires': 4},
                                { 'gates_num': [12, 3, 1, 0],
                                   'identity_factors': [1, 1, 0, 0],
                                   'noise_level': [12, 6, 0, 0],
                                   'noise_profile': {
                                     'noise_prob': [
                                         0.001,
                                         0.001,
                                         0.0],
                                     'status': True,
                                     'type':
                                         'dephasing'},
                                   'odd_wires': 4},
                                { 'gates_num': [20, 5, 1, 0],
                                   'identity_factors': [2, 2, 0, 0],
                                   'noise_level': [20, 10, 0, 0],
                                   'noise_profile': {
                                     'noise_prob': [
                                         0.001,
                                         0.001,
                                         0.0],
                                     'status': True,
                                     'type':
                                         'dephasing'},
                                   'odd_wires': 4}
                                ]
    }
}

```

```

0.0],
0.0,
'status': True,
'type':
'dephasing'},
'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
'identity_factors': [3, 3, 0, 0],
'noise_level': [28, 14, 0, 0],
'noise_profile': {
'noise_prob': [
0.001,
0.001,
0.0,
'status': True,
'type':
'dephasing'},
'odd_wires': 4}},
'run_time_sec': 12.644636869430542}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0780449993030362,
0.10783439075300198,
0.5845642021744096,
0.7814268466054133,
0.8443273792280553,
2.8661173620125435,
2.866459808493013,
2.866459808493013,
2.866459808493013,
2.866459808493013,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.787331067871434,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
7.013227786440474,
7.016905064536492,
8.483022393696176,
8.483022393696176,
8.483022393696176,
8.483022393696176,
10.30807680489275,
10.30807680489275,
10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,

```

0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,
-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,

```

-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,

```



```
'value': 0},

0.001,
0.001,

0.0],

'noise_prob': [

0.0,

'status': True,
'type':

'odd_wires': 4]],
'run_time_sec': 13.415238857269287}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0}},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,
8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
```


-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,

[illegible]

```

'0.5 X 1 X 2 + '
'0.5 Z 1 Z 2 + '
'0.5 X 2 X 3 + '
'0.5 Z 2 Z 3 + '
'0.5 X 3 X 4 + '
'0.5 Z 3 Z 4 + '
'0.5 X 4 X 5 + '
'0.5 Z 4 Z 5 + '
'0.5 X 5 X 6 + '
'0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.517540187131846],
                              [12, 6, 0, 0, -5.995430553311077],
                              [20, 10, 0, 0, -4.811503569399131],
                              [28, 14, 0, 0, -3.882991739348091]],
  'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                      'identity_factors': [0, 0, 0, 0],
                      'noise_level': [4, 2, 0, 0],
                      'noise_profile': {
'noise_on_init_param': { 'status': False,
                          'value': 0},

                              'noise_prob': [
0.001,
0.001,

0.0],

                              'status': True,
                              'type':
'dephasing'},

                              'odd_wires': 4},
{ 'gates_num': [12, 3, 1, 0],
  'identity_factors': [1, 1, 0, 0],
  'noise_level': [12, 6, 0, 0],
  'noise_profile': {
                              'noise_prob': [
0.001,
0.001,

0.0],

                              'status': True,
                              'type':
'dephasing'},

                              'odd_wires': 4},
{ 'gates_num': [20, 5, 1, 0],
  'identity_factors': [2, 2, 0, 0],
  'noise_level': [20, 10, 0, 0],
  'noise_profile': {
                              'noise_prob': [
0.001,
0.001,

0.0],

                              'status': True,
                              'type':
'dephasing'},

                              'odd_wires': 4},
{ 'gates_num': [28, 7, 1, 0],
  'identity_factors': [3, 3, 0, 0],
  'noise_level': [28, 14, 0, 0],
  'noise_profile': {
                              'noise_prob': [
0.001,
0.001,

0.0],

                              'status': True,
                              'type':
'dephasing'},

                              'odd_wires': 4}],
  'run_time_sec': 13.281006097793579}},

```

```
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.009858867127598406,
                              0.6371504690778502,
                              1.0666846377193393,
                              1.104415373659906,
                              1.464838116920491,
                              1.4648381169204907,
                              1.9080243917751187,
                              2.1981829040528402,
                              2.6297267537087063,
                              2.658272884291024,
                              3.752650276988564,
                              3.752985814574736,
                              4.407595284592648,
                              4.407595284592648,
                              5.031509128956631,
                              5.031509128956631,
                              5.031509128956631,
                              5.033233798634394,
                              5.033233798634395,
                              5.036830700020435,
                              5.036830700020434,
                              7.933675292188165,
                              7.950623161195741,
                              7.950659720589115,
                              8.023568123130133,
                              8.309873389450049,
                              8.310972029308223,
                              8.959490059358307,
                              9.904746541347446,
                              9.949991834284726,
                              -0.41951749885307754,
                              0.029035276468521477,
                              0.030870805732646677,
                              -0.17295615718073906,
                              0.5816925184728065,
                              0.9382868815479958,
                              0.3165320552518337,
                              0.42890233892866725,
                              0.013440093351710147,
                              1.2501583763834956,
                              -0.35093579496718846,
                              1.139236973962438,
                              -0.013100397435237714,
                              -0.1455411128269196,
                              0.09889932710883725,
                              -0.821320047619297,
                              -0.4558166636467179,
                              1.1031921527651727,
                              -2.1974313327571964,
                              1.4792834343123176,
                              -0.30822227350872,
                              0.588013825131612,
                              0.515130316597666,
                              0.22678729305502876,
                              0.10858837903610288,
                              0.5318167388137908,
                              0.8532518222915644,
```

-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,

```

0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'redundant',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': None,
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z '
                                '1 + 1 X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + '
                                '1 Z 3 + 1 X 4 X 5 + 1 Z 4 + 1 X 5 X '
                                '6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + '
                                                         '0.5 Z 0 Z 1 + '
                                                         '0.5 X 1 X 2 + '
                                                         '0.5 Z 1 Z 2 + '
                                                         '0.5 X 2 X 3 + '
                                                         '0.5 Z 2 Z 3 + '
                                                         '0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + '
                                                         '0.5 X 4 X 5 + '
                                                         '0.5 Z 4 Z 5 + '
                                                         '0.5 X 5 X 6 + '
                                                         '0.5 Z 5 Z 6']],
'output': { 'data_points': [ [4, 2, 0, 0, -7.407013825547437],

```

```

[12, 6, 0, 0, -6.114813208156221],
[20, 10, 0, 0, -5.082084674797013],
[28, 14, 0, 0, -4.249248629692988]],
'noise_details': [ { 'gates_num': [4, 1, 1, 0],
                    'identity_factors': [0, 0, 0, 0],
                    'noise_level': [4, 2, 0, 0],
                    'noise_profile': {

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

'noise_on_init_param': { 'status': False,
                        'value': 0},

                                'noise_prob': [

0.001,
0.001,

                                0.0,

                                'status': True,
                                'type':

                                'odd_wires': 4},
                                { 'gates_num': [28, 7, 1, 0],
                                'identity_factors': [3, 3, 0, 0],
                                'noise_level': [28, 14, 0, 0],
                                'noise_profile': {

                                'run_time_sec': 12.752790689468384}}],
'vqe': [ { 'config': { 'ansatz': { 'gateset': 1,
                                'layer': 30,
                                'ugate': { 'coefficients': { 'bn': [ 0,
                                0,
                                0,
                                0,
                                0,
                                0,
                                0],
                                'cn': [ 0.5,
                                0.5,

```

```

0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': {'value': 'random'},
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'}},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': True,
                      'type': 'png'},
            'file_name_prefix': 'xy_noisefree_time_evo'},
'redundant': { 'identity_factors': [[0, 0, 0, 0], [1, 1, 1, 1]]},
'run': 'vqe',
'state': 'dmatrix',
'vqe': { 'iteration': 10,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': True}},
'zne': { 'data_points': None,
         'degree': 1,
         'method': 'richardson',
         'sampling': 'default'}},
'others': { 'observable_string': '1 X 0 X 1 + 1 Z 0 + 1 X 1 X 2 + 1 Z 1 + 1 '
                                'X 2 X 3 + 1 Z 2 + 1 X 3 X 4 + 1 Z 3 + 1 X '
                                '4 X 5 + 1 Z 4 + 1 X 5 X 6 + 1 Z 5 + 1 Z 6',
            'time_evolution_gate_hamiltonian_string': [ '0.5 X 0 X 1 + 0.5 Z '
                                                         '0 Z 1 + 0.5 X 1 X 2 '
                                                         '+ 0.5 Z 1 Z 2 + 0.5 '
                                                         'X 2 X 3 + 0.5 Z 2 Z '
                                                         '3 + 0.5 X 3 X 4 + '
                                                         '0.5 Z 3 Z 4 + 0.5 X '
                                                         '4 X 5 + 0.5 Z 4 Z 5 '
                                                         '+ 0.5 X 5 X 6 + 0.5 '
                                                         'Z 5 Z 6']},
'output': { 'exact_sol': -8.56677223350563,
            'initial_cost_history': [ 2.490874640234538,
                                     3.123425159431485,
                                     3.646429251689861,
                                     2.764268013304591,
                                     3.17022836931859,
                                     3.7735896776807554,
                                     3.622216535636285,
                                     2.438144835449103,
                                     3.477758051251201,
                                     4.254627215438117],
            'noise_details': { 'gates_num': [4, 1, 1, 0],
                              'identity_factors': [0, 0, 0, 0],
                              'noise_level': [4, 2, 0, 0],
                              'noise_profile': { 'noise_on_init_param': {

```

```

'status': False,
'value': 0},

```

```

'noise_prob': [ 0.001,
                0.001,

```



```
0.0,
0.0],
'status': True,
'type': 'dephasing'},
'odd_wires': 4},
'optimized_minimum_cost': [ -7.597273562260196,
-7.556050750105993,
-7.517540187131898,
-7.665909699744024,
-7.524810336843006,
-7.40701382554743,
-7.601751527683981,
-7.561979391804892,
-7.59036798324113,
-7.588052513021082],
'optimized_parameters': [ [ 0.00487815438629808,
0.3836991336608517,
0.3836991336608516,
1.1027183519022283,
1.430822963194853,
1.7336849694382164,
1.9079110557306378,
1.9079110557306376,
2.407942958541018,
2.4384441996231097,
2.7198331200698176,
2.7363529901218957,
3.8707803400773457,
3.8707803400773457,
4.283774948106429,
4.283774948106428,
4.283774948106429,
4.283774948106429,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
```

0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,
-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,

0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559],
[0.0780449993030362,
0.10783439075300198,
0.5845642021744096,
0.7814268466054133,
0.8443273792280553,
2.8661173620125435,
2.866459808493013,
2.866459808493013,
2.866459808493013,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.786930144034059,
4.787331067871434,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
6.292785833628836,
7.013227786440474,
7.016905064536492,
8.483022393696176,
8.483022393696176,
8.483022393696176,
8.483022393696176,
10.30807680489275,
10.30807680489275,
10.987173174448158,
-0.09982429964200479,
-0.1011988902651152,
-0.11924685050397593,
0.11938936982148393,
0.26480809900884805,
0.9860195530322333,
0.32479711291812374,
-0.24554677857208176,
-0.049000425450729294,
2.064341791548283,
-0.10644000592090332,
1.3304809249936382,
0.46278363868645017,
0.14042287528230984,
0.2133289095055383,
-0.6423967247784159,
-0.42996733188660946,
-0.3318304058008573,
0.6204697000624867,
-0.5183204366214588,
0.6244473440442605,
0.0183714264553683,
-0.3522016603760132,
-0.9892342164188427,
-1.502610568052502,
0.3280599415277172,
-0.48553815029820807,
-0.010371641322887242,
1.573878039413477,
-0.020575326946359336,

-0.14841518088300737,
0.26016829146348547,
0.15593168727347026,
-0.5856941876551979,
0.12654079580108715,
-0.007993830462140899,
0.5218497886375917,
1.4495850566219788,
-1.1997007030306313,
0.005867405286106471,
-1.158661564042903,
0.7480115816132973,
0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,

-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,
1.5813078296578018,
0.45463950449288687,
1.5804878072389892],
[2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,
8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,

-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,
0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,

0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109],
[0.8953934585022342,
1.0441540460805852,
1.2122882144067701,
1.3902546362390118,
2.4452392605971704,
2.4476376114298906,
3.59167707372567,
3.592168933497119,
4.752173032506677,
4.770483368888809,
5.226089600509016,
5.227272839220446,
5.277877376458118,
6.286147534970271,
6.330458303415314,
7.343951335722083,
7.407616509271585,
7.5311385683234535,
7.532344729415073,
7.683350056306,
7.684095349251904,
8.372928051556965,
8.450880672852195,
8.893521328096941,
8.933105958948394,
10.009729102620259,
10.02381465220654,
10.426876716424587,
10.426876716424587,
12.52667527736335,
-0.5078346121107161,
0.5290912698971372,
0.08473868887373436,
2.3815253901669893,
-0.9125459057391813,
1.622892785810718,
0.18455439491180806,
1.6293359147694093,
0.19824902915765533,
-0.3582104166740486,
-0.3615784742236456,
1.130851060616608,
0.40734929063355524,
-0.21774506648190217,
0.7068030490484047,
0.4700157215887062,
-1.0774614500962085,
0.15964544330271405,
1.8155397001056257,
-0.2537621787194128,
1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,

0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,
1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,

0.986915389457714,
0.40188944483333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684],
[1.0344655484192336e-17,
0.36119418156962224,
0.3611941815696223,
0.36119418156962224,
0.8440759344131337,
1.1727045787919286,
1.5692279974524073,
1.570888864611923,
1.592214192172409,
1.7598342563277258,
3.059056443368036,
3.0591149843524144,
4.3676699123871625,
4.3676699123871625,
5.510828301667605,
5.51914482396091,
6.113847931320903,
6.113847931320903,
6.533178927747682,
6.533178927747682,
6.755940863606281,
6.837142696961756,
7.694830693654466,
7.694830693654467,
7.694830693654466,
7.873525558910739,
7.873525558910739,
7.922034998277628,
7.922034998277628,
12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,
-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,

0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,
0.2314394697747134,
-0.6381970494622153,
-0.869070575517897,
-0.3854227231439233,
-0.05974868627606693,
-1.3618983878421893,
-0.14735213954174853,
-0.16144336826377106,
-0.5726725646368419,
0.14863429745149237,
-0.7763964261749154,
-0.3515427279763936,
0.09267593493455233,
0.9941259428822948,
0.6113266248291309,
-2.249730610329486,
-1.4853322232412747,
-0.6624750796271123,
1.5218612703215841,
-0.060867877516889796,
0.4881957659663681,
-0.5823031021435859,
0.01548453122295137,
-0.8946631440087691,
0.2724176161972246,
1.1142419366260101,
1.1351916054213225,
0.24996433759527287,
0.10925434181359692,
0.35878338934390347,
-0.4145708548690295,
-1.0045262071213437,
-1.2832451981705593,
1.5127219636109106,
1.0543608408458283,
-0.025449240007986165,
-0.4042653654410071,
-0.5770570175675142,
-0.3798602211137218,
-0.20337950006519365,
1.1741528813917048,
0.4416785611413231,
1.2685661107772639,
-0.29162960462044624,
0.30563636100934116,
-0.34543251279098935,
-1.452238106816157,
-0.35828114927859783,
-1.6519696012854037,
0.9758818868953008,
1.8739098589464835,
0.08690130739368318,
-0.0063915911623125795,
0.058888399162031244,

-0.2901105978964291,
-0.0016858525368986894,
0.021900786471813132,
1.743664682334748,
1.2798318789150203,
-1.0536291022271305,
0.3537083031530898,
-1.4149627812472323,
1.0773219987831761,
0.29151315320582294,
-0.0298469457137327,
2.156594522363883,
1.5651984449575558,
0.9412318447665341,
1.574999269983593],
[0.009858867127598406,
0.6371504690778502,
1.0666846377193393,
1.104415373659906,
1.464838116920491,
1.4648381169204907,
1.9080243917751187,
2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,
0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,

2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,
-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,

0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012],
[0.08803864939944098,
0.2533857257734638,
0.4067196832055002,
0.4411063797089926,
1.0492461473989636,
1.0543168722856693,
2.21020815530381,
2.218381560687012,
2.367234970755274,
2.658389587322544,
3.2990226425967433,
3.5549209440787726,
4.0112101619321985,
4.138810393646779,
4.424339957187965,
6.02518256832302,
6.064174786863594,
6.069208430366886,
6.069208430366886,
6.941565801297405,
7.124864639242438,
7.129421087787242,
7.45818921038985,
7.468062221399231,
8.693928169186727,
8.75723930487613,
8.77623189944677,
8.812275470753457,
8.81486926281639,
11.019616289520474,
-0.611476740771433,
0.05158007641927867,
0.011569665901033028,
-0.054784977759570794,
0.634383931417704,
0.04908784681904958,
0.14144925064556785,
0.11185153756258273,
-0.11524892510531,
1.6968309568925728,
0.22683917566025713,
0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,

0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,
-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,

-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548],
[0.029287487040924362,
0.6764270956551205,
1.6620807976810363,
1.7056653351603042,
1.8485187197716686,
1.9397654142515774,
1.9426163313124116,
1.9760071204347747,
2.0652508647096677,
3.338212066261205,
3.3382549495561102,
3.507308503786301,
3.668597772988933,
3.6686655840136484,
3.7105300968471533,
3.737148430160667,
5.361211694679849,
5.40430867517309,
5.45325014669751,
5.461795374069968,
5.465563084002408,
5.492346359399902,
7.18245326721486,
7.245488992311329,
7.248698529808956,
8.18726919118503,
8.197326336426727,
8.213328813441153,
8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,

0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,

1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,
0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623],
[0.25473970784448047,
1.2025511746035562,
1.5481560170840756,
1.8324395977107744,
1.8439662803769572,
2.9830338898217192,
2.9830338898217192,
3.241693555484232,
3.634822278674692,
3.7522709686014135,
3.7522709686014135,
4.94705318367592,
4.95555446952993,
5.343973550433924,
5.3577622539956655,
6.390003257794682,
6.419263094824833,
8.039405153723413,
8.039804264859526,
8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,

-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,
0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,

-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956],
[0.1021834784589592,
0.9196384424899031,
0.9261694014921312,
0.983882928321485,
1.8913212439431486,
1.9426406276748993,
1.9426406276748993,
1.9426406276748993,
2.7354895381565307,
2.7475159984062567,
4.89472071800884,
4.89472071800884,
4.89472071800884,
4.951228136281907,
4.9774006951115535,
4.9774006951115535,
5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
-2.0374813859941323,

1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,
2.03737481794317,

0.7140742582546448,
-0.19665741775049655,
1.2805953485019392,
0.4766070853794039,
-0.1513782125549316,
0.2843851110065978,
-0.1577669274159728,
0.8159467256345032,
-0.6277355568432926,
0.4234084643525468,
0.7727132433737902,
-0.4733506778299185,
-0.016913095665733176,
1.2972634551629,
-0.4021788475841492,
1.4329035678455102,
0.04601713995115883,
0.17730750124782801,
0.6576144017394324,
0.8689823804207295,
0.07224948080875138,
0.46191559703995305,
0.12867490356288674,
-0.15878998029069671,
-1.0100069249851762,
-0.3314550091774262,
1.2493786629984123,
0.054568399455378225,
-0.30606643031087594,
-0.2448294164035933,
-0.27657766716916143,
0.2520940808708988,
1.092294514572833,
-0.39854325103719523,
0.7986417176519885,
-1.9442836367150043,
0.18693532185287462,
-0.8481482218221922,
-0.800305275467376,
-0.25555849012226245,
-0.20196885248203056,
0.4303789056434112,
-0.44835057571793363,
0.41691106496327845,
0.2642809626397743,
-0.8458681092967723,
-0.9876315167850924,
0.0776426669241819,
-1.285333652803171,
-0.4405835440158052,
-0.15224019288145826,
0.34671310216370815,
0.6471060776233684,
0.1360566769916084,
1.8091109536241932,
-0.5351913579644869,
0.055471984934165774,
-0.15157508478154533,
0.18330677524394517,
-0.8453242452252082,
0.2773204752174777,
-1.0714782618703085,
1.0229381545897953,
-0.41950407512958304,
-0.06937524205549918,
0.5060372963806762,
1.7035954854533657,
-0.026758189375660553,
1.4694599193724123,
0.7123580060713516,
2.7286178796506597,
0.28137612888314545,
0.5117902798374474,
-0.0974671610655493,
1.6368001090464388,

```

-0.20411846091471758,
0.5651863825669846,
0.25886809843206293,
0.5237365109643947,
-0.25021271472984985,
-1.3425637271689042,
1.15966418435163,
1.7389325851162865,
0.11263724490733788,
0.25093603513444857,
-0.013747376171993678,
-0.6509298615906746,
0.797396830947522,
0.504371040265238,
0.9933896490146416,
0.026780494246964354,
0.8037474690897719,
1.0394664682790256,
0.4736648609591909,
-0.8809394805637577,
0.31134361989806336,
1.5414914535843482,
0.643706845834748,
0.9181473047558538,
0.8813297845914521,
-0.29471651584940234,
1.6328484590945802,
0.12994830764432042,
0.8133205428193304,
-0.648285090973953,
0.09867171549157837,
1.5745657968642301,
0.1391985422471169,
-1.5484800820323548]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noise_free_time_evo_dephasing_ric4_sample#7'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.6017515276839065],
                          [12, 6, 0, 0, -6.187268174258091],
                          [20, 10, 0, 0, -5.079662736732888],
                          [28, 14, 0, 0, -4.206145082489222]],
        'degree': 0,

```


8.039804264859526,
8.097462604506834,
8.099750106093039,
8.685653778392616,
8.685653778392616,
8.685653778392616,
8.686674341764832,
8.842213669739554,
10.173486136133235,
10.173486136133235,
12.284287827852737,
-0.5275758730734342,
0.9066548612174815,
0.21500588132446097,
-1.0687833608508335,
0.6112018055607131,
1.4602447412527104,
0.5461149477957931,
0.7262216027427579,
0.29207423930862947,
1.0780988617567295,
0.030019770603227844,
0.6811845529236524,
0.16899250416157743,
0.3817265788697317,
0.8504511882214915,
0.2943749516217996,
2.5678685545322057,
-0.09707287225998511,
-0.05255145791734958,
0.2596628656770196,
1.4709186207405707,
0.36806659135242686,
1.1820332803522335,
-1.8688297897861523,
-0.023758258884169974,
-0.1539594509982686,
0.8699600311365924,
0.026062392574156816,
2.0579809580202704,
0.36704866549095816,
-0.42819223169421283,
-0.5093566943293824,
-1.9075487684328258,
-1.3810384294539255,
-1.5948684983369932,
0.9548136506191117,
0.48748532474959005,
-0.002608217350405503,
-0.6436639802673768,
-0.33250098360488384,
-2.1988529165637916,
-0.0529849319723307,
-0.5597615748640121,
-0.025876758800784605,
0.8744089624915572,
-0.01781512318958427,
-2.0833539306578586,
0.5255825940884734,
0.46948249968376826,
-0.8684311132761416,
0.4113822821242297,
0.04468218652182579,
0.43057683465524366,
0.7216954140083459,
0.7582295452144445,
0.5846573887278302,
-0.18467679793302294,
-0.12433171940725242,
0.09401401799941657,
-0.08879246907567143,
0.1383532845398174,
0.07738434728909922,
-0.39286999755540336,
1.063795934955782,

```

0.41908397165229244,
-0.42414511531979027,
-2.445836161760281,
0.06366727223695554,
-0.3742486117994384,
1.233072060683239,
0.011579473267681002,
0.897691867494392,
-0.26951728948975057,
-0.019652950430447964,
-0.4197888468028097,
-1.7260581579681544,
0.6073036000153095,
0.36957541784377107,
-0.42577280485039204,
0.6126629161697821,
0.9569327834926988,
0.3625998069074113,
-0.7312242625337203,
-0.4619176906289592,
2.085762453480289,
0.21634273093322726,
0.886234282246128,
-0.05320153003959402,
-0.2037315780251997,
0.6751812622879115,
-1.295572290327989,
-0.591508167438854,
1.592465236726996,
1.2158946657842995,
0.12039615756015172,
1.7755162594667067,
-0.04317991191957587,
0.7866056357593854,
-0.7365139362251757,
1.2881216616453464,
2.390829218577945,
1.450290921931147,
0.37551612317672206,
-0.735409730758913,
-1.1873527149276413,
-0.7214628808906888,
-0.5216606444510401,
0.08722322131599758,
0.980472902064755,
-0.04237803470525226,
-1.6462159894114976,
0.7010629237242704,
1.8950387777212538,
1.148330504456061,
1.3553409367626532,
0.056982119725580566,
1.044218720078282,
1.5556158851408661,
1.5239616241776535,
1.595603065170956]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],

```

```

        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
                                'status': False,
                                'type': 'png'},
                    'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#9'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                                [1, 1, 0, 0],
                                                [2, 2, 0, 0],
                                                [3, 3, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -7.590367983241113],
                                    [12, 6, 0, 0, -6.1449349379730975],
                                    [20, 10, 0, 0, -5.02259982066239],
                                    [28, 14, 0, 0, -4.142640637847531]],
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
                    'run_time_sec': 4.482269287109375e-05,
                    'zne_values': { 'degree': 0,
                                    'extrapolated_value': -8.459471851815815,
                                    'others': { 'beta_coefficients': [ 2.1875,
                                                                        -2.1875,
                                                                        1.3125,
                                                                        -0.3125],
                                                'cost_richardson_zne': 11.390625,
                                                'sorted_expectation_vals': [
-7.590367983241113,
-6.1449349379730975,
-5.02259982066239,
-4.142640637847531],

                                'sorted_noise': [6, 18, 30, 42]},
                                'sampled data': [ [ 4,
                                                        2,
                                                        0,
                                                        0,
                                                        -7.590367983241113],
                                                    [ 12,
                                                        6,
                                                        0,
                                                        0,
                                                        -6.1449349379730975],
                                                    [ 20,
                                                        10,
                                                        0,
                                                        0,
                                                        -5.02259982066239],
                                                    [ 28,
                                                        14,
                                                        0,
                                                        0,
                                                        -4.142640637847531]],
                                'sampling': 'default'}}},
        { 'config': { 'ansatz': { 'gateset': 1,
                                   'layer': 30,
                                   'ugate': { 'coefficients': { 'bn': [ 0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0,
                                                                        0],
                                                                    'cn': [ 0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5,
                                                                        0.5],
                                                                    }
                                   }
        }

```

```
        'r': 0},
        'time': {'max': 10.0, 'min': 0.0},
        'type': 'xy-iss'}},
'init_param': { 'value': [ 0.0780449993030362,
                           0.10783439075300198,
                           0.5845642021744096,
                           0.7814268466054133,
                           0.8443273792280553,
                           2.8661173620125435,
                           2.866459808493013,
                           2.866459808493013,
                           2.866459808493013,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.786930144034059,
                           4.787331067871434,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           6.292785833628836,
                           7.013227786440474,
                           7.016905064536492,
                           8.483022393696176,
                           8.483022393696176,
                           8.483022393696176,
                           8.483022393696176,
                           10.30807680489275,
                           10.30807680489275,
                           10.987173174448158,
                           -0.09982429964200479,
                           -0.1011988902651152,
                           -0.11924685050397593,
                           0.11938936982148393,
                           0.26480809900884805,
                           0.9860195530322333,
                           0.32479711291812374,
                           -0.24554677857208176,
                           -0.049000425450729294,
                           2.064341791548283,
                           -0.10644000592090332,
                           1.3304809249936382,
                           0.46278363868645017,
                           0.14042287528230984,
                           0.2133289095055383,
                           -0.6423967247784159,
                           -0.42996733188660946,
                           -0.3318304058008573,
                           0.6204697000624867,
                           -0.5183204366214588,
                           0.6244473440442605,
                           0.0183714264553683,
                           -0.3522016603760132,
                           -0.9892342164188427,
                           -1.502610568052502,
                           0.3280599415277172,
                           -0.48553815029820807,
                           -0.010371641322887242,
                           1.573878039413477,
                           -0.020575326946359336,
                           -0.14841518088300737,
                           0.26016829146348547,
                           0.15593168727347026,
                           -0.5856941876551979,
                           0.12654079580108715,
                           -0.007993830462140899,
                           0.5218497886375917,
                           1.4495850566219788,
                           -1.1997007030306313,
                           0.005867405286106471,
                           -1.158661564042903,
                           0.7480115816132973,
```

0.039953849295646895,
0.33138583333477056,
1.0961072721311436,
0.3042562670537851,
-1.240206197921376,
-0.16045508549607168,
-0.5385893713450153,
-0.014846874284287876,
-0.7173023530564685,
-0.041597699134906435,
0.32942338240374697,
0.15384741699350016,
-0.47063220004456435,
-0.1563046091307352,
0.42744170528430053,
-1.5243501126097334,
-1.145929016946796,
0.4680727384105657,
0.20540075247290163,
-0.49186440834908374,
-0.6221962598330377,
-0.18127207464165512,
-1.8237294859972677,
1.1777148994276196,
0.3060307920386465,
-0.08673998548611989,
0.6722807750634033,
0.7846377428452906,
-0.6030919785490115,
1.1894017334307887,
-0.40982754613077327,
1.8398702010321104,
-0.7064951591521436,
0.8167551876243924,
1.8413642104037675,
1.622557234491075,
-0.9603021225146625,
-0.059150342376347045,
-0.327562154081115,
1.1625450309385352,
-0.12130326289754916,
0.737304357051249,
0.11654441570931674,
-0.017799365215141246,
-0.7451751922261942,
0.9187105365737189,
0.3319278443156427,
0.5255755558222307,
1.007075658812707,
0.012486492197275819,
2.4607863731936233,
-0.02469551189301725,
0.022845961123708635,
0.6087761936259497,
1.64463021199776,
1.7660512783228122,
1.14772631645334,
0.051807444127151325,
-0.16122523854491633,
-0.002232791349175812,
0.4216643195629175,
0.6480059932443213,
-0.6016181032648462,
-0.7416926571351954,
-0.20144259421747251,
-0.03417100956282172,
0.3936545435805671,
-0.0024467113956832326,
0.8859812383776746,
-0.054343875816376515,
0.21951825373100647,
0.2713945224676663,
0.4702326999273568,
-0.00376868439302198,
-1.440934502366767,

```

        1.5813078296578018,
        0.45463950449288687,
        1.5804878072389892]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-iss_noisefree_time_evo_dephasing_ric4_sample#2'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                      [1, 1, 0, 0],
                                      [2, 2, 0, 0],
                                      [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'veqe': { 'iteration': 1,
          'optimization': { 'algorithm': 'SLSQP',
                            'constraint': True,
                            'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.556050750105972],
                           [12, 6, 0, 0, -6.124528497687961],
                           [20, 10, 0, 0, -5.005182740239412],
                           [28, 14, 0, 0, -4.1231804854463325]],
         'degree': 0,
         'method': 'richardson',
         'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.389617919921875e-05,
            'zne_values': { 'degree': 0,
                             'extrapolated_value': -8.412263372026647,
                             'others': { 'beta_coeffiecents': [ 2.1875,
                                                                  -2.1875,
                                                                  1.3125,
                                                                  -0.3125],
                                          'cost_richardson_zne': 11.390625,
                                          'sorted_expectation_vals': [-7.556050750105972,
                                                                        -6.124528497687961,
                                                                        -5.005182740239412,
                                                                        -4.1231804854463325]},
                             'sorted_noise': [6, 18, 30, 42]}},
'sampled_data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.556050750105972],
                  [ 12,
                    6,
                    0,
                    0,
                    -6.124528497687961],
                  [ 20,
                    10,
                    0,
                    0,
```

```

-5.005182740239412],
[ 28,
 14,
 0,
 0,
-4.1231804854463325]],
'sampling': 'default']}},
{ 'config': { 'ansatz': { 'gateset': 1,
                          'layer': 30,
                          'ugate': { 'coefficients': { 'bn': [ 0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0,
                                                                0],
                                                                'cn': [ 0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5,
                                                                0.5],
                                                                'r': 0},
                          'time': {'max': 10.0, 'min': 0.0},
                          'type': 'xy-iss'}},
  'init_param': { 'value': [ 0.8953934585022342,
                              1.0441540460805852,
                              1.2122882144067701,
                              1.3902546362390118,
                              2.4452392605971704,
                              2.4476376114298906,
                              3.59167707372567,
                              3.592168933497119,
                              4.752173032506677,
                              4.770483368888809,
                              5.226089600509016,
                              5.227272839220446,
                              5.277877376458118,
                              6.286147534970271,
                              6.330458303415314,
                              7.343951335722083,
                              7.407616509271585,
                              7.5311385683234535,
                              7.532344729415073,
                              7.683350056306,
                              7.684095349251904,
                              8.372928051556965,
                              8.450880672852195,
                              8.893521328096941,
                              8.933105958948394,
                              10.009729102620259,
                              10.02381465220654,
                              10.426876716424587,
                              10.426876716424587,
                              12.52667527736335,
                              -0.5078346121107161,
                              0.5290912698971372,
                              0.08473868887373436,
                              2.3815253901669893,
                              -0.9125459057391813,
                              1.622892785810718,
                              0.18455439491180806,
                              1.6293359147694093,
                              0.19824902915765533,
                              -0.3582104166740486,
                              -0.3615784742236456,
                              1.130851060616608,
                              0.40734929063355524,
                              -0.21774506648190217,
                              0.7068030490484047,
                              0.4700157215887062,
                              -1.0774614500962085,
                              0.15964544330271405,
                              1.8155397001056257,
                              -0.2537621787194128,

```

1.1180987998831535,
0.40746041865641885,
1.1108616066191834,
0.00610488557648816,
0.30678493019567926,
0.01091028791907567,
-0.7445142550792123,
-1.2352196719449153,
-0.653571772816555,
-0.4611953330557266,
0.08565752991537581,
0.0813437096448839,
-2.27279485915513,
-0.03473677255954619,
-0.4894636538833787,
0.4669511043718073,
-0.2371490248626185,
1.1636545881809435,
0.21860613447462532,
0.0497921238669219,
1.375807329118683,
0.021126067943046542,
-1.0713987364132078,
0.344762452911964,
-1.5512909361457807,
-0.310330728513221,
-1.689871687312018,
0.32869389154852596,
0.3906815561030999,
-0.34984892733723044,
-1.6456823763368127,
0.1093639311548509,
1.4684526520797045,
-0.09093723787377864,
-0.6803953498331798,
2.088957585805826,
-0.04413049439524755,
0.3277143811391025,
-0.24948682321783303,
0.1768906258817653,
0.13682670740442265,
-0.6541066319750621,
0.3150602528479564,
-0.459363269883666,
-0.047846043993038684,
1.1576193080471304,
0.1972943390200383,
-0.2014172390066315,
-0.1228043364674638,
-0.143826640846101,
-0.26341202303603034,
0.2670898249561223,
-0.028361072275259608,
1.014167829240247,
-0.011636582116980779,
0.1843617184800121,
0.36292859138408246,
0.8430675903929425,
0.23366733094759262,
-0.6385979893598965,
-0.007183689544316126,
1.7853025731914869,
-0.034132835655650305,
-0.1455957248203186,
0.05585584859557191,
0.06537395729569952,
-0.4023181809949219,
-0.05370753609469685,
-0.6787322938276873,
-0.5690319733909538,
0.42060663092242057,
0.06217371011548838,
0.37745819786193147,
-0.18256214854019348,
0.42531442251314333,


```

1.3857409574918338,
-1.1344275258378198,
-1.4248113082896838,
-1.0998402046367368,
-0.335917335859939,
1.3386757647812868,
-0.10182363734031222,
-1.1208849597916206,
-0.4211161214521572,
-0.013464091767170082,
0.986915389457714,
0.4018894448333316,
-0.06024988836486815,
-0.726084972369868,
0.04754575551114563,
0.06802926538764557,
1.2790185155667642,
1.550006917751972,
-0.07696390135313262,
0.989634739410609,
0.050547565895975255,
0.9863776720391318,
1.5872064544307674,
0.08812181275390576,
1.5342669559266684]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#4'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.665909699744035],
                          [12, 6, 0, 0, -6.335722541155832],
                          [20, 10, 0, 0, -5.2782113804278685],
                          [28, 14, 0, 0, -4.431307763135173]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 4.0531158447265625e-05,
            'zne_values': { 'degree': 0,
                            'extrapolated_value': -8.45265317024353,
                            'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                1.3125,
                                                                1.3125,
                                                                1.3125]
                                }
                            }
            }

```

-7.665909699744035,
-6.335722541155832,
-5.2782113804278685,
-4.431307763135173],

-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [

'sorted_noise': [6, 18, 30, 42]],
'sampled data': [[4,
2,
0,
0,
-7.665909699744035],
[12,
6,
0,
0,
-6.335722541155832],
[20,
10,
0,
0,
-5.2782113804278685],
[28,
14,
0,
0,
-4.431307763135173]],
'sampling': 'default']}]},

{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [0,
0,
0,
0,
0,
0],
'cn': [0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [2.067283500702386e-18,
0.00305317943967044,
0.6707334318791758,
0.6819425533370418,
2.0804317671049177,
2.0811966167088074,
3.3186336689124984,
3.3186336689124984,
4.0962902377799315,
4.0962902377799315,
4.0962902377799315,
4.235998445520904,
4.662355223605685,
4.662355223605685,
4.801118955358666,
4.801118955358666,
5.157394229481825,
6.5093436750150495,
6.5217715636056575,
7.126394528770992,
7.126394528770992,
7.5433952316895745,
7.5433952316895745,
8.461929873684097,
8.461929873684097,
8.516782749532563,
8.558798125131,
8.770317023984102,

8.869770466030348,
10.891182667941633,
-0.17551436837428686,
-0.09405623650981064,
1.4401112083844008,
-0.21984615875458338,
-1.2243534933440952,
-0.21317591292102225,
1.1730311609183601,
0.18285933627515502,
-0.16374038568220986,
0.49758703398713866,
-0.11662974810194524,
-1.8302240243419805,
-0.8876433539714791,
-0.5042106139218809,
0.2510944840638433,
0.2904423394991934,
-0.15052766025181108,
0.0485323125786155,
-0.23225254624760458,
-1.4505258693023013,
0.29319886843755527,
0.6416318622903661,
0.6810789378910713,
-0.03223238490139622,
-0.5600259701866226,
-0.09073514112069092,
-0.4866640178361377,
-0.8112756521867465,
-0.4472812339981313,
0.18546628616530797,
-0.3017400389598454,
0.020143029065560876,
1.220801874402924,
-0.8250485337987721,
-1.821298771084876,
0.4989152401958621,
0.01544357202836929,
1.1688745785443748,
-0.09626627763726886,
0.13498746834242364,
0.14236935407805887,
0.5138324701079404,
-0.70567129454201,
-0.45238670541244236,
0.4557602919843048,
-0.10639871186720319,
-0.1449779391068916,
0.6567235413177127,
1.6434203590388357,
0.4690727212457002,
0.7340394449185292,
-1.682257123392106,
0.6408454201508185,
-0.4354599267026243,
-1.0013539627315173,
-0.8014884208507198,
0.5593496069794457,
1.9739390283257456,
-0.35414021048847627,
-0.3024020548165168,
-0.12571822676098343,
-0.18350047821225085,
-0.4467766557524969,
-0.07944513367196834,
1.2149561188150322,
0.12080967529323339,
-0.3148883019098932,
0.9426497867827067,
0.6497759547016642,
0.06908207744966537,
-0.6478169981402983,
-0.2709025359476109,
0.6292918380932265,

```

0.5252354303643885,
0.07327033677741535,
-0.026619196060195742,
0.7594641713204215,
0.05031173306613426,
-1.4807162551726014,
0.3333838257074943,
1.040750341033686,
-1.207078213716876,
0.7239720352372947,
-0.010449420692833968,
1.6597294512570255,
-0.06437370840521775,
-0.8579848526364434,
-0.4834755930024723,
-0.7735260668724492,
0.3047381111486677,
0.6952642407616384,
0.06498018057930618,
0.607263159576155,
0.7008934040607981,
0.20712882246163866,
-0.008112248274554014,
0.4872667554021213,
-0.27713620153995255,
-0.20235667223955767,
-0.25319225213553376,
-0.05387656440785283,
-0.9166617023895225,
0.9544506975384591,
1.0672534352193421,
-0.22074936862583866,
0.7612079928609474,
-0.5987709069277743,
0.5862500142251926,
1.3948287010656306,
0.12274721485756479,
1.4572577859766789,
0.6279493492392955,
1.3018581073498696,
1.4856691921094258,
0.7199280321280211,
0.06861566603887055,
1.5494687184965839,
1.5976087131016672,
2.3502768013298017,
1.5631880040310109]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#3'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],

```

```

[2, 2, 0, 0],
[3, 3, 0, 0]],
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
        'optimization': { 'algorithm': 'SLSQP',
                          'constraint': True,
                          'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.517540187131846],
                          [12, 6, 0, 0, -5.995430553311077],
                          [20, 10, 0, 0, -4.811503569399131],
                          [28, 14, 0, 0, -3.882991739348091]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 3.695487976074219e-05,
            'zne_values': { 'degree': 0,
                          'extrapolated_value': -8.431278340273012,
                          'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                          'cost_richardson_zne': 11.390625,
                          'sorted_expectation_vals': [
-7.517540187131846,
-5.995430553311077,
-4.811503569399131,
-3.882991739348091],
'sorted_noise': [6, 18, 30, 42]],
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.517540187131846],
                  [ 12,
                    6,
                    0,
                    0,
                    -5.995430553311077],
                  [ 20,
                    10,
                    0,
                    0,
                    -4.811503569399131],
                  [ 28,
                    14,
                    0,
                    0,
                    -3.882991739348091]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
                        'layer': 30,
                        'ugate': { 'coefficients': { 'bn': [ 0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0,
                                                            0],
                                                    'cn': [ 0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5,
                                                            0.5],
                                                    'r': 0},
                        'time': {'max': 10.0, 'min': 0.0},
                        'type': 'xy-iss'}},
'init_param': { 'value': [ 1.0344655484192336e-17,
                            0.36119418156962224,
                            0.3611941815696223,
                            0.36119418156962224,
                            0.8440759344131337,
                            1.1727045787919286,

```

1.5692279974524073,
1.570888864611923,
1.592214192172409,
1.7598342563277258,
3.059056443368036,
3.0591149843524144,
4.3676699123871625,
4.3676699123871625,
5.510828301667605,
5.51914482396091,
6.113847931320903,
6.113847931320903,
6.533178927747682,
6.533178927747682,
6.755940863606281,
6.837142696961756,
7.694830693654466,
7.694830693654467,
7.694830693654466,
7.873525558910739,
7.873525558910739,
7.922034998277628,
7.922034998277628,
12.310276979046629,
-0.6972922025249428,
-0.0508578639309505,
0.03511585194544882,
-0.05290285417260422,
0.6885525491076263,
0.23164759975262209,
0.06930481892124933,
-0.05051704727786026,
0.2745998381310896,
2.8540854853592244,
0.1762956340846527,
-0.46176266603345323,
0.4487337371763968,
1.96119949254268,
-0.028430732456242037,
1.020183549895116,
0.09052188212177692,
0.6283703459809612,
-0.2008669437139055,
-0.333213682394554,
0.49298533108167947,
-0.073860646808665,
-0.13593481376514907,
-0.597217576988091,
0.38422303935103047,
-0.1405570093523461,
-0.17552659439894247,
0.802391437181995,
0.41745069672968155,
-0.3651927864609047,
0.40673923582512855,
1.8402265216806437,
0.0347958050087225,
1.3731019473127588,
-0.031644945604438944,
0.07785925108599989,
-0.44699302887370956,
0.3657440768540896,
0.10260821539336823,
-0.7165979369194562,
0.5076396918302244,
0.07362014039826455,
-0.1816691032245074,
-1.2219405377945243,
0.30583520143801574,
0.23528029219292396,
1.4689187491327858,
0.06809628133224771,
-0.766861690837204,
0.012120501277695121,
-1.045003198738065,

```
0.2314394697747134,  
-0.6381970494622153,  
-0.869070575517897,  
-0.3854227231439233,  
-0.05974868627606693,  
-1.3618983878421893,  
-0.14735213954174853,  
-0.16144336826377106,  
-0.5726725646368419,  
0.14863429745149237,  
-0.7763964261749154,  
-0.3515427279763936,  
0.09267593493455233,  
0.9941259428822948,  
0.6113266248291309,  
-2.249730610329486,  
-1.4853322232412747,  
-0.6624750796271123,  
1.5218612703215841,  
-0.060867877516889796,  
0.4881957659663681,  
-0.5823031021435859,  
0.01548453122295137,  
-0.8946631440087691,  
0.2724176161972246,  
1.1142419366260101,  
1.1351916054213225,  
0.24996433759527287,  
0.10925434181359692,  
0.35878338934390347,  
-0.4145708548690295,  
-1.0045262071213437,  
-1.2832451981705593,  
1.5127219636109106,  
1.0543608408458283,  
-0.025449240007986165,  
-0.4042653654410071,  
-0.5770570175675142,  
-0.3798602211137218,  
-0.20337950006519365,  
1.1741528813917048,  
0.4416785611413231,  
1.2685661107772639,  
-0.29162960462044624,  
0.30563636100934116,  
-0.34543251279098935,  
-1.452238106816157,  
-0.35828114927859783,  
-1.6519696012854037,  
0.9758818868953008,  
1.8739098589464835,  
0.08690130739368318,  
-0.0063915911623125795,  
0.058888399162031244,  
-0.2901105978964291,  
-0.0016858525368986894,  
0.021900786471813132,  
1.743664682334748,  
1.2798318789150203,  
-1.0536291022271305,  
0.3537083031530898,  
-1.4149627812472323,  
1.0773219987831761,  
0.29151315320582294,  
-0.0298469457137327,  
2.156594522363883,  
1.5651984449575558,  
0.9412318447665341,  
1.574999269983593}},  
'noise_profile': { 'noise_on_init_param': { 'status': False,  
                                              'value': 0},  
                    'noise_prob': [0.001, 0.001, 0.0, 0.0],  
                    'status': True,  
                    'type': 'dephasing'},  
'nqubits': 7,
```

```

        'observable': { 'coefficients': { 'bn': [ 1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0,
                                                    1.0],
                                            'cn': [ 0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5,
                                                    0.5],
                                            'r': 1},
                        'def': 'ising'},
        'output': { 'draw': { 'fig_dpi': 100,
                                'status': False,
                                'type': 'png'},
                    'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#5'},
        'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                              [1, 1, 0, 0],
                                              [2, 2, 0, 0],
                                              [3, 3, 0, 0]]},
        'run': 'zne',
        'state': 'dmatrix',
        'vqe': { 'iteration': 1,
                  'optimization': { 'algorithm': 'SLSQP',
                                    'constraint': True,
                                    'status': False}},
        'zne': { 'data_points': [ [4, 2, 0, 0, -7.524810336842993],
                                   [12, 6, 0, 0, -6.145141411469215],
                                   [20, 10, 0, 0, -5.055735320587114],
                                   [28, 14, 0, 0, -4.18905102035353]],
                  'degree': 0,
                  'method': 'richardson',
                  'sampling': 'default'}},
        'output': { 'exact_sol': -8.566772233505624,
                    'run_time_sec': 4.410743713378906e-05,
                    'zne_values': { 'degree': 0,
                                    'extrapolated_value': -8.34459993866525,
                                    'others': { 'beta_coefficients': [ 2.1875,
                                                                        -2.1875,
                                                                        1.3125,
                                                                        -0.3125],
                                                'cost_richardson_zne': 11.390625,
                                                'sorted_expectation_vals': [
-7.524810336842993,
-6.145141411469215,
-5.055735320587114,
-4.18905102035353],

                                'sorted_noise': [6, 18, 30, 42]},
                                    'sampled data': [ [ 4,
                                                         2,
                                                         0,
                                                         0,
                                                         -7.524810336842993],
                                                         [ 12,
                                                          6,
                                                          0,
                                                          0,
                                                          -6.145141411469215],
                                                         [ 20,
                                                          10,
                                                          0,
                                                          0,
                                                          -5.055735320587114],
                                                         [ 28,
                                                          14,
                                                          0,
                                                          0,
                                                          -4.18905102035353]],
                                'sampling': 'default'}}},
        { 'config': { 'ansatz': { 'gateset': 1,
                                    'layer': 30,

```



```
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.1021834784589592,
0.9196384424899031,
0.9261694014921312,
0.983882928321485,
1.8913212439431486,
1.9426406276748993,
1.9426406276748993,
1.9426406276748993,
2.7354895381565307,
2.7475159984062567,
4.89472071800884,
4.89472071800884,
4.89472071800884,
4.951228136281907,
4.9774006951115535,
4.9774006951115535,
5.053403895048273,
5.053403895048273,
5.101252532365191,
7.988961801418946,
7.988961801418945,
7.988961801418946,
7.988961801418946,
8.978507898154866,
8.993364745936525,
9.010251752506093,
9.010251752506093,
9.808175975385067,
9.808175975385067,
10.987620898296862,
-0.17474125502348634,
0.15377532976778396,
0.10530559022128577,
-0.002734731613738592,
0.18566397098861895,
-0.21248706612806187,
0.00824191786492345,
-0.17814151074061357,
0.01796252087548974,
0.34324522632459664,
0.0074801226415391366,
0.7599717323032058,
-0.10515906185801696,
1.0828859909757265,
0.414437884618902,
1.5228608411556923,
0.33559085686809753,
0.015737214192853767,
-0.5136276162084019,
-1.8563551585814209,
0.5712467952174463,
0.8929866894843974,
0.13811028968068004,
-0.2763065211700681,
-0.7071702673229635,
-0.18135433485295524,
-0.27765639095770533,
0.7217433560748705,
-0.8828879013995732,
```

-2.0374813859941323,
1.36473854657663,
0.034109249352092696,
-0.6149254344617949,
0.03372845053867827,
0.503938133787832,
-0.6883442402732416,
0.2648421898076424,
0.6333527739880076,
-0.04525869325144416,
-0.005623930623010625,
1.128020889356044,
-1.3504741675340033,
0.4563972098057263,
0.05375760079980123,
-0.21911598163890048,
-0.06488839113013266,
-0.01181730848543227,
-0.41291804178803715,
-0.5389032519971739,
-0.479835182738992,
-0.19788219223707681,
0.9600571509635428,
-0.0030559949283493204,
1.5758921704264812,
-0.03282558866751487,
-1.7314093119971763,
-0.6690974950145957,
1.411191021979108,
0.7413019294926171,
-0.9351691242609317,
0.7225594125732181,
-0.24544096955774433,
-0.2757214458340262,
-0.1293511632733071,
-1.1157165888503884,
1.77231166364723,
1.0062423711696755,
-1.1433484209574243,
0.3276730678194393,
0.1915280099572572,
-0.011019641383667301,
0.5201943477386407,
-0.20687953385151048,
0.5111065633642609,
0.4748895808260117,
1.6382441107389658,
-0.28105766356439466,
1.4035885688096705,
-1.3528862990074717,
-0.05065799754978554,
-0.13653460387745367,
0.29789868618191495,
-0.0840154743722884,
0.11164933802065877,
-1.428254211684401,
1.6939675422871827,
-2.3120430894530672,
-0.39217472669836195,
-0.03433336202746711,
0.3550969915171978,
-0.5536295186541139,
-0.1651780940587393,
-1.566346421602003,
0.08989465754235938,
0.32012869379205766,
0.8313639447915222,
0.07625815825831544,
2.49413024893377,
0.7216117355438889,
0.017072826297347257,
0.27952417304038707,
-0.017587044505758563,
0.3405641489362918,
0.15232410125752438,

```

2.03737481794317,
-0.5582524700932,
0.9121080021707879,
-0.07173216146529754,
-1.3421014045147113,
0.028627573515451674,
1.9279053503307635,
0.06198165932557796,
-1.6072305184801001,
1.1842841983660186,
-0.33449540725075716,
-0.012221452614921814,
-1.0370098958383984,
1.563143862815049,
1.531218321029045,
1.5872676744332948]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#10'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.588052513021028],
                          [12, 6, 0, 0, -6.143369929824418],
                          [20, 10, 0, 0, -4.998571859003288],
                          [28, 14, 0, 0, -4.085121357543176]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 9.036064147949219e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.444268291452158,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-7.588052513021028,
-6.143369929824418,
-4.998571859003288,
-4.085121357543176],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,

```

```

2,
0,
0,
-7.588052513021028],
[ 12,
6,
0,
0,
-6.143369929824418],
[ 20,
10,
0,
0,
-4.998571859003288],
[ 28,
14,
0,
0,
-4.085121357543176]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}}},
'init_param': { 'value': [ 0.009858867127598406,
0.6371504690778502,
1.0666846377193393,
1.104415373659906,
1.464838116920491,
1.4648381169204907,
1.9080243917751187,
2.1981829040528402,
2.6297267537087063,
2.658272884291024,
3.752650276988564,
3.752985814574736,
4.407595284592648,
4.407595284592648,
5.031509128956631,
5.031509128956631,
5.031509128956631,
5.033233798634394,
5.033233798634395,
5.036830700020435,
5.036830700020434,
7.933675292188165,
7.950623161195741,
7.950659720589115,
8.023568123130133,
8.309873389450049,
8.310972029308223,
8.959490059358307,
9.904746541347446,
9.949991834284726,
-0.41951749885307754,
0.029035276468521477,
0.030870805732646677,
-0.17295615718073906,
0.5816925184728065,
0.9382868815479958,
0.3165320552518337,

```

0.42890233892866725,
0.013440093351710147,
1.2501583763834956,
-0.35093579496718846,
1.139236973962438,
-0.013100397435237714,
-0.1455411128269196,
0.09889932710883725,
-0.821320047619297,
-0.4558166636467179,
1.1031921527651727,
-2.1974313327571964,
1.4792834343123176,
-0.30822227350872,
0.588013825131612,
0.515130316597666,
0.22678729305502876,
0.10858837903610288,
0.5318167388137908,
0.8532518222915644,
-0.6728171258281335,
0.6989815794052712,
0.1656152058660693,
2.2712111306791654,
-0.15051151312079816,
0.7977470630871717,
-0.1251622231807751,
-1.8891635494392358,
0.051451390647974285,
-0.7348344848883321,
0.7395913896376409,
-0.06856262751164309,
-0.17117407536263385,
-0.18593999781702203,
0.10811754652223245,
0.4196541060235856,
-0.05456992109076927,
-0.342869030991957,
2.1674650608531283,
-0.5070172257051728,
-0.0658073986126462,
-0.5652644350797837,
-0.04661359801412543,
-0.8211476277064503,
0.2688028104686384,
-0.12556704820535647,
-0.023840075766175,
-0.18959105165277507,
-0.026652605847716424,
-0.0368848436080881,
-0.17033806404624055,
-0.24172965402850155,
1.8740805145949957,
3.3954947367399346,
0.685069809753845,
0.18877298628713768,
-0.8116148922898279,
0.3642216955021626,
0.776870193177977,
-0.2915626595335885,
0.48528749335602656,
0.3052797687185898,
0.8094152496869139,
-0.8933636925729215,
-0.927850020272857,
0.26040310486864426,
-0.02621475103327714,
-0.3044278549592545,
-0.45445142071133066,
-0.660404261852709,
1.1611135994497148,
-0.08594794341260105,
0.7467695488502767,
-0.5557725464437727,
-0.16927866426422344,

```

-0.043602634805518736,
0.9335958048537285,
-0.6399661361094481,
0.23806458661693586,
1.596804769065565,
0.05733925195491384,
0.7574169294466017,
0.040475318628828375,
-0.8594601463323506,
-0.09619413741349449,
-0.21197930687835795,
0.2757964313499001,
0.06286971719145724,
1.1856485993065717,
-0.08746041433688896,
-0.3598130921460406,
-0.15364860008047934,
1.2858349919806815,
0.18001757541551167,
-0.5562723277178536,
0.3666171831013428,
1.6125643575569677,
-1.3673007848855094,
0.09427109776859108,
0.5265559951107008,
-1.7052545510096164,
2.291596524870159,
0.046069640183881794,
0.6559544004774626,
-0.5991904786957847,
-0.6867617223561892,
-0.1637234892612084,
0.30313993491542573,
1.548639749817063,
-0.9376256579293304,
1.0052059218812999,
0.9493196821469583,
0.13892828355847012]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#6'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.407013825547437],

```

-7.407013825547437,
-6.114813208156221,
-5.082084674797013,
-4.249248629692988],

-7.407013825547437,
-6.114813208156221,
-5.082084674797013,
-4.249248629692988],

4.283774948106428,
4.283774948106429,
4.283774948106429,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
5.706222745811317,
6.386180553174607,
6.421245480123581,
8.31501274226813,
8.345179306265157,
11.444382336838434,
0.4142599738840607,
-0.0179112834331289,
-1.916401049394031,
0.14120666199524137,
0.7700670528228858,
0.4673833851581437,
-1.0389421964094037,
0.4744345335688107,
-0.02892536105987987,
0.7177135899887972,
0.03424375267071561,
-0.8966767104461824,
-0.2754786671261514,
-0.38073768598802404,
-0.05488081631876704,
1.3786960897007534,
0.49169988282833965,
-0.0017740881267394302,
-0.0730349088569547,
1.022071163826922,
0.35709372961339875,
-0.25522274095888026,
0.1780496791518154,
1.1037611427142897,
0.41779504446715204,
0.5262749279658233,
0.06490819244365402,
-1.4552807970719621,
-0.3245263390561476,
1.0554193966046508,
0.022339750983574838,
-0.3669003487168035,
2.663679114021979,
0.10849122591523966,
-0.13414192207644796,
-1.5118683308835437,
-0.30825243618478626,
0.9695383135496236,
0.257959037696186,
0.15694512581472056,
-0.16358187739115868,
0.34099750217619934,
-0.5402166992961691,
0.3593532165312357,
-1.364276002655152,
-0.99924307141965,
-0.3752500200552189,
-0.0018215261709029979,
-0.45075489021511456,
-0.04208315502591536,
0.8249107630283551,
-0.05400603050886625,
-0.5392190802384772,
-0.8386187523158375,
-0.6091096268131945,
0.024668710490207954,
0.05602431119852906,
0.02580658599429192,
-0.5620121364690127,
0.9463151560365789,


```

-1.359537213829514,
0.16416256013395814,
-0.7008538515531048,
0.10281262767437246,
0.16226030695338217,
0.1045845472418919,
-3.460449108622277,
-1.3581056924942707,
-0.04860072573676619,
0.8583304742037813,
0.32574507917022166,
0.12486068474919,
0.3848484786860077,
0.13263416178943582,
-2.0565442214631187,
-0.5561084451963657,
0.6933115579235511,
-0.7248122710788838,
0.47336028605018116,
1.303902278490238,
0.0901454631126506,
-0.7336636885785707,
-0.10464603275499512,
0.5790815509668216,
-0.11216362879580807,
0.39567579154864924,
2.758332677620952,
0.04270654862566299,
-0.1724323823943382,
0.45437353083833626,
-0.3336534023367508,
0.7720982719279817,
0.571475106459852,
1.1706470777141371,
-0.35007907187953646,
-0.39678725288598204,
-0.45268800610896537,
0.3874330317234988,
-0.39501247513147103,
-0.615405673461264,
2.9358919100517546,
0.02513265040947631,
-0.7687653885525301,
0.11593850937998196,
-0.21520127573910922,
0.8374533643808371,
0.9843281584680158,
-0.006536614273180726,
0.6186165381537709,
0.02480411410111467,
-1.8247765182338995,
2.1817137165821427,
-0.06885103221382978,
-0.6211668118311473,
-1.5784593997581706,
-0.0038216687406707125,
1.435433592411287,
1.5778673594641464,
2.169764508045927,
-1.5507143059542559]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                'cn': [ 0.5,
                                         0.5,

```

```

0.5,
0.5,
0.5,
0.5],
'r': 1},
'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
'status': False,
'type': 'png'},
'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#1'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
[1, 1, 0, 0],
[2, 2, 0, 0],
[3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
'optimization': { 'algorithm': 'SLSQP',
'constraint': True,
'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.597273562260126],
[12, 6, 0, 0, -6.262099522076846],
[20, 10, 0, 0, -5.200592923966883],
[28, 14, 0, 0, -4.350606799333773]],
'degree': 0,
'method': 'richardson',
'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
'run_time_sec': 4.4345855712890625e-05,
'zne_values': { 'degree': 0,
'extrapolated_value': -8.386906800815655,
'others': { 'beta_coeffieicients': [ 2.1875,
-2.1875,
1.3125,
-0.3125],
'cost_richardson_zne': 11.390625,
'sorted_expectation_vals': [
-7.597273562260126,
-6.262099522076846,
-5.200592923966883,
-4.350606799333773],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
2,
0,
0,
-7.597273562260126],
[ 12,
6,
0,
0,
-6.262099522076846],
[ 20,
10,
0,
0,
-5.200592923966883],
[ 28,
14,
0,
0,
-4.350606799333773]],
'sampling': 'default'}}},
{ 'config': { 'ansatz': { 'gateset': 1,
'layer': 30,
'ugate': { 'coefficients': { 'bn': [ 0,
0,
0,
0,
0,
0],
'cn': [ 0.5,
0.5,
0.5,
0.5,
0.5,
0.5]

```

```
0.5,
0.5,
0.5,
0.5],
'r': 0},
'time': {'max': 10.0, 'min': 0.0},
'type': 'xy-iss'}},
'init_param': { 'value': [ 0.029287487040924362,
0.6764270956551205,
1.6620807976810363,
1.7056653351603042,
1.8485187197716686,
1.9397654142515774,
1.9426163313124116,
1.9760071204347747,
2.0652508647096677,
3.338212066261205,
3.3382549495561102,
3.507308503786301,
3.668597772988933,
3.6686655840136484,
3.7105300968471533,
3.737148430160667,
5.361211694679849,
5.40430867517309,
5.45325014669751,
5.461795374069968,
5.465563084002408,
5.492346359399902,
7.18245326721486,
7.245488992311329,
7.248698529808956,
8.18726919118503,
8.197326336426727,
8.213328813441153,
8.251719559085661,
10.791930409527371,
-0.48557603875244904,
-0.06546231446775812,
-1.4810159853023965,
-0.6439509610243448,
-0.4889611776104492,
0.10983589949376514,
-1.2155949975003408,
1.1338047187824702,
0.18557769650103798,
-0.36145572824608646,
-0.7821715666301856,
0.8512861656756193,
0.30536456746903484,
-0.30807040011779185,
0.12092796067845869,
-0.24545333188928314,
-0.129976073158145,
0.023137142660462955,
-0.1650766547725121,
1.2066757915428714,
0.30346492466779185,
-0.39001235903392717,
0.22721995462271521,
0.5382230474507628,
0.00019763985679924906,
0.06828428498679436,
-0.20088260327468863,
-0.4092889734740989,
0.03350490310414833,
0.2806734861806061,
0.1113395749536363,
0.230636273744993,
0.8251946739675019,
1.1608144596385723,
0.6723758755359734,
-0.10317602477852598,
-0.4420709064762177,
-1.0960363173936596,
```

-0.9468941300070383,
0.5033961390519779,
-0.2243236772482122,
0.5892197397332068,
0.05991690407088531,
0.701285681572503,
0.08910649730701763,
0.8460679218665452,
1.1623340763406136,
1.0865973894420775,
0.5351551478532627,
0.9235153288380824,
-0.5582651446845596,
-0.34222728577250333,
1.5143962112065452,
0.7564033664211318,
0.8123992700678001,
-0.9664797348096636,
0.27401672810110755,
-0.18494852404663462,
-0.9910946352688285,
0.05182985166611149,
0.5660348103142446,
0.4119220568486742,
-0.2551932204664784,
-0.7776730862016209,
0.34610880550840434,
0.7576760112497428,
-0.8577033748778926,
-1.4968938568032009,
0.5269521491450873,
0.2373061672114889,
0.7288903206551578,
0.1710119085151455,
-0.176136081545097,
-0.5935327549888737,
-0.1515533123757876,
-0.6771002165072852,
0.4717499430890327,
-0.6420368402390211,
-0.5643843520072841,
-0.11277697804122554,
-1.464860995022597,
0.2046291054044903,
-0.7612462049107216,
1.0068778461917443,
-0.6195821495862486,
0.7218912637774252,
-1.3562632264268242,
0.7211536601201256,
0.5104120961355565,
-0.7864014822434545,
-0.20491893468164094,
1.4874102287162618,
1.08746640867374,
-1.142973595398691,
-2.0155867565934926,
-0.750509876349441,
-2.0230361256060485,
1.5097130841243176,
0.9880269111953318,
-1.1665664330230259,
0.7519837029078653,
1.3761701698303868,
0.8852780263768104,
0.16438414494284956,
-0.10719774898963329,
1.5136856555155305,
0.6604135539088288,
1.1059611188120957,
1.025640803866066,
1.5008045081548957,
0.12777319216570615,
0.2672167471405113,
-0.24284995686166302,

```

0.02895262837171797,
-0.0032155077919554156,
0.7332520346543212,
1.651331605401529,
1.592752189260533,
-0.10527970333315,
1.5617475952289623]],
'noise_profile': { 'noise_on_init_param': { 'status': False,
                                             'value': 0},
                   'noise_prob': [0.001, 0.001, 0.0, 0.0],
                   'status': True,
                   'type': 'dephasing'},
'nqubits': 7,
'observable': { 'coefficients': { 'bn': [ 1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0,
                                           1.0],
                                   'cn': [ 0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5,
                                           0.5],
                                   'r': 1},
                'def': 'ising'},
'output': { 'draw': { 'fig_dpi': 100,
                      'status': False,
                      'type': 'png'},
            'file_name_prefix': 'AUTOMATE_xy-
iss_noisefree_time_evo_dephasing_ric4_sample#8'},
'redundant': { 'identity_factors': [ [0, 0, 0, 0],
                                     [1, 1, 0, 0],
                                     [2, 2, 0, 0],
                                     [3, 3, 0, 0]]},
'run': 'zne',
'state': 'dmatrix',
'vqe': { 'iteration': 1,
         'optimization': { 'algorithm': 'SLSQP',
                           'constraint': True,
                           'status': False}},
'zne': { 'data_points': [ [4, 2, 0, 0, -7.561979391804815],
                          [12, 6, 0, 0, -6.098488474706268],
                          [20, 10, 0, 0, -4.939317874668222],
                          [28, 14, 0, 0, -4.015452245150805]],
        'degree': 0,
        'method': 'richardson',
        'sampling': 'default'}},
'output': { 'exact_sol': -8.566772233505624,
            'run_time_sec': 6.008148193359375e-05,
            'zne_values': { 'degree': 0,
                           'extrapolated_value': -8.429412265045487,
                           'others': { 'beta_coeffieicients': [ 2.1875,
                                                                -2.1875,
                                                                1.3125,
                                                                -0.3125],
                                       'cost_richardson_zne': 11.390625,
                                       'sorted_expectation_vals': [
-7.561979391804815,
-6.098488474706268,
-4.939317874668222,
-4.015452245150805],
'sorted_noise': [6, 18, 30, 42]},
'sampled data': [ [ 4,
                    2,
                    0,
                    0,
                    -7.561979391804815],
                  [ 12,
                    6,
                    0,
                    0,
                    -6.098488474706268],

```

```
[ 20,  
 10,  
 0,  
 0,  
 -4.939317874668222],  
[ 28,  
 14,  
 0,  
 0,  
 -4.015452245150805]],  
'sampling': 'default'[][][]]
```